



WORK ORDER NO. 1783264

CONTRACT NO.

## **P1360 Joint Maritime Facility**

At the  
**VARLOCS**

Final Design Submittal - Volume 5 of 5

PREPARED BY:

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Submitted By:

Date:

APPROVED BY:

Specifications:

For Commander, NAVFAC:  
Date: 01/28/2026

*Wm B Forbes*

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document  
Date: 2026.01.28  
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PROJECT NUMBER: 1838720

CONTRACT NUMBER:

TASK ORDER NUMBER:

**P1360 JOINT MAIRITIME FACILITY  
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND  
ATLANTIC**

HAVING PARTICIPATED IN THE DESIGN OF PROJECT NO. 1838720, P1360 JOINT MARITIME FACILITY AND HAVING THOROUGHLY REVIEWED THE COMPLETED PROJECT DOCUMENTS, I DECLARE THAT THE FACILITY DESIGN INCLUDED HEREIN COMPLIES WITH ALL APPLICABLE CANADIAN CODES AND LAWS.

DATE

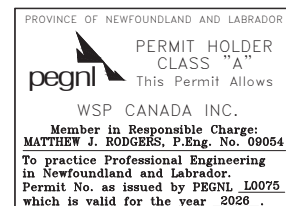
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(PROFESSIONAL SEAL)  
REGISTERED ENGINEER

January 23, 2026

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|          |       |                          |
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| 23 05 93       | 05/25               | TESTING, ADJUSTING, AND BALANCING FOR<br>HVAC                                |
| 23 07 00       | 08/24               | THERMAL INSULATION FOR MECHANICAL<br>SYSTEMS                                 |
| 23 08 00       | 05/23, CHG 1: 08/24 | COMMISSIONING OF MECHANICAL AND<br>PLUMBING SYSTEMS                          |
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| 23 09 23.02    | 08/24               | BACNET DIRECT DIGITAL CONTROL FOR HVAC<br>AND OTHER BUILDING CONTROL SYSTEMS |
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| 23 30 00       | 02/25               | HVAC AIR DISTRIBUTION                                                        |
| 23 64 10       | 05/25               | WATER CHILLERS, VAPOR COMPRESSION TYPE                                       |
| 23 64 26       | 08/09, CHG 7: 02/24 | CHILLED, CHILLED-HOT, AND CONDENSER<br>WATER PIPING SYSTEMS                  |
| 23 81 00       | 05/24               | DECENTRALIZED UNITARY HVAC EQUIPMENT                                         |
| 23 81 23       | 11/20               | COMPUTER ROOM AIR CONDITIONING UNITS                                         |
| 23 83 00.00 20 | 04/06               | ELECTRIC SPACE HEATING EQUIPMENT                                             |

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| 25 05 11.02    | 08/24               | CYBERSECURITY FOR LOW IMPACT EMERGENCY<br>AND BACK-UP ELECTRICAL POWER SYSTEMS |
| 25 05 11.03    | 08/24               | CYBERSECURITY FOR LOW IMPACT<br>ELECTRICITY METERING SYSTEM                    |
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| 26 12 19       | 05/24               | PAD-MOUNTED, LIQUID-FILLED, MEDIUM-VOLTAGE TRANSFORMERS      |
| 26 20 00       | 08/23               | INTERIOR DISTRIBUTION SYSTEM                                 |
| 26 24 13       | 08/21               | SWITCHBOARDS                                                 |
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| 26 32 15       | 05/24               | ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES |
| 26 33 53       | 05/19               | STATIC UNINTERRUPTIBLE POWER SUPPLY (UPS)                    |
| 26 36 23       | 05/20, CHG 1: 08/21 | AUTOMATIC TRANSFER AND STATIC TRANSFER SWITCHES              |
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|----------|-------|--------------------------------------------|

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| 32 11 23    | 05/22               | AGGREGATE BASE COURSE FOR FLEXIBLE AND RIGID PAVING             |
| 32 12 16.16 | 11/20               | ROAD-MIX ASPHALT PAVING                                         |
| 32 13 13.06 | 05/20               | PORTLAND CEMENT CONCRETE PAVEMENT FOR ROADS AND SITE FACILITIES |
| 32 16 19    | 05/18               | CONCRETE CURBS, GUTTERS AND SIDEWALKS                           |
| 32 17 23    | 08/16, CHG 5: 11/18 | PAVEMENT MARKINGS                                               |
| 32 31 13.53 | 11/21               | HIGH-SECURITY FENCES (CHAIN LINK AND ORNAMENTAL) AND GATES      |
| 32 92 19    | 08/17, CHG 1: 08/21 | SEEDING                                                         |

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| 33 08 53    | 08/18 | AVIATION FUEL DISTRIBUTION SYSTEM START-UP              |
| 33 09 52    | 11/18 | FUEL PUMP CONTROL AND ANNUNCIATION SYSTEM (NON-HYDRANT) |

|             |                     |                                        |
|-------------|---------------------|----------------------------------------|
| 33 11 00    | 02/18, CHG 2: 11/23 | WATER UTILITY DISTRIBUTION PIPING      |
| 33 30 00    | 05/18               | SANITARY SEWERAGE                      |
| 33 40 00    | 11/21               | STORMWATER UTILITIES                   |
| 33 46 13    | 05/20               | FOUNDATION DRAINAGE                    |
| 33 52 10    | 11/18, CHG 1: 11/20 | FUEL SYSTEMS PIPING (SERVICE STATION)  |
| 33 52 23.15 | 05/25               | POL SERVICE PIPING WELDING             |
| 33 56 10    | 05/25               | FACTORY-FABRICATED FUEL STORAGE TANKS  |
| 33 57 55    | 11/18, CHG 1: 11/20 | FUEL SYSTEM COMPONENTS (NON-HYDRANT)   |
| 33 71 01    | 05/19, CHG 1: 11/19 | OVERHEAD TRANSMISSION AND DISTRIBUTION |
| 33 71 02    | 08/21               | UNDERGROUND ELECTRICAL DISTRIBUTION    |
| 33 82 00    | 04/06               | TELECOMMUNICATIONS OUTSIDE PLANT (OSP) |

-- End of Project Table of Contents --

## SECTION 32 17 23

PAVEMENT MARKINGS  
08/16, CHG 5: 11/18

## PART 1 GENERAL

## 1.1 UNIT PRICES

## 1.1.1 Measurement

## 1.1.1.1 Surface Preparation

The unit of measurement for surface preparation (cleaning) is the number of square meters of pavement surface prepared for marking and accepted by the Contracting Officer.

## 1.1.1.2 Pavement Striping and Markings

The unit of measurement for pavement markings is the number of square meters of reflective and/or nonreflective striping or markings actually completed and accepted by the Contracting Officer.

## 1.1.1.3 Raised Pavement Markers

The unit of measurement for raised pavement markers is the number actually placed as specified and approved by the Contracting Officer.

## 1.1.1.4 Removal of Pavement Markings on Roads and Automotive Parking Areas

The unit of measurement for removal of pavement markings is the number of square meters of pavement markings removed as specified and accepted by the Contracting Officer.

## 1.1.2 Payment

The quantities of surface preparation, pavement striping or markings, raised pavement markers, and removal of pavement markings determined as specified in paragraph Measurement, will be paid for at the contract unit price. The payment constitutes full compensation for furnishing all labor, materials, tools, equipment, appliances, and doing all work involved in preparing and marking the pavements as shown on the drawings. Remove and replace any striping or markings which required reflective media, but are placed without it, do not meet the stated minimum retro-reflective requirements, or with other defects, at no cost to the Government. Remove and replace striping or markings which do not conform to the required physical characteristics, alignment or location required at no cost to the Government.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

Canadian General Standards Board (CGSB)

CGSB 1-GP-74M

(1979) Standard for: Paint, Traffic, Alkyd



AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| AASHTO M 247 | (2013) Standard Specification for Glass Beads Used in Pavement Markings                                           |
| AASHTO M 249 | (2012; R2016) Standard Specification for White and Yellow Reflective Thermoplastic Striping Material (Solid Form) |

## ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D471       | (2016a) Standard Test Method for Rubber Property - Effect of Liquids                                                                         |
| ASTM D476       | (2015) Dry Pigmentary Titanium Dioxide Pigments                                                                                              |
| ASTM D522/D522M | (2017) Mandrel Bend Test of Attached Organic Coatings                                                                                        |
| ASTM D638       | (2014) Standard Test Method for Tensile Properties of Plastics                                                                               |
| ASTM D695       | (2010) Standard Test Method for Compressive Properties of Rigid Plastics                                                                     |
| ASTM D711       | (2010; R 2015) No-Pick-Up Time of Traffic Paint                                                                                              |
| ASTM D823       | (2018) Standard Practices for Producing Films of Uniform Thickness of Paint, Coatings, and Related Products on Test Panels                   |
| ASTM D1652      | (2011; E 2012) Standard Test Method for Epoxy Content of Epoxy Resins                                                                        |
| ASTM D2074      | (2007; R2013) Standard Test Methods for Total, Primary, Secondary, and Tertiary Amine Values of Fatty Amines by Alternative Indicator Method |
| ASTM D2240      | (2015; R 2021) Standard Test Method for Rubber Property - Durometer Hardness                                                                 |
| ASTM D2621      | (1987; R 2016) Standard Test Method for Infrared Identification of Vehicle Solids from Solvent-Reducible Paints                              |
| ASTM D2697      | (2003; R 2014) Volume Nonvolatile Matter in Clear or Pigmented Coatings                                                                      |
| ASTM D3335      | (1985a; R 2020) Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy                                   |
| ASTM D3718      | (1985a; R 2015) Low Concentrations of Chromium in Paint by Atomic Absorption                                                                 |

## Spectroscopy

|            |                                                                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D3924 | (2016) Standard Specification for Environment for Conditioning and Testing Paint, Varnish, Lacquer, and Related Materials                                                |
| ASTM D3960 | (2005; R 2013) Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings                                                                        |
| ASTM D4060 | (2019) Abrasion Resistance of Organic Coatings by the Taber Abraser                                                                                                      |
| ASTM D4061 | (2013) Standard Test Method for Retroreflectance of Horizontal Coatings                                                                                                  |
| ASTM D4383 | (2012) Standard Specification for Plowable, Raised Retroreflective Pavement Markers                                                                                      |
| ASTM D4505 | (2012; R 2017) Standard Specification for Preformed Retroreflective Pavement Marking Tape for Extended Service Life                                                      |
| ASTM D4541 | (2022) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers                                                                            |
| ASTM D6628 | (2003; R 2015) Standard Specification for Color of Pavement Marking Materials                                                                                            |
| ASTM D7234 | (2012) Standard Test Method for Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers                                              |
| ASTM E1347 | (2006; R 2020) Standard Test Method for Color and Color Difference Measurement by Tristimulus (Filter) Colorimetry                                                       |
| ASTM E1710 | (2011) Standard Test Method for Measurement of Retroreflective Pavement Marking Materials with CEN-Prescribed Geometry Using a Portable Retroreflectometer               |
| ASTM E2177 | (2011) Standard Test Method for Measuring the Coefficient of Retroreflected Luminance (RL) of Pavement Markings in a Standard Condition of Wetness                       |
| ASTM E2302 | (2003; R 2016) Standard Test Method for Measurement of the Luminance Coefficient Under Diffuse Illumination of Pavement Marking Materials Using a Portable Reflectometer |
| ASTM G154  | (2023) Standard Practice for Operating Fluorescent Ultraviolet (UV) Lamp                                                                                                 |

## Apparatus for Exposure of Materials

## INTERNATIONAL CONCRETE REPAIR INSTITUTE (ICRI)

ICRI 03732 (1997) Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, and Polymer Overlays

## MASTER PAINTERS INSTITUTE (MPI)

MPI 32 (2012) Traffic Marking Paint, S.B.

MPI 97 (2012) Traffic Marking Paint, Latex

## SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS-STD-595A (2017) Colors used in Government Procurement

## U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 150/5370-10 (2018; Rev H; Errata 1 2019) Standard Specifications for Construction of Airports

## U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2009; Rev 2012) Manual on Uniform Traffic Control Devices

## U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS TT-B-1325 (Rev D; Notice 1; Notice 2 2017) Beads (Glass Spheres) Retro-Reflective (Metric)

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-03 Product Data

Surface Preparation Equipment List; G

Application Equipment List; G

Exterior Surface Preparation

Safety Data Sheets; G

Reflective media for airfields; G

Reflective media for roads; G

Waterborne Paint; G

Solventborne Paint; G

Thermoplastic compound; G

Raised Pavement Markers Primers and Adhesives; G

#### SD-06 Test Reports

Reflective Media for Airfields; G

Reflective Media for Roads; G

Waterborne Paint; G

Solventborne Paint; G

High Build Acrylic Coating (HBAC); G

Thermoplastic Compound; G

Raised Pavement Markers Primers and Adhesives; G

Test Reports

#### SD-07 Certificates

Qualifications; G

Reflective Media for Airfields

Reflective Media for Roads

Waterborne Paint

Solventborne Paint

Volatile Organic Compound, (VOC); G

Thermoplastic Compound

#### SD-08 Manufacturer's Instructions

Waterborne Paint; G

Solventborne Paint; G

Thermoplastic Compound; G

### 1.4 QUALITY ASSURANCE

#### 1.4.1 Regulatory Requirements

Submit certificate stating that the proposed pavement marking paint meets the Volatile Organic Compound, (VOC) regulations of the local Air Pollution Control District having jurisdiction over the geographical area in which the project is located. Submit Safety Data Sheets for each product.

#### 1.4.2 Qualifications

Submit documentation certifying that pertinent personnel are qualified for

equipment operation and handling of applicable chemicals. The documentation should include experience on five projects of similar size and scope with references for all personnel.

#### 1.5 DELIVERY AND STORAGE

Deliver paint materials, thermoplastic compound materials, and reflective media in original sealed containers that plainly show the designated name, specification number, batch number, color, date of manufacture, manufacturer's directions, and name of manufacturer.

Provide storage facilities at the job site, only in areas approved by the Contracting Officer, for maintaining materials at temperatures recommended by the manufacturer. Make available paint stored at the project site or segregated at the source for sampling not less than 30 days prior to date of required approval for use to allow sufficient time for testing. Notify the Contracting Officer when paint is available for sampling.

#### 1.6 PROJECT/SITE CONDITIONS

##### 1.6.1 Environmental Requirements

##### 1.6.1.1 Weather Limitations for Application

Apply pavement markings to clean, dry surfaces, and unless otherwise approved, only when the air and pavement surface temperature is at least 3 degrees C above the dew point and the air and pavement temperatures are within the limits recommended by the pavement marking manufacturer. Allow pavement surfaces to dry after water has been used for cleaning or rainfall has occurred prior to striping or marking. Test the pavement surface for moisture before beginning work each day and after cleaning. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer. Employ the "plastic wrap method" to test the pavement for moisture as specified in paragraph TESTING FOR MOISTURE.

##### 1.6.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas

Pavement surface must be free of snow, ice, or slush; with a surface temperature of at least 4 degrees C and rising at the beginning of operations, except those involving shot or sand blasting or grinding. Cease operation during thunderstorms, or during rainfall, except for waterblasting and removal of previously applied chemicals. Cease waterblasting where surface water accumulation alters the effectiveness of material removal.

##### 1.6.2 Traffic Controls

Place warning signs conforming to MUTCD near the beginning of the worksite and well ahead of the worksite for alerting approaching traffic from both directions. Place small markers along newly painted lines or freshly placed raised markers to control traffic and prevent damage to newly painted surfaces or displacement of raised pavement markers. Mark painting equipment with large warning signs indicating slow-moving painting equipment in operation.

When traffic must be rerouted or controlled to accomplish the work,

provide necessary warning signs, flag persons, and related equipment for the safe passage of vehicles.

### 1.6.3 Lighting

When night operations are necessary, provide all necessary lighting and equipment. The Government reserves the right to accept or reject night work on the day following night activities by the Contractor.

## PART 2 PRODUCTS

### 2.1 EQUIPMENT

#### 2.1.1 Surface Preparation and Paint Removal

##### 2.1.1.1 Surface Preparation Equipment for Roads and Automotive Parking Areas

Submit a surface preparation equipment list by serial number, type, model, and manufacturer. Include descriptive data indicating area of coverage per pass, pressure adjustment range, tank and flow capacities, and safety precautions required for the equipment operation. Mobile equipment must allow for removal of markings without damaging the pavement surface or joint sealant. Maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition.

##### 2.1.1.1.1 Waterblasting Equipment

Use mobile waterblasting equipment capable of producing a pressurized stream of water that effectively removes paint from the pavement surface without significantly damaging the pavement. Provide equipment, tools, and machinery which are safe and in good working order at all times.

##### 2.1.1.1.2 Grinding or Scarifying Equipment

Use equipment capable of removing surface contaminants, paint build-up, or extraneous markings from the pavement surface without leaving any residue. Clean the surface by hydro blast to remove surface contaminants and ash after a weed torch is used to remove paint.

##### 2.1.1.1.3 Chemical Removal Equipment

Use chemical equipment capable of applying and removing chemicals and paint from the pavement surface, leaving only non-toxic biodegradable residue without scarring or other damage to the pavement or joints and joint seals.

#### 2.1.2 Application Equipment

Submit application equipment list appropriate for the material(s) to be used. Include manufacturer's descriptive data and certification for the planned use that indicates area of coverage per pass, pressure adjustment range, tank and flow capacities, and all safety precautions required for operating and maintaining the equipment. Provide and maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition, or remove them from the work site. Provide mobile and maneuverable application equipment to the extent that straight lines can be followed and normal curves can be made in a true arc.

#### 2.1.2.1 Paint Application Equipment

##### 2.1.2.1.1 Hand-Operated, Push-Type Machines

Provide hand-operated push-type applicator machine of a type commonly used for application of water based paint or two-component, chemically curing paint, thermoplastic, or preformed tape, to pavement surfaces for small marking projects, such as legends and cross-walks, automotive parking areas, or surface painted signs. Provide applicator machine equipped with the necessary tanks and spraying nozzles capable of applying paint uniformly at coverage specified. Hand operated spray guns may be used in areas where push-type machines cannot be used.

##### 2.1.2.1.2 Self-Propelled or Mobile-Drawn Spraying Machines

Provide self-propelled or mobile-drawn spraying machine with suitable arrangements of atomizing nozzles and controls to obtain the specified results. Provide machine having a speed during application capable of applying the stripe widths indicated at the paint coverage rate specified herein and of even uniform thickness with clear-cut edges.

##### 2.1.2.1.2.1 Road Marking

Provide equipment used for marking roads capable of placing the prescribed number of lines at a single pass as solid lines, intermittent lines, or a combination of solid and intermittent lines using a maximum of three different colors of paint as specified.

##### 2.1.2.1.2.2 Hand Application

Provide spray guns for hand application of paint in areas where the mobile paint applicator cannot be used.

#### 2.1.2.2 Thermoplastic Application Equipment

##### 2.1.2.2.1 Thermoplastic Material

Apply thermoplastic material with equipment that is capable of providing continuous uniformity in the dimensions and reflectorization of the marking.

##### 2.1.2.2.2 Application Equipment

- a. Provide application equipment capable of continuous mixing and agitation of the material, with conveying parts which prevent accumulation and clogging between the main material reservoir and the extrusion shoe or spray gun. All parts of the equipment which come into contact with the material must be easily accessible and exposed for cleaning and maintenance. All mixing and conveying parts up to and including the extrusion shoes and spray guns must maintain the material at the required temperature with heat-transfer oil or electrical-element-controlled heat.
- b. Provide application equipment constructed to ensure continuous uniformity in the dimensions of the stripe. Provide an applicator with a means for cleanly cutting off stripe ends squarely and providing a method of applying "skiplines." Provide equipment capable of applying varying widths of traffic markings.

- c. Provide mobile and maneuverable application equipment allowing straight lines to be followed and normal curves to be made in a true arc. Provide equipment used for the placement of thermoplastic pavement markings of two general types: mobile applicator and portable applicator.
- d. Equip the applicator with a pressurized or drop-on type bead dispenser capable of uniformly dispensing reflective glass spheres at controlled rates of flow. The bead dispenser must operate automatically to begin flow prior to the flow of binder to assure that the strip is fully reflectorized.

#### 2.1.2.2.3 Mobile Application Equipment

Provide a truck-mounted, self-contained pavement marking machine that is capable of hot applying thermoplastic by either the extrusion or spray method.

- a. Equip the unit to apply the thermoplastic marking material at temperatures according to the manufacturer's instructions, at widths varying from 75 to 300 mm, with an automatic pressurized or drop-on bead dispensing system, capable of operating continuously, and of installing a minimum of 6 lineal km of longitudinal markings in an 8-hour day.
- b. Equip the mobile unit with a melting kettle which holds a minimum of 2.7 metric tons of molten thermoplastic material; capable of heating the thermoplastic composition to temperatures as recommended by the manufacturer. Use a thermostatically controlled heat transfer liquid. Heating of the composition by direct flame is not allowed. Oil and material temperature gauges must be visible at both ends of the kettle.
- c. Equip mobile units for application of extruded markings with a minimum of two extrusion shoes; located one on each side of the truck, capable of marking simultaneous edge line and centerline stripes; each being a closed, oil-jacketed unit; holding the molten thermoplastic at a temperature as recommended by the manufacturer; and capable of extruding a line of 75 to 200 mm in width; and at a thickness of not less than 3 mm nor more than 5.0 mm, of generally uniform cross section.
- d. Equip mobile units for application of spray markings with a spray gun system capable of marking simultaneous edgeline and centerline stripes. Surround (jacket) the spray system with heating oil to maintain the molten thermoplastic at a temperature of 191 to 218 degrees C, capable of spraying a stripe of 75 to 305 mm in width, and in thicknesses varying from 1.52 mm to 2.49 mm, of generally uniform cross section.
- e. Equip the mobile unit with an electronic programmable line pattern control system, capable of applying skip or solid lines in any sequence, through any and all of the extrusion shoes, or the spray guns, and in programmable cycle lengths. In addition, equip the mobile unit with an automatic counting mechanism capable of recording the number of lineal meters of thermoplastic markings applied to the pavement surface with an accuracy of 0.5 percent.



#### 2.1.2.2.4 Portable Application Equipment

Provide portable hand-operated equipment, specifically designed for placing special markings such as crosswalks, stop bars, legends, arrows, and short lengths of lane, edge and centerlines; and capable of applying thermoplastic pavement markings by the extrusion method. Equip the portable applicator with all the necessary components, including a materials storage reservoir, bead dispenser, extrusion shoe, and heating accessories, capable of holding the molten thermoplastic at the temperature recommended by the manufacturer, and of extruding a line of 75 to 305 mm in width, and in thickness of not less than 3 mm nor more than 5 mm and of generally uniform cross section.

#### 2.1.2.3 Reflective Media Dispenser

Attach the dispenser for applying the reflective media to the thermoplastic dispenser and designed to operate automatically and simultaneously with the applicator through the same control mechanism. The bead applicator must be capable of adjustment and designed to provide uniform flow of reflective media over the full length and width of the stripe at the rate of coverage specified in paragraph APPLICATION.

#### 2.1.2.4 Preformed Tape Application Equipment

Provide and use mechanical application equipment for the placement of preformed marking tape which is a mobile pavement marking machine specifically designed for use in applying pressure-sensitive pavement marking tape of varying widths. Equip the applicator with rollers, or other suitable compaction device to provide initial adhesion of the material with the pavement surface. Use additional tools and devices as needed to properly seat the applied material as recommended by the manufacturer.

### 2.2 MATERIALS

Use epoxy paint thermoplastic preformed tape for roads. Use non-reflectORIZED waterborne or solventborne paint for automotive parking areas. The maximum allowable VOC content of pavement markings is 150 grams per liter. Color of markings are indicated on the drawings and must conform to ASTM D6628 for roads and automotive parking areas and SAE AMS-STD-595A for airfields. Provide materials conforming to the requirements specified herein.

#### 2.2.1 Waterborne Paint

MPI 97, CGSB 1-GP-74M.

#### 2.2.2 Solventborne Paint

MPI 32.

#### 2.2.3 Methacrylate Paint

Formulate methacrylate paint to meet the requirements of FAA AC 150/5370-10, Item P-620.2, Methacrylate.

## 2.2.4 Epoxy Paint

### 2.2.4.1 Formulation

Epoxy pavement marking material will be a two component, 100 percent solids, material formulated to provide simple volumetric mixing ratio of two volumes of component A and one volume of component B unless otherwise recommended by the manufacturer.

### 2.2.4.2 Composition

The component A of both white and yellow must be within the following limits:

| TABLE I     |                                                                       |                         |
|-------------|-----------------------------------------------------------------------|-------------------------|
|             | White                                                                 | Yellow                  |
| Pigments    | Minimum 18 percent by weight<br>Titanium Dioxide (ASTM D476, Type II) | 21-27 percent by weight |
| Epoxy Resin | 75-82 percent                                                         | 73-79 percent           |

The epoxy resin must be free of lead, cadmium, mercury, hexavalent chromium and other toxic heavy metals as defined by the Environmental Protection Agency. Submit a manufacturer's certification of compliance with this requirement.

### 2.2.4.3 Epoxide Value

Determine epoxide epoxy number of the epoxy resin in accordance with ASTM D1652 for white and yellow component A on pigment free basis. The epoxide number must be within plus or minus 50 of the published manufacturer's standard.

### 2.2.4.4 Total Amine Value

Determine the amine number on the curing agent (component B) in accordance with ASTM D2074. The amine number must be within plus or minus 50 of the published manufacturer's standard.

### 2.2.4.5 Toxicity

Upon heating to application temperature, the material must not produce fumes which are toxic or injurious to persons or property.

### 2.2.4.6 Daylight Directional Reflectance

Directional reflectance of white and yellow paint (without glass beads) in accordance with ASTM E1347: White 84 percent Yellow 55 percent.

### 2.2.4.7 Laboratory Drying Time

The epoxy pavement marking material must have a maximum no-pick-up time of 30 minutes when tested in accordance with ASTM D711.

#### 2.2.4.8 Curing

The epoxy material must be capable of fully curing under a constant surface temperature of 7 Degrees C or above.

#### 2.2.4.9 Adhesion to Concrete

The catalyzed epoxy pavement marking material must have a high degree of adhesion to the specified concrete surface (100 percent concrete failure) when tested according to ASTM D7234. The concrete substrate must have a minimum compressive strength of 27.5 MPa. Condition prepared specimens at a temperature of 23.9 plus or minus 1.1 Degrees C for a minimum of 24 hours and a maximum of 72 hours prior to performance of the test.

#### 2.2.4.10 Hardness

Epoxy pavement marking materials must have a Shore D Hardness between 75 and 100 when tested in accordance with ASTM D2240. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 72 hours and a maximum of 96 hours prior to performing the tests.

#### 2.2.4.11 Abrasion Resistance

The wear index for a catalyzed sample must not exceed 82 when tested in accordance with ASTM D4060 using a 1000 gram load, CS-17 wheels and a test duration of 1000 cycles. Run the test on cured samples of material which have been applied at a film thickness of 15 plus or minus 0.5 mils to code S-16 stainless steel plates. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 48 hours prior to performing the tests.

#### 2.2.4.12 Tensile Strength

Epoxy pavement marking materials must have a tensile strength of at least 41,370 kPA when tested in accordance with ASTM D638. Cast the Type IV specimens in a suitable mold and pull at the rate of 6 mm per minute using a suitable dynamic testing machine. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 12 hours and a maximum of 48 hours prior to performing the tests.

#### 2.2.4.13 Compressive Strength

Catalyzed epoxy pavement marking materials must have a compressive strength of at least 82,700 kPA when tested in accordance with ASTM D695. Condition the cast sample at 23.9 plus or minus 1.1 Degrees C for a minimum of 12 hours and a maximum of 48 hours prior to performing the tests. The rate of compression of these samples must not exceed 6 mm per minute.

### 2.2.5 Thermoplastic Compound

#### 2.2.5.1 Composition Requirements

Thermoplastic compound must conform to AASHTO M 249. Formulate the binder component as an alkyd resin.

#### 2.2.5.2 Primer

- a. Asphalt concrete primer: Provide thermosetting adhesive primer with

a solids content of pigment reinforced synthetic rubber and synthetic plastic resin dissolved or dispersed in a volatile organic solvent for asphaltic concrete pavements. The solids content must not be less than 10 percent by weight at 21 degrees C and 60 percent relative humidity. A wet film thickness of 0.13 mm, plus or minus 0.03 mm, must dry to a tack-free condition in less than 5 minutes.

- b. Portland cement concrete primer: Provide an epoxy resin primer for Portland cement concrete pavements, of the type recommended by the manufacturer of the thermoplastic composition.

#### 2.2.6 High Build Acrylic Coating (HBAC)

Formulate High Build Acrylic Coating (HBAC) to meet the requirements of Table II.

| TABLE II - REQUIREMENTS FOR HIGH BUILD ACRYLIC COATINGS (HBAC)                                     |                                                             |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| TEST                                                                                               | MINIMUM REQUIREMENT (AND MAXIMUM WHERE INDICATED)           |
| Resin System (ASTM D2621)                                                                          | Waterborne 100 percent Acrylic                              |
| Percent Volume Solids (ASTM D2697)                                                                 | 58 percent                                                  |
| Volatile Organic Compound, max. (ASTM D3960)                                                       | 150 g/l                                                     |
| White (SAE AMS-STD-595A)                                                                           | 37925                                                       |
| Yellow (SAE AMS-STD-595A)                                                                          | 33538                                                       |
| Shore D Hardness (ASTM D2240)                                                                      | 45                                                          |
| 1/8 inch Mandrel Bend at 5 mils Dry Film Thickness (DFT, one-week cure (ASTM D522/D522M, Method B) | No visual defects at bend (Conditions at ASTM D3924)        |
| Adhesion to Concrete and Asphaltic Pavements (ASTM D4541)                                          | 0.97 MPa or 100 percent cohesive failure in pavement        |
| Accelerated Weathering, Yellow, 2500 Hours UV Exposure (ASTM G154: see note 1)                     | Max. color loss to 33655 (SAE AMS-STD-595A)                 |
| Water Absorption at 168 Hours Immersion Tap Water (ASTM D471)                                      | 9.0 percent max. weight increase (conditions at ASTM D3924) |

| TABLE II - REQUIREMENTS FOR HIGH BUILD ACRYLIC COATINGS (HBAC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| TEST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MINIMUM REQUIREMENT (AND MAXIMUM WHERE INDICATED)        |
| Application at 1650 microns Wet, One Coat, One-week Cure, (see note 2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | No visual cracking or curling (conditions at ASTM D3924) |
| No Pick-Up at 630 microns (ASTM D711)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Wet 10 minutes max.                                      |
| Lead (ASTM D3335)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.06 percent max.                                        |
| Cadmium (ASTM D3335)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.06 percent max.                                        |
| Chromium (ASTM D3718)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.00 percent                                             |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                          |
| <p>(1) Properly mix and apply yellow paint at 250 microns plus or minus 50 microns DFT over a suitably sized, clean aluminum substrate (ASTM D823), and cure for a minimum of 48 hours: prepare four individual yellow samples. Expose three samples to continuous Ultraviolet (UV) light for 2500 hours, without cycles condensation, in accordance to ASTM G154: use UVA-340 lamps in the testing apparatus. Following exposure, compare the three exposed samples to the "one" non-exposed sample using SAE AMS-STD-595A colors 33538 and 33655 as visual references: evaluate exposed samples for degree of visual color loss. Yellow paint is rated as passing if each exposed sample appears equivalent to the non-exposed sample, and in addition, displays color loss no greater than SAE AMS-STD-595A color 33655.</p>                                                                                                                                                                                                                                                                                   |                                                          |
| <p>(2) Using double-stick, foam mounting tape (or equal) with a nominal thickness of 1625 microns, apply a rectangular mold with inner dimensions of 7.6 cm by 25.5 cm to a clean aluminum sample approximately sized at 15 cm by 30 cm by 0.30 cm. Do not remove the tape's plastic backing. Mix and apply excess paint into mold. Remove excess paint, by squeegee or other appropriate draw down technique, to a uniform thickness equal to the tape's height. Perform paint application and draw down within a period of no more than 60 seconds. Approximately one to two minutes following the draw down, remove tape from sample and allow coating to cure for a minimum period of one week ASTM D3924. Using a micrometer or other appropriate device, measure cured coating thickness (less sample thickness) to confirm resulting coating application was at or above 950 microns DFT. Inspect coating for visual signs of cracking and curling. Following a one week cure, the coating is rated as passing if applied greater than 950 microns DFT and visually free of both cracking and curling.</p> |                                                          |

### 2.2.7 Preformed Tape

Provide adherent reflectorized strip preformed tape in accordance with ASTM D4505 Retroreflectivity Level II, Class 1, 2 or 3, Skid Resistance Level B.

### 2.2.8 Raised Pavement Markers Primers and Adhesives

Use either metallic or nonmetallic markers of the button or prismatic reflector type. Provide permanent color markers as specified for pavement marking, which retain the color and brightness under the action of traffic. Provide button markers with a diameter of not less than 100 mm, spaced not more than 12 m apart on solid longitudinal lines. Make broken centerline marker spacing in segments of 3 m with gaps of 9 m between segments. Provide button markers with rounded surfaces presenting a smooth contour to traffic and not projecting more than 19 mm above level of pavement. Provide plowable pavement markers and adhesive epoxy conforming to ASTM D4383.

### 2.2.9 Reflective Media

#### 2.2.9.1 Reflective Media for Roads

AASHTO M 247, Type 1.

## PART 3 EXECUTION

### 3.1 EXAMINATION

#### 3.1.1 Testing for Moisture

Test the pavement surface for moisture before beginning pavement marking after each period of rainfall, fog, high humidity, or cleaning, or when the ambient temperature has fallen below the dew point. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer or authorized representative.

Employ the "plastic wrap method" to test the pavement for moisture as follows: Cover the pavement with a 300 mm by 300 mm section of clear plastic wrap and seal the edges with tape. After 15 minutes, examine the plastic wrap for any visible moisture accumulation inside the plastic. Do not begin marking operations until the test can be performed with no visible moisture accumulation inside the plastic wrap. Re-test surfaces when work has been stopped due to rain.

#### 3.1.2 Surface Preparation Demonstration

Prior to surface preparation, demonstrate the proposed procedures and equipment. Prepare areas large enough to determine cleanliness and rate of cleaning. Approved demonstration area establishes the standard for the remainder of the work.

#### 3.1.3 Test Stripe Demonstration

Prior to paint application, demonstrate test stripe application within the work area using the proposed materials and equipment. Apply separate test stripes in each of the line widths and configurations required herein using the proposed equipment. Make the test stripes long enough to determine the proper speed and operating pressures for the vehicle(s) and

machinery, but not less than 15 m long.

#### 3.1.4 Application Rate Demonstration

During the Test Stripe Demonstration, demonstrate compliance with the application rates specified herein. Document the equipment speed and operating pressures required to meet the specified rates in each configuration of the equipment and provide a copy of the documentation to the Contracting Officer prior to proceeding with the work.

#### 3.1.5 Retroreflective Value Demonstration

After the test stripes have cured to a "no-track" condition, demonstrate compliance with the average retroreflective values specified herein. Take a minimum of ten readings on each test stripe with a Retroreflectometer with a direct readout in millicandelas per square meter per lux (mcd/m<sup>2</sup>/lx). Perform testing in accordance with ASTM D4061, ASTM E1710, ASTM E2177, and ASTM E2302.

#### 3.1.6 Level of Performance Demonstration

The Contracting Officer will be present at the application demonstrations to observe the results obtained and to validate the operating parameters of the vehicle(s) and equipment. If accepted by the Contracting Officer, the test stripe is the measure of performance required for this project. Do not proceed with the work until the demonstration results are satisfactory to the Contracting Officer.

### 3.2 EXTERIOR SURFACE PREPARATION

Allow new pavement surfaces to cure for a period of not less than 30 days before application of marking materials. Thoroughly clean surfaces to be marked before application of the paint. Remove dust, dirt, and other granular surface deposits by sweeping, blowing with compressed air, rinsing with water, or a combination of these methods as required. Remove rubber deposits, existing paint markings, residual curing compounds, and other coatings adhering to the pavement by water blasting or approved chemical removal method.

- a. For Portland Cement Concrete pavement, grinding, light shot blasting, or light scarification, to a resulting profile equal to ICRI 03732 CSP 2, CSP 3, and CSP 4, respectively, can be used in addition to water blasting on most pavements, to either remove existing coatings, or for surface preparation.
- b. Do not use shot blasting on airfield pavements due to the potential of Foreign Object Damage (FOD) to aircraft. Scrub affected areas, where oil or grease is present on old pavements to be marked, with several applications of trisodium phosphate solution or other approved detergent or degreaser and rinse thoroughly after each application. After cleaning oil-soaked areas, seal with shellac or primer recommended by the manufacturer to prevent bleeding through the new paint. Do not commence painting in any area until pavement surfaces are dry and clean.

#### 3.2.1 Early Painting of Rigid Pavements

Pretreat rigid pavements that require early painting with an aqueous solution containing 3 percent phosphoric acid and 2 percent zinc

chloride. Apply the solution to the areas to be marked.

### 3.2.2 Early Painting of Asphalt Pavements

For asphalt pavement systems requiring painting application at less than 30 days, apply the paint and beads at half the normal application rate, followed by a second application at the normal rate after 30 days.

## 3.3 APPLICATION

Apply pavement markings to dry pavements only.

### 3.3.1 Paint

Apply paint with approved equipment at rate of coverage specified herein. Provide guidelines and templates as necessary to control paint application. Take special precautions in marking numbers, letters, and symbols. Manually paint numbers, letters, and symbols. Sharply outline all edges of markings. The maximum drying time requirements of the paint specifications will be strictly enforced, to prevent undue softening of bitumen, and pickup, displacement, or discoloration by tires of traffic. If there is a deficiency in drying of the markings, painting operations must cease until the cause of the slow drying is determined and corrected.

#### 3.3.1.1 Waterborne Paint

##### 3.3.1.1.1 Roads

Apply paint at a rate of 2.6 plus or minus 0.1 square meter per liter. Apply AASHTO M 247 Type 1 beads at a rate of 834 plus or minus 60 grams of glass spheres per liter.

#### 3.3.1.2 Solventborne Paint

Apply paint at a minimum wet film thickness of 0.381 mm. Apply AASHTO M 247 Type 1 beads at a minimum rate of 715 grams of glass spheres per liter.

#### 3.3.1.3 Methacrylate Paint

Apply paint evenly to the pavement surface at a maximum rate of 1.1 square meters per liter. Apply glass spheres conforming to FS TT-B-1325 uniformly to the wet paint on airfield pavement. Use either Type I (Gradation A), Type III, or Type IV (Gradation A or B) beads. Apply Type I (Gradation A) beads at a minimum rate of 1.8 kilograms of glass spheres per liter. Apply Type III beads at a minimum rate of 2.4 kilograms of glass spheres per liter. Apply Type IV (Gradation A or B) beads at a minimum rate of 1.8 kilograms of glass spheres per liter.

#### 3.3.1.4 Epoxy Paint

Apply paint evenly to the pavement surface at a wet film thickness of 0.508 mm plus or minus 0.025 mm to cover 2.0 plus or minus 0.1 square meters per liter. Apply glass spheres uniformly to the wet paint on road and street pavement at a rate of 834 plus or minus 60 grams of glass spheres per liter.



### 3.3.1.5 High Build Acrylic Coating

Apply High Build Acrylic Coating (HBAC) at a rate of 1.3 square meters per liter. Apply Type IV (Gradation A) beads at a minimum rate of 1.8 kilograms of glass spheres per liter.

### 3.3.2 Thermoplastic Compound

Place thermoplastic pavement markings, free from dirt or tint, upon dry pavement. The temperature must be a minimum of 4.4 degrees C and rising at the time of installation. Apply all centerline, skipline, edgeline, and other longitudinal type markings with a mobile applicator. Place all special markings, crosswalks, stop bars, legends, arrows, and similar patterns with a portable applicator, using the extrusion method.

#### 3.3.2.1 Primer

After surface preparation has been completed, prime the asphalt or concrete pavement surface with spray equipment. Allow primer materials to "set-up" prior to applying the thermoplastic composition. Allow the asphalt concrete primer to dry to a tack-free condition, usually occurring in less than 10 minutes. Allow the Portland cement concrete primer to dry in accordance with the thermoplastic manufacturer recommendations. To shorten the curing time of the epoxy resins, an infrared heating device may be used on the concrete primer. Apply asphalt concrete primer to all asphalt concrete pavements at a wet film thickness of 0.13 mm, plus or minus 0.03 mm (6.5 to 9.82 square meters per liter). Apply Portland cement concrete primer to all concrete pavements (including concrete bridge decks) at a wet film thickness of between 1.0 to 1.3 mm 7.85 to 9.82 square meters per liter.

After the primer has "set-up", apply the thermoplastic at temperatures no lower than 191 degrees C nor higher than 218 degrees C at the point of deposition. Apply all extruded thermoplastic markings at the specified width and at a thickness of not less than 3 mm nor more than 5 mm. Apply all sprayed thermoplastic markings at the specified width and the thickness designated in the contract plans. If the plans do not specify a thickness, apply centerline markings at a wet thickness of 2.3 mm, plus or minus 0.13 mm, and edgeline markings at a wet thickness of 1.5 mm, plus or minus 0.13 mm.

#### 3.3.2.2 Reflective Media

Immediately after installation of the thermoplastic material, mechanically apply drop-on reflective glass spheres conforming to AASHTO M 247 Type 1 at the rate of 0.24 kg per square meter such that the spheres are held by and imbedded in the surface of the molten material. Accomplish drop-on application of the glass spheres to ensure even distribution at the specified rate of coverage. If there is a malfunction of either thermoplastic applicator or reflective media dispenser, discontinue operations until deficiency is corrected.

### 3.3.3 Raised Pavement Markers

Align prefabricated markers carefully at the spacing indicated on the drawings and permanently fix in place by means of epoxy adhesives. To ensure good bond prior to applying adhesive, thoroughly clean all areas where markers are to be set by water blasting and use of compressed air.

### 3.3.4 Preformed Tape

The pavement surface and ambient air temperature must be a minimum of 15 degrees C and rising. Place the preformed markings in accordance with the manufacturer's written instructions.

### 3.3.5 Cleanup and Waste Disposal

Keep the worksite clean and free of debris and waste from the removal and application operations. Dispose of debris at approved sites.

## 3.4 FIELD QUALITY CONTROL

### 3.4.1 Sampling and Testing

As soon as the paint and thermoplastic materials and reflective media are available for sampling, obtain by random selection from the sealed containers, two quart samples of each batch in the presence of the Contracting Officer. Accomplish adequate mixing prior to sampling to ensure a uniform, representative sample. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Clearly identify samples by designated name, specification number, batch number, project contract number, intended use, and quantity involved.

At the discretion of the Contracting Officer, samples provided may be tested by the Government for verification.

### 3.4.2 Material Inspection

Examine material at the job site to determine that it is the material referenced in the report of test results or certificate of compliance. Provide test results substantiating conformance to the specified requirements with each certificate of compliance.

### 3.4.3 Dimensional Tolerances

Apply all markings in the standard dimensions provide in the drawings. New markings may deviate a maximum of 10 percent larger than the standard dimension. The maximum deviation allowed when painting over an old marking is up to 20 percent larger than the standard dimensions.

### 3.4.4 Bond Failure Verification

Inspect newly applied markings for signs of bond failure based on visual inspection and comparison to results from Test Stripe Demonstration paragraph.

### 3.4.5 Reflective Media and Coating Application Verification

Use a wet film thickness gauge to measure the application of wet paint. Use a microscope or magnifying glass to evaluate the embedment of glass beads in the paint. Verify the glass bead embedment with approximately 50 percent of the individual bead spheres embedded and 50 percent of the individual bead spheres exposed.

-- End of Section --

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## SECTION 32 31 13.53

HIGH-SECURITY FENCES (CHAIN LINK AND ORNAMENTAL) AND GATES  
11/21

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A121         | (2022) Standard Specification for Metallic-Coated Carbon Steel Barbed Wire                                                                               |
| ASTM A153/A153M   | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                                                      |
| ASTM A240/A240M   | (2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications |
| ASTM A307         | (2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                                                |
| ASTM A478         | (1997; R 2019) Standard Specification for Chromium-Nickel Stainless Steel Weaving and Knitting Wire                                                      |
| ASTM A563         | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                                                                                   |
| ASTM A780/A780M   | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings                                                         |
| ASTM A824         | (2001; R 2022) Standard Specification for Metallic-Coated Steel Marcellled Tension Wire for Use With Chain Link Fence                                    |
| ASTM A1023/A1023M | (2021) Standard Specification for Stranded Carbon Steel Wire Ropes for General Purposes                                                                  |
| ASTM C94/C94M     | (2025) Standard Specification for Ready-Mixed Concrete                                                                                                   |
| ASTM F567         | (2023) Standard Practice for Installation of Chain Link Fence                                                                                            |
| ASTM F626         | (2014; R 2023) Standard Specification for Fence Fittings                                                                                                 |

|            |                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM F844  | (2019; R 2024) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use                                                             |
| ASTM F883  | (2013; R 2022) Standard Performance Specification for Padlocks                                                                                                 |
| ASTM F900  | (2011; R 2017) Standard Specification for Industrial and Commercial Swing Gates                                                                                |
| ASTM F1043 | (2018; R 2022) Standard Specification for Strength and Protective Coatings on Steel Industrial Fence Framework                                                 |
| ASTM F1083 | (2018; R 2022) Standard Specification for Pipe, Steel, Hot-Dipped Zinc Coated (Galvanized) Welded, for Fence Structures                                        |
| ASTM F1145 | (2005; R 2017) Standard Specification for Turnbuckles, Swaged, Welded, Forged                                                                                  |
| ASTM F1184 | (2023; E 2023) Standard Specification for Industrial and Commercial Horizontal Slide Gates                                                                     |
| ASTM F1664 | (2008; R 2022) Standard Specification for Poly(Vinyl Chloride) (PVC) and Other Conforming Organic Polymer-Coated Steel Tension Wire Used with Chain-Link Fence |
| ASTM F1910 | (1998; R 2018) Standard Specification for Long Barbed Tape Obstacles                                                                                           |
| ASTM F1911 | (2005; R 2019) Standard Practice for Installation of Barbed Tape                                                                                               |

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Fence Installation Drawings; G

### SD-03 Product Data

Fabric

Posts

Post Caps

Chain Link Braces

Line Posts

Sleeves

Rails

Tension Wire

Barbed Wire

Barbed Wire Supporting Arms

Barbed Tape

Latches

Hinges

Stops

Keepers

Rollers

Turnstiles

Padlocks

Wire Ties

Ornamental Fence Systems

Swing Gates

Slide Gates

Fence Fabric Reinforcement

#### SD-07 Certificates

Chain Link Fence

Fabric

Barbed Wire

Gate Hardware and Accessories

Concrete

Gate Operator

#### SD-10 Operation and Maintenance Data

Electro-Mechanical Locks

Gate Operator

Operating and maintenance instructions

### 1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials elevated off of the ground to protect against oxidation caused by ground contact.

## PART 2 PRODUCTS

### 2.1 COMPONENTS

#### 2.1.1 Chain Link Fence Fabric

##### 2.1.1.1 General

Provide Class 2b polyvinyl chloride-coated steel fabric with 92 grams of zinc coating per square meter in accordance with ASTM F668. Provide manufacturer's standard dark greenolive greenbrownblack in color for polyvinyl chloride coating for fabric and all other fence components complying with ASTM F934. Fabricate fence fabric of 9 gauge wire woven in 50 mm diamond mesh. Provide twisted and barbed fabric on the top selvage and knuckled on the bottom selvage.

##### 2.1.1.2 Approval Of Polyvinyl Chloride-Coated Fence Materials

Inspect polyvinyl chloride-coated fence materials for cracking, peeling, and conformance with the specifications prior to installation. Replace any fence materials rejected by the Contracting Officer with approved materials at no additional cost to the Government.

#### 2.1.2 Posts

##### 2.1.2.1 Metal Posts for Chain Link Fence

Provide posts conforming to ASTM F1083, zinc-coated. Provide either Group IA steel pipe posts, Group IC, with PVC polymer overcoat, or Group II, roll-formed steel sections, and be coated conforming to the requirements of ASTM F1043. Provide sizes as shown on the drawings. Use line posts and terminal (corner, gate, and pull) posts of the same designation throughout the fence. Provide gate post for the gate type specified subject to the limitation specified in ASTM F900 or ASTM F1184.

##### 2.1.2.2 Accessories

- a. Provide accessories conforming to ASTM F626. Coat ferrous accessories with polyvinyl chloride-coated, minimum thickness of 0.25 mm. Match color coating of fittings to color coating of the fabric.
- b. Provide truss rods (with turnbuckles or other means of adjustment) for each terminal post.
- c. Provide barbed wire supporting arms of the 45 degree outward angle 3-strand arm type and of the design required for the post furnished. Secure arms by riveting.
- d. Furnish post caps in accordance with manufacturer's standard accessories with coating matching that of fence posts.
- e. Provide 9 gauge tie wire for attaching fabric to rails, braces, and posts and match the material and coating of the fence fabric. Use

tie wires for attaching fabric to tension wire on high security fences made from 1.6 mm stainless steel. Provide double loop tie wires 165 mm in length. Provide miscellaneous hardware coatings which conform to ASTM A153/A153M unless modified.

### 2.1.1.3 Chain Link Braces

ASTM F1083, zinc-coated, Group IA High Strength, steel pipe, size NPS 1-1/4. Use braces and rails that are Group IA High Strength, steel pipe, size NPS 1-1/4 or Group II, formed steel sections, size 42 mm and be zinc coated and polyvinyl chloride-coated, minimum thickness, 0.25 mm conforming to the requirements of ASTM F1043. Group II, formed steel sections, size 42 mm, conforming to ASTM F1043, may be used as braces and rails if Group II line posts are furnished.

### 2.1.1.4 Chain Link Gates

#### 2.1.1.4.1 Gate Assembly

Provide gate assembly conforming to ASTM F900 and/or ASTM F1184 of the type and swing shown. Provide gate frames conforming to strength and coating requirements of ASTM F1083 for Group IA High Strength, steel pipe, nominal pipe size (NPS) 1-1/2. Provide gate frames made of polyvinyl chloride-coated steel pipe (Group IA)(Group IC) a nominal pipe size (NPS) 1-1/2, conforming to ASTM F1043. Use gate fabric that matches the specified chain link fabric.

#### 2.1.1.4.2 Gate Leaves

For gate leaves, more than 2.44 m wide, provide either intermediate members and diagonal truss rods or tubular members as necessary to provide rigid construction, free from sag or twist. For gate leaves less than 2.44 m wide, provide truss rods or intermediate braces. Provide intermediate braces on all gate frames with an electro-mechanical lock. Attach fabric to the gate frame by method standard with the manufacturer. Welding is not an acceptable method for attaching fabric to gate frames.

#### 2.1.1.4.3 Gate Hardware and Accessories

Submit manufacturer's catalog data. Furnish and install latches, hinges, stops, keepers, rollers, and other hardware items as required for the operation of the gate. All items are required to match the material characteristics of the fence system being installed. Provide hinge with stainless steel pin. Arrange latches for padlocking so that the padlock will be accessible from both sides of the gate. Provide stops for holding the gates in the open position. For high security applications, extend each end member of gate frames sufficiently above the top member to carry three strands of barbed wire in horizontal alignment with barbed wire strands on the fence. Coating for steel latches, stops, hinges, keepers, and accessories, must be PVC, minimum thickness of 0.25 mm.

### 2.1.1.5 Ornamental Fence Gates

#### 2.1.1.5.1 Swing Gates

Submit manufacturer's catalog data. Fabricate swing gates by welding 5 sq cm tubular steel ends and rails. Use pickets that match the adjacent fence construction. Reinforce gates to ensure assembly sags no more than 1% of the gate leaf width or 5 cm, whichever is less. Size gate posts to accommodate the weight and width of each gate leaf. Mount gates to posts with weldable steel plates or blocks, pressed steel, or malleable iron



hinges. Hot-dip galvanize all hinges with a minimum zinc weight of 66 g/sq m. Provide hinge with stainless steel pin. Secure all tamper points by welding or peening the threads. Use swing gate latches and drop bar guides manufactured of pressed steel, hot-dipped galvanized with a minimum zinc weight of 366 g/sq m. Finish all gate hardware in the same color/coating as the fence system.

#### 2.1.5.2 Slide Gates

Submit manufacturer's catalog data. Fabricate slide gates by welding 5 sq cm tubular steel ends and rails. Use pickets that match the adjacent fence construction. Select the type and class of slide gate to comply with ASTM F1184. Size gate posts to accommodate the weight and width of each gate leaf in accordance with ASTM F1184, or per manufacturer's recommendations. Specify Type II, Class 2, interior roller design for cantilever slide gates.

#### 2.1.6 Padlocks

Provide padlocks conforming to ASTM F883, Type P01 , Options A, B, and G and , Grade 6. Size 44 mm. . Key all padlocks into master key system as specified in Section 08 71 00 DOOR HARDWARE.

#### 2.1.7 Gate Operator

Provide electric gate operators for sliding gates as follows: Provide electric gate operators with a right angle gearhead instantly reversing motor with magnetic drum-type brake, friction disc clutch, reversing starter with thermal overload protection, and a chain-driven geared rotary-type automatic limit switch. Use only hardened steel machine cut worm and mating bronze gears that operate in a bath of oil. Gate operators with V-belt pulleys are not allowed. Equip gate operators with an emergency release to allow the gate to be operated manually that is also capable of being locked in the engaged or disengaged position. Provide positive stops on the gate tracks as a backup to the limit switches.

#### 2.1.8 Electro-Mechanical Locks

Provide electro-mechanical locking devices for sliding gates and personnel gates that are solenoid actuated such that the deadbolt retracts when the solenoid is energized and remains electrically retracted until the gate is closed. Provide continuous duty type solenoid, rated for 120V ac, 60Hz operation. Ensure the locking device is unlockable by key and keyed on both sides. Monitor status of the electro-mechanical lock by two limit switches (integral to the locking device) wired in series. Ensure one switch monitors the deadlock lever and the other monitors the locking tongue.

### 2.2 MATERIALS

#### 2.2.1 Wire

##### 2.2.1.1 Wire Ties

Submit samples as specified. Provide wire ties constructed of the same material and finish as the fencing fabric.

#### 2.2.1.2 Barbed Wire

Provide barbed wire conforming to ASTM A121 aluminum-coated, Type A, with 12.5 gauge wire with 14 gauge, round, 4-point barbs spaced no more than 125 mm apart.

#### 2.2.1.3 Tension Wire

Provide metallic coated steel marcelled tension wire (No. 7-gauge), complying with ASTM A824. Provide aluminum-coated (aluminized) steel wire with coating that weighs not less than 122 gram per square meter. Provide PVC-coated tension wire of the same class and color as the fencing fabric complying with ASTM F1664.

#### 2.2.2 Barbed Tape

Provide reinforced barbed tape, single coil, for fence toppings fabricated from 430 series stainless steel with a hardness range of Rockwell (30N) 37-45 conforming to the requirements of ASTM A240/A240M. Provide stainless steel strip 0.6 mm thick by 25 mm wide before fabrication. Provide barbs that are a minimum of 30.5 mm in length, in groups of 4, spaced on 102 mm centers. Use stainless steel core wire with a 2.5 mm diameter and a minimum tensile strength of 9.68 MPa and conforming to ASTM A478. The above requirements also apply to reinforced barbed tape, single coil, for ground application. Use 1.3 mm stainless steel twistable wire ties for attaching the barbed tape to the barbed wire and to the fence for ground application.

Ensure long barbed tape obstacles conform to ASTM F1910.

#### 2.2.3 Concrete

ASTM C94/C94M, using 19 mm maximum size aggregate, and having minimum compressive strength of 21 MPa at 28 days. Use grout consisting of one part portland cement to three parts clean, well-graded sand and the minimum amount of water to produce a workable mix.

### 2.3 FENCE FABRIC REINFORCEMENT

Provide galvanized PVC-coated structural wire rope as indicated and in accordance with ASTM A1023/A1023M, 20mm nominal diameter of strand, 19 wire strand, regular lay, extra improved plow steel (EIPS), independent wire rope core (IWRC) and with a minimum breaking strength of 9,175 Newtons. Galvanize cable, Class A, in accordance with ASTM A1023/A1023M.

#### 2.3.1 Wire Rope Accessories

Use hot-dipped galvanized structural steel members in cable anchoring system. Provide hot-dipped galvanized clamps, U-bolts, and associated hardware in accordance with ASTM A153/A153M.

#### 2.3.2 Turnbuckles

Provide turnbuckles that are 30mm x 300mm x 300mm Type I, Grade 1 galvanized, in accordance with ASTM F1145.

#### 2.3.3 Rope Clamps

Provide hot-dipped galvanized rope clamps in accordance with ASTM A153/A153M.

### 2.3.4 Threaded Rods, U-Bolts, and Bolts

Provide all threaded rods, U-bolts conforming to ASTM A307 and install with ASTM F844 and ASTM A563 nuts. Galvanize the entire bolt assembly.

## PART 3 EXECUTION

### 3.1 PREPARATION

Perform complete installation conforming to ASTM F567.

#### 3.1.1 Line and Grade

Install fence to the lines and grades indicated. Clear the area on either side of the fence line to the extent indicated. Space line posts equidistant at intervals not exceeding 3 m. Set terminal (corner, gate, and pull) posts whenever abrupt changes in vertical and horizontal alignment are encountered. Provide continuous fabric between terminal posts; however, ensure runs between terminal posts do not exceed 152.4 m. Repair any damage to galvanized surfaces, including welding, with paint containing zinc dust in accordance with ASTM A780/A780M.

#### 3.1.2 Excavation

Excavate holes to depths indicated. Clear all post holes of loose material and spread waste material where directed. Eliminate ground surface irregularities along the fence line to the extent necessary to maintain a 50 mm clearance between the bottom of the fabric and finish grade.

### 3.2 INSTALLATION

#### 3.2.1 Installation Drawings

Submit complete Fence Installation Drawings for review and approval by the Contracting Officer prior to shipment. Submit drawing details that include, but are not limited to the following information: Fence Installation Drawings, Location of gate, corner, end, and pull posts, Gate Assembly, Turnstiles, and Gate Hardware and Accessories. Install fence system per approved drawings.

#### 3.2.2 Security Fencing

Install new security fencing, remove existing security fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer.

#### 3.2.3 Posts

##### 3.2.3.1 Earth and Bedrock

- a. Set posts plumb and in alignment. Except where solid rock is encountered, set posts in concrete to the depth indicated on the drawings. Where solid rock is encountered with no overburden, set posts to a minimum depth of 457 mm in rock. Where solid rock is covered with an overburden of soil or loose rock, set posts to the minimum depth indicated on the drawing unless a penetration of 457 mm in solid rock is achieved before reaching the indicated depth, in

which case terminate depth of penetration. Grout all portions of posts set in rock.

- b. Set portions of posts not set in rock in concrete from the rock to ground level. Set posts in holes not less than the diameter shown on the drawings. Make diameters of holes in solid rock at least 25 mm greater than the largest cross section of the post. Thoroughly consolidate concrete and grout around each post, free of voids and finished to form a dome. Allow concrete and grout to cure for 72 hours prior to attachment of any item to the posts. Group II line posts may be mechanically driven, for temporary fence construction only, if rock is not encountered. Set driven posts to a minimum depth of 914 mm and protect with drive caps when setting.
- c. Test fence post rigidity by applying a 222.4 newtons force on the post, perpendicular to the fabric, at 1.52 m above ground. Ensure post movement measured at the point where the force is applied is less than or equal to 19 mm from the relaxed position. Test every tenth post for rigidity. When a post fails this test, make further tests on the next four posts on either side of the failed post. Remove, replace, and retest all failed parts at the Contractor's expense.

#### 3.2.4 Rails

Bolt bottom rail to double rail ends and securely fasten double rail ends to the posts. Peen bolts to prevent easy removal. Install bottom rail before chain link fabric.

#### 3.2.5 Fabric

- a. Set fabric height at 2.1 m .
- b. Install chain link fabric on the side of the post indicated. Attach fabric to terminal posts with stretcher bars and tension bands. Space bands at approximately 381 mm intervals. Install fabric and pull taut to provide a smooth and uniform appearance free from sag, without permanently distorting the fabric diamond or reducing the fabric height. Fasten fabric to line posts at approximately 381 mm intervals and fastened to all rails and tension wires at approximately 305 mm intervals.
- c. Cut fabric by untwisting and removing pickets. Accomplish splicing by weaving a single picket into the ends of the rolls to be joined. Install the bottom of the fabric 50 mm plus or minus 13 mm above the ground.
- d. After the fabric installation is complete, exercise the fabric by applying a 222 newton push-pull force at the center of the fabric between posts; use a 133 newton pull at the center of the panel to ensure fabric deflection of not more than 63.5 mm when pulling fabric from the post side of the fence. Every second fence panel is required to meet this requirement. Resecure and retest all failed panels at the Contractor's expense.

#### 3.2.6 Supporting Arms

Install barbed wire supporting arms and barbed wire as indicated on the drawings and as recommended by the manufacturer. Anchor supporting arms to the posts in a manner to prevent easy removal with hand tools Pull

barbed wire taut and attach to the arms with clips or other means that will prevent easy removal.

### 3.2.7 Barbed Tape Installation

Install stainless steel reinforced barbed tape per ASTM F1911 and as detailed on the drawings. Stretch out barbed tape to its manufacturer's recommended length, set on top of the barbed wire and "V" shaped support arms, then secure it to the barbed wire. Secure the barbed tape to the barbed wire at the two points and at every spiral turn of both coils as shown on the drawings. Install stainless steel non-reinforced barbed tape for ground applications in accordance with manufacturer's recommendations.

### 3.2.8 Gate Installation

- a. Install gates at the locations shown. Mount gates to swing as indicated. Install latches, stops, and keepers as required. Install Slide gates as recommended by the manufacturer.
- b. Attach padlocks to gates or gate posts with chains. Weld or otherwise secure hinge pins, and hardware assembly to prevent removal.
- c. Submit 6 copies of operating and maintenance instructions. Outline the step-by-step procedures required for system startup, operation, and shutdown. Include the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Include in the maintenance instructions routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. Also include the general gate layout, equipment layout and simplified wiring and control diagrams of the system as installed.

### 3.2.9 Grounding

- a. Ground fencing as indicated on drawings and specified.
- b. Ground fences crossed by overhead powerlines in excess of 600 volts and ground all electrical equipment attached to the fence.
- c. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 15 m of the fence, and where the fence alignment changes more than 15 degrees. Ensure grounding locations are located no more than 198 m apart. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by powerlines of 600 volts or more at or near the point of crossing and at distances not exceeding 45 m on each side of crossing.
- d. Provide ground conductor consisting of 3.3 mm solid copper wire. Use grounding electrodes that measures 19 mm by 3.05 m long and are a copper-clad steel rod. Drive electrodes into the earth so that the top of the electrode is at least 152 mm below the grade. Where driving is impracticable, bury electrodes a minimum of 305 mm deep and radially from the fence. Install the top of the electrode to be less than 610 mm or more than 2.4 m from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Measure total resistance of the fence to ground and ensure it is not greater than 25 ohms.

### 3.2.10 Cable Reinforcement Installation

Comply with the contract drawings. Install 1 strands of 1.9 cm steel aircraft cables mounted to the chain link fence posts utilizing u-bolts and nuts. Peen the ends of the U-bolts to prevent removal. Tighten cables with galvanized steel turnbuckles with swaged fittings to the point there is no visible sag. Install 1 m x 1 m x.5 m concrete deadman anchors which are a minimum of 1 m underground and installed in undisturbed surrounding soils. Space dead man anchors no more than 61 m apart and on each side of gate openings. Connect steel aircraft cables to dead man anchors using swaged end fittings and turnbuckles attached to 3.2 cm galvanized threaded rod which is embedded into concrete anchor and held in place with two 10 cm x 10 cm x 0.66 cm thick steel plates welded to the threaded rod. Place deadman anchors within 3 m of last post and on each side of gate openings.

## 3.3 CLOSEOUT ACTIVITIES

### 3.3.1 Cleanup

Remove waste fencing materials and other debris from the work site each workday.

-- End of Section --

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## SECTION 32 92 19

## SEEDING

**08/17, CHG 1: 08/21**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|            |                                                                      |
|------------|----------------------------------------------------------------------|
| ASTM C602  | (2023) Agricultural Liming Materials                                 |
| ASTM D4427 | (2018) Standard Classification of Peat Samples by Laboratory Testing |
| ASTM D4972 | (2018) Standard Test Methods for pH of Soils                         |

## U.S. DEPARTMENT OF AGRICULTURE (USDA)

|              |                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------|
| AMS Seed Act | (1940; R 1988; R 1998) Federal Seed Act                                                                      |
| DOA SSIR 42  | (2022) Kellogg Soil Survey Laboratory Methods Manual, Soil Survey Investigations Report, No. 42, Version 6.0 |

## 1.2 DEFINITIONS

## 1.2.1 Stand of Turf

95 percent ground cover of the established species.

## 1.3 RELATED REQUIREMENTS

Section 31 00 00 EARTHWORK and Section 32 05 33 LANDSCAPE ESTABLISHMENT applies to this section for pesticide use and plant establishment requirements, with additions and modifications herein.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-03 Product Data

Wood Cellulose Fiber Mulch; G

Fertilizer; G

Include physical characteristics, and recommendations.



## SD-06 Test Reports

## Topsoil Composition Tests; G (reports and recommendations)

Submit a minimum of (3) 1-pint soil samples for agricultural suitability soils testing (Standard test A05-N) at a qualified soils testing laboratory. Collect soil samples within planting areas after the completion of grading. Provide one copy of the soil reports and recommendations to the contracting officer prior to carrying out corrective measures per the analysis.

## SD-07 Certificates

State Certification and Approval for Seed from vendor for each grass-seed monostand or seed mixture stating the botanical and common name and percentage of purity, germination and weed seed. Include the year of production and date of packaging. Product certificates for soil amendments and fertilizers, signed by product manufacturer.

## SD-08 Manufacturer's Instructions

## Erosion Control Materials

## 1.5 DELIVERY, STORAGE, AND HANDLING

## 1.5.1 Delivery

## 1.5.1.1 Seed Protection

Deliver to the site in original sealed, labeled, undamaged containers. Protect from drying out and from contamination during delivery, on-site storage, and handling.

## 1.5.1.2 Soil Amendment Delivery

Deliver to the site in original, unopened containers bearing manufacturer's chemical analysis, name, trade name, trademark, and indication of conformance to province and federal laws.

## 1.5.2 Storage

## 1.5.2.1 Seed, Soil Amendment Fertilizer and Lime Storage

Store in cool, dry locations away from contaminants.

## 1.5.2.2 Topsoil

Prior to stockpiling topsoil, treat growing vegetation with application of appropriate specified non-selective herbicide. Clear and grub existing vegetation three to four weeks prior to stockpiling topsoil.

## 1.5.2.3 Handling

Do not drop or dump materials from vehicles.

## 1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS

### 1.6.1 Restrictions

Do not plant when the ground is frozen, snow covered, or muddy. Plant when soil temperature range is between 10 and 18 degrees Celsius or during regular seeding season from August 15 to October 1 and from March 1 to May 15. Planting between May 15 and August 15 is acceptable provided it is schedule-driven, and repairs are made during the regular fall season. Coordinate planting periods with maintenance periods to provide required maintenance from date of substantial completion. Proceed with planting only when existing and forecasted weather conditions permit.

## 1.7 TIME LIMITATIONS

### 1.7.1 Seed

Apply seed within twenty-four hours after seed bed preparation.

## PART 2 PRODUCTS

### 2.1 SEED

#### 2.1.1 Classification

Provide Province-certified and Endophyte-enhance seed of the latest season's crop delivered in original sealed packages, bearing producer's guaranteed analysis for percentages of mixtures, purity, germination, weedseed content, and inert material. Label in conformance with AMS Seed Act and applicable province seed laws. Wet, moldy, or otherwise damaged seed will be rejected.

#### 2.1.2 Planting Dates

| <u>Planting Season</u> | <u>Planting Dates</u>  |
|------------------------|------------------------|
| Season 1               | August 15 to October 1 |
| Season 2               | March 1 to May 15      |
| Temporary Seeding      | As conditions allow    |

#### 2.1.3 Seed Mixture

| <u>Common Name</u>  | <u>Minimum</u> | <u>Maximum</u> |
|---------------------|----------------|----------------|
| Creeping Red Fescue | 30%            | 40%            |
| Chewings Fescue     | 15%            | 20%            |
| Hard Fescue         | 10%            | 15%            |
| Sheeps Fescue       | 10%            | 15%            |

| <u>Common Name</u> | <u>Minimum</u> | <u>Maximum</u> |
|--------------------|----------------|----------------|
| Perennial Ryegrass | 10%            | 15%            |

Temporary seeding must later be replaced by Season 1 plantings for a permanent stand of turf. Some seeds may be replaced by autochthonous wild species. Provide recommendations from a local nursery or extension office.

## 2.2 TOPSOIL

### 2.2.1 On-Site Topsoil

Surface soil stripped and stockpiled on site and modified as necessary to meet the requirements specified for topsoil in paragraph COMPOSITION. When available topsoil must be existing surface soil stripped and stockpiled on-site in accordance with Section 31 00 00 EARTHWORK.

### 2.2.2 Off-Site Topsoil

Conform to requirements specified in paragraph COMPOSITION. Additional topsoil must be furnished by the Contractor.

### 2.2.3 Composition

Containing from 5 to 10 percent organic matter as determined by the topsoil composition tests of the Organic Carbon, 6A, Chemical Analysis Method described in DOA SSIR 42. Maximum particle size, 19 mm, with maximum 3 percent retained on 6 mm screen. The pH must be tested in accordance with ASTM D4972. Topsoil must be free of sticks, stones, roots, and other debris and objectionable materials. Other components must conform to the following limits:

|               |                 |
|---------------|-----------------|
| Silt          | 25-50 percent   |
| Clay          | 10-30 percent   |
| Sand          | 20-35 percent   |
| pH            | 5.5 to 7.0      |
| Soluble Salts | 500 ppm maximum |

## 2.3 SOIL CONDITIONERS

Add conditioners to topsoil as required to bring into compliance with "composition" standard for topsoil as specified herein.

### 2.3.1 Lime

Commercial grade pelletized limestone containing a calcium carbonate equivalent (C.C.E.) as specified in ASTM C602 of not less than 80 percent.

### 2.3.2 Aluminum Sulfate

Commercial grade.

### 2.3.3 Sulfur

100 percent elemental

### 2.3.4 Iron

100 percent elemental

### 2.3.5 Peat

Natural product of peat moss derived from a freshwater site and conforming to ASTM D4427. Shred and granulate peat to pass a 12.5 mm mesh screen and condition in storage pile for minimum 6 months after excavation.

### 2.3.6 Sand

Clean and free of materials harmful to plants.

### 2.3.7 Perlite

Horticultural grade.

### 2.3.8 Composted Derivatives

Ground bark, nitrolized sawdust, humus or other green wood waste material free of stones, sticks, and soil stabilized with nitrogen and having the following properties:

#### 2.3.8.1 Particle Size

Minimum percent by weight passing:

|                |    |
|----------------|----|
| 4.75 mm screen | 95 |
| 2.36 mm screen | 80 |

#### 2.3.8.2 Nitrogen Content

Minimum percent based on dry weight:

|                  |     |
|------------------|-----|
| Fir Sawdust      | 0.7 |
| Fir or Pine Bark | 1.0 |

### 2.3.9 Gypsum

Coarsely ground gypsum comprised of calcium sulfate dihydrate 80 percent, calcium 18 percent, sulfur 14 percent; minimum 96 percent passing through 850 micrometers, 100 percent passing thru 970 micrometers screen.

### 2.3.10 Calcined Clay

Calcined clay must be granular particles produced from montmorillonite clay calcined to a minimum temperature of 650 degrees C. Gradation: A minimum 90 percent must pass a 2.36 mm sieve; a minimum 99 percent must be retained on a 0.250 mm sieve; and material passing a 0.150 mm sieve must not exceed 2 percent. Bulk density: A maximum 640 kilogram per cubic meter.

## 2.4 FERTILIZER

### 2.4.1 Granular Fertilizer

Organic, granular controlled release fertilizer containing minimum percentages, by weight, of plant food nutrients as recommended by the Topsoil Composition Test recommendations.

### 2.4.2 Hydroseeding Fertilizer

Controlled release fertilizer, to use with hydroseeding and composed of pills coated with plastic resin to provide a continuous release of nutrients for at least 6 months and containing the recommended percentages, by weight, of plant food nutrients as recommended by the Topsoil Composition Test reports and recommendations.

## 2.5 MULCH

Mulch must be free from noxious weeds, mold, and other deleterious materials.

### 2.5.1 Straw

Stalks from oats, wheat, rye, barley, or rice. Furnish in air-dry condition and of proper consistency for placing with commercial mulch blowing equipment. Straw must contain no fertile seed.

### 2.5.2 Hay

Air-dry condition and of proper consistency for placing with commercial mulch blowing equipment. Hay must be sterile, containing no fertile seed.

### 2.5.3 Wood Cellulose Fiber Mulch

Use recovered materials of either paper-based (100 percent post-consumer content) or wood-based (100 percent total recovered content) hydraulic mulch. Processed to contain no growth or germination-inhibiting factors and dyed an appropriate color to facilitate visual metering of materials application. Composition on air-dry weight basis: 9 to 15 percent moisture, pH range from 5.5 to 8.2. Use with hydraulic application of grass seed and fertilizer.

## 2.6 WATER

Source of water must be approved by Contracting Officer and of suitable quality for irrigation, containing no elements toxic to plant life.

## 2.7 EROSION CONTROL MATERIALS

Erosion control material must conform to the following:

### 2.7.1 Erosion Control Blanket

100 percent agricultural straw stitched with a degradable nettings, designed to degrade within 12 months.

### 2.7.2 Erosion Control Fabric

Fabric must be knitted construction of polypropylene yarn with uniform

mesh openings 19 to 25 mm square with strips of biodegradable paper. Filler paper strips must have a minimum life of 6 months.

### 2.7.3 Erosion Control Net

Net must be heavy, twisted jute mesh, weighing approximately 605 grams per meter and 1200 mm wide with mesh openings of approximately 25 mm square.

### 2.7.4 Erosion Control Material Anchors

Erosion control anchors must be as recommended by the manufacturer.

## PART 3 EXECUTION

### 3.1 PREPARATION

#### 3.1.1 EXTENT OF WORK

Provide soil preparation prior to planting (including soil conditioners as required), fertilizing, seeding, and surface topdressing of all newly graded finished earth surfaces, unless indicated otherwise, and at all areas inside or outside the limits of construction that are disturbed by the Contractor's operations.

##### 3.1.1.1 Topsoil

Provide 102 mm of off-site topsoil and existing soil to meet indicated finish grade. After areas have been brought to indicated finish grade, incorporate soil amendments into soil a minimum depth of 100 mm by disking, harrowing, tilling or other method approved by the Contracting Officer. Remove debris and stones larger than 19 mm in any dimension remaining on the surface after finish grading. Correct irregularities in finish surfaces to eliminate depressions. Protect finished topsoil areas from damage by vehicular or pedestrian traffic.

##### 3.1.1.2 Soil Conditioner Application Rates

Apply soil conditioners at rates as determined by laboratory soil analysis of the soils at the job site.

##### 3.1.1.3 Fertilizer Application Rates

Apply fertilizer at rates as determined by laboratory soil analysis of the soils at the job site.

### 3.2 SEEDING

#### 3.2.1 Seed Application Seasons and Conditions

Immediately before seeding, restore soil to proper grade. Do not seed when ground is muddy, frozen, snow covered or in an unsatisfactory condition for seeding. If special conditions exist that may warrant a variance in the above seeding dates or conditions, submit a written request to the Contracting Officer stating the special conditions and proposed variance. Apply seed within twenty four hours after seedbed preparation. Sow seed by approved sowing equipment. Sow one-half the seed in one direction, and sow remainder at right angles to the first sowing.

### 3.2.2 Seed Application Method

Seeding method must be hydroseeding.

#### 3.2.2.1 Hydroseeding

First, mix water and fiber. Wood cellulose fiber, paper fiber, or recycled paper must be applied as part of the hydroseeding operation. Fiber must be added at 11.2 kg per 100 square meter. Then add and mix seed and fertilizer to produce a homogeneous slurry. Seed must be mixed to ensure broadcasting at the rate of 250 kilograms per hectare. When hydraulically sprayed on the ground, material must form a blotter like cover impregnated uniformly with grass seed. Spread with one application with no second application of mulch.

### 3.2.3 Mulching

#### 3.2.3.1 Straw Mulch

Straw mulch must be spread uniformly at the rate of 0.75 metric tons per hectare. Mulch must be spread by hand, blower-type mulch spreader, or other approved method. Mulching must be started on the windward side of relatively flat areas or on the upper part of steep slopes, and continued uniformly until the area is covered. The mulch must not be bunched or clumped. Sunlight must not be completely excluded from penetrating to the ground surface. All areas installed with seed must be mulched on the same day as the seeding. Mulch must be anchored immediately following spreading.

#### 3.2.3.2 Non-Asphaltic Tackifier

Hydrophilic colloid must be applied at the rate recommended by the manufacturer, using hydraulic equipment suitable for thoroughly mixing with water. A uniform mixture must be applied over the area.

### 3.2.4 Erosion Control Material

Install in accordance with manufacturer's instructions, where indicated or as directed by the Contracting Officer.

### 3.2.5 Watering

Start watering areas seeded as required by temperature and wind conditions. Apply water at a rate sufficient to insure thorough wetting of soil to a depth of 50 mm without run off. During the germination process, seed is to be kept actively growing and not allowed to dry out.

## 3.3 PROTECTION OF TURF AREAS

Immediately after turfing, protect area against traffic and other use. Install temporary post and rope barriers. Remove after turf is established and has been mowed at least once in accordance with Section 32 05 22 LANDSCAPE ESTABLISHMENT.

## 3.4 RESTORATION

Restore to original condition existing turf areas which have been damaged

during turf installation operations at the Contractor's expense. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work in adjacent areas is complete.

-- End of Section --



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## SECTION 33 01 50.31

LEAK DETECTION FOR FUELING SYSTEMS  
02/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN PETROLEUM INSTITUTE (API)

API RP 540 (1999; R 2004) Electrical Installations in Petroleum Processing Plants

API RP 2003 (2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents

## ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book

IEEE 1100 (2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 77 (2024; ERTA 1 2023) Recommended Practice on Static Electricity

NFPA 407 (2022; TIA 24-2) Standard for Aircraft Fuel Servicing

NFPA 780 (2023) Standard for the Installation of Lightning Protection Systems

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Leak Detection System; G,  
Electronic Monitoring/Alarm Panel

### SD-03 Product Data

Leak Detection System; G,  
Electronic Monitoring/Alarm Panel

### SD-06 Test Reports

Leak Detection System Test

### SD-07 Certificates

Demonstrations

### SD-08 Manufacturer's Instructions

Leak Detection System

### SD-10 Operation and Maintenance Data

Leak Detection System; G,  
Electronic Monitoring/Alarm Panel; G,

## 1.3 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND EQUIPMENT

Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacturing of such products, that

are of a similar material, design and workmanship, and that have been in satisfactory commercial or industrial use for a minimum 2 years prior to bid opening. Include applications of the equipment and materials under similar circumstances and of similar size. Materials and equipment must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

#### 2.1.1 Nameplates

Attach nameplates to all specified equipment defined herein. List on each nameplate the manufacturer's name, address, component type or style, model or serial number, catalog number, capacity or size, and the system which is controlled. Construct plates of melamine plastic, 3 mm thick, UV resistance, black with white center core, matte finish surface and square corners. Install nameplates in prominent locations with nonferrous screws, nonferrous bolts, or permanent adhesive. Provide nameplates with a minimum size of 25 by 65 mm. Provide normal block style lettering with a minimum 6 mm height. Accurately align all lettering on nameplates. For plastic nameplates, engrave lettering into the white core. Key the nameplates to a chart and schedule for each system. Frame charts and schedule under glass, and locate where directed near each system. Furnish two copies of each chart and schedule. Provide nameplate identifying its function.

#### 2.1.2 Metallic Requirements

Do not construct internal parts and components of equipment, piping, piping components, and valves that could be exposed to fuel during system operation of zinc coated (galvanized) metal. Do not install cast iron bodied valves in piping systems that could be exposed to fuel during system operation.

### 2.2 ELECTRICAL WORK

Provide controllers, integral disconnects, contactors, controls, and control wiring with their respective pieces of equipment. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide switches and devices necessary for controlling and protecting electrical equipment. Provide controllers and contactors that have a maximum of 120-volt control circuits and that have auxiliary contacts for use with the controls provided.

#### 2.2.1 Underground Wiring

Enclose underground electrical wiring in PVC coated conduit. Dielectrically isolate conduit at any steel storage tank connection.

#### 2.2.2 Grounding and Bonding

Perform grounding and bonding in accordance with NFPA 70, NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

### 2.3 LEAK DETECTION SYSTEM

Provide a system, including sensors and detectors, that is intrinsically safe for use in a Class 1, Division 1, Group D environment as defined by

NFPA 70. Provide system compatible with the fuel to be handled. Furnish sensors that distinguish and report the difference between hydrocarbons and water. Provide electronic output and transmission from sensors and detectors. Provide sensors that have a minimum probability of detection of 95 percent and a maximum probability of false alarm of 5 percent. Provide sensors and detectors that are compatible with the electronic monitoring/alarm panel. Reuse sensors after an alarm condition is sensed. Submit shop drawings for the leak detection system that include the following.

- a. Wiring schematics for all parts of the system showing each operating device and listing their normal ranges of operating values (including pressures, temperatures, voltages, currents, speeds, etc.).
- b. Single line diagrams of the entire system, including any required alterations to the existing piping and tank(s).
- c. Diagrams for posting that include distance markings such that alarm indications can be correlated to leak location in plan view. Include a piping and wiring display map with schematic diagrams from the leak detection system manufacturer. Frame diagrams under glass or laminated plastic and be posted where indicated by the Contracting Officer.

#### 2.3.1 Aboveground Vaulted Storage Tanks

Furnish system that continuously and automatically monitors the interstitial space of a vaulted tank for breaches in the integrity of the primary tank and the exterior vaulted shell. Provide a manual access port that can be used to stick the interstice. In addition, provide either (1) a visual method that can automatically monitor the interstitial space such as a Pop-Up gauge, or (2) an electronic method such as an electronic capacitance type liquid sensor.

#### 2.3.2 Underground Piping associated with tanks less than or equal to 50,000 gallons

Provide Leak Detection System associated with tanks less than or equal to 50,000 gallons that continuously and automatically monitors for piping leaks using an automatic line leak detector. Detect a minimum leak rate of 0.003 L/s at 69 kPa line pressure within 1 hour. Detect leaks against a minimum 1.8 m of head pressure. Detect leaks from any portion of the underground product piping.

#### 2.3.3 Monitoring Wells

Continuously and automatically monitor each monitoring well with a vapor sensor. Provide vapor sensor to detect vapors of the fuel to be handled as well as sense the presence of liquid.

#### 2.4 ELECTRONIC MONITORING/ALARM PANEL FOR PIPELINES ASSOCIATED WITH TANKS GREATER THAN 50,000 GALLONS

Perform continuous integrity checks on the status of each sensor's connections and wiring. Include a battery backup (rechargeable) that can operate the complete leak detection system during a power failure for a minimum period of 48 hours. Submit shop drawings of the panel layout along with panel mounting and support details. Provide panel that is

compatible with and connected to the following:

- a. Tank interstitial sensors and detectors.
- b. Automatic line leak detectors.
- c. Monitoring well sensors and detectors.
- d. Digital tank gauge system as defined in Section 33 56 10  
FACTORY-FABRICATED FUEL STORAGE TANKS.

#### 2.4.1 Panel Housing

Provide panel housing that is a NEMA 4 rated enclosure in accordance with NEMA 250. Provide panel housing consisting of a hinged door to swing left or right (doors must not swing up or down).

#### 2.4.2 Panel Alarms

Account for the effects of thermal expansion or contraction of the fuel product, vapor pockets, tank or piping deformation, evaporation or condensation, as well as groundwater levels (if applicable) prior to initiating an alarm condition. Panel must produce an audible and visual alarm in the event any of the following occur.

- a. Sensing of a liquid from a sensor or detector.
- b. Failure disconnection/malfunction of a sensor or detector.

##### 2.4.2.1 Audible Alarm

Panel must have internal speakers that produce a buzzer sound of 70 decibels or greater in the event of a detected alarm condition.

##### 2.4.2.2 Visual Alarm

Provide a visual alarm that illuminates in the event of a detected alarm condition. Include either individual lights for each alarm condition or include a single light and a liquid crystal display (LCD) panel that displaces information regarding each alarm condition.

##### 2.4.2.3 Signal Alarm

Panel must have alarm output for external device connection compatible with input of the Automatic Fill Port system control panel and the Automatic Fuel Polisher in Section 33 57 55 FUEL SYSTEMS PIPING (SERVICE STATION).

#### 2.4.3 Acknowledge Switch

Provide panel with a manual acknowledge switch that will deactivate the audible alarm. Do not deactivate subsequent audible alarms unless depressed manually again for each occurrence. Do not extinguish the visual alarms until the alarm condition has been corrected. Switches must

be an integral component located on the front panel and be either a key switch or push button.

## 2.5 FINISHES

### 2.5.1 Factory Coating

Unless otherwise specified, provide equipment and components fabricated from ferrous metal with the manufacturer's standard factory finish. Each factory finish must be capable of withstanding 500 hours exposure to the salt spray test specified in ASTM B117. For test acceptance, the test specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 3 mm on either side of the scratch mark immediately after completion of the test. For equipment and component surfaces subject to temperatures above 50 degrees C, appropriately design the factory coating for the temperature service.

### 2.5.2 Field Painting

Field paint surfaces not otherwise specified as specified in Section 09 90 00 PAINTING, GENERAL. Do not paint stainless steel and aluminum surfaces. Do not coat equipment or components provided with a complete factory coating. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install parts requiring periodic inspection, operation, maintenance, and repair in locations that allow ready access. Install leak detection system and components in accordance with manufacturer's installation instructions.

#### 3.1.1 Storage Tank Sensors/Detectors

Install interstitial tank sensors and detectors at the tank's low end. Install sensors in accordance with the tank manufacturer's recommendations and do not compromise the tank's secondary containment in any manner. Provide sensors that are easily removed from a tank. Connect metal conduit to steel tanks with dielectric fittings.

#### 3.1.2 Automatic Line Leak Detector

Install detector on discharge side of each submersible pump in accordance with the pump and detector manufacturer's recommendations.

### 3.2 FIELD QUALITY CONTROL

#### 3.2.1 Leak Detection System Test

Activate and test the entire leak detection system in accordance with manufacturer's testing procedures. Use the electronic monitoring/alarm panel to record and present the results.

#### 3.2.2 Storage Tank Tightness Tests

For tanks less than or equal to 50,000 gallons, storage tank tightness tests must be performed in accordance with Section 33 56 10

FACTORY-FABRICATED FUEL STORAGE TANKS. Use the electronic monitoring/alarm panel to record and present the results.

For tanks greater than 50,000 gallons, storage tank tightness test must be performed with a system that is certified to be able to detect leaks at no more than 0.5 gallons per hour.

### 3.2.3 Tank Fill Tests

Perform high liquid level alarm tests on storage tanks in accordance with Section 33 56 10 FACTORY-FABRICATED FUEL STORAGE TANKS. Use the electronic monitoring/alarm panel to record and present the results.

### 3.3 DEMONSTRATIONS

Conduct a training session for designated Government personnel in the operation and maintenance procedures related to the equipment/systems specified herein. Include pertinent safety operational procedures in the session as well as physical demonstrations of the routine maintenance operations. Furnish instructors who are familiar with the installation/equipment/systems, both operational and practical theories, and associated routine maintenance procedures. The training session must consist of a total of 8 hours of normal working time and must start after the system is functionally completed, but prior to final system acceptance. Submit a letter, at least 14 working days prior to the proposed training date, scheduling a proposed date for conducting the onsite training.

-- End of Section --



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## SECTION 33 08 53

AVIATION FUEL DISTRIBUTION SYSTEM START-UP  
08/18

## PART 1 GENERAL

## 1.1 ADMINISTRATIVE REQUIREMENTS

## 1.1.1 System Start-up Plan

Submit a detailed written plan prepared by the System Supplier for implementation of System Start-Up. Submit the plan 90 (OCONUS) days prior to System Start-Up. Include a list of personnel by trade, list of key personnel, safety equipment, list of miscellaneous equipment (such as two-way radios personnel transportation vehicles etc.) and detailed procedures and schedules. The Contractor and System Supplier are responsible for implementing System Start-Up in coordination with ongoing base operations.

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-01 Preconstruction Submittals

System Start-Up Plan; G

Performance Testing Plan; G

## SD-06 Test Reports

Equipment Tests; G

Final Reports; G

Equipment Tests; G

## SD-11 Closeout Submittals

Certification of Entire System

## 1.3 CLOSEOUT SUBMITTALS

## 1.3.1 Final Reports

Submit a final report which includes the final settings of the valves and switches and a copy of the strip chart graphs and excel data and charts on CDR media with an explanation of what the graph indicates and what the system is doing.

## 1.4 QUALITY ASSURANCE

### 1.4.1 Definitions

Subject Matter Expert (SME) is defined as Service Headquarters Subject Matter Experts. SME for this project is Air Force - The Air Force Fuels Facilities Subject Matter Expert (HQ AFCEC/COS) Navy/Marine Corps - NAVFAC POL Facility Subject Matter Expert (NAVFAC EXWC, CI11).

### 1.4.2 Certification of Entire System

Prior to the acceptance of the newly constructed system by the Government, all installed mechanical and electrical equipment must be inspected and approved by the Contracting Officer. Provide the Contracting Officer 60 (overseas) days notice in order to schedule the Installation and SME for participation in the inspection, Performance Testing, and approval. Any deficiencies observed must be corrected by the Contractor without cost to the Government. .

## PART 2 PRODUCTS

### 2.1 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT

The Government will furnish the following materials, equipment and services during the performance of the work under this section.

#### 2.1.1 Aircraft Turbine Fuel

The Contractor will provide the fuel necessary for the System Start-Up. The Contractor must provide an approved copy of the System Start-Up Plan at least 60 days in advance for US locations and 90 days in advance for locations outside the Continental United States to the Government SME before any fuel is provided.

Fuel will not be delivered to the system until the Contractor has satisfactorily completed all work and, in particular, the cleaning and coating of the interior surfaces of the operating storage tanks and the removal of preservatives and foreign matter from those portions coming in contact with the fuel valves, pumps, filter separators and other such equipment.

The Contractor is responsible for reimbursing the Executing Agent (EA) for any loss of fuel and contaminated fuel. The EA will reimburse DLA for lost and contaminated fuel. The Contractor is responsible for the disposal of any contaminated fuel. Some fuel loss, 1 BBL (42 gallons), is expected during Flushing, Cleaning, Equipment Tests, and Performance Testing.

The System Start-Up Plan must specify what grade of fuel, volume (gallons), recommended delivery dates, and phase of System Start-Up (Initial Fuel Receipt, Fuel Filling and Packing, Flushing, Cleaning, Equipment Tests, and Performance Testing). Fuel used for Flushing and Cleaning will possibly become contaminated, the fuel must be isolated and is considered off-specification until the fuel sampling results are provided.

Upon satisfactory completion of the flushing and cleaning operations, the Government will supply the additional quantities of fuel required to complete the other work under this section.

An empty Operating Tank must never be filled at a velocity greater than 1 m/s in the fill line until fuel is in contact with the floater (or 1 m above the fill nozzle).

#### 2.1.2 Utilities

Electric power required for the performance of the work under this section will be furnished at no charge to the Contractor.

### 2.2 MATERIAL AND EQUIPMENT

#### 2.2.1 Contractor Furnished

Provide material, equipment and labor not specified to be Government-furnished and required for proper System Start-Up of the system. Equipment must include but not be limited to the following:

- a. Temporary strainers.
- b. Pipe spools.
- c. Flow meters.
- d. Pressure gages.
- e. Electronic sensors and recorders for pressure and flow recording are included in the PCP, except a sensor and cable or RF will need to be provided by the Contractor for the data from the Hydrant Control Valve and plugged into the PCP. This equipment must be used to monitor and record the system during the "Equipment Test" and "Performance Testing" portions of this Specification Section. Recorded data must be used by the Contractor and equipment factory representatives to achieve final control valve and equipment adjustments. Recorded data must include:
  - (1) Fueling pumps discharge pressures.
  - (2) Supply Venturi flow rates.
  - (3) Hydrant Control Valve pressures.
  - (4) Back Pressure Control Valve upstream pressures.
  - (5) Back Pressure Control Valve downstream pressures.
  - (6) Return Venturi flow rates.
- f. The Contractor must have on hand sufficient filter elements and coalescer cartridges to adequately clean the system. During cleaning operation, provide a flow versus pressure drop graph for each filter separator, as provided in Attachment 1 - Equipment Tests. Change coalescers and cartridges upon reaching a differential pressure of 100 kPa or when pressure drop is less than previous graph or fails to increase properly. Isolate each filter separator, one at a time and use one fueling pump to obtain rated flow rate (38 L/s). A minimum of one complete set of coalescer elements and separator cartridges for each filter separator must be turned over to the Government after new coalescer elements and separator cartridges are installed in each filter separator vessel after completion of Performance Testing.

### 2.2.2 Design Conditions

Use temporary flushing lines and equipment that are equal in strength, stability, and materials to the associated permanent components. However, spools may be carbon steel.

## PART 3 EXECUTION

### 3.1 PREPARATIONS FOR FLUSHING

Upon completion of the system to the satisfaction of the Contracting Officer and the SME, make the following preparations for flushing the system.

#### 3.1.1 Protection of Equipment

The following items must be removed from the system prior to start of flushing operations and, where applicable, replaced with spools of pipe, diameter equal to the item removed.

- a. Control valves, including hydrant pit control valves if flushing outlets into tank trucks.
- b. Sensors which are exposed to the fluid.
- c. Coalescer and separator elements in filter separators.
- d. Venturi Tubes and Pressure Indicating Transmitters.
- e. Meter.

After flushing, the above items must be reinstalled in the system and the spool sections turned over to the Contracting Officer.

#### 3.1.2 Strainers

Temporary 150 um mesh cone type strainers with minimum 300 percent open area must be installed in the suction line ahead of each fueling pump and will be left in place. Any damaged strainers must be replaced by the Contractor at no additional cost to the Government.

#### 3.1.3 Water Draw-off

Remove any accumulated water from Operating Tanks' sumps and bottoms.

### 3.2 FLUSHING

Flushing procedures must precede cleaning procedures. The transfer line, pump house piping, apron loop, supply and return lines to the operating tanks, hydrant laterals, product recovery lines must be flushed with fuel until the fuel being delivered is free of construction debris to the satisfaction of the Contracting Officer. Samples of fuel must be taken and tested by the designated government agency and must be free of gross contamination, maximum of 8.0 mg/gallon solids and free water not to exceed 2 ml per quart.

#### 3.2.1 Fueling System Piping

The flushing of apron system pipelines must be accomplished by pumping fuel from one of the operating tanks through the fueling system piping and back to another tank. Air must be bled from system high points. The procedure must be continued until the fuel being delivered into the tanks is acceptable to the Contracting Officer. After the system has been

flushed to the satisfaction of the Contracting Officer, remove any water remaining in the low point drains and remove any accumulated water from Operating Tank sumps and bottoms by means of the Water Draw-off systems. Cone strainers must be kept clean in order to insure maximum flow rate. In addition, baskets from all strainers must be removed and cleaned.

#### 3.2.1.1 Transfer Line

Flushing of the transfer line must occur during the filling operations. Samples of the incoming fuel must be taken at the point of connection with bulk storage supply line. These samples must be taken at one hour intervals and must be tested by the designated government agency and turned over to the Contracting Officer.

### 3.3 CLEANING

After initial flushing is completed, clean the pump house and apron loop piping in accordance with the procedure specified hereafter. Isolate Operating Tanks from the system and clean as specified per manufacturer's installation instructions.

#### 3.3.1 Preparation for Cleaning

Filter elements must be installed in the filter separators. Adjust filter separator flow control valve. Valves and equipment removed for flushing must be reinstalled. Operating Tanks must be drained, vapor freed and cleaned. Transfer the contents from one operating tank to the other for the purposes of cleaning.

#### 3.3.2 Cleaning Requirements

Cleaning must continue until the Contracting Officer certifies that the fuel passes the color and particle assessment method as defined in T.O. 42B-1-1 or contains 2 milligrams per gallon (mg/gallon) or less of particulate. Fuel must also contain 10 parts per million (ppm) or less of free water. Sampling must be performed by the contractor and testing must be done by an approved independent testing laboratory

#### 3.3.3 Cleaning Procedure

During cleaning procedure periodically bleed air through high point vent and drain water through low point drains.

##### 3.3.3.1 Transfer Line

Continue to receive fuel and circulate it until fuel samples taken at the tanks meet the requirements of paragraph Cleaning Requirements above.

##### 3.3.3.2 Product Recovery Lines

Repeat the process described under initial flushing until samples taken at the connection of the pipe line back to storage meet the requirements.

### 3.4 EQUIPMENT TESTS

After completion of flushing, cleaning, and control valve and electrical components adjusting operations, the tests specified hereinafter must be performed. After cleaning is complete and prior to Performance Testing, Equipment Tests must be performed. Field adjustment of automatic control

valves and automatic pump controls while in operation must be made only by the valve manufacturer's authorized field test engineer. For final adjustment of installed electrical control equipment provide an experienced electrical engineer, factory representative of PCP manufacturer and factory representative of PIT and DPT manufacturers. Both the mechanical and electrical components must be adjusted concurrently. Record required data necessary to prepare Equipment Tests Report.

#### 3.4.1 Equipment Tests

System Supplier must complete and submit the Equipment Tests, see Attachment 1 - Equipment Tests. Submit Equipment Tests prior to Performance Testing (Government approval not required prior to Performance Testing.)

#### 3.4.2 Operating Tank Low Level Alarm

Position valves to transfer fuel between operating tanks. Start one fueling pump and pump sufficient fuel out of the first operating tank to allow the low level alarm (LLA) to stop the fueling pump. This procedure must be repeated for each fueling pump and each tank until the low level alarm stops the fueling pump due to low liquid level in operating tank.

#### 3.4.3 Fuel Delivery

Deliver fuel to each fueling point against a backpressure at the outlet of the hydrant control valve created by the tank trucks and hoses used during the tests. The flow rate must be not less than 38 L/s for a 100 mm valve.

#### 3.4.4 Filter Separator Float Control Valves with Manual Tester

Using the manual float control test level on each Filter Separator, lift the weight from the float ball slowly and observe the Operation and closure of the water slug shut-off feature on the Filter Separator Control Valve.

#### 3.4.5 Overfill Valve

Place fuel transfer pump in the "off" position. Delivery quantity of fuel to Product Recovery Tank to demonstrate capability of valve to close. Place Fuel Transfer Pump in the "Automatic" position to demonstrate capability of valve to open when fuel level drops below set point.

### 3.5 PERFORMANCE TESTING

Performance testing is required to occur after the Contractor has performed the Equipment Tests. Performance testing must demonstrate to the satisfaction of the Contracting Officer and SME these portions of the fueling system are working as specified. Performance testing must consist of repeating the Equipment Tests (indicated in previous paragraphs) and operating the fueling system during actual fueling and defueling operations in the presence of Government Witnesses. The maximum rated capacity of the system must be demonstrated. The Contractor must notify the Contracting Officer 30 calendar days in advance of the test to permit arrangement for the use of Government-furnished items. During the time period of performance testing, no construction activities will be allowed on the project site. The project site must be considered an operational (fuel) zone (verses a construction zone) during this performance testing

period. Personnel, dressed for fuel's operation, will be present to witness testing and participate in Contractor provided training.

### 3.5.1 Final Performance Test

A final performance test must consist of fueling aircraft if the installation has aircraft available.

#### 3.5.1.1 Satisfactory Performance

In the event a portion of the system or any piece of equipment fails to meet the test, make the necessary repairs or adjustments and repeat the Performance Test until satisfactory performance is obtained. The determination of satisfactory performance must be made by the Contracting Officer and the SME.

### 3.5.2 Performance Testing Plan

System Supplier must edit the example Performance Testing plan and submit for approval. An example Performance Testing plan can be found at the end of this Section as Attachment 2 - Performance Testing Plan. Submit plan a minimum of 60 days prior to performance testing.

### 3.5.3 Equipment Tests

System Supplier must provide 10 hard copies of the completed Equipment Tests to the SME at the start of Performance Testing for validation during Performance Testing.

### 3.5.4 Final Acceptance

Fill the system with fuel and operate leak-free prior for acceptance. Anything wet with fuel is considered to be leaking.

#### 3.5.4.1 Operating Tank High Liquid Level Shut-Off Valve Test and Adjustments

During the final filling of operating tanks, check the tank automatic high liquid level shut-off valve for proper functioning at least three times by lowering the fuel level and refilling again. Adjust valve to achieve a safe fill level.

#### 3.5.4.2 Tank Level Indicator Adjustments

Also during the final filling of operating tanks, adjust and calibrate the tank level indicators including the final setting of the high high level (HHLA) and high level (HLA) alarms. Since the HHLA is at a point higher than the High Liquid Level Shut-Off Valve float set point, an artificial method of simulating HHL must be used.

#### 3.5.4.3 Water Draw-Off System Test

During the performance testing, fill Water Draw-off Systems from Operating Tank sump to ensure proper operation. After filling system, allow time for fuel/water mixture to separate. Verify liquid separation through system's sight glasses. Proper operation includes capability to drain separated water and capability to pump separated fuel back to a full Operating Tank.



-- End of Section --

## SECTION 33 09 52

## FUEL PUMP CONTROL AND ANNUNCIATION SYSTEM (NON-HYDRANT)

**11/18**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C37.90 (2005; R 2011) Standard for Relays and Relay Systems Associated With Electric Power Apparatus

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

## INTERNATIONAL SOCIETY OF AUTOMATION (ISA)

ISA 18.1 (2024) Annunciator Sequences and Specifications

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NEMA IA 2 (2005) Programmable Controllers - Parts 1 thru 8

NEMA ICS 1 (2022) Standard for Industrial Control and Systems: General Requirements

NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V

NEMA ICS 4 (2015) Application Guideline for Terminal Blocks

NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

## U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15 Radio Frequency Devices (47 CFR 15)

## UL SOLUTIONS (UL)

|         |                                                                                              |
|---------|----------------------------------------------------------------------------------------------|
| UL 508  | (2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment                 |
| UL 1012 | (2010; Reprint Apr 2016; Rev Mar 2021) UL Standard for Safety Power Units Other than Class 2 |
| UL 1449 | (2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices                     |

## 1.2 ADMINISTRATIVE REQUIREMENTS

- a. Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section, with the additions and modifications specified herein.
- b. Programmable Logic Controllers (PLCs) receive information from pressure transmitters and other devices to control the pumps and control valves.
- c. The control system must be furnished by a single supplier. See Section 33 57 55 FUEL SYSTEM COMPONENTS (NON-HYDRANT) for other required components of the control system. The control system supplier must be responsible for providing a fully functional control system, in accordance with the drawings and specifications, including the field devices. Installation must be in accordance with NFPA 70.
- d. Submit six copies of Operation and Maintenance Manuals, within 7 calendar days following the completion of factory tests. Installation, Operation, and Maintenance manuals for all system components supplied must be furnished following the completion of shop tests and must include:
  - (1) Pump Control Panel including interior and exterior system components layout. Complete guide outlining step-by-step procedures for system startup and operation.
  - (2) All documents previously submitted and approved with all comments and field changes annotated. Complete description of the sequence of operation including that described in PART 3 and any subsystems not controlled by the PLC (e.g. alarm annunciator safety/circuit).
  - (3) Complete listing of all programming of the PLCs, laptop computer, and Personal Computer.
  - (4) Complete relay ladder logic diagrams, PLC input/output diagrams and control power distribution diagrams for the complete control system.
  - (5) Complete troubleshooting guide, which lists possible operational problems and corrective action to be taken, including all as-built conditions.
- e. Submit documents demonstrating the accuracy and completeness of the list of material and components, that items proposed comply fully with contract requirements, and are otherwise suitable for the application indicated. Documents must consist of all data or drawings published

by the manufacturer of individual items listed including manufacturer's descriptive and technical literature, performance data, catalog cuts, and installation instructions. Submit additional material if the listed items are not adequate to identify intent or conformance to technical requirements. Provide typed and electronic copies of these lists for approval. Any delays associated with resubmittals of incomplete or ambiguous initial submittals will be the Contractor's responsibility.

- f. Documents must be bound in a suitable binder adequately marked or identified on the spine and front cover. A table of contents page must be included and marked with pertinent contract information and contents of the manual. Tabs must be provided to separate different types of documents, such as catalog ordering information, drawings, instructions, and spare parts data. Index sheets must be provided for each section of the manual when warranted by the quantity of documents included under separate tabs or dividers.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Shop Drawing; G

#### SD-03 Product Data

Tools and Spare Parts

Pump Control Panel (PCP) and Components; G

Laptop Computer; G

Personal Computer (PC); G

Laser Printer; G

FCC Computer; G

Programmable Logical Controller (PLC) Hardware and Software; G

Control Wiring Data Lists; G

Graphics Display Screen; G

#### SD-06 Test Reports

Certified Pump Control Panel (PCP) Shop Test Report

Record of Test

#### SD-07 Certificates

Experience and Qualifications; G

Testing Plan; G

Training Plan for Instructing Personnel; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

#### 1.4 TOOLS AND SPARE PARTS

Provide the following:

- a. Any special tools necessary for operation and maintenance of the system components providing supplier name, current cost, catalog order number, and a recommended list of spare parts to be stocked.
- b. One spare set of fuses of each type and size.
- c. Recommended manufacturer list of spare parts, including part number, current unit price, and source of supply.
- d. One spare power supply module.
- e. One spare I/O module for discrete devices and one for analog devices.
- f. Two PLC RAM back-up batteries.
- g. Two complete sets of spare ink cartridges for the laser printer.
- h. Minimum of 10 spare lamps for the Alarm Annunciator.
- i. Minimum of 10 spare lamps of each type of non-LED lamps used on the Pump Control Panel.

#### 1.5 EXPERIENCE AND QUALIFICATIONS

Submit the following data for approval:

- a. Certification stating that the manufacturer has manufactured, installed, and successfully completed at least five PLC-based systems for automatic cycling of pumps based upon varying dispensing demands ranging from 0 to 182 L/s utilizing multiple pumps. At least two of the five PLC-based systems must be for dispensing jet fuel into a pressurized, constant pressure, flow demand aircraft hydrant system.
- b. Certification that the control systems have successfully operated over the last 2-years and are currently in service.
- c. Project names, locations, and system description of these installations. Include user point-of-contact and current telephone numbers.

#### 1.6 WARRANTY

The Pump Control and Annunciation System including devices, hardware and software must be warranted for a period of one year from the date of acceptance of the system by the Government. This warranty service must include parts and labor service for system components supplied under this specification. Upon notification by the Government of system or component

failure, respond at the site with necessary parts within 48-hours of notification.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND SYSTEM COMPONENTS

#### 2.1.1 Pump Control Panel (PCP) and Components

NEMA ICS 1, NEMA ICS 6, NEMA 250, and UL 508. The PCP enclosure must be a freestanding NEMA Type 12, smooth, gasketed enclosure constructed of 12 gauge steel. All seams must be continuously welded and there must be no drilled holes or knockout prior to delivery to the job site. The pump control panel dimensions must be a maximum of 2.3 m high, maximum 1.8 m wide, and a maximum of 610 mm deep and must have removable lifting eyes. The interior surfaces of the panel must be properly cleaned, primed, and spray painted with white high-gloss enamel. Exterior surfaces must have standard factory finish. Access for the PCP must be front only and must consist of hinged doors having 3-point latching mechanisms. The doors must open approximately 120 degrees. Rack mounting angles, swing-out panels and other component mounting hardware must be installed such that servicing of one component must not require removal or disconnection of other components. No clearance will be required between the back of the panel and the room walls. Terminal facilities must be arranged for entrance of external conductors from the top or bottom of the enclosure.

#### 2.1.2 Ventilation System

Provide one supply fans, single phase, 115 volt. Each fan must supply a minimum of 47 L/s. The supply and exhaust grill must contain a filter that is easily removed from the exterior of the enclosure. Also provide two thermostats with an adjustable set point range of 21 degrees C to 60 degrees C. Locate the thermostats near the top in the interior of the PCP.

#### 2.1.3 Ground Bar

The control panel must have a tin-plated copper equipment ground bar. The bar must have a minimum of twenty grounding screws.

#### 2.1.4 Standard Indicator Lights

NEMA ICS 1, NEMA ICS 2, and UL 508. Lights must be heavy duty, NEMA 13, 22.5 mm mounting hole, round indicating lights operating at 120 volts ac/dc or 24 volts ac/dc. Long life bulbs must be used. Indicator lights must have a legend plate with words as shown on drawings. Lens color as indicated on the drawings. Lights must be "push to test (lamp)" type. LED type lamps of comparable size and color may be substituted for standard indicator lights.

#### 2.1.5 Selector Switches

NEMA ICS 1, NEMA ICS 2, and UL 508. Non-illuminated lever operated selector switches must be heavy duty, NEMA 13, round, and utilize a 22.5 mm mounting hole. They must have the number of positions as indicated on the drawings. Switches must be rated 600 volt, 10 amperes continuous. Legend plates must be provided with each switch with words as indicated on the drawings.

### 2.1.6 Pushbuttons

NEMA ICS 1, NEMA ICS 2, and UL 508. Non-illuminated pushbuttons must be heavy duty, NEMA 13, round, utilize a 22.5 mm mounting hole, and have the number and type of contacts as indicated on the drawings or elsewhere in the specifications. The emergency stop switch must be a red mushroom head, 38 mm diameter, momentary contact type. Pushbuttons must be rated 600 volt, 10 amperes continuous. Legend plates must be provided with each switch with words as indicated on the drawings.

### 2.1.7 Relays

IEEE C37.90, NEMA ICS 2, UL 508.

### 2.1.8 Nameplates

Nameplates must be made of laminated plastic with black outer layers and a white core. Edges must be chamfered. Nameplates must be fastened with black-finished round-head drive screws or approved nonadhesive metal fasteners.

### 2.1.9 Transient Voltage Surge Suppression Devices

IEEE C62.41 for Category "B" transients, UL 1449.

### 2.1.10 Terminal Blocks

NEMA ICS 4. Terminal blocks for conductors exiting the PCP must be two-way type with double terminals, one for internal wiring connections and the other for external wiring connections. Terminal blocks must be made of bakelite or other suitable insulating material with full deep barriers between each pair of terminals. A terminal identification strip must form part of the terminal block and each terminal must be identified by a number in accordance with the numbering scheme on the approved wiring diagrams.

### 2.1.11 Circuit Breakers

UL 508. Individual, appropriately sized, terminal block mounted, circuit breakers must be provided for all 120 volt PCP mounted system components and for the 120 volt terminal boards shown on the drawings.

### 2.1.12 Uninterruptible Power Supplies

UL 1012. Input voltage must be 120 volts (nominal), 1 phase, 60 Hertz. Output voltage regulation must be plus/minus 5.0 percent for the following conditions:

- a. 20 to 100 percent load on output.
- b. Input voltage variation of minus 15 to 10 percent.
- c. Constant load power factor between 80 and 100 percent.

Response time must be 1.5 cycles or less. Battery capacity must be such as to provide an orderly shut down of operating programs or as a minimum 10 minutes.

### 2.1.13 Miscellaneous Power Supplies

UL 1012. Certain field devices may require power other than 120 VAC (i.e. 24 VDC). The power supplies must be convection cooled, have fully isolated independent outputs, have constant voltage, have short circuit and overvoltage protection, and have automatic current limiting.

### 2.1.14 Alarm Annunciator

UL 508 and ISA 18.1. The Alarm Annunciator must provide visual annunciation, local and remote monitoring, constant or flashing visual and audible alarm as specified herein. The annunciator must be completely solid state with no moving parts. The annunciator must be furnished with cabinet and hardware appropriate for flush mounting on the control panel. A power supply either integral or separately mounted must operate on 120 volts, 60 Hertz. The annunciator must have windows arranged in a matrix configuration (rows and columns). Each window must be at least 24 mm high by 40 mm wide and must have rear illuminated translucent engraved nameplate. Lettering must be at least 4 mm high. System lamp voltage must be 24 to 28 volts dc. The cells must be individually addressable and not hardwired.

### 2.1.15 Alarm Horns

UL 508. The alarm horns must consist of 3-vibrating horns and 2-resonating horns. One vibrating horn is to be mounted in the PCP, and two vibrating and two resonating horns must be mounted outside of the pumphouse as shown on the drawings. The exterior horns must each produce 100 db at 3 m and must be provided in a weather proof housing. The PCP horn must produce 70 db at 3 m.

### 2.1.16 Laptop Computer

#### 2.1.16.1 Hardware

The following are the minimum hardware requirements for the laptop computer:

- a. Latest Core i3 CPU operating at 3.9 GHz or faster
- b. 8 GB RAM
- c. 1 TB hard drive
- d. 16X Read Write DVD drive
- e. 381 mm 1080p LED - Backlit Display
- f. Keyboard
- g. Pointing device (e.g. mouse, track ball)
- h. 120VAC and Battery power supply
- i. All cables and connectors for interfacing with PLC and personal computer
- j. HDMI Port



- k. Two USB 3.0 communications ports
- l. Provide a carrying case for the Laptop Computer

#### 2.1.16.2 Software

The following is the minimum software to be loaded on the laptop. The software must be the most current versions and compatible with each other to make a complete and usable system. All software needs to be fully site licensed (provide with software key) and come with all disks to allow a full restore or reload of software in the event of a hard drive crash.

- a. Operating system (e.g. the latest commercially available MS Operating system)
- b. Software for programming the PLCs
- c. Software for programming the personal computer

#### 2.1.17 Personal Computer (PC)

##### 2.1.17.1 Hardware

The following are the minimum hardware requirements for the personal computer:

- a. Latest Core i3 CPU operating at 2.4 GHZ or faster
- b. 8 GB RAM
- c. 1 TB hard drive
- d. 16X Read Write DVD drive
- e. Color 432 mm 1080p LED-Backlit flat screen monitor
- f. Keyboard
- g. Pointing device (e.g. mouse)
- h. Parallel communication port
- i. Serial communication port compatible with PLC (e.g. RS-232-C, RS-485)
- j. 120 VAC operating power
- k. All cables and connectors interfacing with PLC and Laser Printer
- l. Provide a modem capable of remote troubleshooting of the system. The modem will not be permanently connected to the System
- m. Two USB 3.0 communications ports
- n. Two HDMI ports

##### 2.1.17.2 Software

The following is the minimum software to be loaded on the personal computer. The software must be the most current versions and compatible

with each other to make a complete and usable system. All software needs to be fully site licensed (provide with software key) and come with all disks to allow a full restore or reload of software in the event of a hard drive crash.

- a. Operating system (e.g. the latest commercially available MS Operating System).
- b. Software for programming the PLCs.
- c. The personal computer must communicate with the PLCs to display system status and change system set points. The personal computer must have run-time graphical software to display the graphical screens described later and to change set points.
- d. Software for recording, tracking, trending, and printing out the pressures, flows, and operational status of all monitored components of the fueling system on a real time basis.
- e. MS Office Professional with Excel to allow the trending data described above to be imported to Excel where it can be studied, manipulated, graphed, and easily sent electronically.

#### 2.1.18 Laser Printer

Provide color laser jet alarm/report printer. The unit must print in black at a minimum speed of twelve pages per minute. It must print in color at a minimum speed of ten pages per minute. As a minimum print color graphs of various system pressures, issue flow, and return flow vs. time in seven colors. Provide two sets of spare replacement ink cartridges.

#### 2.1.19 FCC Computer

##### 2.1.19.1 Hardware

The FCC computer must be a copy of the personal computer so that upon failure of the personal computer it could be relocated to the pumphouse to assume the personal computers duties. The normal duties of the FCC computer must be to serve as a remote monitor only of the screens that are available on the personal computer. The following are the minimum hardware requirements for the FCC computer:

- a. Latest Core i3 CPU operating at 3.9 GHZ or faster
- b. 8 GB RAM
- c. 1 TB hard drive
- d. 16X Read Write DVD drive
- e. Color 432 mm 1080p LED-Backlit flat screen monitor
- f. Keyboard
- g. Pointing device (e.g. mouse)
- h. Parallel communication port

- i. Serial communication port compatible with PLC (e.g. RS-232-C, RS-485)
- j. 120 VAC operating power
- k. All cables and connectors interfacing with PLC and Laser Printer
- l. Provide a modem capable of remote troubleshooting of the system. The modem will not be permanently connected to the System
- m. Two USB 3.0 communications ports
- n. Two HDMI ports

#### 2.1.19.2 Software

The following is the minimum software to be loaded on the FCC computer. The FCC computer must be capable of replacing the Personal computer in the pumphouse if the personal computer fails. It will be set up initially to serve only as a remote monitor of the system while located at the FCC. Should the personal computer fail, the FCC computer will be relocated to the pumphouse and then assume the role of the personal computer. The computer software must have a built in command to tell the computer whether it is serving as the personal computer or as the remote monitor only. The software must be the most current versions and compatible with each other to make a complete and usable system. All software needs to be fully site licensed (provide with software key) and come with all disks to allow a full restore or reload of software in the event of a hard drive crash.

- a. Operating system (the latest commercially available MS Operating System).
- b. Software to tell the computer which mode it is to operate in, i.e. (personal computer or remote monitor).
- c. Software to run as a remote monitor.
- d. Software for programming the PLCs.
- e. The personal computer must communicate with the PLCs to display system status and change system set points. The personal computer must have run-time graphical software to display the graphical screens described later and to change setpoints.
- f. Software for recording, tracking, trending, and printing out the pressures, flows, and operational status of all monitored components of the fueling system, on a real time basis.
- g. MS Office Professional with Excel to allow the trending data described above to be imported to Excel where it can be studied, manipulated, graphed, and easily sent electronically.

### 2.2 PROGRAMMABLE LOGICAL CONTROLLER (PLC) HARDWARE AND SOFTWARE

#### 2.2.1 General

- a. NEMA IA 2. PLC must be able to receive discrete and analog inputs and through its programming it must control discrete and analog output functions, perform data handling operations and communicate with

external devices and remote I/O racks. PLC must be a modular, field expandable design allowing the system to be tailored to the process control application. The capability must exist to allow for expansion to the system by the addition of hardware and user software. At a minimum the PLC must include mounting backplanes, power supply modules, CPU module, communication modules, and I/O modules.

- b. Design and test PLC provided for use in the high electrical noise environment of an industrial plant. PLC module must comply with the FCC Part 15 Part A for radio noise emissions. The programmable controller processor must be able to withstand conducted susceptibility tests as outlined in NEMA ICS 2, IEEE C37.90.
- c. PLC must function properly at temperatures between 0 and 50 degrees C, at 5 to 95 percent relative humidity non-condensing and have storage temperatures between minus 40 to 60 degrees C at 5 to 95 percent relative humidity non-condensing.
- d. PLC must have manufacturer's standard system status indicators (e.g. power supply status, system fault, run mode status, back-up battery status).

#### 2.2.2 Central Processing Unit Module

The CPU must be a modular self-contained unit that will provide time of day, scanning, application (ladder rung logic) program execution, storage of the application program, storage of numerical values related to the application process and logic, I/O bus traffic control, peripheral and external device communications and self-diagnostics.

#### 2.2.3 Power Supply Module

- a. The power supply module must be plugged into the backplane not separately mounted. The power supply must be wired to utilize 120 VAC, 60 Hz power, the system must function properly within the range of minus 10 to plus 15 percent of nominal voltage. The power supply must provide an output to the backplane at a wattage and voltage necessary to support the attached modules. A single main power supply module must have the capability of supplying power to the CPU module and local communication and I/O modules. Auxiliary power supplies must provide power to remote racks.
- b. Each power supply must have an integral on/off disconnect switch to the module. If the manufacturers standard power supply does not have an on/off disconnect switch a miniature toggle type switch must be installed near the PLC and clearly labeled as to its function.
- c. The power supply must monitor the incoming AC line voltage for proper levels and have provisions for both over current and over voltage protection. If the voltage level is detected as being out of range the system must have adequate time to complete a safe and orderly shutdown.

#### 2.2.4 Program Storage/Memory Requirements

- a. The PLC must have the manufacturers standard nonvolatile executive memory for the operating system. The PLC must also have EEPROM (Electrically Erasable Programmable Read Only Memory) for storage of the user program and battery backup RAM for application memory. The

EEPROM must be loaded by use of the laptop computer or the personal computer.

- b. Submit a calculation of the required amount of EEPROM and RAM (random access memory) needed for this application plus an extra 50 percent.
- c. The number of times a normally open (N.O.) and normally closed (N.C.) contact of an internal output can be programmed must be limited only by the memory capacity to store these instructions.

#### 2.2.5 Input/Output (I/O) Modules

- a. Provide all required I/O modules (analog input, analog output, discrete input, discrete output, and isolated discrete output) to manipulate the types of inputs and outputs as shown on the drawings and to comply with the sequence of operations. Also provide a minimum of 20 percent (round up for calculation) spare input and output points of each type provided, but not less than 2 of each type.
- b. I/O modules must be a self-contained unit housed within an enclosure to facilitate easy replacement. All user wiring to I/O modules must be through a heavy-duty terminal strip. Pressure-type screw terminals must be used to provide fast, secure wire connections. The terminal block must be removable so it is possible to replace any input or output module without disturbing field wiring.
- c. During normal operation, a malfunction in any remote input/output channel must affect the operation of only that channel and not the operation of the CPU or any other channel.
- d. Isolation must be used between all internal logic and external power circuits. This isolation must meet the minimum specification of 1500 VRMS. Provide optically isolated I/O components which are compatible with field devices.
- e. Each I/O module must contain visual indicators to display ON/OFF status of individual input or output points.
- f. Discrete output modules must be provided with self-contained fuses for overload and short circuit protection of the module.
- g. All input/output modules must be color coded and titled with a distinctive label.

#### 2.2.6 Interfacing

The PLC must have communication ports and communication modules using the manufacturers standard communication architecture for connections of the Personal computer, and Laptop Computer.

#### 2.2.7 Program Requirements

- a. The programming format must be ladder diagram type as defined by NEMA IA 2.
- b. There must be a means to indicate contact or output status of the contact or output on the monitor (of the personal computer) or LCD screen (of the laptop computer). Each element's status must be shown independently, regardless of circuit configuration.

- c. The program must be full featured in its editing capabilities (e.g. change a contact from normally open to normally closed, add instructions, change addresses).

#### 2.2.8 Diagnostics

The CPU must continuously perform self-diagnostic routines that will provide information on the configuration and status of the CPU, memory, communications and I/O. The diagnostic routines must be regularly performed during normal system operation. A portion of the scan time of the controller should be dedicated to perform these housekeeping functions. In addition, a more extensive diagnostic routine should be performed at power up and during normal system shutdown. The CPU must log I/O and system faults in fault tables, which must be accessible for display. When a fault affects I/O or communication modules the CPU must shut down only the hardware affected and continue operation by utilizing healthy system components. All faults must be annunciated on the alarm annunciator.

### PART 3 EXECUTION

#### 3.1 PUMP CONTROL PANEL (PCP) AND COMPONENTS

##### 3.1.1 General

- a. Where two or more system components performing the same function are required, they must be exact duplicates produced by the same manufacturer. All display instruments of each type must represent the same outward appearance, having the same physical size and shape, and the same size and style of numbers, characters, pointers, and lamp lenses.
- b. The PCP must include all required resident software programs and hardware to provide the specified sequence of operation. All software optical discs, including programming manuals, must be turned over to the Government at the completion of start-up so modification can be done in the field with no outside assistance.
- c. It is intended that process controlling devices except field devices, and motor controllers be attached to or mounted within the PCP enclosure and all interconnecting wiring installed prior to shipment to the job site. This is to allow shop testing of the system and to decrease field labor requirements.
- d. The PCP must be shipped fully assembled in one piece after the completion of the shop tests and all defects corrected.

##### 3.1.2 Wiring

Wiring methods and practices must be in conformance with NEMA ICS 1, NEMA ICS 2, NEMA ICS 4, and NEMA ICS 6 recommendations as applicable. All wiring to instruments and control devices must be made with stranded wire, and wiring must be permanently labeled with conductor/wire numbers within one-inch of termination points. Labels must be tubular heat-shrinkable wire markers that remain legible after exposure to industrial fluids and abrasion. Position markers so that wire numbers can be read without disturbing or disconnecting wiring. Use of individual character-markers placed side-by-side is not acceptable. Numbers must match approved shop

drawings. All wiring must be neatly laced from point of entry into enclosures to termination points with nylon lacing cord or plastic lacing ties. Lacing within wiring channels is not required. Where the PCP contains intrinsically safe wiring, barriers or other devices, the panel must be arranged so that all intrinsically safe wiring is separated from non-intrinsically safe wiring by approved methods. Wiring channels containing intrinsically safe wiring must be blue and be labeled as "Intrinsically Safe Wiring Only" on the cover. All conduit penetrations into the PCP containing intrinsically safe wiring must be separated from non-intrinsically safe conduit penetrations by a minimum of 50 mm. Provide typed Control Wiring Data Lists within each terminal cabinet and the PCP. The data lists must include: conductor identification number, wire gauge, wire insulation type, "FROM" terminal identification, "TO" terminal identification, and remarks. The preliminary lists generated by the Contractor will be submitted for approval to the Contracting Officer and will be updated to As-Built conditions by the Contractor. The As-Built data lists must be placed in a document holder within each enclosure.

### 3.1.3 Certified Pump Control Panel (PCP) Shop Test Report

The manufacturer must shop test the PCP, Personal computer, and laptop computer. The procedure must include simulation of field components and must provide for fully testing the pump control and annunciator system as a unit before delivery to the project site. The test must, reveal system defects, including, but not limited to, functional deficiencies, operating program deficiencies, algorithm errors, timing problems, wiring errors, loose connections, short circuits, failed components and misapplication of components. The test must be performed prior to shipment to the site and problems detected must be corrected. The final testing and correction sequence must be repeated until no problems are revealed and then two additional successful tests must be performed. Submit certified test report within 15-days after completion of the test. The report must include a statement that the Pump Control Panel performs as specified. Notify the Contracting Officer and the Service Headquarters 30-days prior to the final shop testing date. The Contracting Officer may require a Government witness at the final test before the PCP is shipped to the site.

### 3.1.4 Ventilation System

Thermostat T-1, must control fan F-1 . T-1 must be set at 27 degrees C to maintain interior air temperature to minus 7 degrees C above ambient. Thermostat T-2 , set at 38 degrees C, must provide a non-critical PCP HIGH TEMPERATURE alarm to the alarm annunciator.

### 3.1.5 Grounding

The PCP ground bar must be connected to the building counterpoise via a #10 AWG conductor. Within the enclosure all I/O racks, processor racks, and power supplies must be grounded to meet the manufacturer's specifications.

### 3.1.6 Indicator Lights, Switches, and Pushbuttons

Indicator lights, switches, and pushbuttons must be mounted through the PCP enclosure and must be arranged to allow easy vision and operation of each device. Each device must have a nameplate and legend plate as indicated on the drawings. Nameplate wordings must be as indicated on the

drawings.

### 3.1.7 Surge Protective Devices

Surge protective devices (SPD) must be installed in the PCP to minimize effects of nearby lightning strikes, switching on and off of motors and other inductive loads. SPD must be provided for each control circuit ladder. Each ladder may contain any combination of the following devices: PLCs, power supplies (e.g., 24 volt), fans, relays, lights, switches. SPD must also be provided for PLC I/O originating outside of the building.

### 3.1.8 Terminal Blocks

As a minimum, any PCP device that connects to a field device (devices not located in the PCP) must be connected to a terminal block. A connection diagram similar to the drawings must be provided to the field Contractor for field connections to the PCP.

### 3.1.9 Circuit Breakers

As a minimum, any 120 volt PCP device i.e. (fans, lights, power receptacles, 24 VDC power supplies, PLC CPUs, PLC I/O racks) must be provided with an individual circuit breaker. Additionally 120 volt terminal boards connecting to field devices (devices not located in the PCP) must be protected by a 120 volt circuit breaker.

### 3.1.10 Uninterruptible Power supplies

The PCP must contain two uninterruptible power supplies (UPS) (each connected to a dedicated circuit). As shown on the drawings one UPS must supply the PLC System, and the second UPS must supply the miscellaneous device power. The UPSs output capacity must be sufficient to drive all the system components connected plus 25 percent. The UPSs must be mounted on shelves near the bottom of the PCP but not rest on the floor of the PCP.

### 3.1.11 Power Supplies

Provide and install all 120 VAC and 24 VDC power supply. Interconnecting wiring between UPSs and PLC power supplies must be completely installed prior to shipment to the job site.

### 3.1.12 Alarm Annunciator and Horns

Signals must be initiated by hardwired field contacts or by PCP outputs as required. The annunciator must energize alarm horns, both an integral panel mounted vibrating horn and remote horns, and flash the appropriate annunciator lamp. The minimum number of windows must correspond to the number of alarm points, plus 15 percent spare. The drawings indicate panel layout and the alarms to be annunciated.

#### 3.1.12.1 Non-critical Alarms

Non-critical alarm windows must be white with black lettering and must sound the PCP mounted vibrating horn and the exterior mounted vibrating horns.



### 3.1.12.2 Critical Alarms

Critical alarm windows must be red with white lettering and must sound the PCP mounted vibrating horn and the exterior mounted resonating horns. Critical alarms must also cancel all automatic pump starts in the PLC.

### 3.1.12.3 Alarm Sequence

Alarm sequence for each alarm must be as follows (ISA 18.1 sequence 'A').

- a. For a normal condition, visual indicator and horns will be off.
- b. For an alarm condition, visual indicator will flash, and horns will sound (this condition will be locked in).
- c. Upon acknowledgment of the alarm condition, visual indicator will be steady on and the horns will be off.
- d. If, after acknowledgment of an alarm condition, another alarm condition is established, the new alarm will cause the appropriate window to flash and the horn to sound.
- e. When condition returns to normal after acknowledgment, the visual indicator and the horn will be off.

### 3.1.13 Personal Computer

The personal computer must be a stand alone, desk top mounted unit. The personal computer must download system parameters from the PLCs for display. The personal computer must also upload new set point values that the operator has changed using the personal computer keyboard, after a password has been entered.

#### 3.1.13.1 Screen Number 1

This must be a general opening screen. As a minimum it must display the name and location of the installation (e.g. Selfridge Air National Guard Base, Michigan), name of the project (e.g., Aviation Fuel Farm) and screen navigation information.

#### 3.1.13.2 Screen Number 2

Screen number two must list the systems parameters monitored by the PCP and their current value. The values must be continuously updated, a 2 second delay maximum between updates will be acceptable.

| Parameter | Value |
|-----------|-------|
|           |       |
|           |       |
|           |       |

Only one of the words separated by a slash (/) must be displayed. The xxxxx.x HOURS is the fuel pumps elapsed run time and the value must not be lost when the lead PLC is switched. The pump and valve status words must be color coded to match the colors used on the graphic display screen.

#### 3.1.13.3 Screen Number 3

The following table must be displayed. The table lists the set points that can be adjusted using the operator interface. A password must be entered before the "current value" can be adjusted. The value entered can only be a number within the "set point range". The "default value" is the value held in the program that is loaded into EEPROM memory (This screen may require more than one display screen).

| SET POINT DESCRIPTION | SET POINT RANGE | DEFAULT VALUE | CURRENT VALUE |
|-----------------------|-----------------|---------------|---------------|
|                       |                 |               |               |
|                       |                 |               |               |
|                       |                 |               |               |
|                       |                 |               |               |
|                       |                 |               |               |

#### 3.1.13.4 Screen Number 4

This screen must be a duplicate of the Graphic Display Drawing showing a schematic of the process flow. This screen must be referred to as the graphical display. Many operating parameters must be displayed here as required in later paragraphs of this specification.

#### 3.1.13.5 Screen Number 5

This screen must be a duplicate of the Alarm Annunciator and it must be superimposed over the current active screen on the personal computer monitor when an alarm is activated.

#### 3.1.13.6 Screen Number 6

This screen must be a screen designed solely for assisting the testing team during initial start up to watch all of the significant parameters of the systems operation simultaneously on one screen. This screen must include the system parameters i.e. (flows, pressures, status) from screen 2, the set points from screen 3, and timers for all of the actions that

will take place following a delay function.

#### 3.1.13.7 Screen Number 7

This screen must be an alarm history screen. This screen must be referred to as the Alarm History Display. This screen must be capable of storing and displaying all alarms that have occurred in the system for at least a period of 30-days.

#### 3.1.13.8 Screen Number 8

This screen must be a screen designed solely for displaying the parameters and process involved in the Tightness Test as described in this specification and on the drawings. The following values must be displayed concurrently against time: Pressure (as sensed by PIT3). The system will be able to produce graphs on the screen of this data and be able to print the data in color on the laser printer.

#### 3.1.14 Laptop Computer

The Laptop computer must be used to create, edit, and load the ladder logic program into the PLCs and the operator interface graphics control program into the personal computer. The Laptop must also be used to monitor the PLCs memory and ladder logic program. The computer must be stored in a lockable cabinet located within the Pump Control Panel.

### 3.2 PROGRAMMABLE LOGICAL CONTROLLER (PLC) HARDWARE AND SOFTWARE

#### 3.2.1 General

The basic operation of the PLC system is (Reference "Control System Block Diagram" on the drawings):

- a. The CPU and its associated I/O rack (I/O-1) sends system outputs to appropriate devices and receive input signals from field devices (PITs-1, DPTs-1, DPTs-3, flow switches, valve limit switches).

#### 3.2.2 Programs

- a. Provide two copies of all working programs (i.e. PLC logic, personal computer) on read only optical discs as well as a printed program listing.
- b. Provide rung comments (documentation) in the ladder logic program. Each device, on the ladder, must be identified as to the type of device, i.e. limit switch XX, flow indicator XX, motor starter XX. Rung comments must be provided for input and output rungs. The programmer must also provide a comment describing the function of each rung or group of rungs that accomplish a specific function.

### 3.3 GRAPHICS DISPLAY SCREEN

#### 3.3.1 General

The graphic display screen shall be capable of being displayed on the personal computer monitor.

### 3.3.2 Display Presentation

The Graphic Display shall depict the process fuel flow schematically as indicated on the drawings. Red, green, and amber symbols shall be integrated with the process schematic to provide current equipment status graphically. The symbols shall be located immediately adjacent to related equipment symbol.

### 3.3.3 Process Schematic

The process schematic graphic representation shall utilize conventional symbols when possible. Symbols and flow lines shall be sized and spaced so as to provide a clear representation of the system process. The Graphic Display shall be suitable for supervised field modification when future items are added. Minor changes may be incorporated to allow proper line width and spacing. Component arrangement, piping routing, and location of valves shall match the flow diagram. The Graphic Display layout shall be approved by the Government.

### 3.3.4 Digital Flow and Pressure Indicators

The graphics display screen shall have digital displays for the flows and pressures as indicated on the drawings.

## 3.4 INSTALLATION

Installation must conform to the manufacturer's drawings, written recommendations and directions.

### 3.4.1 Shop Drawing

The shop drawing must be clear and readable and preferably drawn using a computer aided drafting package. At the conclusion of the project the diagram drawings must be redrafted to include all as-built conditions. These updated drawings must be included in the O&M Manuals and appropriate section of the drawings placed in a data pocket located in each of the enclosures. The shop drawing at a minimum must show:

- a. Overall dimensions, front, side and interior elevation views of the PCP showing size, location and labeling of each device.
- b. Overall dimensions, front elevation of the GDP showing graphical layout and size, location and labeling of each device.
- c. Power ladder diagram indicating power connections between SDP, power conditioners, PLCs, power supplies and field and panel devices. Any terminal block connection numbers used must be indicated.
- d. Control ladder diagram indicating control connections between field and devices and PLC I/O modules. Terminal block connection numbers and PLC terminal numbers must be indicated.
- e. Communication connections between PLCs and I/O racks. Communication channel numbers must be indicated.
- f. Bill of materials.
- g. Written control sequence covering all inputs, outputs, and control scheme.

### 3.4.2 System Start-Up and Testing

- a. At PCP start-up and testing provide personnel, onsite, to provide technical assistance, program fine tuning, and to start-up and test the system. Start-up and testing must be coordinated with the overall fueling system start-up test specified in Section 33 08 53 AVIATION FUEL DISTRIBUTION SYSTEM START-UP. Prior to this test, all connections must have been made between the PCP, the personal computer, the motor control center, and all field devices. In addition, wiring must have been checked for continuity and short circuits. Adjust set point values, timing values, and program logic as required to provide a functional hydrant fuel control system. Once the system has been fine tuned and passed the system test, the new system default values, must be loaded into the PLC EEPROM and the personal computer screens adjusted to indicate the new values.
- b. A step-by-step Testing Plan of the PCP must be submitted. The test must be designed to show that every device (lights, switches, personal computer display screens, alarms) on the PCP and personal computer is in working order and that the PLC program controls the system per specifications. The test must be performed in conjunction with Section 33 08 53 AVIATION FUEL DISTRIBUTION SYSTEM START-UP. The plan must include a place for the Contractor and government representative to initial each step of the plan after satisfactory completion and acceptance of each step. The complete initialed Record of Test must be certified by the Contractor and then submitted.

### 3.4.3 Training Plan for Instructing Personnel

- a. Upon completion of the system start-up a competent technician regularly employed by the PCP manufacturer must hold a training class for the instruction of Government personnel in the operation and maintenance of the system. Provide both classroom type theory instruction and hands-on instruction using operating system components provided. The period of instruction must be a minimum of three 8-hour working days. The training must be designed to accommodate 8 operators, 4 maintenance personnel, and 2 programmers. The Contracting Officer must receive written notice a minimum of 14-days prior to the date of the scheduled classes.
- b. Furnish a written lesson plan and training schedule for Government approval at least 60-days prior to instructing operating, maintenance and programming personnel. Concurrently submit above to the Service Headquarters or officially designated alternate for their input into the review process. Approval of lesson plan will be based on both Government and Service Headquarters' concurrence. This plan must be tailored to suit the requirements of the Government. The training must be divided into three separate classes. Each class must be tailored to a specific group of personnel. The groups are: 1) Operators, those that will use the control system on a day-to-day basis; 2) Maintenance personnel, those that will perform routine and non-routine maintenance and trouble shooting of the control system; 3) Programmers, those that will make changes to and trouble shoot the PLC and personal computer programs. The training program must provide:
  - (1) a detailed overview of the control system including the complete step-by-step procedures for start-up, operation and shut-down of the control system.

- (2) a general overview of programmable logic controllers.
  - (3) the maintenance of system components installed.
  - (4) the programming of the PLC and Personal Computer.
  - (5) troubleshooting of the system.
- c. Complete approved Operation and Maintenance manuals for this specification and 26 20 00 INTERIOR DISTRIBUTION SYSTEM (specifically pertaining to the motor control center and its relay ladder diagrams) must be used for instructing operating personnel. Training must include both classroom and hands-on field instruction. The class must be recorded in DVD format.
  - d. Provide training courses in DVD format covering system overview, operation, maintenance, trouble shooting, and programming. These DVDs must be produced onsite by the Contractor using the supplied Pump Control Panel as the teaching aid, or commercially produced DVDs by the PLC manufacturer or third party who specializes in training on PLC systems. Along with the DVDs, provide workbooks, which follow along with the DVDs.

### 3.5 PLC CONTROL SYSTEM SEQUENCE OF OPERATION

The following describes general functions of the fueling system components.

#### 3.5.1 Abbreviations

- a. PCP: Pump Control Panel.
- b. PC: Personal Computer.
- c. UPS: Uninterruptible Power Supply.

#### 3.5.2 Operating Tanks

##### 3.5.2.1 Level Control

Each operating tank has four level float switches to measure low-low, low, high and high-high levels.

##### 3.5.2.1.1 Low-Low Level

When the low-low level float is activated the associated tank's graphic display low-low level light must light up. If the selected tank's level drops below the low-low sensor the alarm annunciator's low-low level alarm sequence activates, issue pumps running in automatic mode must be disabled. If all tanks are at low-low level, no issue pumps in automatic mode must be enabled.

##### 3.5.2.1.2 Low Level

When the low level float is activated the associated tank's graphic display low level light must light up and the alarm annunciator's non-critical low level alarm sequence activates.

#### 3.5.2.1.3 High Level

When the high level float is activated the associated tank's graphic display high level light must light up and the alarm annunciator's non-critical high level alarm sequence activates.

#### 3.5.2.1.4 High-High Level

When the high-high level float is activated the associated tank's graphic display high-high level light must light up, the alarm annunciator's critical high-high level alarm sequence activates, the pump control panel must de-energize the solenoid on the tank's high level shutoff valve to force it closed.

#### 3.5.2.2 Level Control

Each operating tank has level switches to monitor low-low, low, high, and high-high fuel levels.

##### 3.5.2.2.1 Low-Low Level

When the low-low level elevation is attained the associated tank's GDP low-low level light must light up. If the selected tank's level drops below the low-low sensor the alarm annunciator's low-low level alarm sequence activates, issue pumps running in automatic mode must be disabled. If all tanks are at low-low level, no issue pumps in automatic mode must be enabled.

##### 3.5.2.2.2 Low Level

When the low level elevation is attained the associated tank's GDP low level light must light up and the alarm annunciator's non-critical low level alarm sequence activates.

##### 3.5.2.2.3 High Level

When the high level elevation is attained the associated tank's GDP high level light must light up and the alarm annunciator's non-critical high level alarm sequence activates.

##### 3.5.2.2.4 High-High Level

When the high-high level elevation is attained, the associated tank's GDP high-high level light must light up and the alarm annunciator's critical high-high level alarm sequence activates. The pump control panel must de-energize the solenoid on the tank's high level shutoff valve to force it closed.

#### 3.5.2.3 Outlet Valve

Each operating tank's outlet valve has two limit switches to indicate valve position. The closed limit switch is DPDT. The closed limit switch must close when the valve is fully closed. When the closed limit switch is closed the associated tank's valve graphic display closed light must activate. When the valve is fully open, the open limit switch is closed. At this time the associated tank's valve graphic display open light must activate.

### 3.5.3 Product Recovery Tank

#### 3.5.3.1 Fuel Transfer Pump (FTP)

The pump's motor controller has a status relay to indicate the on/off status of the pump. When status relay is open the pump's graphic display off light must activate. When the status relay is closed the pump's graphic display on light must light up. The status relay state must also be used to start and stop the pumps elapsed run time timer.

#### 3.5.3.2 Overfill Valve (OV)

The tank's overfill valve has a limit switch to indicate valve position. The switch is SPST. The switch must close when the valve is fully closed. When the limit switch is closed, the tank's graphic display valve closed light must light up and the alarm annunciator's non-critical alarm sequence activates. When the limit switch is open the tank's graphic display valve open light must light up.

#### 3.5.3.3 High Level Alarm

The tank has a high level alarm float switch. The switch is SPST. When the high level alarm float is activated the tank's graphic display high level light must light up and the alarm annunciator's critical alarm sequence activates.

#### 3.5.3.4 High-High Level Alarm

The tank has a high-high level alarm float switch. When the high-high level alarm float is activated the tank's graphic display high-high level light must light up and the alarm annunciator's critical alarm sequence activates.

#### 3.5.3.5 Leak Detection

The tank has a leak detection system. When the leak alarm is activated the alarm annunciator's non-critical alarm sequence activates.

### 3.5.4 Fueling Pumps (FP)

There are three fueling pumps with a maximum of two pumps running at one time. The lead pump selector switch must select the pump starting sequence. Each pump's motor controller has a status relay to indicate the on/off status of the pump. When status relay is open the associated pump's graphic display off light must activate and screen number 2 must indicate on. When the status relay is closed the associated pump's graphic display on light must activate and screen number 2 must indicate off. The status relay state must also be used to start and stop the pumps elapsed run time timer and must be displayed on screen number 2.

### 3.5.5 Flow Switch, Fueling Pump

On the discharge side of each pump is a flow switch to indicate positive flow (fail safe feature). If the PLC has given a signal to start a pump and the flow switch has not closed before the set point timer expires or if the flow switch opens after the pump has been running then the pump must be in a failure state and it must be disabled (taken out of the starting sequence), the alarm annunciator's non-critical alarm sequence must also be activated, and the next pump in the start sequence started.



After the PLC has stopped all of the pumps, any failed pump must be added back into the start sequence.

### 3.5.6 Transmitters

#### 3.5.6.1 Differential Pressure Transmitter (DPT)

The DPT's measure flow in L/s. The net flow is sent to the personal computer display. The issue rate, return rate and net flow must be displayed on the personal computer.

### 3.5.7 Safety Circuit

#### 3.5.7.1 Emergency Stop Status

The emergency stop circuit status relay (ER1) N.O. contact must be connected to I/O-1, I/O-2 and UPS#2 as indicated on the Terminal Block Connection drawing. When the circuit is activated the alarm annunciator's critical alarm sequence is activated and any calls to start fueling pumps must be canceled and no additional pump start signals must be sent until the circuit has been reset. The fueling pumps will actually be stopped by an emergency stop circuit status relay (ER2) N.O. contact in the fuel pump motor control circuit located in the motor control center.

#### 3.5.7.2 Emergency Shutoff Valves (ESO) Status

The ESO status relay (ER2) N.O. contact must be connected to I/O and UPS#2 as indicated on the Terminal Block Connection drawing. When the ESO status relay is closed the Filter Separator control valves (FSV) closed lights must light up.

#### 3.5.7.3 Circuit Power Status

When the safety circuit power status relay is closed the PCP emergency circuit power on light must light up.

### 3.5.8 Pump Control Panel

#### 3.5.8.1 CPU Faults

The PCP mounted CPU on light is connected to SYS. The associated CPU light must light when no system faults are detected. When a fault is detected by the CPU the faulted CPU's on light must be turned off and the alarm annunciator's non-critical alarm sequence must be activated.

#### 3.5.8.2 Mode Select Switch

The 2-position switch selects what mode of fueling is active: issue, or off. The screen number 2 status must indicate the active mode.

#### 3.5.8.3 Lead Pump Selector Switch

The 3-position switch selects which pump must be the lead pump. The switch position must fix the starting sequence for all pumps. The sequences must be 1-2-3, 2-3-1, and 3-1-2, 1-2-3. The off sequence must be the reverse of the start sequence, therefore, first on will be last off. A maximum of two pumps will be allowed to run at one time. If a pump fails to start or fails during operation, that pump will be disabled and the next pump in the sequence started. The screen number 2 status

display must indicate the lead pump.

#### 3.5.8.4 PCP Temperature Alarm

The alarm thermostat when activated must activate the alarm annunciator's non-critical alarm sequence.

### 3.6 OPERATING PROGRAM REQUIREMENTS

The control system's logic program must be stored on a EEPROM chip. Default values of operator adjustable parameters must be permanently stored on the chip with the capability of resetting the values in RAM to the values within the range specified below. The default values can be changed through the use of the personal computer (after the correct password has been entered). After loss of power and battery failure the adjustable settings must revert back to the default values located on the chip. The default values shown here must be reset to the values determined during the system start up and test.

| SET POINT DESCRIPTION                                                                                                                                                 | SET POINT RANGE  | DEFAULT VALUE |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------|
| pump starting pressure                                                                                                                                                | 345 to 552 kPa   | 65 psi        |
| Timer to enable start-up of pump                                                                                                                                      | 0 to 120 seconds | 15 seconds    |
| System pressure to stop pump                                                                                                                                          | 345 to 552 kPa   | 75 psi        |
| Timer to establish fueling pump failure                                                                                                                               | 5 to 30 seconds  | 15 seconds    |
|                                                                                                                                                                       |                  |               |
| Should the operator enter a value not within the range for that parameter, the personal computer must indicate "INVALID ENTRY" and revert back to the previous value. |                  |               |

| SET POINT DESCRIPTION                                                                                                                                                                 | SET POINT RANGE | DEFAULT VALUE |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|
| A number inside braces, {x}, in the following paragraphs indicates that the number may be changed by the operator via the operator interface within the Set Point Range listed above. |                 |               |

### 3.7 AUTOMATIC MODE - ISSUE CONDITION

- a. The lead pump will start when a Pump Control Station start pushbutton is depressed.
- b. Depressing an additional Pump Control Station pushbutton will bring on additional pumps.
- c. Depress the Pump Control Station stop pushbutton will disable the lead pump.
- d. Depressing an additional Pump Control Station stop pushbuttons will disable the pumps in the order they were brought on.

### 3.8 TIGHTNESS TEST MODE

This mode must be used in conjunction with the Tightness Monitoring Panel provided by Section 33 57 55 FUEL SYSTEM COMPONENTS (NON-HYDRANT) to perform tightness tests. Placing the selector switch to "TIGHTNESS TEST" the PCP will send a signal to the Tightness Monitoring Panel telling it that it is ready to perform the tests. At this time it will also operate MOV valves, closing and opening as required to run tests. The PCP will then receive signals from the Tightness Monitoring Panel to prepare for High Pressure Test, run High Pressure Test, Prepare for Low Pressure Test, run Low Pressure Test, prepare for 2nd High Pressure Test, run 2nd High Pressure Test, and when the test is over. The following PCP actions will occur after the corresponding signal:

#### Prepare for High Pressure Test:

- a. Automatically start the pump to obtain pressure.
- b. When the pump is started, a 15 second timer must start. If the timer expires before the flow switch closes the alarm annunciator's associated non-critical alarm sequence must activate.
- c. If the pump's flow switch opens after the pump successfully started the alarm annunciator's associated non-critical alarm sequence must activate.
- d. If the pump fails to establish flow, automatically start the lead fueling pump to obtain pressure.
- e. When a fueling pump is started, a 15 second timer must start. If the timer expires before the flow switch closes the alarm annunciator's associated non-critical alarm sequence must activate.
- f. If a fueling pumps flow switch opens after the pump successfully started the alarm annunciator's associated non-critical alarm sequence

must activate.

- g. MOV will be opened.

Run High Pressure Test:

- a. MOV will actuate as determined by the Tightness Test system manufacturer.
- b. Fueling pump will be shut off.

Prepare for Low Pressure Test:

- a. MOV will actuate as determined by the Tightness Test system manufacturer.
- b. The system will remain in this status until such time as the PCP receives a Run Low Pressure test signal from the Tightness Monitoring Panel. Note: The Tightness Monitoring Panel will monitor the loop pressure until it reaches the 345 kPa value. It will then instruct the PCP to run the Low pressure test.

Run Low Pressure Test:

- a. MOV will actuate as determined by the Tightness Test system manufacturer.
- b. The system will remain in this status until such time as the PCP receives a Prepare for 2nd High Pressure test signal from the Tightness Monitoring Panel. Note: The Tightness Monitoring Panel will wait for a ten minute settling period to expire, then it will monitor the loop pressure for two minutes. Upon finishing this test it will instruct the PCP to prepare for 2nd High Pressure Test.

Prepare for 2nd High Pressure Test:

- a. Automatically start the pump to obtain pressure.
- b. When a fueling pump is started, a 15 second timer must start. If the timer expires before the flow switch closes the alarm annunciator's associated non-critical alarm sequence must activate.
- c. If a fueling pumps flow switch opens after the pump successfully started the alarm annunciator's associated non-critical alarm sequence must activate.
- d. MOV will actuate as determined by the Tightness Test system manufacturer.
- e. When the pump is started, a 15 second timer must start. If the timer expires before the flow switch closes the alarm annunciator's associated non-critical alarm sequence must activate.
- f. If the pump's flow switch opens after the pump successfully started the alarm annunciator's associated non-critical alarm sequence must activate.
- g. If the pump fails to establish flow, automatically start the lead fueling pump to obtain pressure.

- h. When a fueling pump is started, a 15 second timer must start. If the timer expires before the flow switch closes the alarm annunciator's associated non-critical alarm sequence must activate.
- i. If a fueling pumps flow switch opens after the pump successfully started the alarm annunciator's associated non-critical alarm sequence must activate.
- j. MOV will actuate as determined by the Tightness Test system manufacturer.
- k. Pump will continue to run until such time as the run 2nd High Pressure test signal is received. Note: The Tightness Monitoring Panel is monitoring the Loop pressure and when it is satisfied that it is high enough it will instruct the PCP to Run the 2nd High Pressure test.

Run 2nd High Pressure Test:

- a. MOV will actuate as determined by the Tightness Test system manufacturer.
- b. Fueling pump will be shut off.
- c. The PCP will leave the system as is until such time as the PCP selector switch is placed into a different mode.

3.9 OFF MODE

- a. Automatic starting of fueling pumps must be disabled. All other functions (graphic display, alarm annunciator, personal computer, control valve solenoids) must be active to allow manual control of the fueling pumps using the Hand-Off-Auto or Hand-Auto switch.
- b. The second, fueling pumps maybe started or stopped manually as needed by the operator.

3.10 MANUAL OPERATION OF FUELING PUMPS

- a. If the PLC system is still active see paragraph OFF MODE.
- b. If the PLC system has no power or the CPUs have faulted (CPU light on PCP off) the pumping system will be in a completely manual mode. The safety circuit will need power so that the ESO solenoids on the non-surge check valves will be open and fuel can flow. The solenoids on the other solenoid controlled valves will be de-energized so the valves will have to be manually opened or enabled for the system to run. Other valves may need to be opened or closed manually by the operators for the system to work properly.

-- End of Section --

## SECTION 33 11 00

## WATER UTILITY DISTRIBUTION PIPING

**02/18, CHG 2: 11/23**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## CSA GROUP (CSA)

|              |                                                                                      |
|--------------|--------------------------------------------------------------------------------------|
| CSA A257.2   | (2019) Standards for concrete pipe and manhole sections                              |
| CSA B51      | (2024) Boiler, pressure vessel, and pressure piping code                             |
| CSA B125.1   | (2018) Plumbing supply fittings                                                      |
| CSA B125.3   | (2022) Plumbing fittings                                                             |
| CSA B137     | Thermoplastic pressure piping                                                        |
| CSA B137.1   | (2020) Polyethylene (PE) pipe, tubing, and fittings for cold-water pressure services |
| CSA B137.3   | (2020) Rigid polyvinylchloride (PVC) pipe and fittings for pressure applications     |
| CSA B137.4   | (2020) Polyethylene (PE) piping systems for gas services                             |
| CSA B137.12  | (2023) Polyamide (PA) piping systems for gas services                                |
| CSA B131     | (1999) External Zinc-Coating of Ductile Iron Pipe (Adopted ISO 8179:1985)            |
| CSA B181.2   | PVC Drain, Waste, and Vent Pipe and Pipe Fittings                                    |
| CSA B483.1   | Drinking water treatment systems                                                     |
| CAN/ULC-S560 | (2021) Standard For High-Performance And Aviation Foam Liquid Concentrates           |

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

|              |                                                                                            |
|--------------|--------------------------------------------------------------------------------------------|
| AASHTO HB-17 | (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges |
|--------------|--------------------------------------------------------------------------------------------|

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                 |                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------|
| ASME B1.20.1    | (2013; R 2018) Pipe Threads, General Purpose (Inch)                                                                    |
| ASME B1.20.3    | (1976; R 2003; R 2008; R 2013; R 2018; R 2023) Dryseal Pipe Threads (Inch)                                             |
| ASME B16.1      | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250                                            |
| ASME B16.18     | (2021) Cast Copper Alloy Solder Joint Pressure Fittings                                                                |
| ASME B16.26     | (2018) Standard for Cast Copper Alloy Fittings for Flared Copper Tubes                                                 |
| ASME B18.2.2    | (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series) |
| ASME B18.5.2.1M | (2006; R 2011) Metric Round Head Short Square Neck Bolts                                                               |
| ASME B18.5.2.2M | (1982; R 2010) Metric Round Head Square Neck Bolts                                                                     |

## AMERICAN WATER WORKS ASSOCIATION (AWWA)

|                  |                                                                                  |
|------------------|----------------------------------------------------------------------------------|
| AWWA B300        | (2024) Hypochlorites                                                             |
| AWWA B301        | (2024) Liquid Chlorine                                                           |
| AWWA C104/A21.4  | (2022) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water         |
| AWWA C110/A21.10 | (2021) Ductile-Iron and Gray-Iron Fittings                                       |
| AWWA C111/A21.11 | (2023) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings          |
| AWWA C115/A21.15 | (2020) Flanged Ductile-Iron Pipe With Ductile-Iron or Gray-Iron Threaded Flanges |
| AWWA C151/A21.51 | (2017) Ductile-Iron Pipe, Centrifugally Cast                                     |
| AWWA C153/A21.53 | (2019) Ductile-Iron Compact Fittings for Water Service                           |
| AWWA C219        | (2017) Bolted Sleeve-Type Couplings for Plain-End Pipe                           |
| AWWA C500        | (2019) Metal-Seated Gate Valves for Water Supply Service                         |
| AWWA C502        | (2018) Dry-Barrel Fire Hydrants                                                  |
| AWWA C503        | (2021) Wet-Barrel Fire Hydrants                                                  |

|           |                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| AWWA C504 | (2023) Standard for Rubber-Seated Butterfly Valves                                                                                        |
| AWWA C508 | (2017) Swing-Check Valves for Waterworks Service, 2 In. Through 48-In. (50-mm Through 1,200-mm) NPS                                       |
| AWWA C509 | (2023) Resilient-Seated Gate Valves for Water Supply Service                                                                              |
| AWWA C512 | (2015) Air-Release, Air/Vacuum, and Combination Air Valves for Water and Wastewater Service                                               |
| AWWA C515 | (2020) Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service                                                                |
| AWWA C520 | (2019) Knife Gate Valves, Sizes 2 In. (50 mm) Through 96 In. (2,400 mm)                                                                   |
| AWWA C550 | (2017) Protective Interior Coatings for Valves and Hydrants                                                                               |
| AWWA C600 | (2017) Installation of Ductile-Iron Mains and Their Appurtenances                                                                         |
| AWWA C605 | (2021) Underground Installation of Polyvinyl Chloride (PVC) and Molecularly Oriented Polyvinyl Chloride (PVC0) Pressure Pipe and Fittings |
| AWWA C651 | (2014) Standard for Disinfecting Water Mains                                                                                              |
| AWWA C655 | (2009) Field Dechlorination                                                                                                               |
| AWWA C800 | (2021) Underground Service Line Valves and Fittings                                                                                       |
| AWWA C900 | (2016) Polyvinyl Chloride (PVC) Pressure Pipe, and Fabricated Fittings, 4 In. Through 60 In. (100 mm Through 1,500 mm)                    |
| AWWA C901 | (2020) Polyethylene (PE) Pressure Pipe and Tubing, 3/4 In. (19mm) Through 3 In. (76 mm), for Water Service                                |
| AWWA C906 | (2021) Polyethylene (PE) Pressure Pipe and Fittings, 4 In. through 65 In. (100 mm Through 1,650 mm), for Waterworks                       |
| AWWA C909 | (2022) Molecularly Oriented Polyvinyl Chloride (PVC0) Pressure Pipe, 4 In. (100 mm) and Larger                                            |
| AWWA M23  | (2020) Manual: PVC Pipe - Design and Installation - Third Edition                                                                         |



|                           |                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| AWWA M41                  | (2009; 3rd Ed) Ductile-Iron Pipe and Fittings                                                                                      |
| ASTM INTERNATIONAL (ASTM) |                                                                                                                                    |
| ASTM A48/A48M             | (2003; R 2021) Standard Specification for Gray Iron Castings                                                                       |
| ASTM A307                 | (2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                          |
| ASTM A536                 | (2024) Standard Specification for Ductile Iron Castings                                                                            |
| ASTM A563M                | (2007; R 2013) Standard Specification for Carbon and Alloy Steel Nuts (Metric)                                                     |
| ASTM B32                  | (2020) Standard Specification for Solder Metal                                                                                     |
| ASTM B61                  | (2015; R 2021) Standard Specification for Steam or Valve Bronze Castings                                                           |
| ASTM B62                  | (2017) Standard Specification for Composition Bronze or Ounce Metal Castings                                                       |
| ASTM B88M                 | (2020) Standard Specification for Seamless Copper Water Tube (Metric)                                                              |
| ASTM C94/C94M             | (2025) Standard Specification for Ready-Mixed Concrete                                                                             |
| ASTM C1433                | (2020) Standard Specification for Precast Reinforced Concrete Monolithic Box Sections for Culverts, Storm Drains, and Sewers       |
| ASTM D1784                | (2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds |
| ASTM D1785                | (2021) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120                              |
| ASTM D2241                | (2015) Standard Specification for Poly(Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series)                                      |
| ASTM D2466                | (2023) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40                                    |
| ASTM D2467                | (2015) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80                                    |

|            |                                                                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D2683 | (2020) Standard Specification for Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Tubing                                                                                |
| ASTM D2855 | (2020) Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets |
| ASTM D3035 | (2015) Polyethylene (PE) Plastic Pipe (DR-PR) Based on Controlled Outside Diameter                                                                                                                              |
| ASTM D3139 | (2019) Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals                                                                                                                                       |
| ASTM D3350 | (2021) Polyethylene Plastics Pipe and Fittings Materials                                                                                                                                                        |
| ASTM F477  | (2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe                                                                                                                  |
| ASTM F714  | (2024) Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Outside Diameter                                                                                                              |
| ASTM F1483 | (2023) Standard Specification for Oriented Poly(Vinyl Chloride), PVC0, Pressure Pipe                                                                                                                            |
| ASTM F1674 | (2011) Standard Test Method for Joint Restraint Products for Use with PVC Pipe                                                                                                                                  |
| ASTM F3190 | (2021) Standard Practice for Heat Fusion Equipment (HFE) Operator Qualification on Polyethylene (PE) and Polyimide (PA) Pipe and Fittings                                                                       |

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH  
(FCCCHR)

|               |                                                                        |
|---------------|------------------------------------------------------------------------|
| FCCCHR List   | (continuously updated) List of Approved Backflow Prevention Assemblies |
| FCCCHR Manual | (10th Edition) Manual of Cross-Connection Control                      |

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

|           |                                                   |
|-----------|---------------------------------------------------|
| MSS SP-80 | (2019) Bronze Gate, Globe, Angle and Check Valves |
|-----------|---------------------------------------------------|

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|         |                                         |
|---------|-----------------------------------------|
| NFPA 24 | (2025) Standard for the Installation of |
|---------|-----------------------------------------|

Private Fire Service Mains and Their  
Appurtenances

NSF INTERNATIONAL (NSF)

|             |                                                                |
|-------------|----------------------------------------------------------------|
| NSF/ANSI 14 | (2023) Plastics Piping System Components and Related Materials |
| NSF/ANSI 61 | (2022) Drinking Water System Components - Health Effects       |

U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| UFC 3-600-01 | (2016; with Change 6, 2021) Fire Protection Engineering for Facilities |
|--------------|------------------------------------------------------------------------|

UNDERWRITERS LABORATORIES (UL)

|        |                                                                                      |
|--------|--------------------------------------------------------------------------------------|
| UL 246 | (2011; Reprint Jul 2020) UL Standard for Safety Hydrants for Fire-Protection Service |
| UL 262 | (2004; Reprint Jul 2023) Gate Valves for Fire-Protection Service                     |
| UL 312 | (2022) UL Standard for Safety Check Valves for Fire-Protection Service               |

## 1.2 DEFINITIONS

### 1.2.1 Water Mains

Water mains include water piping having diameters 100 through 350 mm, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

### 1.2.2 Water Service Lines

Water service lines include water piping from a water main to a building service at a point approximately 1.5 m from building or the point indicated on the drawings, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

### 1.2.3 Additional Definitions

For additional definitions refer to the definitions in the applicable referenced standard.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Connections; G

SD-03 Product Data

Pipe, Fittings, Joints and Couplings; G

Ball And Socket Joint; G

Valves; G

Valve Boxes; G

Fire Hydrants; G

Pipe Restraint; G

Corporation Stops; G

Backflow Preventer; G

Precast Concrete Thrust Blocks; G

Disinfection Procedures; G

Fusion Joining

#### SD-06 Test Reports

Backflow Preventer Tests; G

Bacteriological Samples; G

Post-Construction Fusion Report; G

Hydrostatic Sewer Test

Leakage Test

Hydrostatic Test

#### SD-07 Certificates

Pipe, Fittings, Joints and Couplings

Lining

Lining for Fittings

Valves

Fire Hydrants

Backflow Prevention Training Certificate

Backflow Tester Certification

Backflow Certificate

Fusion Technician Qualifications; G

## SD-08 Manufacturer's Instructions

Ductile-Iron Piping

PVC Piping

Copper Pipe For Service Lines

## 1.4 QUALITY CONTROL

## 1.4.1 Regulatory Requirements

Use CAN/CSA B483.1 or CAN/CSA B137 NSF/ANSI 61 or NSF/ANSI 14 materials for potable water systems to comply with lead free content requirements as defined by the U.S. Safe Drinking Water Act effective January 2014.

Comply with CAN/CSA B137 NFPA 24 for materials, installation, and testing of fire main piping and components.

## 1.4.2 Qualifications

## 1.4.2.1 Backflow Preventers

## 1.4.2.1.1 Backflow Preventer Certificate

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval will not be acceptable.

## 1.4.2.1.1.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with any company participating in any other phase of this Contract.

## 1.4.2.1.1.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

## 1.4.2.2 Fusion Technician Qualifications

Submit a certificate from the manufacturer of the fusible pipe that shows the fusion technician is fully qualified to install fusible pipe of the types and sizes being used. Qualification must be current as of the actual date of fusion performance on the project.

## 1.4.2.2.1 Fusion Technician Qualification on Polyethylene (PE) Pipe and Fittings

Provide certification for PE Pipe heat fusion in accordance with CSA B137.3 ASTM F3190.

## 1.5 DELIVERY, STORAGE, AND HANDLING

### 1.5.1 Delivery and Storage

Inspect materials delivered to site for required pipe markings and damage. Unload and store with minimum handling and in accordance with manufacturer's instructions to prevent cuts, scratches and other damage. Store materials on site in enclosures or under protective covering. Store plastic piping, jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes, fittings, valves, fire hydrants, and other accessories free of dirt and debris or other contaminants.

### 1.5.2 Handling

Handle pipe, fittings, valves, fire hydrants, and other accessories in accordance with applicable AWWA standard, manufacturer's instructions and in a manner to ensure delivery to the trench in sound undamaged condition. Avoid injury to coatings and linings on pipe and fittings; make repairs if coatings or linings are damaged. Do not place other material, hooks, or pipe inside a pipe or fitting after the coating has been applied. Inspect the pipe for defects before installation. Carry, do not drag pipe to the trench. Use of pinch bars and tongs for aligning or turning pipe will be permitted only on the bare ends of the pipe. Clean the interior of pipe and accessories of foreign matter before being lowered into the trench and keep them clean during laying operations by plugging. Replace defective material without additional expense to the Government. Store rubber gaskets, not immediately installed, under cover or out of direct sunlight.

Handle ductile iron pipe, fittings, and accessories in accordance with CSA B131 AWWA C600 and CSA B131 AWWA M41. Handle PVC pipe, fittings, and accessories in accordance with AWWA C605.

## PART 2 PRODUCTS

### 2.1 MATERIALS

All materials are intended for potable water use unless otherwise indicated. Comply with CAN/CSA B483.1 NSF/ANSI 61 or CAN/CSA B137 NSF/ANSI 14 for all potable water pipe, fittings and other applicable materials. Provide pipe, fittings and other applicable materials bearing CAN/CSA B483.1 NSF/ANSI 61 or CAN/CSA B137 NSF/ANSI 14 markings for potable water service.

Provide all materials in accordance with CSA B137.3, CSA B137.4 AWWA C800 and as indicated herein. Provide valves and fittings with pressure ratings equivalent to the pressure ratings of the pipe.

#### 2.1.1 Pipe, Fittings, Joints And Couplings

Submit manufacturer's standard drawings or catalog cuts, except submit both drawings and cuts for push-on and rubber-gasketed bell-and-spigot joints. Include information concerning gaskets with submittal for joints and couplings.

### 2.1.1.1 Ductile-Iron Piping

#### 2.1.1.1.1 Pipe and Fittings

Pipe, CSA B131 AWWA C151/A21.51, Pressure Class 350 . Fittings, CSA B137.3; CSA B137.4; CSA B131 AWWA C110/A21.10 or CSA B131 AWWA C153/A21.53; fittings with push-on joint ends are to meet the same requirements as fittings with mechanical-joint ends, except for the factory modified bell design. Provide fittings with pressure ratings equivalent to that of the pipe. Provide compatible pipe ends and fittings for the specified joints. Provide cement-mortar lining, AWWA C104/A21.4, standard thickness on pipe and fittings.

#### 2.1.1.1.2 Joints and Jointing Material

Provide push-on joints or mechanical joints for pipe and fittings.

- a. Push-On Joints: Shape of pipe ends and fitting ends, gaskets, and lubricant for joint assembly as recommended in CSA B131 AWWA C111/A21.11.
- b. Mechanical Joints: Dimensional and material requirements for pipe ends, glands, bolts and nuts, and gaskets as recommended in CSA B131 AWWA C111/A21.11.
- c. Flanged Joints: Bolts, nuts, and gaskets for flanged connections as recommended in Appendix A of AWWA C115/A21.15. Provide CSA B131 AWWA C115/A21.15 ductile iron flanges and conform to ASME B16.1, Class 125. Provide CSA B51 ASTM A536 epoxy coated steel set screw flanges. Gasket and lubricants for set screw flanges, in accordance with mechanical-joint gaskets specified in CSA B131 AWWA C111/A21.11.
- d. Insulating Joints: Designed to prevent metal-to-metal contact at the joint between adjacent sections of piping. Provide flanged type joint with insulating gasket, insulating bolt sleeves, and insulating washers. Provide full face dielectric type gaskets, as recommended in the Appendix to CSA B131 AWWA C115/A21.15. Bolts and nuts, as recommended in the Appendix to CSA B131 AWWA C115/A21.15.
- e. Sleeve-Type Mechanical Coupled Joints: As specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS.

### 2.1.1.2 Plastic Piping

#### 2.1.1.2.1 PVC and PVC0 Piping

##### 2.1.1.2.1.1 PVC Piping

CSA B137.3 AWWA C900 plain end or gasket bell end pipe meeting or exceeding CSA B137 ASTM D1784 cell class 12454, with a minimum Pressure Class 150 (DR27.5), 235 (DR 18), with ductile iron outside diameter (DIOD).

##### 2.1.1.2.1.2 PVC0 Piping

CSA B137.12 AWWA C909, CSA B137.3 ASTM F1483 plain end or gasket bell end pipe meeting or exceeding ASTM D1784 cell class 12454, Pressure CSA B137.3 Class 165 PVC0 pressure pipe, with ductile iron outside diameter (DIOD).

#### 2.1.1.2.1.3 Fittings for PVC and PVC0 Pipe

Ductile iron fittings, CSA B137.3; CSA B137.4; CSA B131 AWWA C110/A21.10 or compact fittings in accordance with CSA B131 AWWA C153/A21.53, with cement-mortar lining for fittings, AWWA C104/A21.4, standard thickness. Fittings with push-on joint ends are to conform to the same requirements as fittings with mechanical-joint ends, except for the factory modified bell design compatible for use with PVC pipe as specified.

Fittings from material that meets or exceeds ASTM D1784 cell class 12454 and is the same material as the pipe with elastomeric gaskets, in conformance with CSA B137.3 (PVC), CSA B137.12 (PVC0) AWWA C605 and CSA B137.3 AWWA C900.

#### 2.1.1.2.1.4 Joints and Jointing Material for PVC and PVC0 Piping

- a. Push-on joints: Use jointing material in accordance with CSA B137.3/ CSA B137.4 ASTM D3139 and CSA B131 AWWA C111/A21.11 between pipes, pipes and metal fittings, valves, and other accessories or compression-type joints/mechanical joints. Provide each joint connection with an elastomeric gasket compatible for the bell or coupling used. Gaskets for push-on joints for pipe, CSA B181.2 ASTM F477. Gaskets for push-on joints and compression-type joints/mechanical joints for joint connections between pipe and metal fittings, valves, and other accessories, CSA B131 AWWA C111/A21.11, respectively, for push-on joints and mechanical joints.
- b. Mechanical Joint: Use mechanically coupled joints having a sleeve-type mechanical coupling, as specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS, as an optional jointing method for plain-end PVC pipe, subject to the limitations specified for mechanically coupled joints using a sleeve-type mechanical coupling as specified for compression-type joints in CSA B137.3, CSA B137.4 ASTM D3139. Provide jointing material in accordance with CSA B131 AWWA C111/A21.11 between pipe and sleeve-type mechanical couplings.

#### 2.1.1.2.2 PVC Piping for Service Lines

##### 2.1.1.2.2.1 Pipe and Fittings

Provide CSA B137.3 ASTM D1784 cell class 12454 pipe and fittings of the same PVC material.

- a. CSA B137.3 ASTM D1785, Schedule 40 with ASTM D2466 CSA B137.3 Schedule 40 or ASTM D2467 Schedule 80 fittings.
- b. CSA B137.3 ASTM D2241 pipe and fittings with SDR as necessary to provide 1000 kPa minimum pressure rating with CSA B137.3 ASTM D2466 Schedule 40 or CSA B137.3 ASTM D2467 Schedule 80 fittings.

##### 2.1.1.2.2.2 Joints and Connections

Fittings may be joined by the solvent-cement method or threading.

##### 2.1.1.2.2.3 Solvent Joining

Provide solvent joints in accordance with ASTM D2855.



#### 2.1.1.2.3 Polyethylene (PE) Pipe

CSA B137.1 AWWA C906, CSA B137.1 ASTM F714, CSA B137.4 PE4710, minimum cell class PE 445574C, oxidative resistance classification CC3 with a minimum Pressure Class 160 (DR13.5) and ductile iron outside diameter (DIOD).

##### 2.1.1.2.3.1 Fittings For PE Pipe

Ductile iron fittings, CSA B137.3; CSA B137.4; CSA B131 AWWA C110/A21.10 or compact ductile iron fittings in accordance with CSA B131 AWWA C153/A21.53, with cement-mortar lining for ductile iron fittings, CSA A257.2 AWWA C104/A21.4, standard thickness. Fittings with push-on joint ends are to conform to the same requirements as fittings with mechanical-joint ends.

CSA B137.1 AWWA C906, PE4710, CSA B137.4 ASTM D3035 minimum cell class PE 445574C, oxidative resistance classification CC3 with minimum Pressure Class 250, molded CSA B137.4 ASTM D2683 meeting or exceeding the requirements in CSA B137.1 AWWA C906 for caps, reducers, couplings, elbows, and tees.

##### 2.1.1.2.3.2 Joints and Jointing Materials

Mechanical Joint: CSA B131 AWWA C111/A21.11 DIOD Mechanical joint adapter and gaskets for mechanical joints for joint connections between pipe and metal fittings, valves, and other accessories.

#### 2.1.1.2.4 Polyethylene (PE) Piping and Tubing for Service Lines

##### 2.1.1.2.4.1 PE Service Line Pipe And Tubing

CSA B137.1 AWWA C901, PE4710, CSA B137.4 CSA B137.1 ASTM D3035, ASTM D3350 minimum cell class PE 445574C, oxidative resistance classification CC3 with a minimum Pressure Class 250 (SDR 9).

##### 2.1.1.2.4.2 PE Service Line Fittings

CSA B137.1 AWWA C901, PE4710, CSA B137.4 CSA B137.1 ASTM D3350 minimum cell class PE 445574C, oxidative resistance classification CC3 with a minimum Pressure Class 250, molded CSA B137.4 ASTM D2683 caps, reducers, couplings, elbows, and tees or compatible fittings in accordance with this specification.

#### 2.1.1.3 Copper Pipe For Service Lines

##### 2.1.1.3.1 Copper Tubing and Associated Fittings

Provide CSA B137.1 ASTM B88M, Type K copper tubing. Provide AWWA C800 fittings. CSA B137.3, CSA B137.4 AWWA C800 includes CSA B51 and CSA B137 ASME B1.20.3, CSA B51 ASME B1.20.1, CSA B51, CSA B125.1, CSA B125.3 ASME B16.18 solder-type joint fittings.

#### 2.1.2 Valves

Provide a protective interior coating in accordance with CSA B137.3, CSA B137.4 AWWA C550.

#### 2.1.2.1 Gate Valves 80 mm Size and Larger on Buried Piping

CSA B137.3, CSA B137.4 AWWA C500, AWWA C509, AWWA C515, or CAN/ULC-S560 UL 262 and:

- a. CSA B137.3, CSA B137.4 AWWA C500: nonrising stem type with double-disc gate and mechanical-joint ends or push-on joint ends compatible for the adjoining pipe
- b. CSA B137.3, CSA B137.4 AWWA C509 or CAN/ULC-S560 AWWA C515: nonrising stem type with mechanical-joint ends or resilient-seated gate valves 80 to 300 mm in size
- c. CAN/ULC-S560 UL 262: inside-screw type with operating nut, double-disc or split-wedge type gate, designed for a hydraulic working pressure of 1200 kPa, and have mechanical-joint ends or push-on joint ends as appropriate for the pipe to which it is joined.

Match materials for CAN/ULC-S560 UL 262 gate valves to the reference standards specified in CSA B137.3, CSA B137.4 AWWA C500. Gate valves open by counterclockwise rotation of the valve stem. Stuffing boxes have O-ring stem seals, except for those valves for which gearing is specified, in which case use conventional packing in place of O-ring seal. Stuffing boxes are bolted and constructed so as to permit easy removal of parts for repair. Use gate valves with special ends for connection to cement piping or sleeve-type mechanical coupling in lieu of mechanical-joint ends and push-on joint ends. Provide valve ends and gaskets for connection to cement piping or to sleeve-type mechanical couplings that conform to the requirements specified respectively for the joint or coupling. Provide gate valves on 50 mm service lines with threaded ends. Gate valves on 50 mm service lines have ends compatible with joining to the pipe used; push-on joint ends or mechanical-joint ends for joining to ductile-iron pipe; with CSA B131 AWWA C111/A21.11 gaskets and pipe ends. Provide all valves from one manufacturer.

#### 2.1.2.2 Gate Valves 75 mm Size and Larger in Valve Pit(s) and Aboveground Locations

CSA B137.3, CSA B137.4 AWWA C500, AWWA C509, AWWA C515, or CAN/ULC-S560 UL 262 and:

- a. CSA B137.3, CSA B137.4 AWWA C500: outside-screw-and-yoke rising-stem type with double-disc gates and flanged ends
- b. CSA B137.3, CSA B137.4 AWWA C509 or AWWA C515: outside-screw-and-yoke rising-stem type with flanged ends
- c. CAN/ULC-S560 UL 262: outside-screw-and-yoke type, with double-disc or split-wedge type gate and flanged ends, and designed for a hydraulic working pressure of 1200 kPa

#### 2.1.2.3 Knife Gate Valves

Provide CSA B137.3, CSA B137.4 AWWA C520 ductile iron knife gate valves with bonnet.

#### 2.1.2.4 Check Valves

Provide a protective interior coating in accordance with CSA B137.3,

CSA B137.4 AWWA C550. Swing-check type, CSA B137.3, CSA B137.4 AWWA C508 or UL 312 and:

- a. AWWA C508: Iron or steel body and cover and flanged ends
- b. UL 312: Cast iron or steel body and cover, flanged ends, and designed for a minimum working pressure of 1000 kPa.

#### 2.1.2.5 Rubber-Seated Butterfly Valves

Provide rubber-seated butterfly valves and wafer type valves that match the performance requirements of AWWA C504. Wafer type valves not meeting laying length requirements are acceptable if supplied and installed with a spacer, providing the specified laying length. Meet all tests required by AWWA C504. Flanged-end valves are required in a pit. Provide a union or sleeve-type coupling in the pit to permit removal. Direct-bury mechanical-end valves 80 through 250 mm in diameter. Provide a valve box, means for manual operation, and an adjacent pipe joint to facilitate valve removal. Provide valve operators that restrict closing to a rate requiring approximately 60 seconds, from fully open to fully closed.

#### 2.1.2.6 Pressure Reducing Valves

Maintain a constant downstream pressure regardless of fluctuations in demand. Using pressure reducing valves capable of providing kPa operating pressure on the inlet side, with outlet pressure set for kPa. Provide hydraulically-operated, pilot controlled, globe or angle type valves that are capable of being actuated either by diaphragm or piston. Provide diaphragm-operated, adjustable, spring-loaded type pilot controls made of lead-free bronze with stainless steel working parts, designed to permit flow when controlling pressure exceeds the spring setting. Construct the bodies of bronze, cast iron or cast steel with lead-free bronze trim; the valve stem of stainless steel; the seat of lead-free bronze; and the valve discs and diaphragms of synthetic rubber. Provide threaded ends.

#### 2.1.2.7 Air Release, Air/Vacuum, and Combination Air Valves

Provide AWWA C512 air release and combination air valves that release air and prevent the formation of a vacuum. Provide valves with an iron body, lead-free bronze trim and stainless steel float that automatically releases air when the lines are being filled with water and admits air into the line when water is being withdrawn in excess of the inflow.

#### 2.1.2.8 Water Service Valves

##### 2.1.2.8.1 Gate Valves Smaller than 75 mm in Size on Buried Piping

Gate valves smaller than 75 mm size on Buried Piping MSS SP-80, Class 150, solid wedge, nonrising stem, with flanged or threaded end connections, a union on one side of the valve, and a handwheel operator.

##### 2.1.2.8.2 Gate Valves Smaller Than 75 mm Size in Valve Pits

MSS SP-80, Class 150, solid wedge, inside screw, rising stem. Provide valves with flanged or threaded end connections, a union on one side of the valve, and a handwheel operator.

#### 2.1.2.8.3 Check Valves Smaller than 50 mm in Size

Provide check valves with a minimum working pressure of 1000 kPa or as indicated with a clear waterway equal to the full nominal diameter of the valve. Valves open to permit flow when inlet pressure is greater than the discharge pressure, and close tightly to prevent return flow when discharge pressure exceeds inlet pressure. Cast the size of the valve, working pressure, manufacturer's name, initials, or trademark on the body of each valve.

Provide valves for screwed fittings, made of lead-free bronze and in conformance with MSS SP-80, Class 150, Types 3 and 4 compatible for the application.

#### 2.1.2.9 Valve Boxes

Provide a valve box for each gate valve on buried piping. Construct adjustable valve boxes manufactured from cast iron or precast concrete of a size compatible for the valve on which it is used. Provide cast iron valve boxes with a minimum cover and wall thickness of 5 mm and conforming to ASTM A48/A48M, Class 35B. Coat the cast-iron box with a heavy coat of bituminous paint. Cast the word "WATER" on the lid. The minimum diameter of the shaft of the box is 135 mm. Provide ASTM C1433 precast concrete valve box. Provide precast concrete boxes installed in locations subjected to vehicular traffic to withstand AASHTO load designation as outlined in AASHTO HB-17.

#### 2.1.2.10 Valve Pits

Construct the valve pits at locations indicated or as required above and in accordance with the details shown.

### 2.1.3 Fire Hydrants

#### 2.1.3.1 Fire Hydrants

Provide fire hydrants where indicated. Paint fire hydrants with at least one coat of primer and two coats of enamel paint. Paint barrel and bonnet colors in accordance with UFC 3-600-01. Stencil fire hydrant number and main size on the fire hydrant barrel using black stencil paint.

Provide a protective epoxy interior coating conforming to AWWA C550 on those portions of the fire hydrant continuously in contact with sea water or salt water.

##### 2.1.3.1.1 Dry-Barrel Type Fire Hydrants

Provide Dry-barrel type fire hydrants, AWWA C502 or UL 246, "Base Valve" with 150 mm inlet, 135 mm valve opening, one 115 mm pumper connection, and two 65 mm hose connections. Provide

Provide mechanical-joint or push-on joint end inlet ; with end matching requirements as specified in AWWA C502 or AWWA C503 or UL 246 for size and shape of operating nut, cap nuts, and threads on hose and pumper connections. Provide fire hydrants with frangible sections as mentioned in AWWA C502 . Provide fire hydrant with special couplings joining upper and lower sections of fire hydrant barrel and upper and lower sections of fire hydrant stem that break from a force imposed by a moving vehicle.

#### 2.1.4 Disinfection

Chlorinating materials are to conform to: Chlorine, Liquid: AWWA B301; Hypochlorite, Calcium and Sodium: AWWA B300.

### 2.2 ACCESSORIES

#### 2.2.1 Pipe Restraint

##### 2.2.1.1 Thrust Blocks

Use ASTM C94/C94M concrete having a minimum compressive strength of 15 MPa at 28 days or use concrete of a mix not leaner than one part cement, two and one half parts sand, and five parts gravel, having the same minimum compressive strength.

##### 2.2.1.2 Precast Thrust Blocks

Provide precast concrete thrust blocks.

##### 2.2.1.3 Joint Restraint

Provide restrained joints in accordance with NFPA 24, Chapter 10 and in accordance with ASTM F1674.

Provide mechanical joint restraint .

#### 2.2.2 Protective Enclosures

Provide Freeze-Protection Enclosures that are insulated and designed to protect aboveground water piping, equipment, or specialties from freezing and damage, with heat source to maintain minimum internal temperature of degrees C when external temperatures reach as low as degree C.

##### 2.2.2.1 Housing

Reinforced and insulated aluminum or fiberglass construction; with anchoring devices for attaching housing to concrete base, access doors with locking devices, sized to allow access and service of the protected unit, drain openings, and an electric heating cable or heater with self-limiting temperature control.

#### 2.2.3 Sleeve-Type Mechanical Couplings

Use AWWA C219 couplings to join plain-end piping by compression of a ring gasket at each end of the adjoining pipe sections. The coupling consists of one middle ring flared or beveled at each end to provide a gasket seat; two follower rings; two resilient tapered rubber gaskets; and bolts and nuts to draw the follower rings toward each other to compress the gaskets. Provide true circular middle ring and the follower rings sections free from irregularities, flat spots, and surface defects; provide for confinement and compression of the gaskets. For ductile iron pipe, use ASTM A536 ductile iron. Use gaskets for resistance to set after installation and to meet the requirements specified for gaskets for mechanical joint in AWWA C111/A21.11. Provide track-head type bolts ASTM A307, Grade A, with ASTM A563M, Grade A nuts or round-head square-neck type ASME B18.5.2.2M or ASME B18.5.2.1M bolts with ASME B18.2.2 hex nuts. Provide 16 mm diameter bolts. Shape bolt holes in follower

rings to hold fast to the necks of the bolts used. Do not use mechanically coupled joints using a sleeve-type mechanical coupling as an optional method of jointing except where pipeline is adequately anchored to resist tension pull across the joint. Provide a tight flexible joint with mechanical couplings under reasonable conditions, such as pipe movements caused by expansion, contraction, slight settling or shifting in the ground, minor variations in trench gradients, and traffic vibrations. Match coupling strength to that of the adjoining pipeline.

#### 2.2.4 Insulating Joints

Provide a rubber-gasketed insulating joint or dielectric coupling between pipe of dissimilar metals which will effectively prevent metal-to-metal contact between adjacent sections of piping.

#### 2.2.5 Bonded Joints

For all ferrous pipe, provide a metallic bond at each joint, including joints made with flexible couplings, caulking, or rubber gaskets, of ferrous metallic piping to effect continuous conductivity. Provide Size 1/0 copper conductor thermal weld type bond wire designed for direct burial and shaped to stand clear of the joint.

#### 2.2.6 Dielectric Fittings

Install dielectric fittings between threaded ferrous and nonferrous metallic pipe, fittings and valves, except where corporation stops join mains to prevent metal-to-metal contact of dissimilar metallic piping elements and compatible with the indicated working pressure.

#### 2.2.7 Tracer Wire for Nonmetallic Piping

Provide a continuous bare copper or aluminum wire not less than 2.5 mm in diameter in sufficient length over each separate run of nonmetallic pipe.

#### 2.2.8 Water Service Line Appurtenances

##### 2.2.8.1 Corporation Stops

Ground key type; lead-free bronze, ASTM B61 or ASTM B62; compatible with the working pressure of the system and solder-joint, or flared tube compression type joint. Threaded ends for inlet and outlet of corporation stops, AWWA C800; coupling nut for connection to flared copper tubing, ASME B16.26.

##### 2.2.8.2 Curb or Service Stops

Ground key, round way, inverted key type; made of lead-free bronze, ASTM B61 or ASTM B62; and compatible with the working pressure of the system. Provide compatible ends for connection to the service piping. Cast an arrow into body of the curb or service stop indicating direction of flow.

##### 2.2.8.3 Service Clamps

Provide single or double flattened strap type service clamps used for repairing damaged cast-iron, steel or PVC pipe with a pressure rating not less than that of the pipe being repaired. Provide clamps with a galvanized malleable-iron body with cadmium plated straps and nuts and a

rubber gasket cemented to the body.

#### 2.2.8.4 Goosenecks

Manufacture goosenecks from Type K copper tubing; provide joint ends for goosenecks compatible with connecting to corporation stop and service line. Where multiple gooseneck connections are required for an individual service, connect goosenecks to the service line through a compatible lead-free brass or bronze branch connection; the total clear area of the branches to be at least equal to the clear area of the service line.

#### 2.2.8.5 Curb Boxes

Provide a curb box for each curb or service stop manufactured from cast iron, size capable of containing the stop where it is used. Provide a round head. Cast the word "WATER" on the lid. Factory coat the box with a heavy coat of bituminous paint.

### PART 3 EXECUTION

#### 3.1 PREPARATION

##### 3.1.1 Connections to Existing System

Perform all connections to the existing water system in the presence of the Contracting Officer.

##### 3.1.2 Operation of Existing Valves

Do not operate valves within or directly connected to the existing water system unless expressly directed to do so by the Contracting Officer.

##### 3.1.3 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK.

#### 3.2 INSTALLATION

Install all materials in accordance with the applicable reference standard, manufacturers instructions and as indicated herein.

##### 3.2.1 Piping

###### 3.2.1.1 General Requirements

Install pipe, fittings, joints and couplings in accordance with the applicable referenced standard, the manufacturer's instructions and as specified herein.

###### 3.2.1.1.1 Termination of Water Lines

Terminate the work covered by this section at a point approximately 1.5 m from the building, unless otherwise indicated.

Do not lay water lines in the same trench with gas lines, fuel lines, electric wiring, or any other utility. Do not install copper tubing in the same trench with ferrous piping materials. Where nonferrous metallic pipe (i.e., copper tubing) crosses any ferrous piping, provide a minimum vertical separation of 300 mm between pipes.

### 3.2.1.1.2 Pipe Laying and Jointing

Remove fins and burrs from pipe and fittings. Before placing in position, clean pipe, fittings, valves, and accessories, and maintain in a clean condition. Provide proper facilities for lowering sections of pipe into trenches. Under no circumstances is it permissible to drop or dump pipe, fittings, valves, or other water line material into trenches. Cut pipe cleanly, squarely, and accurately to the length established at the site and work into place without springing or forcing. Replace a pipe or fitting that does not allow sufficient space for installation of jointing material. Blocking or wedging between bells and spigots is not permitted. Lay bell-and-spigot pipe with the bell end pointing in the direction of laying. Grade the pipeline in straight lines; avoid the formation of dips and low points. Support pipe at the design elevation and grade. Secure firm, uniform support. Wood support blocking is not permitted. Lay pipe so that the full length of each section of pipe and each fitting rests solidly on the pipe bedding; excavate recesses to accommodate bells, joints, and couplings. Provide anchors and supports for fastening work into place. Make provision for expansion and contraction of pipelines. Keep trenches free of water until joints have been assembled. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Do not lay pipe when conditions of trench or weather prevent installation. Provide a minimum of 760 mm depth of cover over top of pipe.

### 3.2.1.1.3 Tracer Wire

Install a continuous length of tracer wire for the full length of each run of nonmetallic pipe. Attach wire to top of pipe in such manner that it will not be displaced during construction operations.

### 3.2.1.1.4 Connections to Existing Water Lines

Make connections to existing water lines after coordination with the facility and with a minimum interruption of service on the existing line. Make connections to existing lines under pressure in accordance with the recommended procedures of the manufacturer of the pipe being tapped and as indicated.

### 3.2.1.1.5 Sewer Manholes

No water piping is to pass through or come in contact with any part of a sewer manhole.

### 3.2.1.1.6 Water Piping Parallel With Sewer Piping

- a. Normal Conditions: Lay water piping at least 3.0 m horizontally from sewer or sewer manhole whenever possible. Measure the distance from outside edge to outside edge of pipe or outside edge of manhole. When local conditions prevent horizontal separation install water piping in a separate trench with the bottom of the water piping at least 450 mm above the top of the sewer piping.
- b. Unusual Conditions: When local conditions prevent vertical separation, construct sewer piping of AWWA compliant ductile iron water piping and perform hydrostatic sewer test, without leakage, prior to backfilling. When local conditions prevent vertical separation, test the sewer manhole in place to ensure watertight construction.



### 3.2.1.1.7 Water Piping Crossing Sewer Piping

- a. Normal Conditions: Provide a separation of at least 450 mm between the bottom of the water piping and the top of the sewer piping in cases where water piping crosses above sewer piping.
- b. Unusual Conditions: When local conditions prevent a vertical separation described above, construct sewer piping passing over or under water piping of AWWA compliant ductile iron water piping and perform hydrostatic sewer test, without leakage, prior to backfilling. Construct sewer crossing with a minimum 6.1 m length of the AWWA compliant ductile iron water piping, centered at the point of the crossing so that joints are equidistant and as far as possible from the water piping. Protect water piping passing under sewer piping by providing a vertical separation of at least 450 mm between the bottom of the sewer piping and the top of the water piping; adequate structural support for the sewer piping to prevent excessive deflection of the joints and the settling on or damage to the water piping.

### 3.2.1.1.8 Penetrations

Provide ductile-iron or Schedule 40 steel wall sleeves for pipe passing through walls of valve pits and structures. Fill annular space between walls and sleeves with rich cement mortar. Fill annular space between pipe and sleeves with mastic.

### 3.2.1.1.9 Flanged Pipe

Only install flanged pipe aboveground or with the flanges in valve pits.

### 3.2.1.2 Ductile-Iron Piping

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of AWWA C600 for pipe installation, joint assembly, valve-and-fitting installation, and thrust restraint.

- a. Jointing: Make insulating joints with the gaskets, sleeves, washers, bolts, and nuts previously specified for this type joint. Assemble insulating joints as specified for flanged joints, except that bolts with insulating sleeves are to be full size for the bolt holes. Ensure that there is no metal-to-metal contact between dissimilar metals after the joint has been assembled.
- b. Allowable Deflection: Follow AWWA C600 and AWWA M41 for the maximum allowable deflection. If the alignment requires deflection in excess of the above limitations, provide special bends or a sufficient number of shorter lengths of pipe to achieve angular deflections within the limit set forth.

### 3.2.1.3 PVC Water Main Pipe

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of AWWA C605 for laying of pipe, joining PVC pipe to fittings and accessories, setting of fire hydrants, valves, and fittings; and with the recommendations for pipe joint assembly and appurtenance installation in AWWA M23, Chapter 7,

"Installation."

- a. Jointing: Make push-on joints with the elastomeric gaskets specified for this type joint, using either elastomeric-gasket bell-end pipe or elastomeric-gasket couplings. For pipe-to-pipe push-on joint connections, use only pipe with push-on joint ends having factory-made bevel; for push-on joint connections to metal fittings, valves, and other accessories, cut spigot end of pipe off square and re-bevel pipe end to a bevel approximately the same as that on ductile-iron pipe used for the same type of joint. Use a lubricant recommended by the pipe manufacturer for push-on joints. Assemble push-on joints for pipe-to-pipe joint connections in accordance with the requirements of AWWA C605 for laying the pipe and the recommendations in AWWA M23, Chapter 7, "Installation," for pipe joint assembly. Assemble push-on joints for connection to fittings, valves, and other accessories in accordance with the requirements of AWWA C605 for joining PVC pipe to fittings and accessories and with the requirements of AWWA C600 for joint assembly. Make compression-type joints/mechanical joints with the gaskets, glands, bolts, nuts, and internal stiffeners previously specified for this type joint; assemble in accordance with the requirements of AWWA C605 for joining PVC pipe to fittings and accessories, with the requirements of AWWA C600 for joint assembly, and with the recommendations of Appendix A to AWWA C111/A21.11. Cut off spigot end of pipe for compression-type joint/mechanical-joint connections and do not re-bevel. Assemble joints made with sleeve-type mechanical couplings in accordance with the recommendations of the coupling manufacturer using internal stiffeners as previously specified for compression-type joints.
- b. Joint Offset: Construct joint offset in accordance AWWA C605. Do not exceed the minimum longitudinal bending as indicated by AWWA C605.
- c. Fittings: Install in accordance with AWWA C605.

#### 3.2.1.4 Metallic Piping for Service Lines

Install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the applicable requirements of AWWA C600 for pipe installation, unless otherwise specified.

##### 3.2.1.4.1 Screwed Joints

Make screwed joints up tight with a stiff mixture of graphite and oil, inert filler and oil, or graphite compound; apply to male threads only or with PTFE Tape, for use with threaded pipe. Threads are to be full cut; do not leave more than three threads on the pipe exposed after assembling the joint.

##### 3.2.1.4.2 Joints for Copper Tubing

Cut copper tubing with square ends; remove fins and burrs. Replace dented, gouged, or otherwise damaged tubing with undamaged tubing. Make solder joints using ASTM B32, 95-5 tin-antimony or Grade Sn96 solder. Use solder and flux containing less than 0.2 percent lead. Before making joint, clean ends of tubing and inside of fitting or coupling with wire brush or abrasive. Apply a rosin flux to the tubing end and on recess inside of fitting or coupling. Insert tubing end into fitting or coupling for the full depth of the recess and solder. For compression joints on flared tubing, insert tubing through the coupling nut and flare tubing.

#### 3.2.1.4.3 Flanged Joints

Make flanged joints up tight, avoid undue strain on flanges, valves, fittings, and accessories.

#### 3.2.1.5 Trenchless Piping

##### 3.2.1.5.1 Butt Fusion

Fusible pipe will be fused by qualified fusion technicians, as required by manufacturer of the fusion equipment. Record and log each fusion joint by an electronic monitoring device (data logger) connected to the fusion machine. Log fusion data and create Post-Construction Fusion Report with software specifically developed for the pipe material being fused. Software must record the parameters required by the fusion equipment manufacturer and these specifications. Manual log data not logged by the data logger and be included in the Post-Construction Fusion Report. Assemble fusible PVC pipe lengths in the field with butt-fused joints. Follow the manufacturer's fusion equipment procedures.

##### 3.2.1.5.1.1 PVC Pipe

For butt fused PVC Pipe, provide joints meeting the requirements of ASTM F1674.

##### 3.2.1.5.2 Post-Construction Fusion Report

Include the following data for each fusible connection in the report:

- a. Pipe Size and Thickness
- b. Machine Size
- c. Fusion Technician Identification
- d. Job Identification
- e. Fusion Joint Number
- f. Fusion, Heating, and Drag Pressure Settings
- g. Heat Plate Temperature
- h. Time Stamp
- i. Heating and Cool Down Time of Fusion
- j. Ambient Temperature

##### 3.2.1.5.3 Installation Ductile Iron Ball and Socket Joint

Install pipe and fittings in accordance with AWWA C600 and AWWA M41 for pipe installation, joint assembly, and thrust restraint.

- a. Allowable Deflection: Meet the applicable requirements of AWWA C600, AWWA M41 and in accordance with pipe manufacturer's instructions for the maximum allowable deflection.

### 3.2.1.6 Fire Protection Service Lines for Sprinkler Supplies

Connect water service lines used to supply building sprinkler systems for fire protection to the water main in accordance with NFPA 24.

### 3.2.1.7 Water Service Piping

#### 3.2.1.7.1 Location

Connect water service piping to the building service where the building service has been installed. Where building service has not been installed, terminate water service lines approximately 1.5 m from the building line at the points indicated; close such water service lines with plugs or caps.

#### 3.2.1.7.2 Water Service Line Connections to Water Mains

Connect 50 mm water service lines to the main by a corporation stop and gooseneck and install a service stop below the frostline .

### 3.2.2 Disinfection

Disinfection of systems supplying non-potable water is not required.

Prior to disinfection, provide disinfection procedures, proposed neutralization and disposal methods of waste water from disinfection as part of the disinfection submittal. Disinfect new water piping and existing water piping affected by Contractor's operations in accordance with AWWA C651. Disinfect new water piping using the AWWA C651 continuous-feed method of chlorination . Ensure a free chlorine residual of not less than 10 mg/L after 24 hour holding period and prior to performing bacteriological tests.

### 3.2.3 Flushing

Perform bacteriological tests prior to flushing. Flush solution from the systems with domestic water until maximum residual chlorine content is within the range of 0.2 to 0.5 mg/L, the residual chlorine content of the distribution system, or acceptable for domestic use. Use AWWA C655 neutralizing chemicals.

### 3.2.4 Pipe Restraint

#### 3.2.4.1 Concrete Thrust Blocks

Install concrete thrust blocks where indicated.

#### 3.2.4.2 Restrained Joints

Install restrained joints in accordance with the manufacturer's instructions where indicated. Provide metal harness fabricated by the pipe manufacturer and furnished with the pipe.

### 3.2.5 Valves

#### 3.2.5.1 Gate Valves

Install gate valves, AWWA C500 and UL 262, in accordance with the

requirements of AWWA C600 for valve-and-fitting installation and with the recommendations of the Appendix ("Installation, Operation, and Maintenance of Gate Valves") to AWWA C500. Install gate valves, AWWA C509 or AWWA C515, in accordance with the requirements of AWWA C600 for valve-and-fitting installation and with the recommendations of the Appendix ("Installation, Operation, and Maintenance of Gate Valves") to AWWA C509 or AWWA C515. Install gate valves on PVC water mains in accordance with the recommendations for appurtenance installation in AWWA M23, Chapter 7, "Installation." Make and assemble joints to gate valves as specified for making and assembling the same type joints between pipe and fittings.

#### 3.2.5.2 Check Valves

Install check valves in accordance with the applicable requirements of AWWA C600 for valve-and-fitting installation. Make and assemble joints to check valves as specified for making and assembling the same type joints between pipe and fittings.

#### 3.2.5.3 Air Release, Air/Vacuum, and Combination Air Valves

Install pressure vacuum assemblies of type, size, and capacity indicated. Include valves and test cocks. Install according to the requirements of plumbing and health department and authorities having jurisdiction. Do not install pressure vacuum breaker assemblies in vault or other space subject to flooding.

#### 3.2.6 Fire Hydrants

Install fire hydrants in accordance with AWWA C600 for fire hydrant installation and as indicated. Make and assemble joints as specified for making and assembling the same type joints between pipe and fittings. Install fire hydrants with the 115 mm connections facing the adjacent paved surface. If there are two paved adjacent surfaces, install fire hydrants with the 115 mm connection facing the paved surface where the connecting main is located.

### 3.3 FIELD QUALITY CONTROL

#### 3.3.1 Tests

Notify the Contracting Officer a minimum of five days in advance of hydrostatic testing. Coordinate the proposed method for disposal of waste water from hydrostatic testing. Perform field tests, and provide labor, equipment, and incidentals required for testing. Provide documentation that all items of work have been constructed in accordance with the Contract documents.

##### 3.3.1.1 Hydrostatic Test

Test the water system in accordance with the applicable AWWA standard specified below. Where water mains provide fire service, test in accordance with the special testing requirements given in the paragraph SPECIAL TESTING REQUIREMENTS FOR FIRE SERVICE. Test ductile-iron water mains in accordance with the requirements of AWWA C600 for hydrostatic testing. The amount of leakage on ductile-iron pipelines with mechanical-joints is not to exceed the amounts given in AWWA C600; no leakage will be allowed at joints made by any other methods. Test water service lines in accordance with requirements of AWWA C600 for hydrostatic testing. No leakage will be allowed at copper pipe joints, copper tubing

joints (soldered, compression type, brazed), flanged joints. Do not backfill utility trench or begin testing on any section of a pipeline where concrete thrust blocks have been provided until at least 7 days after placing of the concrete.

#### 3.3.1.2 Hydrostatic Sewer Test

The hydrostatic pressure sewer test will be performed in accordance with the applicable AWWA standard for the piping material or AWWA C600.

#### 3.3.1.3 Leakage Test

For leakage test, use a hydrostatic pressure not less than the maximum working pressure of the system. Leakage test may be performed at the same time and at the same test pressure as the pressure test.

#### 3.3.1.4 Bacteriological Testing

Perform bacteriological tests in accordance with AWWA C651 Option A . Analyze samples by a certified laboratory, and submit the results of the bacteriological samples.

#### 3.3.1.5 Backflow Preventer Tests

After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

#### 3.3.1.6 Special Testing Requirements for Fire Service

Test water mains and water service lines providing fire service or water and fire service in accordance with NFPA 24. The additional water added to the system must not exceed the limits given in NFPA 24

#### 3.3.1.7 Tracer Wire Continuity Test

Test tracer wire for continuity after service connections have been completed and prior to final pavement or restoration. Verify that tracer wire is locatable with electronic utility locating equipment. Repair breaks or separations and re-test for continuity.

### 3.4 SYSTEM STARTUP

Water mains and appurtenances must be completely installed, disinfected, flushed, and satisfactory bacteriological sample results received prior to permanent connections being made to the active distribution system. Obtain approval by the Contracting Officer prior to the new water piping being placed into service.

### 3.5 CLEANUP

Upon completion of the installation of water lines and appurtenances, remove all debris and surplus materials resulting from the work.

-- End of Section --

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## SECTION 33 30 00

SANITARY SEWERAGE  
05/18

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A48/A48M   | (2003; R 2021) Standard Specification for Gray Iron Castings                                                                                |
| ASTM A123/A123M | (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                                             |
| ASTM A536       | (2024) Standard Specification for Ductile Iron Castings                                                                                     |
| ASTM C94/C94M   | (2025) Standard Specification for Ready-Mixed Concrete                                                                                      |
| ASTM C150/C150M | (2024) Standard Specification for Portland Cement                                                                                           |
| ASTM C260/C260M | (2024) Standard Specification for Air-Entraining Admixtures for Concrete                                                                    |
| ASTM C270       | (2019a; E 2019) Standard Specification for Mortar for Unit Masonry                                                                          |
| ASTM C443M      | (2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets (Metric)                                      |
| ASTM C478       | (2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections                                                     |
| ASTM C478M      | (2018) Standard Specification for Precast Reinforced Concrete Manhole Sections (Metric)                                                     |
| ASTM C923M      | (2008b; R 2013) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals (Metric) |
| ASTM C969M      | (2019) Standard Practice for Infiltration and Exfiltration Acceptance Testing of Installed Precast Concrete Pipe Sewer Lines (Metric)       |



|             |                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C972   | (2024) Standard Test Method for<br>Compression-Recovery of Tape Sealant                                                                                          |
| ASTM C990M  | (2009; R 2019) Standard Specification for<br>Joints for Concrete Pipe, Manholes, and<br>Precast Box Sections Using Preformed<br>Flexible Joint Sealants (Metric) |
| ASTM C1214M | (2013) Standard Test Method for Concrete<br>Pipe Sewerlines by Negative Air Pressure<br>(Vacuum) Test Method (Metric)                                            |
| ASTM C1244M | (2020) Standard Test Method for Concrete<br>Sewer Manholes by the Negative Air<br>Pressure (Vacuum) Test Prior to Backfill<br>(Metric)                           |
| ASTM D412   | (2016; R 2021) Standard Test Methods for<br>Vulcanized Rubber and Thermoplastic<br>Elastomers - Tension                                                          |
| ASTM D624   | (2000; R 2020) Standard Test Method for<br>Tear Strength of Conventional Vulcanized<br>Rubber and Thermoplastic Elastomers                                       |
| ASTM D2321  | (2020) Standard Practice for Underground<br>Installation of Thermoplastic Pipe for<br>Sewers and Other Gravity-Flow Applications                                 |
| ASTM D3034  | (2016) Standard Specification for Type PSM<br>Poly(Vinyl Chloride) (PVC) Sewer Pipe and<br>Fittings                                                              |
| ASTM D3212  | (2020) Standard Specification for Joints<br>for Drain and Sewer Plastic Pipes Using<br>Flexible Elastomeric Seals                                                |
| ASTM D3753  | (2019) Glass-Fiber-Reinforced Polyester<br>Manholes and Wetwells                                                                                                 |
| ASTM D4101  | (2017) Standard Classification System and<br>Basis for Specification for Polypropylene<br>Injection and Extrusion Materials                                      |
| ASTM F477   | (2014; R 2021) Standard Specification for<br>Elastomeric Seals (Gaskets) for Joining<br>Plastic Pipe                                                             |
| ASTM F949   | (2020) Standard Specification for<br>Poly(Vinyl Chloride) (PVC) Corrugated<br>Sewer Pipe with a Smooth Interior and<br>Fittings                                  |

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                |                                                   |
|----------------|---------------------------------------------------|
| 29 CFR 1910.27 | (Nov 2016) Scaffolds and Roope Descent<br>Systems |
|----------------|---------------------------------------------------|

## UNI-BELL PVC PIPE ASSOCIATION (UBPPA)

UBPPA UNI-B-6 (1998) Recommended Practice for  
Low-Pressure Air Testing of Installed  
Sewer Pipe

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-01 Preconstruction Submittals

Contractor's License; G

## SD-02 Shop Drawings

Installation Drawings; G

## SD-03 Product Data

Precast Concrete Manholes

Frames, Covers, and Gratings

Gravity Pipe

Pressure Pipe

Precast Concrete Septic Tanks; G

## SD-06 Test Reports

Precast Concrete Sewer Manhole Test; G

Hydrostatic Sewer Test; G

Infiltration Tests And Exfiltration Tests; G

Negative Air Pressure Test; G

Low-Pressure Air Tests; G

Tests For Pressure Lines; G

Concrete Pipe Test; G

## SD-07 Certificates

Portland Cement

Gaskets

Pre-Installation Inspection Request; G

Post-Installation Inspection; G

### 1.3 QUALITY CONTROL

#### 1.3.1 Installer Qualifications

Install specified materials by a licensed underground utility Contractor licensed for such work in the state where the work is to be performed. Verify installing Contractor's License is current and state certified or state registered.

### 1.4 DELIVERY, STORAGE, AND HANDLING

#### 1.4.1 Delivery and Storage

Check upon arrival; identify and segregate as to types, functions, and sizes. Store off the ground in a manner affording easy accessibility and not causing excessive rusting or coating with grease or other objectionable materials.

##### 1.4.1.1 Piping

Inspect materials delivered to site for damage; store with minimum of handling. Store materials on site in enclosures or under protective coverings. Store plastic piping and jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes and fittings free of dirt and debris.

##### 1.4.2 Handling

Handle pipe, fittings, and other accessories in such manner as to ensure delivery to the trench in sound undamaged condition. Carry, do not drag, pipe to trench. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install the plastic pipe in accordance with the manufacturer's recommendation and discard those materials if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

#### 2.1.1 Sanitary Sewer Gravity Pipeline

Provide new and modify existing exterior sanitary gravity sewer piping and appurtenances. Provide each system complete and ready for operation. The exterior sanitary gravity sewer system includes equipment, materials, installation, and workmanship as specified herein more than 1.5 m outside of building walls.

### 2.2 MATERIALS

Provide materials conforming to the respective specifications and other requirements specified below. Submit manufacturer's product specification, standard drawings or catalog cuts.

## 2.2.1 Gravity Pipe

### 2.2.1.1 PVC Gravity Sewer Piping

#### 2.2.1.1.1 PVC Gravity Pipe and Fittings

ASTM D3034, SDR 35, or ASTM F949 with ends suitable for elastomeric gasket joints.

#### 2.2.1.1.2 PVC Gravity Joints and Jointing Material

Provide joints conforming to ASTM D3212. Gaskets are to conform to ASTM F477.

## 2.2.2 Cement Mortar

Provide cement mortar conforming to ASTM C270, Type M with Type II cement.

## 2.2.3 Portland Cement

Submit certificates of compliance stating the type of cement used in manufacture of concrete pipe, fittings, septic tanks, and precast manholes. Provide portland cement conforming to ASTM C150/C150M, Type II for concrete used in concrete pipe, concrete pipe fittings, septic tanks, and manholes and type optional for cement used in concrete cradle, concrete encasement, and thrust blocking. Use air-entraining admixture conforming to ASTM C260/C260M with Type V cement.

## 2.2.4 Portland Cement Concrete

Provide portland cement concrete conforming to ASTM C94/C94M, compressive strength of 28 MPa at 28 days, except for concrete cradle and encasement or concrete blocks for manholes. Concrete used for cradle and encasement is to have a compressive strength of 17 MPa minimum at 28 days. Protect concrete in place from freezing and moisture loss for 7 days.

## 2.2.5 Precast Concrete Manholes

Provide precast concrete manholes, risers, base sections, and tops conforming to ASTM C478M.

## 2.2.6 Glass-Fiber-Reinforced Polyester Manholes

Glass-Fiber-Reinforced Polyester Manholes are to conform to ASTM D3753.

## 2.2.7 Gaskets and Connectors

Provide gaskets for joints between manhole wastewater tanks sections conforming to ASTM C443M. Resilient connectors for making joints between manhole wastewater tanks and pipes entering manhole are to conform to ASTM C923M or ASTM C990M.

## 2.2.8 External Preformed Rubber Joint Seals

An external preformed rubber joint seal is an accepted method of sealing cast iron covers to precast concrete sections to prevent ground water infiltration into sewer systems. All finished and sealed manholes constructed in accordance with paragraph entitled "Manhole Construction" are to be tested for leakage in the same manner as pipelines as described

in paragraph entitled "Leakage Tests." The seal is to be multi-section with a neoprene rubber top section and all lower sections made of Ethylene Propylene Diene Monomer (EPDM) rubber with a minimum thickness of 1.5 mm. Each unit is to consist of a top and bottom section and have mastic on the bottom of the bottom section and mastic on the top and bottom of the top section. The mastic is to be a non-hardening butyl rubber sealant and seal to the cone/top slab of the manhole/catch basin and over the lip of the casting. Extension sections are to cover up to two more adjusting rings. Properties and values are listed in the following table:

| Properties, Test Methods and Minimum Values for Rubber used in Preformed Joint Seals |                   |        |          |              |
|--------------------------------------------------------------------------------------|-------------------|--------|----------|--------------|
| Physical Properties                                                                  | Test Methods      | EPDM   | Neoprene | Butyl Mastic |
| Tensile, kPa                                                                         | ASTM D412         | 12,684 | 15,132   | --           |
| Elongation, percent                                                                  | ASTM D412         | 553    | 295      | 350          |
| Tear Resistance, N/mm                                                                | ASTM D624 (Die B) | 49     | 29       | --           |
| Rebound, percent, 5 minutes                                                          | ASTM C972 (mod.)  | --     | --       | 11           |
| Rebound, percent, 2 hours                                                            | ASTM C972         | --     | --       | 12           |

#### 2.2.9 Precast Concrete Septic Tanks

Provide precast concrete septic tanks risers, base sections, and tops conforming to ASTM C478M and be manufactured in accordance with the latest edition of Standard CAN/CSA - B66-21; base and first riser are to be monolithic.

#### 2.2.10 Frames, Covers, and Gratings for Manholes

Submit certification on the ability of frame and cover to carry the imposed live load. Frame and cover are to be cast gray iron, ASTM A48/A48M, Class 35B, cast ductile iron, ASTM A536, Grade 65-45-12, or reinforced concrete, ASTM C478M. Frames and covers are to be circular without vent holes. Size are to be for 24 inch opening. Stamp or cast the words "Sanitary Sewer" into covers so that it is plainly visible.

#### 2.2.11 Manhole Steps

Zinc-coated steel conforming to 29 CFR 1910.27 with a plastic or rubber coating pressure-molded to the steel is to be used. Provide plastic coating conforming to ASTM D4101, copolymer polypropylene. Rubber is to conform to ASTM C443M, except shore A durometer hardness is to be 70 plus or minus 5. Aluminum steps or rungs will not be permitted. Steps are not required in manholes less than 1.2 m deep.

#### 2.2.12 Manhole Ladders

Provide a steel ladder where the depth of a manhole exceeds 3.6 m. The

ladder is not to be less than 406 mm in width, with 19 mm diameter rungs spaced 305 mm apart. The two stringers are to be a minimum 10 mm thick and 51 mm wide. Galvanize ladders and inserts after fabrication in conformance with ASTM A123/A123M.

### PART 3 EXECUTION

#### 3.1 PREPARATION

##### 3.1.1 Installation Drawings

Submit Installation Drawings showing complete detail, both plan and side view details with proper layout and elevations.

#### 3.2 INSTALLATION

Backfill after inspection by the Contracting Officer. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer.

##### 3.2.1 Connections to Existing Lines

Obtain approval from the Contracting Officer before making connection to existing line. Conduct work so that there is minimum interruption of service on existing line.

##### 3.2.2 General Requirements for Installation of Pipelines

These general requirements apply except where specific exception is made in the following paragraphs entitled "Special Requirements."

###### 3.2.2.1 Location

Terminate the work covered by this section at a point approximately 1.5 m from the building.

###### 3.2.2.1.1 Sanitary Piping Installation Parallel with Water Line

###### 3.2.2.1.1.1 Normal Conditions

Install sanitary piping or manholes at least 3 m horizontally from a water line whenever possible. Measure the distance from edge-to-edge.

###### 3.2.2.1.1.2 Unusual Conditions

When local conditions prevent a horizontal separation of 3 m, the sanitary piping or manhole may be laid closer to a water line provided that:

- a. The top (crown) of the sanitary piping is to be at least 450 mm below the bottom (invert) of the water main.
- b. Where this vertical separation cannot be obtained, construct the sanitary piping with AWWA-approved ductile iron water pipe pressure and conduct a hydrostatic sewer test without leakage prior to backfilling.

- c. The sewer manhole is to be of watertight construction and tested in place.

#### 3.2.2.1.2 Installation of Sanitary Piping Crossing a Water Line

##### 3.2.2.1.2.1 Normal Conditions

Lay sanitary sewer piping by crossing under water lines to provide a separation of at least 450 mm between the top of the sanitary piping and the bottom of the water line whenever possible.

##### 3.2.2.1.2.2 Unusual Conditions

When local conditions prevent a vertical separation described above, use the following construction:

- a. Construct sanitary piping passing over or under water lines with AWWA-approved ductile iron water pressure piping and conduct a hydrostatic sewer test without leakage prior to backfilling.
- b. Protect sanitary piping passing over water lines by providing:
  - (1) A vertical separation of at least 450 mm between the bottom of the sanitary piping and the top of the water line.
  - (2) Adequate structural support for the sanitary piping to prevent excessive deflection of the joints and the settling on and breaking of the water line.
  - (3) That the length, minimum 6.1 m, of the sanitary piping be centered at the point of the crossing so that joints are equidistant and as far as possible from the water line.

##### 3.2.2.1.3 Sanitary Sewer Manholes

No water piping must pass through or come in contact with any part of a sanitary sewer manhole.

##### 3.2.2.2 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK.

##### 3.2.2.3 Pipe Laying and Jointing

Inspect each pipe and fitting before and after installation; replace those found defective and remove from site. Provide proper facilities for lowering sections of pipe into trenches. Lay nonpressure pipe with the bell or groove ends in the upgrade direction. Adjust spigots in bells and tongues in grooves to give a uniform space all around. Blocking or wedging between bells and spigots or tongues and grooves will not be permitted. Replace by one of the proper dimensions, pipe or fittings that do not allow sufficient space for installation of joint material. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Provide batterboards not more than 7.50 m apart in trenches for checking and ensuring that pipe invert elevations are as indicated. Laser beam method may be used in lieu of batterboards for the same purpose. Construct branch connections by use of regular fittings or solvent cemented saddles as approved. Provide saddles for PVC pipe conforming to Table 4 of ASTM D3034.

### 3.2.3 Special Requirements

#### 3.2.3.1 Installation of PVC Piping

Install pipe and fittings in accordance with paragraph entitled "General Requirements for Installation of Pipelines" of this section and with the requirements of ASTM D2321 for laying and joining pipe and fittings. Make joints with the gaskets specified for joints with this piping and assemble in accordance with the requirements of ASTM D2321 for assembly of joints. Make joints to other pipe materials in accordance with the recommendations of the plastic pipe manufacturer.

#### 3.2.4 Concrete Work

Cast-in-place concrete is included in Section 03 30 00 CAST-IN-PLACE CONCRETE. Support the pipe on a concrete cradle, or encased in concrete where indicated or directed.

#### 3.2.5 Manhole Construction

Construct base slab of cast-in-place concrete or use precast concrete base sections. Make inverts in cast-in-place concrete and precast concrete bases with a smooth-surfaced semi-circular bottom conforming to the inside contour of the adjacent sewer sections. For changes in direction of the sewer and entering branches into the manhole, make a circular curve in the manhole invert of as large a radius as manhole size will permit. For cast-in-place concrete construction, either pour bottom slabs and walls integrally or key and bond walls to bottom slab. No parging will be permitted on interior manhole walls. For precast concrete construction, make joints between manhole sections with the gaskets specified for this purpose; install in the manner specified for installing joints in concrete piping. Parging will not be required for precast concrete manholes. Perform cast-in-place concrete work in accordance with the requirements specified under paragraph entitled "Concrete Work" of this section. Make joints between concrete manholes and pipes entering manholes with the resilient connectors specified for this purpose; install in accordance with the recommendations of the connector manufacturer. Where a new manhole is constructed on an existing line, remove existing pipe as necessary to construct the manhole. Cut existing pipe so that pipe ends are approximately flush with the interior face of manhole wall, but not protruding into the manhole. Use resilient connectors as previously specified for pipe connectors to concrete manholes.

#### 3.2.6 Miscellaneous Construction and Installation

##### 3.2.6.1 Connecting to Existing Manholes

Connect pipe to existing manholes such that finish work will conform as nearly as practicable to the applicable requirements specified for new manholes, including all necessary concrete work, cutting, and shaping. Center the connection on the manhole. Holes for the new pipe are to be of sufficient diameter to allow packing cement mortar around the entire periphery of the pipe but no larger than 1.5 times the diameter of the pipe. Cut the manhole in a manner that will cause the least damage to the walls.



### 3.2.6.2 Metal Work

#### 3.2.6.2.1 Workmanship and Finish

Perform metal work so that workmanship and finish will be equal to the best practice in modern structural shops and foundries. Form iron to shape and size with sharp lines and angles. Do shearing and punching so that clean true lines and surfaces are produced. Make castings sound and free from warp, cold shuts, and blow holes that may impair their strength or appearance. Give exposed surfaces a smooth finish with sharp well-defined lines and arises. Provide necessary rabbets, lugs, and brackets wherever necessary for fitting and support.

#### 3.2.6.2.2 Field Painting

After installation, clean cast-iron frames, covers, gratings, and steps not buried in concrete to bare metal, remove mortar, rust, grease, dirt, and other deleterious materials and apply a coat of bituminous paint. Do not paint surfaces subject to abrasion.

### 3.2.7 Installations of Wye Branches

Install wye branches in an existing sewer using a method which does not damage the integrity of the existing sewer. Do not cut into piping for connections except when approved by the Contracting Officer. When the connecting pipe cannot be adequately supported on undisturbed earth or tamped backfill, support on a concrete cradle as directed by the Contracting Officer. Provide and install concrete required because of conditions resulting from faulty construction methods or negligence without any additional cost to the Government. Do not damage the existing sewer when installing wye branches in an existing sewer.

## 3.3 FIELD QUALITY CONTROL

The Contracting Officer will conduct field inspections and witness field tests specified in this section. Be able to produce evidence, when required, that each item of work has been constructed in accordance with the drawings and specifications.

### 3.3.1 Tests

Perform field tests and provide labor, equipment, and incidentals required for testing.

#### 3.3.1.1 Hydrostatic Sewer Test

When unusual conflicts are encountered between sanitary sewer and waterlines a hydrostatic pressure sewer test will be performed in accordance with the applicable AWWA standard for the piping material or AWWA C600 with a minimum test pressure of 200 kPa.

#### 3.3.1.2 Leakage Tests for Nonpressure Lines

Test lines for leakage by either infiltration tests and exfiltration tests, negative air pressure tests or by low-pressure air tests. When necessary to prevent pipeline movement during testing, place additional backfill around pipe sufficient to prevent movement, but leaving joints uncovered to permit inspection. When leakage or pressure drop exceeds the allowable amount specified, make satisfactory correction and retest pipeline section

in the same manner. Correct visible leaks regardless of leakage test results.

#### 3.3.1.2.1 Infiltration Tests and Exfiltration Tests

##### 3.3.1.2.1.1 Precast Concrete Pipe Sewer Lines

Test leakage of precast concrete pipe in accordance with ASTM C969M. The allowable leakage limit is located in ASTM C969M. Make calculations in accordance with the Appendix to ASTM C969M.

##### 3.3.1.2.2 Negative Air Pressure Test

###### 3.3.1.2.2.1 Concrete Pipe

Test concrete pipe test in accordance with ASTM C1214M. The allowable vacuum loss is located in ASTM C1214M. Make calculations in accordance with the Appendix to ASTM C1214M.

###### 3.3.1.2.2.2 Precast Concrete Manholes

Test precast concrete sewer manhole test in accordance with ASTM C1244M. The allowable vacuum drop is located in ASTM C1244M. Make calculations in accordance with the Appendix to ASTM C1244M.

##### 3.3.1.2.3 Low-Pressure Air Tests

###### 3.3.1.2.3.1 PVC Pipelines

Test PVC pipe in accordance with UBPPA UNI-B-6. The allowable pressure drop is located in UBPPA UNI-B-6. Make calculations in accordance with the Appendix to UBPPA UNI-B-6.

#### 3.3.2 Inspection

Check each straight run of pipeline for gross deficiencies by holding a light in a manhole; the light must show a practically full circle of light through the pipeline when viewed from the adjoining end of line.

##### 3.3.2.1 Pre-Installation Inspection

Prior to connecting the new service, perform pre-installation inspection after trenching and layout is complete. Submit pre-installation inspection request for field support at least 14 days in advance. The Installation's Utilities Field Support personnel will perform the pre-installation inspection.

##### 3.3.2.2 Post-Installation Inspection

Perform a post-installation inspection after connection has been made and before the connection is buried. Submit post-installation inspection request for field support at least 14 days in advance. The Installation's Utilities Field Support personnel will perform the post-connection inspection. During the post-installation inspection the Contractor will be responsible for performing a dye test .

-- End of Section --

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## SECTION 33 40 00

## STORMWATER UTILITIES

**11/21**

## PART 1 GENERAL

## 1.1 UNIT PRICES

## 1.1.1 Pipe Culverts and Storm Drains

The length of pipe installed will be measured along the centerlines of the pipe from end to end of pipe without deductions for diameter of manholes. Pipe will be paid for at the contract unit price for the number of linear meters of culverts or storm drains placed in the accepted work.

## 1.1.2 Box Culverts

The length of box culvert installed will be measured along the centerline of the box from end to end of the box culvert. Box Culvert will be paid for at the contract unit price for the number of linear meters of box culverts placed in the accepted work.

## 1.1.3 Storm Drainage Structures

The quantity of manholes and inlets will be measured as the total number of manholes and inlets of the various types of construction, complete with frames and gratings or covers and, where indicated, with fixed side-rail ladders, constructed to the depth of 3 meters in the accepted work. The depth of manholes and inlets will be measured from the top of grating or cover to invert of outlet pipe. Manholes and inlets constructed to depths greater than the depth specified above will be paid for as units at the contract unit price for manholes and inlets, plus an additional amount per linear meter for the measured depth beyond a depth of 3 meters.

## 1.1.4 Walls and Headwalls

Walls and headwalls will be measured by the number of cubic meters of reinforced concrete, plain concrete, or masonry used in the construction of the walls and headwalls. Wall and headwalls will be paid for at the contract unit price for the number of walls and headwalls constructed in the completed work.

## 1.1.5 Flared End Sections

Flared end sections will be measured by the unit. Flared end sections will be paid for at the contract unit price for the various sizes in the accepted work.

## 1.1.6 Sheeting and Bracing

Payment will be made for that sheeting and bracing ordered to be left in place, based on the number of square meters of sheeting and bracing remaining below the surface of the ground.

## 1.1.7 Rock Excavation

Payment will be made for the number of cubic meters of material acceptably

excavated, as specified and defined as rock excavation in Section 31 00 00 EARTHWORK, measured in the original position, and computed by allowing actual width of rock excavation with the following limitations: maximum rock excavation width, 750 mm for pipe of 300 mm or less nominal diameter; maximum rock excavation width, 400 mm greater than outside diameter of pipe of more than 300 mm nominal diameter. Measurement will include authorized overdepth excavation. Payment will also include all necessary drilling and blasting, and all incidentals necessary for satisfactory excavation and disposal of authorized rock excavation. No separate payment will be made for backfill material required to replace rock excavation; include this cost in the unit price bid per cubic meter for rock excavation. In rock excavation for manholes and other appurtenances, 300 mm will be allowed outside the wall lines of the structures.

#### 1.1.8 Backfill Replacing Unstable Material

Payment will be made for the number of cubic meters of select granular material required to replace unstable material for foundations under pipes or drainage structures, which will constitute full compensation for this backfill material, including removal and disposal of unstable material and all excavating, hauling, placing, compacting, and all incidentals necessary to complete the construction of the foundation satisfactorily.

#### 1.1.9 Concrete Ditch Lining

Payment will be made for the number of linear meters of concrete ditch lining including any steel reinforcing accepted in the completed work measured along the centerline of the ditch.

### 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AASHTO M 294 (2021) Standard Specification for Corrugated Polyethylene Pipe, 300- to 1500-mm (12- to 60-in.) Diameter

#### ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M (2003; R 2021) Standard Specification for Gray Iron Castings

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A536 (2024) Standard Specification for Ductile Iron Castings

ASTM A716 (2018; R 2022) Standard Specification for

## Ductile Iron Culvert Pipe

|                 |                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------|
| ASTM A760/A760M | (2015, R 2020) Standard Specification for Corrugated Steel Pipe, Metallic-Coated for Sewers and Drains |
| ASTM B26/B26M   | (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings                                 |
| ASTM C32        | (2023) Standard Specification for Sewer and Manhole Brick (Made from Clay or Shale)                    |
| ASTM C55        | (2023) Standard Specification for Concrete Building Brick                                              |
| ASTM C62        | (2023) Standard Specification for Building Brick (Solid Masonry Units Made from Clay or Shale)         |
| ASTM C76        | (2022a) Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe            |
| ASTM C76M       | (2022a) Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe (Metric)   |
| ASTM C139       | (2023) Standard Specification for Concrete Masonry Units for Construction of Catch Basins and Manholes |
| ASTM C231/C231M | (2024) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method           |
| ASTM C270       | (2019a; E 2019) Standard Specification for Mortar for Unit Masonry                                     |
| ASTM C425       | (2021) Standard Specification for Compression Joints for Vitrified Clay Pipe and Fittings              |
| ASTM C443       | (2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets          |
| ASTM C443M      | (2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets (Metric) |
| ASTM C478/C478M | (2022) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections                |
| ASTM C655       | (2019a) Standard Specification for Reinforced Concrete D-Load Culvert, Storm Drain, and Sewer Pipe     |
| ASTM C655M      | (2019a) Standard Specification for                                                                     |

|                   |                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | Reinforced Concrete D-Load Culvert, Storm Drain, and Sewer Pipe (Metric)                                                                                            |
| ASTM C923/C923M   | (2020) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals                                           |
| ASTM C990         | (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants                      |
| ASTM C990M        | (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants (Metric)             |
| ASTM C1433        | (2020) Standard Specification for Precast Reinforced Concrete Monolithic Box Sections for Culverts, Storm Drains, and Sewers                                        |
| ASTM C1433M       | (2022) Standard Specification for Precast Reinforced Concrete Monolithic Box Sections for Culverts, Storm Drains, and Sewers (Metric)                               |
| ASTM D1751        | (2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)    |
| ASTM D1752        | (2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction            |
| ASTM D2321        | (2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications                                          |
| ASTM D3212        | (2020) Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals                                                         |
| ASTM F477         | (2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe                                                                      |
| ASTM F894         | (2024) Standard Specification for Polyethylene (PE) Large Diameter Profile Wall Sewer and Drain Pipe                                                                |
| ASTM F2764/F2764M | (2023) Standard Specification for 6 to 60 in. 150 to 1500 mm Polypropylene (PP) Corrugated Double and Triple Wall Pipe and Fittings for Non-Pressure Sanitary Sewer |

### Applications

|                   |                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM F2881/F2881M | (2021; E 2021) Standard Specification for 12 to 60 in. (300 to 1500 mm) Polypropylene (PP) Dual Wall Pipe and Fittings for Non-Pressure Storm Sewer Applications |
| ASTM F3219        | (2019) Standard Specification for 3 to 30 in. (75 to 750 mm) Polypropylene (PP) Corrugated Single Wall Pipe and Fittings                                         |

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-06 Test Reports

Leakage Test; G

#### SD-07 Certificates

Hydrostatic Test on Watertight Joints; G

Frame and Cover or Gratings; G

#### SD-08 Manufacturer's Instructions

Placing Pipe and Box Culvert; G

#### SD-11 Closeout Submittals

Post-Installation Inspection Report; G

LID Verification Report; G

### 1.4 DELIVERY, STORAGE, AND HANDLING

#### 1.4.1 Delivery and Storage

Inspect materials delivered to site for damage and unload and store materials with minimal handling. Do not store materials directly on the ground. Keep the inside of pipes and fittings free of dirt and debris. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install plastic pipe in accordance with the manufacturer's recommendations and discard if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.



#### 1.4.2 Handling

Handle materials in a manner that ensures delivery to the trench in sound, undamaged condition. Carry pipe to the trench.

### PART 2 PRODUCTS

#### 2.1 PIPE FOR CULVERTS AND STORM DRAINS

Pipe sizes for culverts and storm drains are indicated on the drawings.

##### 2.1.1 Concrete Pipe

###### 2.1.1.1 Reinforced Culvert and Storm Drain Pipe

Manufactured in accordance with and conforming to ASTM C76M ASTM C76, Class IV, or ASTM C655M ASTM C655, 50 D-Load.

##### 2.1.2 Ductile Iron Culvert Pipe

Provide ductile iron culvert pipe conforming to ASTM A716.

##### 2.1.3 Polyethylene (PE) Pipe

###### 2.1.3.1 Corrugated PE Pipe

AASHTO M 294, Type S. Provide pipe walls having the following properties:

| Nominal Size (mm) | Minimum Wall Area (square mm/m) | Minimum Moment of Inertia of Wall Section (mm to the 4th/mm) |
|-------------------|---------------------------------|--------------------------------------------------------------|
| 300               | 3200                            | 390                                                          |
| 375               | 4000                            | 870                                                          |
| 450               | 4900                            | 1020                                                         |
| 600               | 6600                            | 1900                                                         |
| 750               | 8300                            | 2670                                                         |
| 900               | 9500                            | 3640                                                         |
| 1050              | 9900                            | 8900                                                         |
| 1200              | 10,900                          | 8900                                                         |
| 1350              | 12,000                          | 13,110                                                       |
| 1500              | 13,650                          | 13,110                                                       |

###### 2.1.3.2 Profile Wall PE Pipe

ASTM F894, RSC 160. Provide pipe walls having the following properties:

| Nominal Size (mm) | Minimum Wall Area<br>(square mm/m) | Minimum Moment (mm to the 4th/mm) |                    |
|-------------------|------------------------------------|-----------------------------------|--------------------|
|                   |                                    | Cell Class 334433C                | Cell Class 335434C |
| 450               | 6300                               | 850                               | 620                |
| 525               | 8800                               | 1150                              | 840                |
| 600               | 9900                               | 1330                              | 970                |
| 675               | 12,500                             | 2050                              | 1490               |
| 750               | 12,500                             | 2050                              | 1490               |
| 825               | 14,800                             | 2640                              | 2160               |
| 900               | 17,100                             | 3310                              | 2700               |
| 1050              | 16,500                             | 4540                              | 3720               |
| 1200              | 18,700                             | 5540                              | 4540               |

#### 2.1.4 Polypropylene(PP) Pipe

Provide double wall and triple wall pipe meeting the requirements of ASTM F2764/F2764M or ASTM F2881/F2881M, Class I .

### 2.2 PIPE JOINTS

Provide joints that have been tested for and meet the requirements of paragraph HYDROSTATIC TEST ON WATERTIGHT JOINTS.

#### 2.2.1 Concrete Pipe

##### 2.2.1.1 Rubber Gasket Joints

Provide rubber gasket joints of a design and physical requirements conforming to ASTM C443M. Provide rubber gaskets that meet the oil resistant gasket requirements of ASTM C443M ASTM C443.

##### 2.2.1.2 Preformed Flexible Sealant Joints

Provide joints made with preformed flexible joint sealant conforming to ASTM C990M.

#### 2.2.2 Ductile Iron Pipe

Provide push-on type joints with rubber gaskets.

#### 2.2.3 Corrugated PE Plastic Pipe

Provide water tight joints conforming to the requirements in AASHTO M 294. Make water tight joints using a PE coupling and rubber gaskets as recommended by the pipe manufacturer. Provide rubber gaskets conforming to ASTM F477.

#### 2.2.4 Dual Wall and Triple Wall PP Pipe

Provide two gaskets conforming to ASTM F477 on the spigot. Gaskets must be installed by the pipe manufacturer and be covered with a removable, protective wrap to ensure the gaskets are free from debris. Use a joint lubricant available from the manufacturer on the gasket and bell during assembly. ASTM F2881/F2881M for 300 to 1500 mm pipe ASTM F3219 for 300 to 750 mm pipe ASTM F2764/F2764M for 750 to 1500 mm pipe diameters must have a reinforced bell with a polymer composite band installed by the manufacturer. Provide fittings conforming to ASTM F2881/F2881M ASTM F3219 ASTM F2764/F2764M. Utilize a spun-on, welded or integral bell and spigot with gaskets meeting ASTM F477 for bell and spigot connections.

#### 2.3 PRECAST REINFORCED CONCRETE BOX CULVERT

Manufacture precast reinforced concrete box culverts in accordance with and conforming to ASTM C1433M ASTM C1433.

#### 2.4 MISCELLANEOUS MATERIALS

##### 2.4.1 Concrete

Unless otherwise specified, provide concrete and reinforced concrete conforming to the requirements for MPa concrete under Section 03 30 00 CAST-IN-PLACE CONCRETE. Provide air content by volume of concrete mixture, based on measurements made immediately after discharge from the mixer, of 5 to 7 percent when maximum size of coarse aggregate exceeds 37.5 mm. Determine air content in accordance with ASTM C231/C231M. Provide a minimum concrete covering over steel reinforcing of not less than 25 mm thick for covers and not less than 40 mm thick for walls and flooring. For concrete deposited directly against the ground, provide a covering thickness of at least 75 mm between steel and ground. Provide expansion-joint filler material conforming to ASTM D1751, or ASTM D1752, or provide be resin-impregnated fiberboard conforming to the physical requirements of ASTM D1752.

##### 2.4.2 Mortar

Mortar is not allowed for pipe joints. Provide mortar for pipe connections to drainage structures and brick or block construction conforming to ASTM C270, Type M, except that the maximum placement time will be 1 hour. Provide a sufficient quantity of water in the mixture to produce a stiff workable mortar but in no case may the quantity exceed 19 liters of water per sack of cement. Use water that is clean and free of harmful acids, alkalis, and organic impurities. Use the mortar within 30 minutes after the ingredients are mixed with water.

##### 2.4.3 Precast Concrete Segmental Blocks

Provide precast concrete segmental block conforming to ASTM C139, not more than 200 mm thick, not less than 200 mm long, and of such shape that joints can be sealed effectively and bonded with cement mortar.

##### 2.4.4 Brick

Provide brick conforming to ASTM C62, Grade SW; ASTM C55, Grade S-I or S-II; or ASTM C32, Grade MS. Provide mortar for jointing and plastering consisting of one part portland cement and two parts fine sand. Lime may

be added to the mortar in a quantity not more than 25 percent of the volume of cement. Provide joints that are completely filled and that are smooth and free from surplus mortar on the inside of the structure. Plaster brick structures with 13 mm of mortar over the entire outside surface of the walls. Lay brick in stretcher courses with a header course every sixth course for square or rectangular structures. Lay brick radially with every sixth course a stretcher course for round structures.

#### 2.4.5 Precast Reinforced Concrete Manholes

Provide precast reinforced concrete manholes conforming to ASTM C478/C478M. Provide joints between precast concrete risers and tops that are made with flexible watertight, rubber-type gaskets meeting the requirements of paragraph PIPE JOINTS.

#### 2.4.6 Frame and Cover or Gratings

Provide frame and cover or gratings made of cast gray iron, ASTM A48/A48M, Class 35B; cast ductile iron, ASTM A536, Grade 65-45-12; or cast aluminum, ASTM B26/B26M, Alloy 356.0-T6. Stamp or cast the word "Storm Sewer" into covers so that it is plainly visible.

#### 2.4.7 Steel Ladder

Provide a steel ladder where the depth of the storm drainage structure exceeds 3.66 m. Provide ladders not less than 406 mm in width, with 19 mm diameter rungs spaced 305 mm apart. Provide two stringers that are a minimum 10 mm thick and 63 mm wide. Galvanize ladders and inserts after fabrication in conformance with ASTM A123/A123M.

#### 2.4.8 Resilient Connectors

Provide flexible, watertight connectors conforming to ASTM C923/C923M for connecting pipe to manholes and inlets.

### 2.5 TESTS, INSPECTIONS, AND VERIFICATIONS

#### 2.5.1 Hydrostatic Test on Watertight Joints

Perform a hydrostatic test on the watertight joint types as proposed. This test will be conducted at the plant or by an independent laboratory. Only one sample joint of each type needs testing; however, if the sample joint fails because of faulty design or workmanship, an additional sample joint may be tested.

##### 2.5.1.1 Concrete, Clay, PVC, PE, SRPE and PP Pipe

Provide joints in reinforced and nonreinforced concrete pipe meeting the performance requirements in ASTM C990M ASTM C990 or ASTM C443M ASTM C443. Provide joints in clay pipe meeting the test requirements in ASTM C425. Provide joints in PVC, PE, SRPE, and PP plastic pipe meeting the test requirements in ASTM D3212.

##### 2.5.1.2 Corrugated Steel and Aluminum Pipe

Perform a hydrostatic pressure test on the proposed joining system in accordance with ASTM A760/A760M. The joining system must not leak when subjected to an internal hydrostatic pressure of 69 kPa for a 10 minute period

### PART 3 EXECUTION

#### 3.1 EXCAVATION FOR PIPE CULVERTS, BOX CULVERTS, STORM DRAINS, AND DRAINAGE STRUCTURES

Excavate trenches, excavate for appurtenances and backfill for culverts and storm drains, in accordance with the applicable portions of Section 31 00 00 EARTHWORK and the requirements specified below.

##### 3.1.1 Trenching

Excavate trenches to the width indicated on the drawings or as specified herein. Trench width should permit satisfactory jointing and thorough tamping of the bedding material under and around the pipe. Place sheeting and bracing, where required, within the trench width as specified, without any overexcavation.

##### 3.1.2 Removal of Rock

Replace rock in either ledge or boulder formation with suitable materials to provide a compacted earth cushion. Provide a compacted earth cushion between unremoved rock and the pipe with a thickness of at least 200 mm or 13 mm for each meter of fill over the top of the pipe, whichever is greater, but not more than three-fourths the nominal diameter of the pipe. Maintain the cushion under the bell as well as under the straight portion of the pipe where bell-and-spigot pipe is used. Provide a compacted earth cushion between unremoved rock and the box culvert of at least 200 mm in thickness for concrete box culverts. Excavate rock as specified and defined in Section 31 00 00 EARTHWORK.

##### 3.1.3 Removal of Unstable Material

Where wet or otherwise unstable soil incapable of properly supporting the pipe or box culvert, as determined by the Contracting Officer, is unexpectedly encountered in the bottom of a trench, remove such material to the depth required and replace with select granular material to the proper grade. Compact select granular material as specified in paragraph FINAL BACKFILL. When removal of unstable material is due to the fault or neglect of the Contractor while performing shoring and sheeting, water removal, or other specified requirements, perform such removal and replacement at no additional cost to the Government.

#### 3.2 BEDDING AND INITIAL BACKFILL

Provide a firm bedding foundation of uniform density throughout the entire length of the pipe or box culvert.

##### 3.2.1 Concrete Pipe

Use select granular material conforming to Section 31 00 00 EARTHWORK for haunch and bedding material. Compact haunch and outer bedding to at least 90 percent laboratory maximum density and place in layers not exceeding 150 mm loose thickness for compaction by hand-operated compactors and 200 mm 8 inches for other than hand-operated machines. Loosely place middle bedding and do not compact. After the pipe has been properly bedded, place haunch material, at a moisture content that will facilitate compaction, evenly along both sides of the pipe and thoroughly compact each layer with mechanical tampers or rammers to the springline of the

pipe. Thoroughly compact the haunch material under the haunches of the pipe. For bell and spigot pipe, form a depression in bedding material for bells so entire barrel of pipe is uniformly supported. Minimize the length, depth, and width of bell depressions to that required for properly making the particular type of joint.

#### 3.2.1.1 Trenches

After the pipe has been properly bedded and haunch material placed to the midpoint (springline) of the pipe, backfill and compact the remainder of the trench by spreading and rolling or compacting by mechanical rammers or tampers in layers not exceeding 150 mm. Test for density as necessary to ensure conformance to the compaction requirements specified below. Leave untreated sheeting in place beneath structures or pavements.

#### 3.2.1.2 Fill Sections

For pipe placed in fill sections, uniformly spread fill material longitudinally on both sides of the pipe in layers not exceeding 150 mm in compacted depth, and compact by rolling parallel with pipe or by mechanical tamping or ramming. Prior to commencing normal filling operations, the crown width of the fill at a height of 300 mm above the top of the pipe must extend a distance of not less than twice the outside pipe diameter on each side of the pipe or 4 m, whichever is less. After the backfill has reached at least 300 mm above the top of the pipe, place and thoroughly compact the remainder of the fill in layers not exceeding 200 mm.

#### 3.2.2 Ductile Iron Pipe

Provide bedding for ductile iron pipe as shown on the drawings.

#### 3.2.3 Precast Reinforced Box Culvert

Use granular material a minimum of 150 mm in depth for bedding precast concrete box culverts in trenches with soil foundation. Provide granular bedding in trenches with rock foundation that is 13 mm in depth per 300 mm of depth of fill. The minimum depth of bedding will be 200 mm up to a maximum depth of 600 mm. Loosely place the granular bedding. Provide uniform support along the entire length of box culvert.

##### 3.2.3.1 Trenches

After the box culvert has been properly bedded, place selected material from excavation or borrow, at a moisture content that will facilitate compaction, along both sides of box culvert in layers not exceeding 150 mm in compacted depth. Bring the backfill up evenly on both sides of box culvert for the full length box culvert. Thoroughly compact each layer with mechanical tampers or rammers. Continue this method of filling and compacting until the fill has reached an elevation equal to the top of the box culvert. Backfill and compact the remainder of the trench by spreading and rolling or by compacting with mechanical rammers or tampers in layers not exceeding 150 mm. Test density as necessary to ensure conformance to the compaction requirements specified below. Where it is necessary, in the opinion of the Contracting Officer, that sheeting or portions of bracing used be left in place, the contract will be adjusted accordingly. Leave untreated sheeting in place beneath structures or pavements.

### 3.2.3.2 Fill Sections

Use backfill material and placement and compaction procedures for box culvert placed in fill sections as specified below. Uniformly spread the fill material longitudinally on both sides of the box in layers not exceeding 150 mm in compacted depth. Compacted by rolling parallel with pipe or by using mechanical tamping or ramming. Prior to commencing normal filling operations, the width of the fill at a height of 300 mm above the top of the box must extend a distance of not less than twice the outside width of the box culvert on each side of the box or 4 m, whichever is less. After the backfill has reached at least 300 mm above the top of the box, place and thoroughly compact the remainder of the fill in layers not exceeding 150 mm.

### 3.3 PLACING PIPE AND BOX CULVERT

Submit printed copies of the pipe or box culvert manufacturer's recommended pipe or box culvert installation procedures prior to installation. Thoroughly examine each section of pipe or box culvert before being laid; do not use defective or damaged pipe. Protect plastic pipe, excluding SRPE pipe, from exposure to direct sunlight prior to laying, if necessary to maintain adequate pipe stiffness and meet installation deflection requirements. Lay pipelines to the grades and alignment indicated. Provide proper facilities for lowering sections of pipe into trenches. Place lifting lugs in vertically elongated corrugated steel or aluminum pipe in the same vertical plane as the major axis of the pipe. Do not lay pipe in water or when trench conditions or weather are unsuitable for such work. Divert drainage or dewater trenches during construction as necessary. Deflection of installed flexible pipe must not exceed the following limits:

| TYPE OF PIPE         | MAXIMUM ALLOWABLE DEFLECTION (percent) |
|----------------------|----------------------------------------|
|                      |                                        |
|                      |                                        |
| Ductile Iron Culvert | 3                                      |
|                      |                                        |

#### 3.3.1 Concrete, Clay, PVC, Ribbed PVC, Ductile Iron Pipe

Lay pipe proceeding upgrade with spigot ends of bell-and-spigot pipe and tongue ends of tongue-and-groove pipe pointing in the direction of the flow.

#### 3.3.2 PE, SRPE, and Dual Wall and Triple Wall PP Pipe

Lay on a bed shaped to line and grade and joint sections together in accordance with manufacturer's guidelines.

#### 3.3.3 Multiple Culverts

Where multiple lines of pipe are installed, adjacent sides of pipe must be

at least half the nominal pipe diameter or 1 meter apart, whichever is less.

### 3.3.4 Precast Reinforced Concrete Box Culvert

Proceed upgrade with laying of sections and point tongue ends of tongue-and-groove box culvert section in the direction of flow.

## 3.4 JOINTING

### 3.4.1 Concrete and Clay Pipe

#### 3.4.1.1 Plastic Sealing Compound Joints for Tongue-and-Grooved Pipe and Box Culverts

Follow the recommendation of the particular manufacturer in regard to sealing compound special installation requirements. When lubricants, primers, or adhesives are used, only apply on surfaces that are dry and clean. Affix sealing compounds to the pipe or box culvert not more than 3 hours prior to installation of the pipe or box culvert. Protect sealing compounds from the sun, blowing dust, and other deleterious agents at all times. Inspect sealing compounds before installation of the pipe or box culvert, and remove and replace any loose or improperly affixed sealing compound. Align the pipe or box culvert with the previously installed pipe or box culvert, and pull the joint together.

#### 3.4.1.2 Flexible Watertight Joints

Use lubricants, cements, adhesives, and other special installation requirements for gaskets and jointing materials as recommended by the manufacturer. When lubricants, cements, or adhesives are used, only apply on surfaces that are clean and dry. Affix gaskets and jointing materials to the pipe not more than 24 hours prior to the installation of the pipe, and protect from the sun, blowing dust, and other deleterious agents at all times. Inspect gaskets and jointing materials before installing the pipe; remove and replace any loose or improperly affixed gaskets and jointing materials. Align the pipe with the previously installed pipe, and push the joint home. If the gasket becomes visibly dislocated when joining sections of pipe, remove the pipe and remake the joint.

## 3.5 DRAINAGE STRUCTURES

### 3.5.1 Manholes and Inlets

Construct manholes of precast reinforced concrete. Construct inlets of precast reinforced concrete. Provide manholes and inlets complete with frames and covers or gratings; and with fixed galvanized steel ladders as indicated. The wall along the line where steel ladders are installed must be vertical for its entire length. Adequately anchor ladders to the wall by means of steel inserts spaced not more than 1.83 m vertically, and install to provide at least 150 mm of space between the wall and the rungs. Make pipe connections to concrete manholes and inlets with flexible, watertight connectors.

### 3.5.2 Walls and Headwalls

Construct headwalls as indicated.



### 3.6 UNDERGROUND STORMWATER RETENTION/DETENTION SYSTEM

Install pipe and collection chambers as recommended by the manufacturer. Place foundation and embedment stone as recommended by the manufacturer of the pipe collection chambers. Begin compaction of initial fill after 300 mm of material have been placed over the pipe chambers. Compact initial fill in 150 mm thick layers to 90 percent maximum density. Use roller with a gross vehicle weight not exceeding 53 kN and a dynamic force not exceeding 89 kN.

### 3.7 FINAL BACKFILL

Backfill trenches with satisfactory material deposited in layers of a maximum of 200 mm loose thickness and compacted to 90 percent of maximum density for cohesive soils and 95 percent of maximum density for cohesionless soils in accordance with Section 31 00 00 EARTHWORK. Testing is the responsibility of the Contractor and will be performed at no additional cost to the Government. Unless otherwise specified, determine field in-place density of final backfill at a frequency of one test per 15 linear meters, or fraction thereof, of each lift of backfill. Submit test results in accordance with Section 31 00 00 EARTHWORK. Do not displace or damage pipe or box when compacting final backfill by rolling or operating heavy equipment parallel with the pipe or box. Movement of construction machinery over a culvert or storm drain at any stage of construction will be at the Contractor's risk. Repair or replace any damaged pipe. Protect concrete pipes with a minimum of 1 meter of cover prior to permitting heavy construction equipment to pass over them during construction. Provide the minimum cover for construction loads over corrugated steel pipes as specified in Section 26, Division II of AASHTO HB-17. Provide minimum cover for construction loads over plastic pipes as specified in ASTM D2321.

### 3.8 FIELD QUALITY CONTROL

#### 3.8.1 Tests

Testing is the responsibility of the Contractor. Perform all testing and retesting at no additional cost to the Government.

#### 3.8.2 Inspection

##### 3.8.2.1 Low Impact Development Inspection

Inspect Low Impact Development (LID) features indicated on the design portion of the LID Verification Report. Certify LID features were constructed according to plans and specifications or by submitting as-built drawings in accordance with UFGS 01 78 00 Closeout Submittals. When as-built drawings show deviations to the LID features, document the deviations on the LID Verification Report.

#### 3.8.3 Repair of Defects

##### 3.8.3.1 Leakage Test

When leakage exceeds the maximum amount specified, correct source of excess leakage by replacing damaged pipe and gaskets and retest.

### 3.8.3.2 Deflection Testing

When deflection readings are in excess of the allowable deflection of average inside diameter of pipe are obtained, remove pipe which has excessive deflection and replace with new pipe. Retest 30 days after completing backfill, leakage testing and compaction testing.

### 3.8.3.3 Inspection

Replace pipe or repair defects indicated in the Post-Installation Inspection Report.

#### 3.8.3.3.1 Concrete Pipe

Replace pipes having cracks with a width greater than 2.5 mm.

#### 3.8.3.3.2 Flexible Pipe

Replace pipes having cracks or splits.

## 3.9 PROTECTION

Protect storm drainage piping and adjacent areas from superimposed and external loads during construction.

## 3.10 WARRANTY PERIOD

Pipe segments found to have defects during the warranty period must be replaced with new pipe and retested.

-- End of Section --

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## SECTION 33 46 13

## FOUNDATION DRAINAGE

05/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO M 252 (2009; R 2017) Standard Specification for  
Corrugated Polyethylene Drainage Pipe

AASHTO M 288 (2021) Standard Specification for  
Geosynthetic Specification for Highway  
Applications

AASHTO M 294 (2021) Standard Specification for  
Corrugated Polyethylene Pipe, 300- to  
1500-mm (12- to 60-in.) Diameter

ASTM INTERNATIONAL (ASTM)

ASTM D2751 (2005) Standard Specification for  
Acrylonitrile-Butadiene-Styrene (ABS)  
Sewer Pipe and Fittings

ASTM D3034 (2016) Standard Specification for Type PSM  
Poly(Vinyl Chloride) (PVC) Sewer Pipe and  
Fittings

ASTM D3212 (2020) Standard Specification for Joints  
for Drain and Sewer Plastic Pipes Using  
Flexible Elastomeric Seals

ASTM D4632/D4632M (2015a) Grab Breaking Load and Elongation  
of Geotextiles

ASTM F667/F667M (2016; R 2021) Standard Specification for  
3 through 24 in. Corrugated Polyethylene  
Pipe and Fittings

ASTM F758 (2014; R 2023) Smooth-Wall Poly(Vinyl  
Chloride) (PVC) Plastic Underdrain Systems  
for Highway, Airport, and Similar Drainage

ASTM F949 (2020) Standard Specification for  
Poly(Vinyl Chloride) (PVC) Corrugated  
Sewer Pipe with a Smooth Interior and  
Fittings

## 1.2 SYSTEM DESCRIPTION

### 1.2.1 Extent

Furnish and install foundation drainage as a complete system as shown.

### 1.2.2 Outlet Connections

Connect foundation pipe to the storm drainage system as shown and specified in Section 33 40 00 STORMWATER UTILITIES.

### 1.2.3 Drainage Lines

Construct drainage lines of drain tile, perforated pipe, or porous pipe.

### 1.2.4 Outlet Lines

Construct outlet lines of closed-joint nonperforated, nonporous pipe.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-04 Samples

Materials.

#### SD-07 Certificates

Materials.

## 1.4 DELIVERY, STORAGE, AND HANDLING

Protect materials placed in storage from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Do not expose plastic pipe to direct sunlight for more than 6 months from time of manufacturer to installation.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Provide pipe for foundation drainage system of the type and size indicated. Use appropriate transitions, adapters, or joint details where pipes of different types or materials are connected. Submit two randomly selected samples of each type of pipe and fitting, prior to delivery of materials to the site, and certifications from the manufacturers attesting that materials meet specification requirements.

#### 2.1.1 Plastic Pipe

Provide plastic pipe containing ultraviolet inhibitor to provide protection from exposure to direct sunlight.

#### 2.1.1.1 Corrugated Polyethylene (PE) Drainage Pipe

Furnish ASTM F667/F667M heavy duty for pipe 80 to 150 mm inches in diameter inclusive, ASTM F667/F667M for pipe 200 to 600 mm in diameter; or AASHTO M 252 for pipe 80 to 250 mm in diameter or AASHTO M 294 for pipe 300 to 600 mm in diameter. Furnish pipe manufacturer's standard fittings and conforming to the indicated specification.

#### 2.1.1.2 Acrylonitrile-Butadiene-Styrene (ABS) Pipe

ASTM D2751, with a maximum SDR of 35.

#### 2.1.1.3 Polyvinyl Chloride (PVC) Pipe

ASTM F758, Type PS 46, ASTM D3034, or ASTM F949 with a minimum pipe stiffness of 317 kPa.

#### 2.1.1.4 Circular Perforations in Plastic Pipe

Cleanly cut circular holes no more than 8 mm or less than 5 mm in diameter, and arrange in rows parallel to the longitudinal axis of the pipe. Place perforations approximately 75 mm apart, center-to-center, along rows. Place rows approximately 38 mm apart and arrange in a staggered pattern so that all perforations lie at the midpoint between perforations in adjacent rows. Space rows no more than 155 degrees of circumference. Do not perforate the spigot or tongue end of the pipe for a length equal to the depth of the socket and continue perforations at uniform spacing over the entire length of the pipe. Manufacturer's standard perforated pipe which essentially meets these requirements may be used with prior approval of the Contracting Officer.

#### 2.1.1.5 Slotted Perforations in Plastic Pipe

Cleanly cut circumferential slots so as not to restrict the inflow of water and uniformly spaced along the length and circumference of the tubing. Width of slots exceeding 3 mm or less than 0.79 mm is not permitted. Length of individual slots exceeding 32 mm on 75 mm diameter tubing is not permitted; 10 percent of the tubing inside nominal circumference on 100 to 200 mm diameter tubing; and 65 mm on 250 mm diameter tubing. Symmetrically space rows of slots so that they are fully contained in quadrants of the pipe. Center slots in the valleys of the corrugations of profile wall pipe. Provide water inlet area consisting of a minimum of 1058 square mm/linear meter of tubing. Manufacturer's standard perforated pipe which essentially meets these requirements may be used with prior approval of the Contracting Officer.

#### 2.1.2 Fittings

Provide fittings consisting of compatible materials for pipe, of corresponding weight and quality, and as specified herein.

#### 2.1.3 Cleanouts and Piping Through Walls

Provide cleanout pipe and fittings and piping through walls and footings consisting of cast-iron soil pipe. Each cleanout must have a brass ferrule and a cast-brass screw-jointed plug with socket or raised head for wrench.

#### 2.1.4 Bedding and Pervious Backfill for Foundation Drains

Provide bedding and pervious backfill in accordance with Section 31 00 00 EARTHWORK.

#### 2.1.5 Geotextile

Provide geotextile conforming to AASHTO M 288 and meeting the subsurface drainage requirements. The minimum grab strength will be 160 pounds in accordance with ASTM D4632/D4632M. Provide geotextile with filaments constructed so as to retain their relative position with respect to each other.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 Trenching and Excavation

Perform required trenching and excavation in accordance with Section 31 00 00 EARTHWORK. Keep trenches dry during installation of drainage system. Make changes in direction of drain lines with 1/8 bends. Use wye fittings at intersections.

##### 3.1.2 Bedding

Place graded bedding, minimum 150 mm in depth, in the bottom of trench for its full width and length compacted as specified prior to laying of foundation drain pipe. Rest each section firmly upon the bedding, through the entire length, with recesses formed for bell joints. Except for recesses for bell joints, fully support the lower quadrant of the pipe.

##### 3.1.3 Pipe Laying

Lay drain lines to true grades and alignment with a continuous fall in the direction of flow. Face bells of pipe sections upgrade. Clean interior of pipe thoroughly before being laid. When drain lines are left open for connection to discharge lines, temporarily close the open ends and mark the location with wooden stakes. Lay perforated pipe with perforations facing down. Remove and relay any length that has had its grade or joints disturbed at no additional cost to the Government. Install perforated corrugated polyethylene drainage tubing and plastic piping in accordance with manufacturer's specifications and as specified herein. Do not install tubing and piping with physical imperfections.

##### 3.1.4 Jointing

###### 3.1.4.1 Perforated and Porous Pipes

Lay perforated and porous types of drain pipes with closed joints.

###### 3.1.4.2 ABS Pipe

Join ABS pipe using solvent cement or elastomeric joints and in accordance with ASTM D2751, with dimensions and tolerances in accordance with TABLE II therein.

#### 3.1.4.3 PVC Pipe

Furnish PVC pipe joints in accordance with ASTM D3034, ASTM D3212, or ASTM F949.

#### 3.1.4.4 Corrugated Polyethylene

Furnish corrugated polyethylene (PE) pipe joints in accordance with ASTM F667/F667M or ASTM F667/F667M.

#### 3.1.5 Outlet Lines

The outlet end of drain lines connecting with an open gutter or outfall must be finished as shown.

#### 3.1.6 Cleanouts

Provide cleanouts in locations indicated. Set cleanouts in unpaved areas in 305 by 305 by 102 mm concrete blocks.

#### 3.2 Backfilling

After joints and connections have been inspected and approved, place the specified pervious backfill material a minimum width of 150 mm on each side of the pipe and 300 mm above the top of the pipe. Place the backfill preventing displacement of or injury to the pipe. Place a protective covering, as specified, over the pervious backfill for the full width of the trench before regular backfill is placed. Compact backfill as specified in Section 31 00 00 EARTHWORK.

-- End of Section --



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## SECTION 33 52 10

## FUEL SYSTEMS PIPING (SERVICE STATION)

11/18, CHG 1: 11/20

## PART 1 GENERAL

## 1.1 SUMMARY

This section defines the requirements for pipe, piping components, and valves as related to small fuel distribution systems (non-aviation type). Provide the entire fuel distribution system as a complete and fully operational system. Size, select, construct, and install system components to operate together as a complete system. Substitutions of functions specified herein will not be acceptable. Coordinate the work of the system manufacturer's service personnel during construction, testing, calibration, and acceptance of the system. Design system components and piping specified herein to handle a working pressure of 1900 kPa at 38 deg C. System components specified herein must be compatible with the fuel to be handled.

## 1.1.1 Related Sections

## 1.1.1.1 Welding

Welding activities for pipe and piping components must be in accordance with Section 33 52 23.15 POL SERVICE PIPING WELDING.

## 1.1.1.2 Earthwork

Excavate and backfill piping as specified in Section 31 00 00 EARTHWORK.

## 1.1.1.3 Cathodic Protection

. Cathodic protection for metal components that attach to a tank must be coordinated and compatible with the tank corrosion control system.

## 1.1.1.4 Concrete Manholes

Construct manhole of concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN PETROLEUM INSTITUTE (API)

|             |                                                                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API RP 540  | (1999; R 2004) Electrical Installations in Petroleum Processing Plants                                                                                                                      |
| API RP 1110 | (2013; R 2018) Recommended Practice for the Pressure Testing of Steel Pipelines for the Transportation of Gas, Petroleum Gas, Hazardous Liquids, Highly Volatile Liquids, or Carbon Dioxide |

|              |                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------|
| API RP 2003  | (2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents |
| API STD 600  | (2015) Steel Gate Valves-Flanged and Butt-welding Ends, Bolted Bonnets                           |
| API STD 608  | (2012) Metal Ball Valves - Flanged, Threaded, And Welding End                                    |
| API Spec 5L  | (2018; 46th Ed; ERTA 2018) Line Pipe                                                             |
| API Spec 6D  | (2021; Addendum 1 2025) Specification for Pipeline and Piping Valves                             |
| API Spec 6FA | (1999; R 2006; Errata 2006; Errata 2008; R 2011) Specification for Fire Test for Valves          |
| API Std 594  | (2017) Check Valves: Flanged, Lug, Wafer and Butt-Welding                                        |
| API Std 607  | (2016) Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats             |

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|              |                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ASME B1.1    | (2024) Unified Inch Screw Threads (UN, UNR, and UNJ Thread Form)                                                                     |
| ASME B16.3   | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                                                                         |
| ASME B16.5   | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard                                                 |
| ASME B16.9   | (2024) Factory-Made Wrought Buttwelding Fittings                                                                                     |
| ASME B16.11  | (2022) Forged Fittings, Socket-Welding and Threaded                                                                                  |
| ASME B16.21  | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                                                                                     |
| ASME B16.34  | (2021) Valves - Flanged, Threaded and Welding End                                                                                    |
| ASME B16.39  | (2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300                                                   |
| ASME B18.2.1 | (2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series) |
| ASME B18.2.2 | (2022) Nuts for General Applications:                                                                                                |

## Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

|                       |                                                                                |
|-----------------------|--------------------------------------------------------------------------------|
| ASME B31.3            | (2024) Process Piping                                                          |
| ASME B40.200          | (2008; R 2013) Thermometers, Direct Reading and Remote Reading                 |
| ASME BPVC SEC VIII D1 | (2023) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1 |

## AMERICAN WATER WORKS ASSOCIATION (AWWA)

|           |                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| AWWA C209 | (2019) Cold-Applied Tape Coatings for the Exterior of Special Sections, Connections and Fitting for Steel Water Pipelines                         |
| AWWA C215 | (2022) Extruded Polyolefin Coatings for Steel Water Pipe                                                                                          |
| AWWA C216 | (2022) Heat-Shrinkable Cross-Linked Polyolefin Coatings for the Exterior of Special Sections, Connections, and Fittings for Steel Water Pipelines |
| AWWA C217 | (2023) Microcrystalline Wax and Petrolatum Tape Coating Systems for Steel Water Pipe and Fittings                                                 |

## AMERICAN WELDING SOCIETY (AWS)

|         |                                 |
|---------|---------------------------------|
| AWS BRH | (2007; 5th Ed) Brazing Handbook |
|---------|---------------------------------|

## ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                                                                                                     |
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                                         |
| ASTM A105/A105M | (2023) Standard Specification for Carbon Steel Forgings for Piping Applications                                                                               |
| ASTM A182/A182M | (2024) Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service |
| ASTM A193/A193M | (2024a) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications      |
| ASTM A194/A194M | (2024) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both        |

|                 |                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A234/A234M | (2024) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service                             |
| ASTM A269/A269M | (2024) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service                                                     |
| ASTM A307       | (2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                                                       |
| ASTM A312/A312M | (2022a) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes                                                   |
| ASTM A358/A358M | (2024a) Standard Specification for Electric-Fusion-Welded Austenitic Chromium-Nickel Stainless Steel Pipe for High-Temperature Service and General Applications |
| ASTM A403/A403M | (2025) Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings                                                                            |
| ASTM A563       | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                                                                                          |
| ASTM A733       | (2016; R 2022) Standard Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples                                          |
| ASTM B117       | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                                                               |
| ASTM B247       | (2020) Standard Specification for Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings                                             |
| ASTM B687       | (1999; R 2023) Standard Specification for Brass, Copper, and Chromium-Plated Pipe Nipples                                                                       |
| ASTM D229       | (2019) Standard Test Methods for Rigid Sheet and Plate Materials Used for Electrical Insulation                                                                 |
| ASTM D3308      | (2012; R 2017) Standard Specification for PTFE Resin Skived Tape                                                                                                |
| ASTM F436       | (2011) Hardened Steel Washers                                                                                                                                   |
| ASTM F844       | (2019; R 2024) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use                                                              |
| ASTM F1172      | (1988; R 2024) Standard Specification for                                                                                                                       |

## Fuel Oil Meters of the Volumetric Positive Displacement Type

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book
- IEEE 1100 (2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment
- IEEE C62.41.2 (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

## MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

- MSS SP-58 (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

## NACE INTERNATIONAL (NACE)

- NACE SP0185 (2024) Extruded Polyolefin Resin Coating Systems with Soft Adhesives for Underground or Submerged Pipe
- NACE SP0188 (2024) Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 30 (2024; TIA 24-1) Flammable and Combustible Liquids Code
- NFPA 30A (2024; ERTA 1 2023) Code for Motor Fuel Dispensing Facilities and Repair Garages
- NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code
- NFPA 77 (2024; ERTA 1 2023) Recommended Practice on Static Electricity
- NFPA 407 (2022; TIA 24-2) Standard for Aircraft Fuel Servicing
- NFPA 780 (2023) Standard for the Installation of Lightning Protection Systems

## SOCIETY FOR PROTECTIVE COATINGS (SSPC)

- SSPC PA 1 (2024) Shop, Field, and Maintenance Coating of Metals

## SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE J514

(2012) Hydraulic Tube Fittings

## U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-PRF-13789

(1999; Rev E; Notice 1 2008; Notice 2  
1016; Notice 3 2021) Strainers, Sediment:  
Pipeline, Basket Type

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Grounding and Bonding

Pipe Supports

## SD-03 Product Data

Insulating Flange Kits; G

Flange Protectors; G

Fuel Piping Flange Bolts, Nuts, and Washers; G

Carbon Steel Pipe; G

Stainless Steel Pipe; GJoint Compound; G

Flexible Connector; G

Strainer; G

Thermometers; G

Pressure Gauge; G

Flexible Ball Joint; G

Bellows Expansion Joint; G

Flow Meter; G

Ball Valves; G

Plug (Double Block and Bleed) Valves; G

Swing Type Check Valves; G

Wafer Type Check Valve; G

Globe Valve; G

Thermal Relief Valve; G

Pressure\Vacuum Relief Valve; G

Foot Valve; G

Tank Overfill Prevention Valve (Gravity Fill); G

Tank Overfill Prevention Valve (Pumped Fuel Receipt); G

Anti-Siphon Valves; G

Pipeline Markers; G

#### SD-06 Test Reports

Exterior Coating Holiday Test

Preliminary Pneumatic Test

Final Pneumatic Test

Hydrostatic Test

Exterior Containment Piping Tests

#### SD-07 Certificates

Contractor Qualifications; G

Licensed Personnel

Pipeline Inventory; G

Demonstrations

#### SD-08 Manufacturer's Instructions

Flexible Ball Joint

Bellows Expansion Joint

#### SD-10 Operation and Maintenance Data

Insulating Flange Kits; G

Flange Protectors; G

Strainer; G

Thermometers; G

Flexible Ball Joint; G

Bellows Expansion Joint; G



Flow Meter; G

Ball Valves; G

Plug (Double Block and Bleed) Valves; G

Swing Type Check Valves; G

Wafer Type Check Valve; G

Globe Valve; G

Thermal Relief Valve; G

Pressure\Vacuum Relief Valve; G

Foot Valve; G

Tank Overfill Prevention Valve (Gravity Fill); G

Tank Overfill Prevention Valve (Pumped Fuel Receipt); G

Anti-Siphon Valves; G

#### 1.4 QUALITY ASSURANCE

##### 1.4.1 Contractor Qualifications

Each installation Contractor must have successfully completed at least 3 projects of the same scope and the same size, or larger, within the last 6-years; demonstrate specific installation experience in regard to the specific system installation to be performed; have taken, if applicable, manufacturer's training courses on the installation of piping; and meet the licensing requirements in the state. Experience must include the erection of piping systems in compliance with the requirements of ASME B31.3, NFPA 30, and NFPA 30A. Submit a letter listing prior projects, the date of construction, a point of contact for each prior project, the scope of work of each prior project, and a detailed list of work performed providing in the letter evidence of prior manufacturer's training and state licensing.

##### 1.4.2 Regulatory Requirements

###### 1.4.2.1 Licensed Personnel

Pipe installers must be licensed/certified when the state, city or locality requires licensed installers.

##### 1.4.3 Design Data

###### 1.4.3.1 Pipeline Inventory

Fuel system volume must be calculated using as constructed pipe lengths, internal diameters, fittings, and components. Totals must be provided for all items containing fuel with the exception of tanks which is covered by other specifications. A certified pipeline inventory with sizes, lengths, quantity, and volumes must be provided for the systems in this project.

### 1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect system components and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

### 1.6 PROJECT/SITE CONDITIONS

Fuel required for the testing, flushing and cleaning efforts, as specified in this section, will be provided and delivered by the Contracting Officer. Fuel used in the system will remain the property of the Government. Fuel shortages not attributable to normal handling losses must be reimbursed to the Government.

## PART 2 PRODUCTS

### 2.1 ELECTRICAL WORK

#### 2.1.1 General

Motors, manual or automatic motor control system components except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 2.1.2 Grounding and Bonding

Ground and bond in accordance with NFPA 70, NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

### 2.2 MATERIALS AND SYSTEM COMPONENTS

Internal parts and components of system components, piping, piping components, and valves that could be exposed to fuel during system operation must not be constructed of zinc coated (galvanized) metal, brass, bronze, or other copper bearing alloys. Do not install cast iron bodied valves in piping systems that could be exposed to fuel during system operation.

#### 2.2.1 Standard Products

Provide materials and system components that are standard products of a manufacturer regularly engaged in the manufacturing of such products; that are of a similar material, design and workmanship; and that have been in satisfactory commercial or industrial use for a minimum 2-years prior to bid opening. The 2-year period must include applications of the system components and materials under similar circumstances and of similar size. Materials and system components must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.

### 2.2.2 Nameplates

Attach nameplates to all specified system components, thermometers, gauges, and valves defined herein. List on each nameplate the manufacturer's name, address, contract number, component type or style, model or serial number, catalog number, capacity or size, and the system that is controlled. Construct plates of melamine plastic, 3 mm thick, UV resistance, black with white center core, matte finish surface and square corners. Install nameplates in prominent locations with nonferrous screws, nonferrous bolts, or permanent adhesive. Minimum size of nameplates must be 25 by 65 mm. Lettering must be the normal block style with a minimum 6 mm height. Accurately align all lettering on nameplates.

For plastic nameplates, engrave lettering into the white core.

### 2.2.3 Flange Gaskets, Non-Electrically Isolating

ASME B16.21, composition ring, using a Nitrile Rubber such as Buna-N and NBR, or a fluoro rubber such as FKM, FPM and Viton®. The gasket must be 3 mm thick. Gaskets must be resistant to the effects of aviation and non-aviation hydrocarbon fuels and manufactured of fire-resistant materials. Use fluoro rubber gaskets for biofuel blend fluids. Full-face gaskets must be used for flat-face flanged joints. Ring gaskets must be used for raised-face flanged joints. Gaskets must be of one piece factory cut material. Select a gasket suitable for the working and test pressure of the fluid.

## 2.3 FLANGED END CONNECTIONS

### 2.3.1 Flanges

Provide flanged end connections on system components, fittings, piping, piping components, adapters, couplers, and valves that conform to ASME B16.5, Class 150.

#### 2.3.1.1 Carbon Steel

Carbon steel flanges must conform to ASTM A105/A105M.

#### 2.3.1.2 Stainless Steel

Stainless steel flanges must conform to ASTM A182/A182M, Grade F304, forged type.

#### 2.3.1.3 Aluminum

Aluminum flanges must conform to ASTM B247, Alloy 6061-T6.

### 2.3.2 Insulating Flange Kits, (Electrically Isolating)

Provide ASTM D229 electrical insulating material of 1,000 ohms minimum resistance or 500 Volts per mil (VPM) minimum dielectric strength; material must be resistant to the effects of aviation and non-aviation hydrocarbon fuels. Provide full face insulating gaskets between flanges with fluoroelastomer (FKM), commonly referred to as Viton, O-ring sealing surfaces. Provide full surface 0.76 mm thick wall thickness, spiral-wound mylar insulating sleeves between the bolts and the holes in flanges; bolts may have reduced shanks of a diameter not less than the diameter at the root of threads. Provide 3.18 mm thick high-strength phenolic insulating washers next to flanges and provide flat circular stainless steel washers

over insulating washers and under bolt heads and nuts. Provide bolts 12 mm longer than standard length to compensate for the thicker insulating gaskets and the washers under bolt heads and nuts. Above grade flanges separated by electrically insulating flange kits must be provided with weatherproof lightning surge arrester devices. The surge arrester must bolt across flanges separated by insulating gasket kits per detail on contract drawings. Provide with flange protector as described in this section. The arrester must have the following features:

- a. Weatherproof NEMA 6P enclosure.
- b. Bidirectional and bipolar protection.
- c. Constructed of solid state components, no lights, fuses or relays and used without required maintenance or replacement.
- d. Withstand unlimited number of surges at 50,000 Amperes.
- e. Maximum clamping voltage of 700 Volts based on a IEEE C62.41.2 8x20 microsecond wave form at 50,000 Amperes peak measured at the device terminals (zero lead length).
- f. A UL listed arrester for installation in Class 1, Division 1 or Class 1, Division 2, Group D, hazardous areas.

Install the mounting bracket and leads on the flange side of the bolt insulating sleeve and washer, and size in accordance with this schedule:

| Line Size                                                                                                               | Bolt Size |
|-------------------------------------------------------------------------------------------------------------------------|-----------|
| 50.8 mm                                                                                                                 | 16 mm     |
| 63.5 mm                                                                                                                 | 16 mm     |
| 76 mm                                                                                                                   | 16 mm     |
| 102 mm                                                                                                                  | 16 mm     |
| 152 mm                                                                                                                  | 19 mm     |
| 203 mm                                                                                                                  | 19 mm     |
| 254 mm                                                                                                                  | 22 mm     |
| 305 mm                                                                                                                  | 22 mm     |
| 356 mm                                                                                                                  | 25 mm     |
| 406 mm                                                                                                                  | 25 mm     |
| Note: Make allowance for the 0.79 mm thickness of the insulating sleeve around the bolts when sizing the mounting lugs. |           |

### 2.3.3 Flange Protectors

Protectors must protect the bolts, studs, nuts, and gaskets of a flanged

end connection from corrosion or damage due to exposure to the environment. Protectors must be weather and ultraviolet (UV) resistant and constructed with stainless steel bands and rubber lining. Protectors must allow for quick and easy removal and re-installation by maintenance personnel. Provide grease filled bolt caps. Corrosion Prevention grease must be non-expansive and designed for the service. Provide protectors that allow visual inspection of the flange gasket without requiring removal.

#### 2.3.4 Fuel Piping Flange Bolts, Nuts, and Washers

- a. Bolts and nuts for pipe flanges, flanged fittings, valves and accessories must conform to ASME B18.2.1 and ASME B18.2.2, except as otherwise specified.
- b. Bolts must be of sufficient length to obtain full bearing on the nuts and must project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque.
- c. Bolts must be regular hexagonal bolts conforming to ASME B18.2.1 with material conforming to ASTM A193/A193M, Class 2, Grade B8, stainless steel, when connections are made where a stainless steel flange is involved, and Grade B7, chromium molybdenum alloy, when only carbon steel flanges are involved. Bolts and nuts chosen must have sufficient strength to seat gasket types chosen. Bolts must be threaded in accordance with ASME B1.1, Class 2A fit, Coarse Thread Series, for sizes one inch and smaller and Eight-Pitch Thread Series for sizes larger than one inch.
- d. Nuts must conform to ASME B18.2.2, hexagonal, heavy series with material conforming to ASTM A194/A194M, Grade 8, stainless steel for stainless steel bolts, and Grade 7, chromium molybdenum alloy for chromium molybdenum alloy bolts. Nuts must be threaded in accordance with ASME B1.1, Class 2B fit, Coarse Thread Series for sizes one inch and smaller and Eight-Pitch Thread Series for sizes larger than one inch.
- e. Provide washers under bolt heads and nuts. Use chromium molybdenum alloy washers dimensioned to ASTM F436 flat circular for chromium molybdenum bolts. Stainless steel washer dimensioned similar to ASTM F436 flat circular, use material the same as the bolt.
- f. Use torque wrenches to tighten all flange bolts to the torque recommended by the gasket manufacturer. Tighten in the pattern recommended by the gasket manufacturer. Use anti-seize compound on stainless steel bolts.

#### 2.4 PIPE

Pipe must meet the material, fabrication and operating requirements of ASME B31.3, except as modified herein.

##### 2.4.1 Carbon Steel Pipe

Provide carbon steel pipe that complies with one of the following:

- a. Pipe must conform to ASTM A53/A53M, Type E or S, Grade B, seamless or electric welded. Pipe smaller than 65 mm must be Schedule 80. Pipe

65 mm and larger must be Schedule 40.

- b. Pipe must conform to API Spec 5L, Product Specification Level (PSL) 1, Grade B, seamless or electric welded.

End connections for pipe or fittings smaller than 65 mm must be forged, socket weld type conforming to ASTM A182/A182M and ASME B16.11, unless indicated otherwise. End connections for pipe or fittings 65 mm and larger must be butt weld type conforming to ASTM A234/A234M, Grade WPB and ASME B16.9 of the same wall thickness as the adjoining pipe. Where threaded end connections are indicated, provide connections that conform to ASME B16.3, Class 150 or ASME B16.11.

#### 2.4.2 Stainless Steel Pipe

Provide stainless steel pipe that complies with one of the following:

- a. Pipe must conform to ASTM A312/A312M, Type TP304L, seamless only. Pipe smaller than 200 mm must be Schedule 40S. Pipe 200 mm or larger must be Schedule 10S.
- b. Pipe must conform to ASTM A358/A358M, Grade 304L, Class 1 or 3, longitudinally welded. Radiographically inspect 100 percent of factory longitudinal welds in accordance with ASME BPVC SEC VIII D1. Minimum pipe wall thickness must be 6 mm for pipe 300 mm and smaller; 8 mm for pipe larger than 300 mm.

##### 2.4.2.1 Fittings 65 mm and Larger

Provide butt welded type fittings that complies with one of the following:

- a. Stainless steel conforming to ASTM A403/A403M, Class WP-S, Grade WP 304L, seamless only and ASME B16.9 of the same thickness as the adjoining pipe.
- b. Stainless steel conforming to ASTM A403/A403M, Class WP-XX, Grade WP 304L, of wall thickness as indicated. Do not fabricate starting material by the fusion welding process without addition of filler metal. Forming will not be allowed using fusion welding process without addition of filler metal. Radiographically inspect all factory longitudinal welds in accordance with ASME BPVC SEC VIII D1.

##### 2.4.2.2 Fittings 50 mm and Smaller

Socket welded type fittings, unless indicated otherwise, must conform to ASME B16.11. Fitting materials must be stainless steel that conforms to ASTM A182/A182M, Type F304L.

##### 2.4.2.3 Control Tubing

Piping must be seamless, fully annealed stainless steel tubing conforming to ASTM A269/A269M, Grade TP316, with a hardness number not exceeding 80 HRB. For 15 mm tubing, provide a minimum 1.3 mm tubing wall thickness.

##### 2.4.2.4 Control Tubing Fittings

Fittings must be the flareless, Type 316 stainless steel type conforming to SAE J514.

## 2.5 PIPING COMPONENTS

Provide piping components that meet the material, fabrication and operating requirements of ASME B31.3, except as modified herein. Pressure design class for piping components must be Class 150 as defined in ASME B16.5.

### 2.5.1 Welded Nipples

Nipples must conform to ASTM A733 or ASTM B687 and be constructed of the same material as the connecting pipe.

### 2.5.2 Steel Couplings

Couplings must conform to API Spec 5L, seamless, extra heavy, wrought steel with recessed ends.

### 2.5.3 Threaded Unions

Unions must conform to ASME B16.39, Class 150. Unions materials must conform to ASTM A312/A312M, Grade 304 or 316. Dielectric unions must conform to dimensional, strength, and pressure requirements of ASME B16.39, Class 150. Steel parts must be galvanized or plated. Union must have a water-impervious insulation barrier capable of limiting galvanic current to one percent of the short-circuit current in a corresponding bimetallic joint. When dry, union must be able to withstand a 600-volt breakdown test.

### 2.5.4 Joint Compound

Joint compounds must be resistant to water and be suitable for use with fuel containing 40 percent aromatics.

### 2.5.5 Flexible Connector

Flexible connectors for fueling pumps must have ANSI Class 300 or 150 flanges to mate directly to the pump and Class 150 flanges to the system flanges. Flanges must be stainless steel and must conform to ASME B16.5. These units must have an inner stainless steel or Inconel, corrugated tube with external stainless steel braid, and all components must be rated for not less than 275 psig @ 100°F. Face by Face dimension must be as recommended by the manufacturer. Use Inconel 625 inner bellows in coastal environments or where chlorides are present in the atmosphere.

For sizes larger than 152 mm 6 inches, connectors must incorporate the use of Lo-corr, multi-ply bellows, without external braid, with bellows rating of 300 psig and overall rating consistent with the flange ANSI class. Flanges must be plate type, Vanstone design, with axial movement control rods.

Fabricate piping to measurements established on the project site and position into place without springing or forcing. Make provisions for absorbing expansion and contraction without undue stress in any part of the system. The use of flexible connectors in permanently mounted pump suction and discharge lines as a method of compensating for piping misalignment is not acceptable.

### 2.5.6 Strainer

Strainer must be single, basket type, arranged in a simplex configuration as indicated in compliance with MIL-PRF-13789, except as specified otherwise. Strainer end connections must be designed in accordance with ASME B16.5, Class 150. Strainer body material must be the same as the material specified for manual valves. Strainers must have removable baskets of 100 mesh wire screen with larger wire mesh reinforcement; wire must be stainless steel, Type 316. Pressure drop for clean strainer must not exceed 3 psig at maximum design flow rate. The ratio of net effective strainer area to the area of the connecting pipe must be not less than three to one. Each strainer must be provided with a suitable drain at the bottom, equipped with a ball valve. The strainer must be equipped with a direct-reading, piston type differential pressure gauge that measures the differential pressure across the basket as per Section 33 57 55 FUEL SYSTEM COMPONENTS (NON-HYDRANT).

### 2.5.7 Thermometers

Analog, dial-type bimetallic actuated type that conforms to ASME B40.200. Thermometer must have a 125 mm diameter dial, a hermetically sealed stainless steel case, a stainless steel stem, a safety glass face, a fixed threaded connection, and a scale range as indicated. Thermometer accuracy must be within one percent of the scale range.

### 2.5.8 Pressure Gauge

See Section 33 57 55 FUEL SYSTEM COMPONENTS (NON-HYDRANT).

### 2.5.9 Pipe Supports

Supports must be the adjustable type conforming to MSS SP-58, except as modified herein. Provide hot-dipped galvanized finish on rods, nuts, bolts, washers, hangers, and supports. Provide miscellaneous metal that conforms to ASTM A36/A36M, standard mill finished structural steel shapes, hot-dipped galvanized.

#### 2.5.9.1 Pipe Protection Shields

Shields must conform to MSS SP-58, Type 40, except material must be Type 316 stainless steel. Provide shields at each slide type pipe hanger and support.

#### 2.5.9.2 Low Friction Supports

Supports must have self-lubricating anti-friction bearing elements composed of 100 percent virgin tetrafluoroethylene polymer and reinforcing aggregates, prebonded to appropriate backing steel members. The coefficient of static friction between bearing elements must be 0.06 from initial installation for both vertical and horizontal loads and deformation must not exceed 51 micrometers under allowable static loads. Bonds between material and steel must be heat cured, high temperature epoxy. Design pipe hangers and support elements for the loads applied. Provide anti-friction material with a minimum of 2.3 mm thick. Provide hot-dipped galvanized steel supports. Provide supports that are factory designed and manufactured.



### 2.5.10 Escutcheon

Escutcheon must be the chrome plated, stamped steel, hinged, split ring type. Inside diameter must closely fit pipe outside diameter. Outside diameter must completely cover the corresponding floor, wall, or ceiling opening. Provided each escutcheon with necessary set screws.

### 2.5.11 Flexible Ball Joint

Flexible ball joints must be carbon steel with electroless nickel-plating to a minimum of 3 mils thickness, capable of 360-degree rotation plus 15-degree angular flex movement, ASME B16.5, Class 150 flanged end connections. Provide either pressure molded composition, PEEK, or polytetrafluoroethylene (PTFM) gaskets designed for continuous operation temperature of 275 degrees F. Joints must be designed for minimum working pressure of ANSI Class 150. Injectable packing will not be allowed.

### 2.5.12 Bellows Expansion Joint

The expansion joints must be for axial compression and extension with capacity as per the design documents. Units must be manufactured by an Expansion Joint Manufacturers Association certified manufacturer. They must incorporate multi-ply, Lo-corr bellows of Inconel 625 if chlorides are present in the atmosphere. Unit must be equipped with travel limit stops, and internal guides vented to reduce the effects of sudden pressure changes. Flanges and housing must be stainless steel or carbon steel to match piping materials. Flanges must conform to ASME B16.5. Dual Expansion Joints must incorporate an intermediate anchor base. Housing must include lifting lug and drain port. Joints must be capable of 10,000 cycles over a period of 20-years.

Cold set joints to compensate for the temperature at the time of installation. Provide initial alignment guides on the connecting piping no more than 4 pipe diameters from the expansion joint. Provide additional alignment guides on the connecting piping no more than 14 pipe diameters from the first guide.

### 2.5.13 Flow Meter

Provide volumetric positive displacement type meter that conforms to ASTM F1172, except as modified herein. Meter must indicate the fuel oil flow rate in L/s. Meter must be provided with overspeed protection and a water escape hole. If meter is not mounted in-line with the piping, then an appropriate pedestal for mounting must be provided. Install meter in accordance with manufacturer's recommendations. Meter must be capable of providing a 4-20 mA analog output signal for the fuel flow rate.

## 2.6 MANUAL VALVES

All portions of a valve coming in contact with fuel must be of noncorrosive material. Valves in stainless steel pipe lines or epoxy lined carbon steel pipe lines must be Type 304 or Type 316 stainless steel or carbon steel internally plated with chromium or nickel or internally electroless nickel plated. Valves in unlined carbon steel pipelines must have carbon steel body. Stem and trim must be stainless steel for all valves. Manually operated valves 150 mm and larger must be worm-gear operated and valves smaller than 150 mm must be lever operated or handwheel operated. Valves smaller than 50 mm must have lever-type handles. Handles installed more than 1.5 m above finished floor must have

chain operators. Valve indicators installed higher than 5 feet must have a position indicator visible from ground level. Sprocket wheel for chain operator must be aluminum.

#### 2.6.1 Ball Valves

Ball valves must be fire tested and qualified in accordance with the requirements of API Std 607 and API STD 608. Seal material for the fire test must be graphite, seal material for the project must be as indicated below. Ball valves must be nonlubricated valves that operate from fully open to fully closed with 90 degree rotation of the ball. Valves 2 inches and larger must conform to applicable construction and dimension requirements of API Spec 6D, ANSI Class 150 and must have flanged ends. Valves smaller than 50 mm must be ANSI class 150 valves with flanged ends, unless noted otherwise. The balls in valves 250 mm and larger full port and 300 mm and larger regular port and larger must have trunnion type support bearings. Except as otherwise specified or indicated, reduced port or full port valves may be provided at the Contractor's option. Balls must be solid, not hollow cavity.

##### 2.6.1.1 Materials

Ball must be stainless steel. Ball valves must have polytetrafluoroethylene (PTFM) or fluoroelastomer (FKM), commonly referred to as Viton seats, body seals and stem seals. Valves 100 mm and smaller must have a locking mechanism.

##### 2.6.1.2 V-Port Ball Valve

Valve must conform to requirements as specified for BALL VALVES paragraph in this section. Valve must be provided with characterized linear v-port for flow rate control, and with infinite position lever bracket with locking bolt for set position.

##### 2.6.1.3 Electric Valve Actuator

Electric valve actuator must be as indicated for Plug (Double Block and Bleed) Valves, electric valve actuator.

#### 2.6.2 Plug (Double Block and Bleed) Valves

API Spec 6D, API Spec 6FA, ANSI Class 150, non-lubricated, resilient, double seated, trunnion mounted, tapered lift plug capable of two-way shutoff. Valve must have tapered plug of steel or ductile iron with chrome or nickel plating and plug supported on upper and lower trunnions. Sealing slips must be steel or ductile iron, with Viton seals which are held in place by dovetail connections. Valve design must permit sealing slips to be replaced from the bottom with the valve mounted in the piping. Valves must operate from fully open to fully closed by rotation of the handwheel to lift and turn the plug. Valves must have weatherproof operators with mechanical position indicators. Indicator shaft must be stainless steel. Minimum bore size must be not less than 65 percent of the internal cross sectional area of a pipe of the same nominal diameter unless bore height of plug equals the nominal pipe diameter and manufacturer can show equal or better flow characteristics of the reduced bore size design.

#### 2.6.2.1 Valve Operation

Rotation of the handwheel toward open must lift the plug without wiping the seals and retract the sealing slips so that during rotation of the plug clearance is maintained between the sealing slips and the valve body. Rotation of the handwheel toward closed must lower the plug after the sealing slips are aligned with the valve body and force the sealing slips against the valve body for positive closure. When valve is closed, the slips must form a secondary fire-safe metal-to-metal seat on both sides of the resilient seal. Plug valves located in Isolation Valve Pits or vaults must be provided with handwheel extensions.

#### 2.6.2.2 Integral Cavity Valves

ANSI Class 150. Provide plug valves with automatic thermal relief valves to relieve the pressure build up in the internal body cavity when the plug valve is closed. Relief valves must open at 172 kPa differential pressure and must discharge to the throat of, and to the upstream side, of the plug valve.

#### 2.6.2.3 Bleed Valves

ANSI Class 150, stainless steel body valve. Provide manually operated bleed valves that can be opened to verify that the plug valves are not leaking when in the closed position.

#### 2.6.2.4 Electric Valve Actuator

The actuator, controls and accessories must be the responsibility of the valve-actuator supplier for sizing, assembly, certification, field-testing and any adjustments necessary to operate the valve as specified. The electric valve actuator must include as an integral unit the electric motor, actuator unit gearing, limit switch gearing, position limit switches, torque switches, drive bushing or stem nut, declutch lever, wiring terminals for power, remote control indication connections and handwheel. The electrically actuated plug valve must be set to open and close completely in 30 to 60 seconds against a differential pressure of 275 PSIG. The actuator settings of torque and limit contacts must be adjustable. The valve actuator must be suitable for mounting in a vertical or horizontal position and be rated for 30 starts per hour. The valve actuator must be capable of functioning in an ambient environment temperature ranging from minus 30 degrees C to 70 degrees C.

- a. The electrical enclosure must be specifically approved by UL or Factory Mutual for installation in Class I, Division 1, Group D locations.
- b. The electric motor must be specifically designed for valve actuator service and must be totally enclosed, non-ventilated construction. The motor must be capable of complete operation at plus or minus 10 percent of specified voltage. Motor insulation must be a minimum NEMA Class F. The motor must be a removable subassembly to allow for motor or gear ratio changes as dictated by system operational requirements. The motor must be equipped with an embedded thermostat to protect against motor overload and also be equipped with space heaters. It must de-energize when encountering a jammed valve.
- c. The reversing starter, control transformer and local controls must be integral with the valve actuator and suitably housed to prevent

breathing or condensation buildup. The electromechanical starter must be suitable for 30 starts per hour. The windings must have short circuit and overload protection. A transformer, if needed, must be provided to supply all internal circuits with 24 VDC or 110 VAC may be used for remote controls.

- d. The actuator gearing must be totally enclosed in an oil-filled or grease-filled gearcase. Standard gear oil or grease must be used to lubricate the gearcase.
- e. The actuator must integrally contain local controls for Open, Close and Stop and a local/remote three position selector switch: Local Control Only, Off, and Remote Control plus Local Stop Only. A metallic handwheel must be provided for emergency operation. The handwheel drive must be mechanically independent of the motor drive. The remote control capability must be to open and close. Rim pull to operate valve manually must not exceed 36 kg.
- f. Position limit switches must be functional regardless of main power failure or manual operation. Four contacts must be provided with each selectable as normally open or normally closed. The contacts must be rated at 5A, 120 VAC, 30 VDC.
- g. Each valve actuator must be connected to a PLC supplied by "others".
- h. The actuator must have a local display of position even when power has been lost.
- i. The actuator must be supplied with a start-up kit comprising installation instruction, electrical wiring diagram and spare cover screws and seals.
- j. The actuator must be performance tested and a test certificate must be supplied at no extra charge. The test should simulate a typical valve load with current, voltage, and speed measured.

#### 2.6.3 Swing Type Check Valves

Swing check valves must conform to API STD 600, regular type, ANSI Class 150 with flanged end connections. Discs and seating rings must be renewable without removing the valve from the line. The disc must be guided and controlled to contact the entire seating surface.

#### 2.6.4 Wafer Type Check Valve

Spring assisted, wafer/tapped lug pattern, butterfly check or globe type with FKM or PTFE seat ring, designed to prevent flow reversal slamming of valve, dual plate, and must conform to ASME B16.34, API Std 594, except face to face dimensions may deviate from standard. Valves must be suitable for installation in any orientation. Valve body and trim material must be as previously indicated herein.

#### 2.6.5 Globe Valve

Valve must conform to ASME B16.34, Class 150.

## 2.6.6 Thermal Relief Valve

### 2.6.6.1 Valve Materials

Valves must have carbon steel bodies (stainless steel on stainless steel pipelines) and bonnets with stainless steel springs and trim. Valves must be Class 150 flanged end connections.

### 2.6.6.2 Thermal Relief Valve (ASME Type)

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, hydraulically operated type with plain caps, and must be labeled in accordance with ASME BPVC (GPM). Valve stems must be fully guided between the closed and fully opened positions. The valves must be factory-set to open at pressures indicated on the drawings. Operating pressure must be adjustable by means of an enclosed adjusting screw. The valves must have a minimum capacity of 20 GPM at 10 percent overpressure. Valves must have a replaceable seat. Relief valves that do not relieve to a zone of atmospheric pressure or tank must be a balanced type relief valve.

### 2.6.6.3 Thermal Relief Valve (Balanced Type)

Thermal relief valves that do not relieve to a zone of atmospheric pressure or atmospheric tank must be a balanced type relief or regulator valve.

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, fully balanced type (back pressure must not affect relief pressure) back pressure regulator/relief valve. Set valve at pressure indicated on drawings. Valve body must have 25 mm (minimum) raised face flange connections unless otherwise indicated. Orifice must have a minimum orifice size of 15 millimeter in diameter. Valve must have bubble-tight piston and seat design with stainless steel piston and Viton seat. Valve must be selected for the nominal flow condition of: pass a minimum of 18 liters per minute, at a differential pressure of 380 kPa, with a nominal set pressure of 345 kPa. Valve must be factory configured to open at required set pressure but must be field adjustable by means of an enclosed adjusting screw.

## 2.6.7 Pressure/Vacuum Relief Valve

Valve must be the pressure/vacuum vent relief type that conforms to NFPA 30. Valve pressure and vacuum relief settings must be set at the factory. Pressure and vacuum relief must be provided by a single valve. Valve body must be constructed of either cast steel or aluminum. Valve trim must be stainless steel. Inner valve pallet assemblies must have a knife-edged drip ring around the periphery of the pallet to preclude condensation collection at the seats. Pallet seat inserts must be of a material compatible with the fuel specified to be stored. Valve intake must be covered with a 40 mesh stainless steel wire screen.

## 2.6.8 Foot Valve

Valve must be the self-activating, double-poppet, shutoff type that prevents fuel flow from reversing. Valve must conform to NFPA 30. Valve body must be constructed of either cast steel or aluminum. Valve must be provided with a minimum 20 mesh stainless steel screen on the intake. Valve seats must be the replaceable type. Valve must be capable of

passing through a 75 mm pipe or tank flange.

#### 2.6.9 Tank Overfill Prevention Valve (Gravity Fill)

Valve must be the two-stage, float-activated, shutoff type that is an integral part of the drop tube used for gravity filling. The first stage must restrict the flow of fuel into the tank to approximately 0.3 L/s when the liquid level rises above 87.5 percent of tank capacity. The second stage must completely stop the flow of fuel into the tank when the liquid level rises above 92.5 percent of tank capacity. Valve must be constructed of the same material as the fill tube.

#### 2.6.10 Tank Overfill Prevention Valve (Pumped Fuel Receipt)

Valve must be the two-stage, float-activated, shutoff type that is an integral part of the drop tube used for pressurized fill systems. The valve must completely stop the flow of fuel into the tank, when the liquid level rises above 90 of tank capacity. Valve must be constructed of the same material as the fill tube.

#### 2.6.11 Anti-Siphon Valves

##### 2.6.11.1 Solenoid Controlled Anti-Siphon Ball Valve

Anti-siphon valves must be solenoid controlled, normally closed, spring loaded valves. Solenoid must be housed in an integral, watertight, explosion proof housing and suitable for installation in Class I, Division I hazardous area locations. Valve body and trim material must be as previously indicated herein.

##### 2.6.11.2 Anti-Siphon Valve

Anti-siphon valves must be normally closed, spring loaded, angle pattern type valves. Valves must be suitable for installation in any orientation and compatible with suction or pressurized systems. Valve must be UL listed. Valve body and trim material must be as previously indicated herein.

### 2.7 ACCESSORIES

#### 2.7.1 Concrete Anchor Bolts

Concrete anchors must conform to ASTM A307, Grade C, hot-dipped galvanized.

#### 2.7.2 Bolts and Studs

Carbon steel bolts and studs must conform to ASTM A307, Grade B, hot-dipped galvanized. Stainless steel bolts and studs must conform to ASTM A193/A193M, Class 2, Grade 8.

#### 2.7.3 Nuts

Carbon steel nuts must conform to ASTM A563, Grade A, hex style, hot-dipped galvanized. Stainless steel nuts must conform to ASTM A194/A194M, Grade 8.

#### 2.7.4 Washers

Provide flat circular washers under each bolt head and each nut. Washer

materials must be the same as the connecting bolt and nut. Carbon steel washers must conform to ASTM F844, hot-dipped galvanized. Stainless steel washers must conform to ASTM A194/A194M, Grade 8.

#### 2.7.5 Polytetrafluoroethylene (PTFE) Tape

Tape must conform to ASTM D3308.

#### 2.7.6 Pipe Sleeves

Provided sleeves constructed of hot-dipped galvanized steel, ductile iron, or cast-iron pipe .

#### 2.7.7 Buried Utility Tape

Provide detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape in minimum 75 mm width rolls, color coded for the utility involved, with warning identification imprinted in bold black letters continuously and repeatedly over entire tape length. Warning identification must be at least 25 mm high and must state as a minimum "BURIED JET FUEL PIPING BELOW". Provide permanent code and letter coloring that is unaffected by moisture and other substances contained in trench backfill material.

#### 2.7.8 Pipeline Markers

Provide pipeline markers constructed of 150 mm diameter, one-half inch thick bronze disk with a 75 mm long bronze headed bolt welded to the back of the disk. Engrave the front of the disk with the words "UNDERGROUND FUEL LINE" in the case of one line and "UNDERGROUND FUEL LINES" in the case of multiple fuel lines.

### 2.8 FINISHES

Ship, store, and handle coating materials as well as apply and cure coatings in accordance with SSPC PA 1.

#### 2.8.1 Exterior Coating, Direct Buried Piping

##### 2.8.1.1 Factory Coating

Provide direct buried pipe and piping components with a factory-applied, adhesive undercoat and continuously extruded plastic resin coating in accordance with NACE SP0185 or AWWA C215; minimum thickness of plastic resin must be 36 mils for pipe sizes 150 mm and larger.

##### 2.8.1.2 Girth Welds

Coat girth welds using one of the following processes.

- a. Heat shrink sleeves in accordance with AWWA C216.
- b. Wax tape coatings in accordance with AWWA C217.
- c. Cold applied tape coatings in accordance with AWWA C209.

#### 2.8.1.3 Damaged Coatings

Repair damaged coating areas using one of the following processes.

- a. Wax tape coatings in accordance with AWWA C217.
- b. Cold applied tape coatings in with AWWA C209.

#### 2.8.1.4 Rock Shield

Provide a minimum 10 mm thick perforated rock shield around buried piping. Rock shield must consist of a polyethylene outer surface bonded to a closed cell foam substrate with uniform perforations intended for use with cathodic protection systems. Rock shield must overlap on itself no less than 150 mm. Secure rock shield tightly to the pipe using either strapping tape or plastic ties. Air filled cell type rock shields are prohibited.

#### 2.8.2 Exterior Coating, Aboveground Piping

Coat the exterior of aboveground steel piping, flanges, fittings, nuts, bolts, washers, valves, and piping components, as defined in this specification, in accordance with Section 09 90 00 PAINTS AND COATINGS.

#### 2.8.3 New System Components

##### 2.8.3.1 Factory Coating

Unless otherwise specified, provide system components fabricated from ferrous metal with the manufacturer's standard factory finish. Each factory finish must withstand 500 hours exposure to the salt spray test specified in ASTM B117. For test acceptance, the test specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 3 mm on either side of the scratch mark immediately after completion of the test. For system component surfaces subject to temperatures above 50 degrees C, the factory coating must be appropriately designed for the temperature service.

##### 2.8.3.2 Field Painting

Painting required for surfaces not otherwise specified must be field painted as specified in Section 09 90 00 PAINTS AND COATINGS. Do not paint aboveground stainless steel and aluminum surfaces. Do not coat system components provided with a complete factory coating. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with ASME B31.3 and NFPA 30, except as modified herein. Safety rules as specified in NFPA 30 must be strictly observed. Never direct bury threaded connections, socket welded connections, unions, flanges, valves, air vents, or drains. Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.



### 3.1.1 Piping

#### 3.1.1.1 General

Thoroughly clean pipe of all scale and foreign matter before the piping is assembled. Cut pipe accurately to measurements established at the jobsite, and worked into place without springing or forcing. Cut pipe square and have burrs removed by reaming. Install pipe to permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers. Cutting or other weakening of the building structure to facilitate piping installation will not be permitted without written approval.

- a. Use reducing fittings for changes in pipe sizes. Install system components and piping into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance. Provide electric isolation fittings between dissimilar metals. Install piping straight and true to bear evenly on supports. Piping must be free of traps, must not be embedded in concrete pavement, and must drain as indicated. Make changes in direction with fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees will not be permitted.
- b. The centerline radius of bends must not be less than 6 diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be accepted. When work is not in progress, securely close open ends of pipe and fittings with an expandable pipe plug so that water, earth, or other substances cannot enter the pipe or fittings. For belowground piping, the full length of each pipe must rest solidly on the underlying pipe bed.

#### 3.1.1.2 Double-Wall Flexible Non-Metallic Piping

Install double-wall flexible non-metallic piping in accordance with manufacturer's instructions.

#### 3.1.1.3 Pipeline Markers

Provide above underground fuel piping spaced every 90 meters, at tees, and at changes in direction. For sections of underground piping less than 90 meters long, place at midpoint. Provide directly above pipe for single lines and between pipes where pipes run in pairs. Provide additional marker over each mid-line fitting connections for steel reinforced flexible pipe. Cast marker into 457 mm diameter, 305 mm thick concrete plug unless it is set in an area with concrete paving in which case it must be cast into the concrete paving.

#### 3.1.1.4 Steel Reinforced Flexible Pipe

Connections between steel pipe and steel reinforced flexible pipe and between separate lengths of steel reinforced flexible pipe must not be made aboveground but must be made either inside a pit or vault, or direct bury them. Where practicable, end-line and mid-line connections must be located inside pit type enclosures of an appropriate size. Where it is not practicable to locate mid-line connections inside pit type enclosures, mid-line connections may be wrapped with a suitable waterproof protective substance and direct buried underground. The location of direct buried

mid-line connections must be indicated on the final drawings and provided with a pipeline marker.

#### 3.1.1.5 Welded Connections

Unless otherwise indicated on the drawings, pipe joints must be welded. Construct branch connections with welding tees or forged welding branch outlets. Do not weld stainless steel pipe to carbon steel pipe.

#### 3.1.1.6 Threaded End Connections

Provide threaded end connections only on piping 50 mm in nominal size or smaller and only where indicated on the drawings. Provide threaded connections with PTFE tape or equivalent thread-joint compound applied to the male threads only. Not more than three threads must show after the joint is tighten.

#### 3.1.1.7 Brazed Connections

Provide brazing in accordance with AWS BRH, except as modified herein. During brazing, fill pipe and fittings with a pressure regulated inert gas, such as nitrogen, to prevent the formation of scale. Before brazing copper joints, clean both the outside of the tube and the inside of the fitting with a wire fitting brush until the entire joint surface is bright and clean. Do not use brazing flux. Remove surplus brazing material at all joints. Support piping prior to brazing and do not be spring or force piping.

#### 3.1.1.8 Existing Piping Systems

No interruptions or isolation of an existing fuel handling service or system must be performed unless the actions are approved by the Contracting Officer. Perform initial cutting of existing fuel pipe with a multiwheel pipe cutter, using a nonflammable lubricant. After cut is made, seal interior of piping with a gas barrier plug. Purge interior of piping with carbon dioxide or nitrogen prior to performing any welding process.

#### 3.1.2 Bolted Connections

For each bolted connection of stainless steel components (e.g., pipes, piping components, valves, and system components) use stainless steel bolts or studs, nuts, and washers. For each bolted connection of carbon steel components, use carbon steel bolts or studs, nuts, and washers. Bolts to project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque. Prior to installing nuts, apply a compatible anti-seize compound to the male threads.

#### 3.1.3 Flanges and Unions

Except where threaded end connections and unions are indicated, provide flanged joints in each line immediately preceding the connection to a system component or material requiring maintenance such as pumps, general valves, control valves, strainers, and other similar items and as indicated. Assemble flanged joints square and tight with matched flanges, gaskets, and bolts. For flanges, provide washers under each bolt head and nut. Torque wrenches must be used to tighten all flange bolts to the torque recommended by the gasket manufacturer. Tightening pattern must be

as recommended by the gasket manufacturer. Use anti-seize compound on threads for stainless steel bolts.

#### 3.1.4 Flange Protectors

Provide flange protectors on each flanged end connection, including valves and system components. Fill the flange cavity of electrically isolating flange connections with corrosion inhibitor type grease. Provide grease filled bolt caps. Caution should be used when installing stainless steel bands to avoid "grounding out" the insulating flanges.

#### 3.1.5 Valves

Install isolation plug or ball valves on each side of each system component, at the midpoint of looped mains, and at any other points indicated or required for draining, isolating, or sectionalizing purpose. Install valves with stems vertically up unless otherwise indicated. Provide individual supports and anchors for each valve.

#### 3.1.6 Air Vents

Provide 40 mm air vents at all high points and where indicated to ensure adequate venting of the piping system.

#### 3.1.7 Drains

Provide 50 mm drains at all low points and where indicated to ensure complete drainage of the piping. Drains must be schedule 120. Drains must be accessible, and must consist of nipples and caps or plugged tees unless otherwise indicated.

#### 3.1.8 Bellows Expansion Joints

Cold set joints to compensate for the temperature at the time of installation. Provide initial alignment guides on the connecting piping no more than 4 pipe diameters from the expansion joint. Provide additional alignment guides on the connecting piping no more than 14 pipe diameters from the first guide.

#### 3.1.9 Thermometers

Provide thermometers with separable sockets. Install separable sockets in pipe lines in such a manner to sense the temperature of flowing fluid and minimize obstruction to flow.

#### 3.1.10 Pipe Sleeves

Provide a pipe sleeve around any pipe that penetrates a wall, floor, or crosses under a roadway. Do not install sleeves in structural members except where indicated or approved. Install pipe sleeves in masonry structures at the time of the masonry construction. Sleeves must be of such size as to provide a minimum of 12 mm all-around clearance between bare pipe and the sleeve. Align sleeve and piping such that the pipe is accurately centered within the sleeve by a nonconductive centering element. Securely anchor the sleeve to prevent dislocation. Closure of the space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush. For wall or floor penetrations, extend each sleeve through its respective wall or floor and cut flush with each

surface. For roadway crossings, pipe sleeves must be continuous for the entire crossing as well as extend a minimum of 150 mm beyond both sides of the crossing. Seal around sleeves that penetrate through valve or fuel related pits with a Buna-N casing seal. Seal around sleeves that penetrate through non-fire-rated walls and floors in accordance with Section 07 92 00 JOINT SEALANTS. Seal around sleeves that penetrate through fire-rated walls and floors as specified in Section 07 84 00 FIRESTOPPING.

### 3.1.11 Escutcheons

Except for utility or equipment rooms, provide finished surfaces where exposed piping pass through floors, walls, or ceilings with escutcheons. Secure escutcheon to pipe or pipe covering.

### 3.1.12 Access Panels

Provide access panels for all concealed valves, vents, controls, and items requiring inspection or maintenance. Access panels must be of sufficient size and located so that the concealed items may be serviced and maintained or completely removed and replaced. .

### 3.1.13 Buried Utility Tape

Bury tape with the printed side up at a depth of 300 mm below the top surface of earth or the top surface of the subgrade under pavements.

### 3.1.14 Framed Instructions

Framed instructions must include system components layout, wiring and control diagrams, piping, valves, control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. Frame under glass or laminated plastic the framed instructions and post where directed by the Contracting Officer. Post the framed instructions before the system performance tests.

## 3.2 PIPE SUPPORTS

Install supports with a maximum spacing as defined in Table 1 below, except where indicated otherwise. In addition to meeting the requirements of Table 1, provide additional supports where concentrated piping loads exist (e.g., valves).

| Table 1. Maximum Support Spacing |                 |    |    |    |     |     |     |     |     |
|----------------------------------|-----------------|----|----|----|-----|-----|-----|-----|-----|
| Nominal Pipe Size<br>(mm)        | 25 and<br>Under | 40 | 50 | 80 | 100 | 150 | 200 | 250 | 300 |
|                                  |                 |    |    |    |     |     |     |     |     |

| Table 1. Maximum Support Spacing |   |      |   |     |      |   |      |      |   |
|----------------------------------|---|------|---|-----|------|---|------|------|---|
| Maximum Support Spacing (m)      | 2 | 2.75 | 3 | 3.5 | 4.25 | 5 | 5.75 | 6.50 | 7 |
|                                  |   |      |   |     |      |   |      |      |   |

### 3.2.1 Seismic Requirements

Support and brace piping and attach valves to resist seismic loads as specified under Section 13 48 73 SEISMIC CONTROL FOR NONSTRUCTURAL COMPONENTS and as shown on the drawings. Structural steel required for reinforcement to properly support piping, headers, and system components but not shown must be provided under this section. Material used for support must be as specified under Section 05 12 00 STRUCTURAL STEEL.

### 3.2.2 Structural Attachments

Provide attachments to building structure concrete and masonry by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Apply inserts and anchors with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Structural steel brackets required to support piping, headers, and system components, but not shown, must be provided under this section. Material used for support must be as specified under Section 05 12 00 STRUCTURAL STEEL.

## 3.3 FIELD QUALITY CONTROLS

### 3.3.1 System Commissioning

System commissioning must conform to Section 33 08 53 AVIATION FUEL DISTRIBUTION SYSTEM START-UP.

### 3.3.2 Tests

Furnish labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein. Perform piping test in accordance with the applicable requirements of ASME B31.3 except as modified herein. To facilitate the tests, various sections of the piping system may be isolated and tested separately. Where piping sections terminate at flanged valve points, close the line by means of blind flanges in lieu of relying on the valve. Provide tapped flanges to allow a direct connection between the piping and the air compressor and pressurizing pump. Use tapped flanges for gauge connections. Taps in the permanent line will not be permitted. Gauges will be subject to testing and approval. Provide provisions to prevent displacement of the piping during testing. Keep personnel clear of the piping during pneumatic testing. Only authorized personnel must be permitted in the area during pneumatic and hydrostatic testing. Isolate system components such as pumps, tanks and meters from the piping system during the testing. Do not exceed the pressure rating of any component in the piping system during the testing. Following satisfactory completion of each test, relieve the test pressure and seal the pipe immediately. Piping to be installed underground must not receive field applied exterior coatings at the joints or be covered by backfill until the piping has passed the final pneumatic

tests described herein.

#### 3.3.2.1 Exterior Coating Holiday Test

Following installation, test the exterior coating of direct buried piping for holidays using high-voltage spark testing in accordance with NACE SP0188. Repair holidays and retest to confirm holiday-free coating. Text must include all existing underground piping exposed for this project.

#### 3.3.2.2 Preliminary Pneumatic Test

Apply a 170 kPa pneumatic test to product piping. Maintain the pressure while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations). If leaks are discovered, repair the leaks accordingly and retest.

#### 3.3.2.3 Final Pneumatic Test

Following the preliminary pneumatic test, apply a 345 kPa pneumatic test to all product piping and hold for a period not less than 2-hours. During the test period, there must be no drop in pressure in the pipe greater than that allowed for thermal expansion and contraction. Disconnect the pressure source during the final test period. If leaks are discovered, repair the leaks accordingly and retest.

#### 3.3.2.4 Hydrostatic Test

Testing must comply with the requirements in ASME B31.3, except as modified herein. Hydrostatically test product piping with the fuel to be handled to the lesser of 1-1/2 times operating pressure or 1896 kPa in accordance with API RP 1110. Maintain the pressure within the piping for 4-hours with no leakage or reduction in gauge pressure. If leaks are discovered, repair the leaks accordingly and retest.

#### 3.3.2.5 Exterior Containment Piping Tests

Apply a minimum pneumatic pressure of 35 kPa to the exterior containment piping. Maintain the pressure for at least 1-hour while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations). Repair leaks discovered in accordance with manufacturer's instructions and retest. Perform testing in compliance with the manufacturer's published installation instructions.

### 3.4 SYSTEM PERFORMANCE TESTS

Conform tests to Section 33 08 53 AVIATION FUEL DISTRIBUTION SYSTEM START-UP.

### 3.5 DEMONSTRATIONS

Conduct a training session for designated Government personnel in the operation and maintenance procedures related to the system components and specified herein. Include pertinent safety operational procedures in the session as well as physical demonstrations of the routine maintenance operations. Furnish instructors who are familiar with the installation/system components/systems, both operational and practical theories, and associated routine maintenance procedures. The training

session must consist of a total of 4 hours of normal working time and must start after the system is functionally completed, but prior to final system acceptance. Submit a letter, at least 14 working days prior to the proposed training date, scheduling a proposed date for conducting the onsite training.

-- End of Section --

## SECTION 33 52 23.15

POL SERVICE PIPING WELDING  
05/25

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN PETROLEUM INSTITUTE (API)

|             |                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| API 570     | (2016; Addendum 1 2017; Addendum 2 2018; ERTA 1 2018) Piping Inspection Code: In-Service Inspection, Rating, Repair, and Alteration of Piping Systems |
| API RP 2009 | (2002; R 2007; 7th Ed) Safe Welding, Cutting, and Hot Work Practices in Refineries, Gasoline Plants, and Petrochemical Plants                         |
| API Std 650 | (2013; Errata 1 2013; Addendum 1 2014; Errata 2 2014; Addendum 2 2016; Addendum 3 2018) Welded Tanks for Oil Storage                                  |

## AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

|                  |                                                                                                     |
|------------------|-----------------------------------------------------------------------------------------------------|
| ANSI/ASNT CP-189 | (2020) ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel        |
| ASNT SNT-TC-1A   | (2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing |

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                  |                                                                                      |
|------------------|--------------------------------------------------------------------------------------|
| ASME B16.5       | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard |
| ASME B16.48      | (2015) Line Blanks                                                                   |
| ASME B31.3       | (2024) Process Piping                                                                |
| ASME BPVC SEC IX | (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications       |
| ASME BPVC SEC V  | (2017) BPVC Section V-Nondestructive Examination                                     |

## AMERICAN WELDING SOCIETY (AWS)

|          |                                      |
|----------|--------------------------------------|
| AWS A2.4 | (2012) Standard Symbols for Welding, |
|----------|--------------------------------------|



## Brazing and Nondestructive Examination

|                    |                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS A3.0M/A3.0     | (2025) Standard Welding Terms and Definitions                                                                                                        |
| AWS A5.1/A5.1M     | (2025) Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding                                                                      |
| AWS A5.3/A5.3M     | (2023) Specification for Aluminum and Aluminum-Alloy Electrodes for Shielded Metal Arc Welding                                                       |
| AWS A5.4/A5.4M     | (2012; R 2022) Specification for Stainless Steel Electrodes for Shielded Metal Arc Welding                                                           |
| AWS A5.9/A5.9M     | (2022) Welding Consumables-Wire Electrodes, Strip Electrodes, Wires, and Rods for Arc Welding of Stainless and Heat Resisting Steels- Classification |
| AWS A5.10/A5.10M   | (2023) Welding Consumables - Wire Electrodes, Wires and Rods for Welding of Aluminum and Aluminum-Alloys - Classification                            |
| AWS A5.18/A5.18M   | (2023) Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding                                                               |
| AWS A5.22/A5.22M   | (2024) Specification for Stainless Steel Flux Cored and Metal Cored Welding Electrodes and Rods                                                      |
| AWS A5.32/A5.32M   | (2021) Welding Consumables-Gases and Gas Mixtures for Fusion Welding and Allied Processes                                                            |
| AWS C5.5/C5.5M     | (2003) Recommended Practices for Gas Tungsten Arc Welding                                                                                            |
| AWS D1.1/D1.1M     | (2025) Structural Welding Code - Steel                                                                                                               |
| AWS D1.2/D1.2M     | (2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum                                                                              |
| AWS D1.6/D1.6M     | (2017) Structural Welding Code – Stainless Steel                                                                                                     |
| AWS D10.4          | (1986; R 2000) Recommended Practices for Welding Austenitic Chromium-Nickel Stainless Steel Piping and Tubing                                        |
| AWS D10.7/D10.7M   | (2008) Guide for the Gas Shielded Arc Welding of Aluminum and Aluminum Alloy Pipe                                                                    |
| AWS D10.10/D10.10M | (2021) Recommended Practices for Local Heating of Welds in Piping and Tubing                                                                         |

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| AWS D10.11M/D10.11 | (2007) Guide for Root Pass Welding of Pipe Without Backing            |
| AWS D10.12M/D10.12 | (2000) Guideline for Welding Mild Steel Pipe                          |
| AWS QC1            | (2016) Specification for AWS Certification of Welding Inspectors      |
| AWS WHB-4.9        | (2010) Welding Handbook, Volume 4 - Materials and Applications Part 1 |
| AWS Z49.1          | (2021) Safety in Welding, Cutting and Allied Processes                |

#### ASTM INTERNATIONAL (ASTM)

|            |                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------|
| ASTM E329  | (2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection |
| ASTM E1316 | (2020a) Standard Terminology for Nondestructive Examinations                                                  |
| ASTM E2700 | (2014) Standard Practice for Contact Ultrasonic Testing of Welds Using Phased Arrays                          |

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                         |
|----------|-----------------------------------------------------------------------------------------|
| NFPA 326 | (2015) Standard for Safeguarding of Tanks and Containers for Entry, Cleaning, or Repair |
|----------|-----------------------------------------------------------------------------------------|

#### U.S. ARMY CORPS OF ENGINEERS (USACE)

|            |                                                        |
|------------|--------------------------------------------------------|
| EM 385-1-1 | (2024) Safety -- Safety and Health Requirements Manual |
|------------|--------------------------------------------------------|

#### U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                               |
|--------------|---------------------------------------------------------------|
| UFC 3-460-01 | (2019; with Change 3, 2023) Design: Petroleum Fuel Facilities |
|--------------|---------------------------------------------------------------|

## 1.2 DEFINITIONS

Except as provided in this Section, definition of welding terms is in accordance with AWS A3.0M/A3.0, and definition of inspection and examination terms is in accordance with ASTM E1316.

### 1.2.1 Closure Weld

The final weld connecting POL service piping systems, components, or sub-assemblies which have been successfully tested in accordance with ASME B31.3. For purposes of this Section, a closure weld can include connection of new work to existing piping and is considered within the scope of ASME B31.3.

### 1.2.2 Complete Joint Penetration (CJP)

A weld condition in which fusion between base and filler has occurred across the entirety of surfaces intended for welding and between all weld passes.

### 1.2.3 Designer of Record

The professional engineer designated by the prime contractor to be in multidisciplinary responsible charge of all POL service piping design and repair.

### 1.2.4 Examination

As used in this Section, a process for determining weld properties, conditions, or characteristics using nondestructive techniques, technologies, and methods, and performed by qualified and certified examiner(s). Nondestructive examination (NDE) is a quality control function performed by the contractor as part of oversight organization duties.

### 1.2.5 Examiner

As used in this Section, specialist with responsibility to use various nondestructive techniques, technologies, and methods to inform the quality of welds. Examiner(s) is a QC Specialist pursuant to Section 01 45 00 QUALITY CONTROL and is part of the contractor quality control organization.

### 1.2.6 Hazardous Area

As used in this Section, any area within 30 meters of active storage tanks, areas within 30 meters of leaking section of fuel pipeline or other vapor sources, areas within 60 meters of the downwind side of potential vapor emission sources (i.e., pressure-vacuum vents, sample ports, open vents on active tanks, weeping flanges or joints).

### 1.2.7 Heat-Affected Zone (HAZ)

The portion of base metal whose mechanical properties or microstructure have been altered by the heat of welding, brazing, soldering, or thermal cutting.

### 1.2.8 Independent

Impartial third party not a part or affiliated with Contractor or subcontractor principal or subsidiary businesses.

### 1.2.9 In-Process Examination

A series of steps prescribed in ASME B31.3 to inspect and examine a closure weld for which a hydrostatic leak test is not possible. Can also be used on other joints. As used in this Section, in-process examination requires visual means aided by liquid penetrant examination (PT) or magnetic particle examination (MT) methods in determination of the weld condition. In this Section, the terms in-process examination and in-process inspection are used synonymously.

### 1.2.10 Inspection

As used in this Section, contractor oversight activity to observe work performed on materials or components to determine acceptability and control the quality of welds. Inspection is performed by qualified and certified inspector(s). In this Section, the weld inspection function is separate and distinct from inspection activities performed by an Owner's Inspector (pursuant to ASME B31.3) and from inspection activity of the Quality Control Manager (pursuant to Section 01 45 00 QUALITY CONTROL).

### 1.2.11 Inspector

As used in this Section, specialist with oversight responsibility to control the quality of welds. Inspector(s) is a QC Specialist pursuant to Section 01 45 00 QUALITY CONTROL and is part of the contractor quality control organization. The contractor inspector role is separate and distinct from both an Owner's Inspector (pursuant to ASME B31.3), and separate and distinct from the Quality Control Manager role (pursuant to Section 01 45 00 QUALITY CONTROL).

### 1.2.12 Hot Work

For work covered by this Section, drilling, boring, flame heating, welding, torch cutting, brazing, carbon arc gouging, grinding, abrasive blasting, or any work which produces heat, by any means, of 200 degrees C or more; or in the presence of flammable material or flammable atmosphere, other ignition sources such as spark or arc producing tools or equipment, static discharges, friction, impact, open flames or embers, nonexplosion-proof lights, fixtures, motors or equipment.

### 1.2.13 Marine Chemist

The holder of a valid Certificate issued by the National Fire Protection Association in accordance with the "Rules for Certification of Marine Chemists" establishing the individual as a Qualified Person pursuant to NFPA 326.

### 1.2.14 Owner's Inspector

Technical representative(s) of the Contracting Officer responsible for assuring that contractor performance meets requirements and contractor quality control organization is effective. In this Section, the Owner's Inspector role (pursuant to ASME B31.3) is separate and distinct from Inspector, Examiner, and Quality Control Manager roles.

### 1.2.15 Piping Engineer

One or more licensed professional engineers, or an engineering firm, acceptable to the Contracting Officer who are knowledgeable and experienced in the engineering disciplines associated with evaluating hydraulic, mechanical, and material characteristics which affect integrity and reliability of POL service piping systems. The piping engineer is the piping or pipeline subject matter expert. More than one individual can serve in the role as long as one is designated to be in responsible charge.

### 1.2.16 POL Service Piping

Pipe, piping, pipeline, fittings, components, assemblies, and appurtenances used for petroleum, oil, or lubricant (POL) conveyance

service under pressure or force of gravity, including modifications to existing systems.

#### 1.2.17 Weld Map

Drawing(s) containing sketches and tables which correlate design, weld plan, shop drawings, welder identification, weld inspection, nondestructive examination, and physical locations.

#### 1.2.18 Timely

As used in this Section, receipt of reporting records by the Contracting Officer no more than 10 days after installation of weld.

#### 1.2.19 Weld Slag

Crystalline residue remaining on a weld surface.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Welding Procedure Specification;

Procedure Qualification Record;

Welder Performance Qualification;

Weld Plan;

#### SD-06 Test Reports

Weld Log;

#### SD-07 Certificates

Weld Inspector Certificate;

Independent Test Organization;

In-Process Weld Inspector Certificates;

Weld Examiner Certificate;

### 1.4 SAFETY

Safety precautions must conform to Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS, EM 385-1-1, API 570, API RP 2009, and AWS Z49.1.

#### 1.4.1 Control of Hazardous Energy

Prior to welding, provide proper lockout and tagout (LOTO) of the piping or pipeline and appurtenances to completely isolate work from fuel, vapors, and sources of energy. Isolate using physical means such as blind

flange compliant with ASME B16.5 or solid-plate line blank compliant with ASME B16.48 to prevent fuel or vapor transfer into the workspace. Use isolation means of sufficient strength to withstand pressure which might be exerted by liquid, gas, or vapor in an active pipeline, piping, or tank. Use gaskets on both sides if a line blank is inserted between two flanges. Do not use a valve as sole means of isolation. If LOTO isolates a segment of active fuel piping from thermal relief, install temporary relief capability.

Execute in accordance with accepted Accident Prevention Plan, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS, and EM 385-1-1. Coordinate LOTO installation, maintenance, inspection, and removal with facility fuel system operator.

#### 1.4.2 Preparation for Welding

Develop written procedures in accordance with EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS, and AWS Z49.1. Test piping for hydrocarbons and purge as necessary. Do not start grinding or welding until vapor-free certification is received from the Marine Chemist and requirements of EM 385-1-1 have been met.

#### 1.4.3 Gas-Free Condition

Degass piping until requirements of Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS, the accepted Accident Prevention Plan, and the certified Marine Chemist are met. Obtain gas-free certification from the Marine Chemist. Maintain the gas-free environment. Display the Marine Chemist certificate on-site and available for review at all times during welding work.

### 1.5 PERFORMANCE

Establish and maintain a system integral with the quality control program for inspection and examination of design and construction. Perform inspection and examination to ensure contract work conforms to requirements. Maintain complete records and make them available to the Government.

Contractor is responsible for the quality of all design, joint preparation, welding, inspection, examination, testing, records maintenance, and reporting. All materials used in welding operations must be clearly identified and recorded. Inspection, examination, and testing defined in this Section are minimum requirements. Provide additional inspection, examination, and testing when necessary to achieve the required quality of work.

### 1.6 GENERAL REQUIREMENTS

This section covers welding of POL service piping systems. Contractor must oversee the quality of design, joint preparation, welding, inspection, and examination. Deviation from applicable codes, approved procedures, and approved drawings is not permitted without prior written approval by the Contracting Officer. Materials or components with welds made offsite are not accepted if the welding does not conform to requirements of this Section, unless otherwise specified. Material with welds will not be accepted unless the welding is specified or indicated on the drawings or otherwise approved. Unless otherwise specified, the choice of weld process is responsibility of the Contractor. See paragraph

DESIGN REQUIREMENTS for limitations on use of closure welds.

The Contracting Officer or Owner's Inspector may conduct inspection, test, or oversight work. Government inspection or testing are for the sole benefit of the Government. They do not:

- a. Relieve Contractor of responsibility for providing adequate quality control measures
- b. Relieve Contractor of responsibility for damage to or loss of the material before acceptance
- c. Constitute or imply acceptance

#### 1.6.1 Procedures

Develop procedures for welding all metals included in the work. Procedures for making transition welds between different P-number materials or between plates or pipes of different thickness must be qualified.

#### 1.6.2 Weld Inspection

This Section contains requirements to inspect all welds. Ensure full compliance with requests of the weld inspector(s) to correct deficiencies in materials and workmanship. It provides minimum requirements to qualify personnel, procedures, and equipment, and contains acceptance criteria. Correct all deficiencies in materials and workmanship in accordance with requirements of this Section.

#### 1.6.3 Nondestructive Examination

This Section contains requirements to examine welds with nondestructive means to detect the presence of defects. It provides minimum requirements to qualify personnel, procedures, and equipment, and contains acceptance criteria. Correct all deficiencies in materials and workmanship in accordance with requirements of this Section.

#### 1.6.4 Records Retention

Maintain current records of test results obtained in weld procedure qualification. Maintain current records of performance test results obtained to qualify welders and welding operators. Maintain welder continuity records. Maintain all records of weld inspection and NDE at the site and readily available for review by the Contracting Officer. Produce all records as part of Statement of Work (SOW) or Project Program completion report requirements.

#### 1.6.5 Timely Reporting

Timely reporting of weld inspection and examination progress is required. A weld is not finished until its associated Weld Log record with affirmative inspection and examination results has been received and is found complete and accurate. Do not submit a request for progress payment for unfinished welding work.

#### 1.6.6 Protect in Place

Protect in place tanks, piping, components, flanges, gaskets, motors,

valves, pumps, impellers, risers, coating, finishes, gauges, alarms, conduit, and conductors. Employ measures to protect people, equipment, and surfaces from exposure to and damage from sparks, heat, radiant energy, and light emission. See paragraph MAINTAINING CLEANLINESS OF PIPING for more requirements.

## 1.7 DESIGN REQUIREMENTS

Consult with experts experienced in fuel piping or pipeline welding. Provide subject matter professional engineering expertise to design welding. Validate site conditions to fully inform procedures, specifications, sketches and shop drawings. Produce work plans, procedures, specifications, sketches, and shop drawings which are complete and usable. Provide means to minimize heat distortion. Identify work which requires special methods such as preheat or postweld heat treatment, dissimilar metal joints or thickness, in-process weld examination, or closure welds. Establish fit-up and edge preparation tolerances.

Product to be conveyed in the piping is Diesel. Unless noted otherwise in SOW or Project Program, Fluid Service Category is Normal per ASME B31.3.

### 1.7.1 Code and Criteria Basis

UFC 3-460-01, API 570, and ASME B31.3 are relevant design codes and criteria. ASME B31.3 incorporates by reference other requirements of ASME BPVC SEC V, ASME BPVC SEC IX, and American Welding Society (AWS). For applicable processes, provide design and plans consistent with standards and guides AWS D10.4, AWS D10.10/D10.10M, AWS D10.11M/D10.11, AWS D10.12M/D10.12, and AWS C5.5/C5.5M.

### 1.7.2 Closure Welds

Restrict use of closure welds, as defined in this Section, to the minimum. Design work to maximize use of hydrostatic test(s) as means of compliance with leak test requirements of ASME B31.3 paragraph 345. Refer to Section 33 52 10 FUEL SYSTEMS PIPING (SERVICE STATION) for hydrostatic test requirements. Designate on drawings or sketches the location of each closure weld and provide notes which state in-process examination requirements in accordance with this Section, API 570, and ASME B31.3. Ad-hoc, field-determined, or unnecessary closure weld designations are not allowed.

Closure welds in the work are only acceptable under the following circumstances.

- a. A closure weld is the only means by which the pipe or assembly can be installed into the system due to a physical constraint
- b. The closure weld cannot be leak-tested in accordance with ASME B31.3, either in-situ or offsite

### 1.7.3 Structural Welds

In accordance with AWS D1.1/D1.1M Structural Welding Code, AWS D1.2/D1.2M Structural Welding Code (aluminum), and AWS D1.6/D1.6M Structural Welding Code (stainless steel). Do not weld attachment studs to any surface of the pressure-containing boundary. A weld which connects a support to the pressure-containing boundary is considered a piping or pipeline weld.



#### 1.7.4 Piping or Pipeline Welds

In accordance with API 570, ASME B31.3 Process Piping Code, and this Section. Restrict process for root pass to gas tungsten arc. Butt joint or groove welds must have CJP. Branch connection welds must be no closer than two times the run piping thickness from an adjacent weld.

#### 1.7.5 Weld Plan

Provide a weld plan which addresses all aspects of welding operations, inspection, and examination to meet requirements of this Section. Include specification and qualifying record for each procedure along with a summary table which lists all qualified welding personnel and the Welding Procedure Specification (WPS) under which they are qualified. For each welding operation, identify a sequence to minimize heat distortion both in-situ and offsite. Provide a process for machine welding if used. Describe how the quality of welding operations will be controlled. Submit Weld Plan in accordance with paragraph SUBMITTALS.

### 1.8 QUALITY ASSURANCE

Provide welding procedure specifications, procedure qualification records, and welder performance qualification records for each procedure and welder. Provide means and methods to control the quality of welding operations. If used, provide a method to track and control the quality of multiple welders on the same weld.

#### 1.8.1 Data Management

Organize repair data in a non-proprietary management system such as a database or spreadsheet. Establish a piping location identification scheme which uniquely identifies each piping weld, its location, its orientation, and its relative position to other piping features. Provide secure, auditable, and organized data. Ensure the system has the capability to track the provenance of each weld. Cloud-based systems are not acceptable. Limit access to the management system to ensure data integrity.

#### 1.8.2 Weld Log

Produce, populate, and maintain a Weld Log capable of uniquely tracking the specific location of every weld in the work. Follow the established location identification scheme. Weld identification on shop drawings must match weld tracking log and weld marking. The log must include the following minimum contents.

- a. Weld map depicting physical location, location on the piping, clock position, and orientation
- b. Type of weld including temporary and tack welds
- c. Applicable WPS
- d. Name or identification number of welding personnel
- e. Date and time of completion of welding or tacking
- f. Date and type of weld inspection, inspector name, acceptance criteria; acceptance determination

- g. Date and type of nondestructive examination, examiner name, acceptance criteria, acceptance determination
- h. Date, name, description of defects found, reason for noncompliance
- i. Corrective action planned or taken, including all steps above leading to an acceptance determination
- j. Date and time of the QCM inspection for each determination that a specific weld (welding, inspection, and examination) is complete, meets requirements, and is acceptable

#### 1.8.2.1 Progress Reporting

Update and populate the log as work progresses. Submit to the Contracting Officer on a regular basis as part of progress documentation. Provide a progress submittal weld log (including records for in-process inspection and examination) for oversight in a timely manner and indicating the QCM determination that a weld is complete, meets requirements, and is acceptable. Notify the Contracting Officer if weld log submittal lags by more than 10 work days.

#### 1.8.3 Weld Symbolology

Provide weld symbolology in accordance with AWS A2.4.

#### 1.8.4 Weld Marking

Neatly mark on the work with a marker or paint pen sufficient information for Government QA of the work. Use block letters approximately 5 cm in height and in legible color contrasting with the work surface. The QCM is responsible to manage and oversee all marking, timeliness, and content. In the event marking cannot be made on the piping itself, use an all weather placard attached to the work at the location of each weld. Minimum information is listed below. Should weld rework be required, neatly place a single line through obsolete marking and provide new marking.

- a. Unique weld identifier (correlated to the design)
- b. Welder identification
- c. Inspector identification, date, and result
- d. Examiner identification, date, and result
- e. QCM determination, date, and result

#### 1.8.5 Shop Drawings

Prepare and submit detail CAD drawings showing location, length, clock position, and type of welds. Indicate preweld and postweld heat treatment, inspection, and NDE. The drawings must identify the WPS to be used at each weld location. Indicate locations of closure welds and in-process examination per the design.

### 1.8.6 Inspection Procedures

Provide weld inspection procedures compliant with API 570, ASME B31.3, and paragraph PROCEDURE STANDARDS. The weld inspector(s) is considered a QC Specialist and must report results directly to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL.

### 1.8.7 Examination Procedures

Provide examination procedures compliant with API 570, ASME B31.3, and paragraph PROCEDURE STANDARDS. The weld examiner(s) is considered a QC Specialist and must report results directly to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. Identify a procedure standard for any method planned to be used which is not listed. Apply standard practice ASTM E2700 when developing procedure for examining welds using phased array ultrasonic method (PAUT). Paragraph SPECIAL REQUIREMENTS FOR PAUT contains additional procedure requirements.

### 1.8.8 Procedure Standards

| Method                        | Procedure Standard                                                          |
|-------------------------------|-----------------------------------------------------------------------------|
| Ultrasonic Examination (UT)   | ASME BPVC SEC V Article 4                                                   |
| PAUT in lieu of RT            | API Std 650 Annex U<br>ASME BPVC SEC V Article 4 and Mandatory Appendix VII |
| PT                            | ASME BPVC SEC V Article 6                                                   |
| MT                            | ASME BPVC SEC V Article 7                                                   |
| Radiographic Examination (RT) | ASME BPVC SEC V Article 2                                                   |
| Visual Examination (VT)       | ASME BPVC SEC V Article 9                                                   |

### 1.8.9 Inspection and Examination Reporting

Provide timely reports in accordance with paragraphs SUBMITTALS and weld log PROGRESS REPORTING. At minimum, reports must consist of the following.

- Records made by the AWS certified inspector for all duties performed per paragraph 4.2 of AWS QC1 using the unique identifier for each weld inspected.
- All NDE examiner reports using the unique identifier for each weld examined.
- Weld map location of each weld, weld procedure, welder identification, results of visual inspection, results of NDE.
- Up-to-date Weld Log.
- Records made by the AWS certified inspector for in-process weld inspection, including visual method report as required in ASME B31.3.

## 1.9 QUALIFICATIONS AND CERTIFICATIONS

Before assigning welders or welding operators to the work, provide the Contracting Officer with names and certification that each individual is performance-qualified as specified. The certification must state the type of welding and positions for which each is qualified, the code and welding procedure specification under which each is qualified, date qualified, and the firm and individual who certified the qualification tests. Provide a summary table showing all welders and the WPS for which they are qualified to weld.

Do not start welding until qualified weld procedures, welders, and welding operators have been provided. Qualify procedures for making transition welds between different materials or between metal of different wall thickness. Qualification testing must be performed by an approved independent test laboratory, or by the Contractor if approved by the Contracting Officer. Cost of such testing must be borne by the Contractor. Notify the Contracting Officer at least one week in advance of the time and place of tests. If the Contracting Officer elects to witness the tests, the qualification tests must be performed at or near the worksite and at the notified time and place.

### 1.9.1 Welding Procedure Specification

Qualify the WPS for every proposed welding procedure. Qualification must conform to the requirements of ASME B31.3 and this Section. Specify back purge gas requirements and end preparation for joints to include cleaning, alignment, and root opening tolerances. Include procedures for weld repairs. Specify interpass temperature control and any weld special treatment requirements as required by the design. Identify weld procedures uniquely and reference on the Weld Tracking Log and shop drawings.

Submit copies of the WPS for each type of welding required in accordance with paragraph SUBMITTALS. Approval of any procedure does not relieve the Contractor of the sole responsibility to produce acceptable welds.

### 1.9.2 Procedure Qualification Record

Perform tests, qualify all procedures including weld repair, and document the results in detail on procedure qualification records. Qualify each proposed welding procedure. Qualify procedures in compliance with ASME B31.3, ASME BPVC SEC IX, and this Section. Use the PQR format QW-483 in ASME BPVC SEC IX. Submit each PQR together with its associated WPS, and in accordance with paragraph SUBMITTALS.

### 1.9.3 Welding Personnel

Conduct tests using qualified procedures to determine welding personnel are capable of producing the minimum requirements of an acceptable weldment. Test all welding personnel for each welding process to be used. Tests conducted by a different employer are not acceptable. Test in accordance with ASME B31.3 and ASME BPVC SEC IX.

#### 1.9.3.1 Welder Performance Qualification

A welder or welding operator may be qualified by volumetric NDE or by bend tests on a test coupon. Qualification by initial production welding is not allowed. Before assigning welding personnel to the work, provide WPQ

records which certify the individual is performance-qualified for the procedure in accordance with ASME B31.3. The certification must state the type of welding and positions for which each is qualified, the code and welding procedure specification under which each is qualified, date qualified, and the firm and individual certifying the qualification tests. Use the WPQ format in ASME BPVC SEC IX QW-484A for welders and QW-484B for welding operators. All welder qualification records must be signed and stamped by a certified welding inspector (CWI) as defined in AWS QC1. Submit each WPQ in accordance with paragraph SUBMITTALS.

#### 1.9.3.2 Renewal of Welder Qualification

Requalification of a welder or welding operator is required under any of the following conditions:

- a. When a welder or welding operator has not used the specific welding procedure for a period of 3 months; the period may be extended to 6 months if the welder or welding operator has been employed on another welding procedure.
- b. When a welder or welding operator has not welded with any procedure during a period of 3 months, all the personal qualifications are considered expired, including any extension by virtue of a. above.
- c. There is specific reason to question the person's ability to make welds that will meet requirements of the specifications.
- d. The welder or welding operator was qualified by an employer, other than firms performing work under this contract, and a qualification test has not been taken within the preceding 12 months.
- e. Renewal of qualification for a specific welding procedure under conditions a., b., and d., above, needs to be made on only a single test joint or pipe of a thickness, position, or material required by the welding procedure specifications to reestablish the welder or welding operator qualification for the previous qualification.

#### 1.9.4 Previously Qualified Procedures and Personnel

Welding procedure specifications, welders, and welding operators previously qualified by test may be accepted for the work without requalification, provided that all of the following conditions are fulfilled.

- a. Copies of each WPS, procedure qualification record (PQR), and welder performance qualification (WPQ) are submitted and approved in accordance with paragraph SUBMITTALS.
- b. Testing was performed by an approved test laboratory, approved technical consultant, or by the Contractor's approved quality control organization.
- c. The welding procedures, welders, and welding operators were qualified in accordance with API 570 and ASME B31.3, and base materials, filler materials, electrodes, equipment, and processes conformed to the applicable requirements of the design and this Section.
- d. The requirements of paragraph RENEWAL OF WELDER QUALIFICATION are met and records showing name of employer and period of employment using

the process for which qualified are submitted as evidence of conformance.

#### 1.9.5 Independent Test Organization

The independent test organization can be a testing laboratory, technical consultant, or NDE testing firm, and must meet principles of ASTM E329. The principal business of the testing organization, testing laboratory, technical consultant, or firm must be NDE, inspection, and testing, and must have no involvement in project procurement, fabrication, construction or installation. The test organization, testing laboratory, technical consultant or NDE testing firm must be a first tier subcontractor. Submit current Independent Test Organization certification which demonstrates compliance with this Section in accordance with paragraph SUBMITTALS.

#### 1.9.6 Inspection and Examination Personnel

Provide a commercially independent organization for all weld examinations. Submit qualifications of all inspection and examination personnel for approval by the Contracting Officer. All inspection and NDE personnel must be qualified in accordance with the following requirements.

##### 1.9.6.1 Prohibited Practices

- a. The contractor quality control manager cannot be the weld inspector or the weld examiner.
- b. Should a weld inspector also be a welder, that individual is disqualified from inspecting or examining a weld or any portion thereof of the inspector's own work.
- c. Should a weld examiner also be a welder, that individual is disqualified from inspecting or examining a weld or any portion thereof of the examiner's own work.

##### 1.9.6.2 Weld Inspector

- a. Qualification: Weld inspector must be qualified in accordance with ASME B31.3 and API 570, and have minimum 5 years of experience inspecting process pipe or pipeline welding on military or commercial fuel systems. The weld inspector(s) is considered a QC Specialist.
- b. Certification: Welding inspectors must be certified as CWI or senior certified welding inspector (SCWI) as defined in AWS QC1 with a ASME BPVC SEC IX endorsement. Provide AWS Weld Inspector Certificate in accordance with paragraph SUBMITTALS.

##### 1.9.6.3 In-Process Weld Inspector

- a. Qualification: In-process weld inspector must be qualified in accordance with ASME B31.3 and API 570, and have minimum 5 years of experience inspecting process pipe or pipeline welding on military or commercial fuel systems. The in-process weld inspector(s) is considered a QC Specialist.
- b. Certification: In-process weld inspector must be certified as CWI or senior certified welding inspector (SCWI) as defined in AWS QC1 with a ASME BPVC SEC IX endorsement, and must be certified as a Level II in visual method compliant with ASNT SNT-TC-1A. Provide both AWS and

ASNT In-Process Weld Inspector Certificates in accordance with paragraph SUBMITTALS.

#### 1.9.6.4 Nondestructive Examiner

- a. Qualification: Examiners must be qualified to perform NDE in accordance with ASME B31.3 and API 570. Examiners must meet minimum requirements for qualification in ANSI/ASNT CP-189. Qualified examiners must have minimum 5 years verifiable experience performing nondestructive examination of fuel pipeline or piping. Experience is defined as work activity performing a specific NDE method under the direction of qualified supervision but does not include time spent in training programs. Examiners performing PAUT must be qualified in accordance with ASME BPVC SEC V Mandatory Appendix VII, and API Std 650 Annex U Personnel Qualifications and Training, with Level III Review. Personnel performing nondestructive examination must not represent nor be an employee of the prime construction contractor, welding subcontractor, fabricator, erector, or manufacturer, and must be employed by an independent test organization.
- b. Certification: Provide examiners certified compliant with ANSI/ASNT CP-189 pursuant to the table EXAMINER CERTIFICATION STANDARDS for the applicable method. Level II Limited certification does not meet the Level II requirement. Provide Weld Examiner Certificate in accordance with paragraph SUBMITTALS.

#### 1.9.6.5 Examiner Certification Standards

| Method             | Standard                                  |
|--------------------|-------------------------------------------|
| UT                 | Level II or III                           |
| PAUT in lieu of RT | API Std 650 Annex U with Level III review |
| PT                 | Level II or III                           |
| MT                 | Level II or III                           |
| RT                 | Level III                                 |
| VT                 | Level II                                  |

#### 1.9.7 Piping Engineer of Record

Professional engineer in responsible charge of the POL service piping design.

- a. Qualification: 5 years of experience in POL facilities engineering, including design, inspection, and construction of pipelines or process piping systems.
- b. Certification: Bachelor of Science degree in Civil or Mechanical Engineering; Licensed as a professional civil or mechanical engineer.

### 1.10 DELIVERY, STORAGE, AND HANDLING

All filler metals, electrodes, and other welding materials must be delivered to the site in manufacturer original package and stored in a dry space until used. Packages must be properly labeled and designed to give maximum protection from moisture, prevent contamination, and to ensure safe handling.

#### 1.10.1 Material Control

Store materials in a controlled access and clean, dry area that is weathertight and is maintained at a temperature recommended by the manufacturer. Materials must not be in contact with the ground or floor, and must be stored on wooden pallets or cribbing.

#### 1.10.2 Partial Issue

When a container of covered electrodes is opened and only a portion of the content is issued, the remaining portion must, within the limits established by AWS D1.1/D1.1M be placed and kept in a storage oven until use.

#### 1.10.3 Damaged Container

Low-hydrogen steel electrodes must be stored in their sealed shipping container. If the seal is damaged during shipment or storage, and the damage is not immediately detected, the covered electrodes in that container must be rebaked in accordance with the manufacturer's instructions prior to issuance, or they must be discarded. If a container is damaged in storage and the damage is witnessed, the electrodes from that container must be immediately placed in a storage oven. The storage oven temperature must be as recommended by the manufacturer or the welding material specification.

#### 1.10.4 Damaged Material

Materials which are damaged must be discarded. Do not use covered electrodes which are oil or water-soaked, dirty, or on which the flux has separated from the wire.

## PART 2 PRODUCTS

### 2.1 WELDING MATERIALS

Welding materials for carbon steel, stainless steel and aluminum must comply with AWS WHB-4.9. Welding equipment, electrodes, welding wire, shielding and backing gas, and fluxes must be capable of producing satisfactory welds when used by a qualified welder or welding operator using qualified welding procedures. All field girth root pass welds must be made with non-covered electrodes or welding wire. All root passes must be made with shielding and backing gas. External welds on the pipe such as attaching pipe supports may be made with covered electrodes or welding wire. Electrodes, welding wire and fluxes must be in accordance with Table 1. Welding materials for aluminum and aluminum alloy must comply with AWS D10.7/D10.7M.



| TABLE I                                                                                                                                                 |         |            |                |                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------|----------------|-------------------------------------------|
| AWS                                                                                                                                                     | Process | Alloy      | Consumable     | Use                                       |
|                                                                                                                                                         |         |            | (Examples) (1) |                                           |
| AWS A5.1/A5.1M                                                                                                                                          | SMAW    | Low Carbon | E7018          | Fill                                      |
| AWS A5.4/A5.4M                                                                                                                                          | SMAW    | Stainless  | E308L, E309L   | Fill                                      |
| AWS A5.3/A5.3M                                                                                                                                          | SMAW    | Aluminum   |                | Fill                                      |
| AWS A5.9/A5.9M                                                                                                                                          | GTAW    | Stainless  | ER308L, ER309L | Root and Fill                             |
| AWS A5.10/A5.10M                                                                                                                                        | GTAW    | Aluminum   |                | Root and Fill                             |
| AWS A5.18/A5.18M                                                                                                                                        | GTAW    | Low Carbon | E70S-3, E70S-6 | Root and Fill                             |
| AWS A5.22/A5.22M                                                                                                                                        | GTAW    | Stainless  | E308LT1-1      | Root (Backing and Shielding Gas Required) |
| AWS A5.32/A5.32M                                                                                                                                        | GTAW    | All        |                | Shielding Gas                             |
| Note(1): The consumable material designations shown are examples only and are not intended to limit the Contractor's selection of consumable materials. |         |            |                |                                           |

### PART 3 EXECUTION

#### 3.1 WELDING OPERATIONS

Perform welding in accordance with the Weld Plan and qualified procedures using qualified personnel. Do not weld when the quality of the completed weld could be impaired by prevailing working or weather conditions. The Contracting Officer must determine when weather or working conditions are unsuitable for welding.

For applicable processes, provide operations consistent with standards and guides AWS D10.4, AWS D10.10/D10.10M, AWS D10.11M/D10.11, AWS D10.12M/D10.12, and AWS C5.5/C5.5M.

- The root pass on stainless steel, aluminum, and carbon steel pipe must be by the GTAW process.
- All joints unless indicated otherwise, in carbon steel, aluminum and stainless steel piping systems must be welded.
- Unless otherwise approved, all girth welds must be CJP groove welds made in accordance with qualified welding procedures.
- Weld preparation must comply with the requirements of ASME B31.3 and the qualified Welding Procedure Specification. The use of "rice paper" as purge blocks is not permitted. Contractor must submit alternate method for approval.
- Backing Rings. The use of backing rings for making or repairing welds

will not be permitted.

### 3.1.1 Identification

Assign each welder or welding operator weld a unique identification mark (number, letter, or symbol). Place identification mark on the work and the weld map. Ensure each identification is neat and legible, traceable to a welder, and associated with a performance qualification record. Apply the unique identification mark adjacent to each weld with permanent weatherproof means using a rubber stamp, felt-tipped marker, paint pen, or other method approved by the Contracting Officer. Do not use fluorescent paint or any method which deforms the metal. Identification by die stamps or electric etcher is not allowed.

### 3.1.2 Base Metal Preparation

Use mechanical grinding of thermal cut ends to remove the HAZ, or to dress metal for proper fit-up. This practice is limited to the HAZ or a maximum 3 mm length of removal. Do not use oxy-fuel cutting on austenitic stainless steel or nonferrous materials. Do not reduce pipe thickness outside of a weld. See paragraph GRINDING for requirements in the event pipe thickness is reduced from nominal.

### 3.1.3 Weld Joint Fit-up

Provide fabrication, assembly, erection, preparation, and fit-up in accordance with API 570 and ASME B31.3. Use bars, jacks, clamps, or other mechanical fixtures to fit, align, and retain joints in position during the welding operation. Align and prepare end joints prior to welding in accordance with Chapter V of ASME B31.3. For socket-welded joints, implement a program to ensure gaps are properly measured and documented to be in conformance with ASME B31.3. Branch connection welds must be no closer than 2 times the run piping thickness from an adjacent weld.

Do not use welded temporary attachments except when it is impractical to use mechanical fixtures. When temporary attachments are used, they must be the same material as the base metal and must be completely removed by careful cutting or grinding after the weld operation is completed. Cut the attachment no closer than 6 mm from the member and remove remaining material by careful grinding. Do not grind into base metal. After the temporary attachment has been removed, inspect the area using VT. For any area with grinding that could reduce metal or pipe wall thickness below nominal, refer to and follow requirements in paragraph GRINDING.

### 3.1.4 Butt Weld Joint Spacing

Butt weld joints must be spaced a minimum of 150 mm apart with measurement taken from the toe of the first weld to the toe of the second weld. Measure at the closest point between the welds when visually inspected. If spacing is not possible due to existing conditions, a reduction in spacing to not less than 50 mm may be made.

### 3.1.5 Preheat and Interpass Temperatures

Meet the requirements specified by ASME B31.3 for preheat temperatures. However, in no case can the preheat be below 10 degrees C for ferritic steel or austenitic stainless steel, or 0 degrees C for nonferrous alloys. Maximum interpass temperatures must not exceed 149 degrees C for austenitic stainless steels, nickel alloys, and copper alloys; and 260

degrees C for carbon steels. Preheat techniques must be such as to ensure the full thickness of weld joint preparation and adjacent base material, at least 75 mm in all directions, is at the specified temperature. Preheating by induction or resistance methods is preferred. When flame heating, only use a neutral flame. Do not use oxy-fuel heating on austenitic stainless steel. However, air-fuel heating is acceptable if controlled to ensure that the surface temperature does not exceed 66 degrees C. Interpass temperatures must be checked on the surface of the component within 25 mm of the weld groove and at the starting location of the next weld pass, and for a distance of 150 mm ahead of the weld, but not on the area to be welded.

### 3.1.6 Production Welding Instructions

- a. Welding must not be done when ambient temperature is lower than minus 18 degrees C.
- b. Welding is not permitted on surfaces that are wet or covered with ice, when snow or rain is falling on the surfaces to be welded. During periods of high winds welding is not permitted unless the welders and the work are properly protected.
- c. Gases for purging and shielding must be welding grade and must have a dew point of minus 40 degrees C or lower.
- d. Back purges are required for austenitic stainless steels and nonferrous alloys welded from one side and must be set up such that the flow of gas from the inlet to the outlet orifice passes across the area to be welded. The oxygen content of the gas exiting from the purge vent must be less than 2 percent prior to welding. The flow rate must be that required by the approved weld procedure specification.
- e. The purge on groove welds must be maintained for at least two passes or 5 mm whichever is greater.
- f. Removable purge dam materials must be made of expandable or flexible plugs, such as plexiglass or dry and clean plywood. Wood dams must be kiln-dried quality. Do not use purge dams made of polyvinyl alcohol.
- g. Any welding process which requires the use of external gas shielding cannot be done in a draft or wind unless the weld area is protected by a shelter. This shelter must be of material and shape appropriate to reduce wind velocity in the vicinity of the weld to a maximum of 8 km/hour.
- h. Tack welds to be incorporated in the final welds must have their ends tapered by grinding or welding technique. Remove tack welds that are cracked or defective and retack the groove prior to welding. Temporary tack welds must be removed, the surface ground smooth, and visually inspected. For low-alloy and hardenable high-alloy steels, the area must be examined by MT.
- i. Grinding of completed welds is to be performed only to the extent required for inspection or NDE, including in-process examination, and to provide weld reinforcement within the requirements of ASME B31.3. If the surface of the weld requires grinding, reducing the weld or base material below the minimum required thickness is not allowed. See paragraph CORRECTION AND REWORK. Minimum weld external

reinforcement must be flush between external surfaces.

- j. Each qualified welder must be assigned unique identification. All welds must be permanently annotated with the mark or symbol of the individual who made the weld.

#### 3.1.7 Postweld Heat Treatment

- a. When required, perform postweld heat treatment in accordance with ASME B31.3. Temperatures for local postweld heat treatment must be measured continuously by thermocouples in contact with the weldment.
- b. Postweld heat treatment of low-alloy steels, when required, must be performed immediately upon completion of welding and prior to the temperature of the weld falling below the preheat temperature. However, postweld heat treatment may be postponed after the completion of the weld, if, immediately after the weld is completed, it is maintained at a minimum temperature of 149 degrees C or the preheat temperature, whichever is greater, for 2-hours per 25 mm of weld thickness.

#### 3.2 EXAMINATIONS, INSPECTIONS AND TESTS

Provide the services of a qualified inspection, testing, or examination organization meeting requirements of paragraph QUALIFICATIONS AND CERTIFICATIONS and approved by the Contracting Officer. Liquid penetrant examination is required pursuant to this Section. Perform inspection, examination, and testing as required by the SOW or Project Program, this Section, the piping engineer, API 570, ASME B31.3, and in accordance with written procedures.

- a. When in-process weld examination is required, perform in accordance with paragraph IN-PROCESS WELD INSPECTION.
- b. Hydrostatic test requirements are found in the SOW or Project Program and Section 33 52 10 FUEL SYSTEMS PIPING (SERVICE STATION).
- c. In all cases where PAUT is used in lieu of RT, permanently record the weld examinations. Have the results reviewed and receive concurrence by an examiner certified to ANSI/ASNT CP-189 Level III and requirements of this Section.
- d. When examination, inspection, and testing indicate a defect in a weld joint, the weld must be repaired by a qualified welder in accordance with paragraph CORRECTION AND REWORK.
- e. When inspection, examination, and tests substantiate defects in a weld, rework the weld with a qualified welder in accordance with paragraph CORRECTION AND REWORK. Perform all inspection, examination, and tests on rework until acceptable results are achieved.
- f. Submit all weld inspection, examination, and test reports to the Contracting Officer.

##### 3.2.1 Quality Control of Welding

- a. Verify base materials and consumable welding materials conform to the

specifications, and that welding filler metals used are as specified for each base material.

- b. Verify welding equipment to be used for the work is appropriate for use with the welding procedure specification and has the capability to meet requirements of the welding procedure.
- c. Verify only approved or qualified welding procedures are used for the work.
- d. Verify that edge preparation and joint geometry meet requirements of the welding procedure and drawings.
- e. Verify specified filler metals are used and that filler metals are maintained in proper condition, per requirements and as recommended by the manufacturer.
- f. Verify work conforms to requirements of the applicable code, standards, drawings, design, or other documents.
- g. Verify the approved WPS pre-heat and post heat procedures are being used.
- h. Prepare clear and concise reports and verify that records of the results of weld inspection and examinations are maintained.

### 3.2.2 Weld Inspection

Perform weld inspection to detect surface and subsurface discontinuities in completed welds. Visually inspect all welds in accordance with requirements of ASME B31.3, ASME BPVC SEC V, and AWS D1.1/D1.1M. Aid the VT by using PT or MT as-needed to determine the condition of a weld.

- a. Provide inspector personnel qualified and certified in accordance with paragraph INSPECTION AND EXAMINATION PERSONNEL.
- b. Visually inspect all tack welds and completed welds.
- c. Submit weld inspection reports to the Contracting Officer within 10 work days of performance.
- d. Provide inspection in accordance with paragraph EXAMINATION AND INSPECTION FREQUENCY.
- e. Weld personnel found making defective welds must be removed from the work or must be required to be requalified in accordance with ASME B31.3 by the Quality Control Manager.
- f. For each method used, provide a qualified procedure in accordance with paragraph INSPECTION PROCEDURES.

#### 3.2.2.1 Weld Inspection Duties

- a. Verify the technique and performance of each welder, welding operator, and tack welder are as specified.
- b. Verify the work inspected is identified and documented in accordance with SOW or Project Program requirements and this Section.

- c. Prepare clear and concise reports and verify that records of the results of weld inspection and examinations are maintained.
- d. Review requirements in paragraph QUALITY CONTROL OF WELDING and determine whether the quality control program is effective.
- e. Perform VT inspection of all welds.

### 3.2.3 In-Process Weld Inspection

For closure welds or welds designated for in-process inspection or examination, inspect in accordance with requirements of ASME B31.3, ASME BPVC SEC V, API 570, and AWS D1.1/D1.1M. Aid visual inspection with PT or MT in accordance with API 570 if needed to determine the condition of a weld. Perform in-process inspection in accordance with paragraph WELD INSPECTION and the following.

- a. Provide in-process inspection in accordance with paragraph EXAMINATION AND INSPECTION FREQUENCY.
- b. Before welding: Inspect for compliance with requirements for joint preparation, cleanliness, preheating, fit-up, joint clearance, and internal alignment in accordance with ASME B31.3. Assess compliance with variables specified in the WPS including filler material, position, and electrode.
- c. During welding: After cleaning, inspect root pass for cracks, slag, and conformance to approved welding procedure. Assess weld condition between passes. When it is accessible, perform visual inspection of the interior side of the root pass by remote means allowed in ASME BPVC SEC V Article 9.
- d. After welding: Inspect for defects, contour and finish, bead reinforcement, undercutting, overlap, weld slag on the interior of the pipe, appearance, and weld size in accordance with ASME B31.3 and AWS D1.1/D1.1M. Perform visual inspection of the interior by remote means allowed in ASME BPVC SEC V Article 9 when it is accessible.
- e. Prepare and provide inspection record of in-process weld inspection including visual method report. Submit in-process weld inspection records and reports to the Contracting Officer within 10 work days of performance.

### 3.2.4 Examination, Inspection, and Tests by the Government

Welding is subject to inspection and tests in the mill, shop, and field. When materials or workmanship do not conform to SOW or Project Program, or requirements of this Section, the work may be rejected at any time before final acceptance of the system containing the weldment. If defective work is identified, provide a corrective action plan and initiate rework procedures. Perform rework, repair, inspection, examination, and testing compliant with SOW or Project Program requirements and this Section. The rework, repair, inspection, examination, and testing, except as noted below, is at no additional cost to the Government.

#### 3.2.4.1 Supplemental NDE, Inspection, and Tests

In addition to requirements of the SOW or Project Program and this Section, the Government Owner's Inspector may perform supplemental

nondestructive examination, inspection, or tests as deemed necessary. The cost of supplemental NDE, inspection, or tests will be borne by the Government. Coordinate safe access to the work for the NDE, inspection, and test personnel and equipment. If defective work is identified, provide a corrective action plan and initiate rework procedures.

#### 3.2.4.2 Destructive Tests

In addition to requirements of the SOW or Project Program and this Section, the Contracting Officer or Owner's Inspector may order destructive tests to be performed by the Contractor. If destructive test results indicate that materials or workmanship do not conform to contract requirements, the cost of destructive tests, corrective action, and repair must be borne by the Contractor. When destructive tests are made, repairs must be made by qualified welders or welding operators using qualified welding procedures which will develop the full strength of the member.

### 3.3 EXAMINATION AND INSPECTION FREQUENCY

Conduct NDE and inspection of all welding, including high point vent and low point drain piping. Determine and report conformance to the paragraph ACCEPTANCE STANDARDS. Provide frequency of NDE and inspection in accordance with the below requirements and paragraph EXAMINATION AND INSPECTION SCHEDULE.

- a. Provide 100 percent RT for welds on underground or inaccessible piping.
- b. Provide 100 percent RT for welds on severe cyclic piping to include aviation hydrant systems and truck fillstands.
- c. Examine each socket weld or socket welded branch connection with either magnetic particle or liquid penetrant methods.
- d. For aboveground welds examined by radiography at frequency less than 100 percent, provide random RT in accordance with ASME B31.3 on no less than 20 percent of welds. Random testing must include representation of welds made by each welding operator or welder and must include representation of each weld type present in the work. The progressive sampling requirements of ASME B31.3 apply.
- e. Where RT is infeasible, notify and provide justification to the Contracting Officer. Justification must be substantive and not be based on convenience. Upon approval by the Contracting Officer, perform UT in lieu of RT in accordance with this Section and API Std 650 Annex U and ASME BPVC SEC V.

#### 3.3.1 Examination And Inspection Schedule

| Weld Type  | Joint Type or Configuration | Frequency  | Method               |
|------------|-----------------------------|------------|----------------------|
| CJP groove | Butt                        | Each pass  | VT                   |
| CJP groove | Butt                        | Final pass | VT, RT               |
| Closure    | Root pass                   | Each       | VT, PT or MT; Note 1 |

| Weld Type             | Joint Type or Configuration | Frequency                           | Method         |
|-----------------------|-----------------------------|-------------------------------------|----------------|
| Closure               | Cover pass                  | Final                               | VT, RT; Note 1 |
| Fillet                |                             | Final                               | VT, MT         |
| Structural attachment | Installation or removal     | Final pass or after removal         | VT, PT or MT   |
|                       | Cutline                     | Each                                | VT; Note 2     |
| Repair                |                             | Note 3                              | Note 3         |
|                       | Pipe interior               | Root or final pass, when accessible | VT             |
| Insert                | Branch connection           | Final pass                          | VT, RT         |
| Non radiographicable  | Branch Connection           | Final pass                          | VT, PT or MT   |
| CJP groove            | Butt; inaccessible          | Final pass                          | VT, RT         |
| Socket                |                             | Final pass                          | VT, PT or MT   |

Note 1. In-process weld examination IAW API 570, ASME B31.3 and this Section.

Note 2. Examine the cutline for laminations, scale, burrs before welding.

Note 3. Examination and inspection requirement for a weld repair is pursuant to that of the original weld and must be repeated until no defect exists.

### 3.4 ACCEPTANCE STANDARDS

#### 3.4.1 Acceptance Criteria

| Method             | Criterion                | Note |
|--------------------|--------------------------|------|
| VT                 | ASME B31.3 Table 341.3.2 | 1    |
| RT                 | ASME B31.3 Table 341.3.2 | -    |
| MT                 | ASME B31.3 344.3         | 2    |
| PT                 | ASME B31.3 344.4         | 3    |
| UT                 | ASME B31.3 344.6         | -    |
| PAUT in lieu of RT | ASME B31.3 Table 341.3.2 | 4    |

Note 1. In addition to paragraph ACCEPTANCE CRITERIA, requirements of paragraph VISUAL also apply.



Note 2. In addition to paragraph ACCEPTANCE CRITERIA, requirements of paragraph MAGNETIC PARTICLE also apply.

Note 3. In addition to paragraph ACCEPTANCE CRITERIA, requirements of paragraph LIQUID PENETRANT also apply.

Note 4. Use acceptance criteria for RT.

#### 3.4.2 Visual

The following indications are unacceptable.

- a. Weld slag on the interior of the pipe.

#### 3.4.3 Magnetic Particle

The following relevant indications are unacceptable.

- a. Any linear indication.
- b. Rounded indication with any dimension greater than 5 mm.
- c. Four or more rounded indications in a line separated by 2 mm or less edge-to-edge.
- d. Ten or more rounded indications in any 3870 square mm of surface with the major dimension of this area not to exceed 150 mm (linear dimension), and with the area calculated in the most unfavorable location relative to the indications being evaluated.

#### 3.4.4 Liquid Penetrant

Indications with a major dimension greater than 2 mm must be considered relevant. The following relevant indications are unacceptable.

- a. Any crack or linear indication.
- b. Rounded indication with any dimension greater than 5 mm.
- c. Four or more rounded indications in a line separated by 2 mm or less edge-to-edge.
- d. Ten or more rounded indications in any 3870 square mm of surface with the major dimension of this area not to exceed 150 mm (linear dimension), and with the area calculated in the most unfavorable location relative to the indications being evaluated.

#### 3.5 CORRECTION AND REWORK

Disqualifying or rejectable inspection findings or examination indications must be completely removed and repaired as specified in ASME B31.3, unless specified otherwise in this Section. Defects discovered between weld passes must be completely removed and repaired before additional weld passes are made. After defect removal is complete and before rewelding, the area must be inspected and examined by the same method(s) which first revealed the defect to ensure the defect has been eliminated. After rewelding, the repaired area must again be inspected and examined by the same method(s) originally used for that area. Any indication of a defect

must be regarded as a defect unless reevaluation by NDE or by surface conditioning shows that no defects are present.

### 3.5.1 Defect Removal

Defective or unsound weld joints must be corrected by removing and replacing the entire weld joint, or for the following defects, corrections must be made as follows.

- a. Excessive Convexity and Overlap: Reduce by careful removal of excess metal.
- b. Excessive Concavity of Weld, Undersized Welds, Undercutting: Clean and deposit additional weld metal.
- c. Excessive Weld Porosity, Inclusions, Lack of Fusion, Incomplete Penetration: Remove defective portions including sound weld metal for a distance of twice the thickness of the base metal or 5 cm, whichever is less, and reweld.
- d. Cracks or linear indications in weld or base metal: Remove crack or linear indication throughout its length, extending removal into sound weld metal for a distance of twice the thickness of the base metal or 5 cm, whichever is less, beyond each end of a crack or linear indication. Confirm complete removal by MT for carbon steel or PT for stainless steel. Once removal is inspected and examined, then rework can commence.
- e. Poor Fit-Up: Cut apart improperly fitted parts, re-fit, and reweld.

#### 3.5.1.1 Methods of Defect Removal

Remove weld metal or portions of base metal by chipping, grinding, sawing, machining, or other mechanical means. It is less preferable but defects also can be removed by thermal cutting techniques. If thermal cutting techniques are used, clean and smooth the cut surfaces by mechanical means to remove the HAZ. In addition, a maximum of 3 mm of metal must be removed by mechanical means from the cut surfaces of stainless steel.

Wherever a defect is removed, and repair by welding is not required, the affected area must be blended into the surrounding surface eliminating sharp notches, crevices, or corners.

#### 3.5.1.2 Grinding

For surfaces which require grinding, exercise caution to avoid reducing wall thickness. After all grinding operations are complete, measure with UT and record remaining metal thickness on areas which were ground. Perform thickness measurements, compare to nominal, and analyze results in accordance with API 570. Report results to the Contracting Officer.

#### 3.5.1.3 Rewelding

Make repair welds using an electrode or filler wire smaller than that used in making the original weld. Rewelding must be done using qualified welding procedures. The surface must be cleaned before rewelding. Repair welds must meet all requirements of this Section.

#### 3.5.1.4 Peening or Caulking

The use of force (peening) or foreign materials to mask, fill, seal, or disguise any welding defect is prohibited.

### 3.6 MAINTAINING CLEANLINESS OF PIPING

Keep the interior and ends of all new pipe and piping affected by work thoroughly cleaned of foreign matter and water before, during, and after installation. Keep piping systems clean during installation by means of plugs, covers, or other effective methods. When work is not in progress, secure ends of piping, components, and fittings so no water, debris, or foreign substance can enter. Inspect piping before placing into position. Clean the interior of each length of pipe after welding. Use a swab with a leather or canvas belt disc to fit the inside diameter of pipe and pull through each length of pipe after welding in place. It is the Contractor's responsibility to ensure the interior of piping is free of dirt, debris, water, and foreign matter, including weld slag, when it is connected into the system.

### 3.7 QUALITY CONTROL DETERMINATION

Perform quality control oversight of the welding, inspection, testing, and NDE work. Provide a determination by the QCM that each weld is complete, has been inspected by the inspector, has been examined by the examiner(s), and meets contract requirements. Mandatory elements of the determination are below. See paragraphs WELD MARKING and WELD LOG.

- a. Review of the weld
- b. Welder, inspector, and NDE examiner markings are complete, accurate, and up to date
- c. Weld inspection and results
- d. Weld examination and results
- e. Weld Log is complete, accurate, and up to date
- f. Reporting to the Contracting Officer is complete, accurate, and up to date

-- End of Section --

## SECTION 33 56 10

FACTORY-FABRICATED FUEL STORAGE TANKS  
05/25

## PART 1 GENERAL

## 1.1 SUMMARY

This section defines the requirements for factory-fabricated fuel storage tanks.

## 1.1.1 Related Sections

## 1.1.1.1 Earthwork

Excavation and backfilling for tanks must be in accordance with Section 31 00 00 EARTHWORK.

## 1.1.1.2 Leak Detection

Provide manual inspection port in secondary containment tank for visual inspection in accordance with UL 142.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN PETROLEUM INSTITUTE (API)

|             |                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------|
| API RP 540  | (1999; R 2004) Electrical Installations in Petroleum Processing Plants                           |
| API RP 2003 | (2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents |

## ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A193/A193M | (2024a) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications |
| ASTM A194/A194M | (2024) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both   |
| ASTM A307       | (2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                                                |
| ASTM A563       | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                                                                                   |

|            |                                                                                                    |
|------------|----------------------------------------------------------------------------------------------------|
| ASTM D3308 | (2012; R 2017) Standard Specification for PTFE Resin Skived Tape                                   |
| ASTM F436  | (2011) Hardened Steel Washers                                                                      |
| ASTM F844  | (2019; R 2024) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use |
| ASTM F1554 | (2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength          |

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

|           |                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| IEEE 142  | (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book |
| IEEE 1100 | (2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment                       |

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------|
| NFPA 30  | (2024; TIA 24-1) Flammable and Combustible Liquids Code                                          |
| NFPA 30A | (2024; ERTA 1 2023) Code for Motor Fuel Dispensing Facilities and Repair Garages                 |
| NFPA 31  | (2024; TIA 23-1) Standard for the Installation of Oil-Burning Equipment                          |
| NFPA 70  | (2023; ERTA 1 2024; TIA 24-1) National Electrical Code                                           |
| NFPA 77  | (2024; ERTA 1 2023) Recommended Practice on Static Electricity                                   |
| NFPA 407 | (2022; TIA 24-2) Standard for Aircraft Fuel Servicing                                            |
| NFPA 704 | (2022) Standard System for the Identification of the Hazards of Materials for Emergency Response |
| NFPA 780 | (2023) Standard for the Installation of Lightning Protection Systems                             |

## SOCIETY FOR PROTECTIVE COATINGS (SSPC)

|           |                                                                                            |
|-----------|--------------------------------------------------------------------------------------------|
| SSPC QP 3 | (2010) Standard Procedure for Evaluating Qualifications of Shop Painting Applicators       |
| SSPC QS 1 | (2015) Standard Procedure for Evaluating a Contractor's Advanced Quality Management System |

## STEEL TANK INSTITUTE (STI)

|                 |                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| STI 020-50-1000 | (2010) ACT-100 Specification for External Corrosion Protection of FRP Composite Steel USTs                |
| STI 700-50-5007 | (2010) Installation Instructions for Shop Fabricated Aboveground Tanks for Flammable, Combustible Liquids |
| STI SP001       | (2018, 6th Ed) SP001 Standard for The Inspection of Aboveground Storage Tanks                             |

## U.S. DEPARTMENT OF DEFENSE (DOD)

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| MIL-STD-161 | (2005; Rev G; Notice 1 2010) Identification Methods for Bulk Petroleum Products Systems Including Hydrocarbon Missile Fuels |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|

## UL SOLUTIONS (UL)

|         |                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------|
| UL 58   | (2018) UL Standard for Safety Steel Underground Tanks for Flammable and Combustible Liquids                   |
| UL 142  | (2006; Reprint Jan 2021) UL Standard for Safety Steel Aboveground Tanks for Flammable and Combustible Liquids |
| UL 1746 | (2007; Reprint Dec 2014) External Corrosion Protection Systems for Steel Underground Storage Tanks            |

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Grounding and Bonding

## SD-03 Product Data

Aboveground Storage Tank (Double Wall Steel); G

Tank Protective Coatings; G

Atmospheric Vent;

Pressure Vent;

Emergency Vent;

Independent Level Alarm System; G

Tank Gauges; G

Tank Anchorage System Materials; G

#### SD-06 Test Reports

Aboveground Storage Tank Tightness Tests; G

Tank Manufacturer's Tests

Tank Fill Tests

Tank Inspection Reports; G

#### SD-07 Certificates

Contractor Qualifications; G

Manufacturer's Certification; G

Permitting

Registration

Licensed Personnel

Demonstrations

STI SP001 Inspector's Certification; G

#### SD-08 Manufacturer's Instructions

Aboveground Storage Tank

Independent Level Alarm System

Tank Gauges

#### SD-10 Operation and Maintenance Data

Aboveground Storage Tank; G

Independent Level Alarm System; G

Tank Gauges; G

### 1.4 QUALITY ASSURANCE

#### 1.4.1 Contractor Qualifications

Each installation Contractor must have successfully completed at least 5 projects of the same scope, and the same size or larger within the last 3 years and demonstrated specific installation experience in regard to the specific system installation to be performed. Each installation Contractor must have taken, if applicable, manufacturer's training courses on the installation of storage tanks and must meet all applicable

licensing requirements in the state. If applicable, state certified installers must be provided by the Contractor. Installers must also be trained and certified by the manufacturer to install the equipment and materials being installed and must be STI certified. Installers must submit certification from the manufacturer. If installing underground storage tanks and piping systems, installation Contractor must have Pollution Liability Insurance. Submit a letter listing prior projects, the date of construction, a point of contact for each prior project, the scope of work of each prior project, and a detailed list of work performed. The letter must also provide evidence of prior manufacturer's training, state licensing, and other related information.

#### 1.4.2 Regulatory Requirements

##### 1.4.2.1 Permitting

Obtain necessary permits in conjunction with the installation of storage tanks as required by federal, state, or local authority.

##### 1.4.2.2 Registration

Obtain and complete all tank registration forms required by federal, state, and local authorities. Submit all completed tank registration forms within 30 days after contract award to the Contracting Officer. The Contracting Officer will ensure the Base Environmental staff for the DoD Installation submits the forms to the proper regulatory agencies.

##### 1.4.2.3 Licensed Personnel

Tank installers must be licensed by the state when the state requires licensed installers.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect system components and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer, upon recommendation by Base Environmental for the DoD Installation. Replace damaged or defective items.

#### 1.6 PROJECT CONDITIONS

Exposed moving parts, parts that produce high operating temperatures and pressures, parts that may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired.

#### 1.7 WARRANTY

All factory-fabricated storage tanks must come with a manufacturer's warranty of a minimum period of 30 years. All warranty paperwork will be completed and submitted by Contractor to both the tank and system component manufacturers, the Contracting Officer, and the Base Environmental for the DoD Installation. This includes all applicable completed manufacturers' equipment installation checklists.



## PART 2 PRODUCTS

### 2.1 ELECTRICAL WORK

Provide controllers, integral disconnects, contactors, controls, and control wiring with their respective pieces of system components. Provide electrical system components, including motors and wiring, in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 2.1.1 Grounding and Bonding

Grounding and bonding must be in accordance with NFPA 70, NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

### 2.2 MATERIALS AND SYSTEM COMPONENTS

Provide materials and system components that are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship. Provide materials and system components that have been in satisfactory commercial or industrial use for a minimum 3 years prior to bid opening. The 3 year period must include applications of the system components and materials under similar circumstances and of similar size. Provide materials and system components that have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 3 year period.

Internal parts and components of system components, piping, piping components, and valves that could be exposed to fuel during system operation must not be constructed of zinc coated (galvanized) metal. Do not install cast iron bodied valves in piping systems that could be exposed to fuel during system operation.

### 2.3 NAMEPLATES

Attach nameplates to all specified system components defined herein. List on each nameplate the manufacturer's name, address, contract number, component type or style, model or serial number, catalog number, capacity or size, and the system that is controlled. Construct plates of melamine plastic, 3 mm thick, UV resistance, black with white center core, matte finish surface and square corners. Install nameplates in prominent locations with nonferrous screws, nonferrous bolts, or permanent adhesive. Minimum size of nameplates must be 25 by 65 mm. Provide manufacturer's storage tank nameplates as required. Lettering must be the normal block style with a minimum 6 mm height. Accurately align all lettering on nameplates. For plastic nameplates, engrave lettering into the white core.

### 2.4 ABOVEGROUND STORAGE TANK

#### 2.4.1 Aboveground Storage Tank (Double Wall Steel)

Provide a factory-assembled unit that includes a factory-fabricated primary storage tank and an integral secondary containment. Tank assembly

must be in accordance with NFPA 30 and be designed and manufactured for a horizontal cylindrical installation. Primary storage tank must be factory-welded, steel in accordance with UL 142. Tank assembly must be mounted on . Tank assembly must have lifting lugs that allow tank relocation. Provide tank assembly with the manufacturer's standard external ladder and platform assembly, except as modified herein.

#### 2.4.1.1 Double Wall Steel Tank

The secondary containment (outer) tank must be a factory-fabricated, steel type that fully-encloses the primary storage (inner) tank. The entire tank assembly must be in accordance with UL 142 and bear the UL 142 label. The interstitial space between the primary tank and the secondary containment tank must be both pressure testable and verifiable. The primary storage tank must be supported within the secondary containment tank reservoir with steel tank saddles, or other similar supports, fabricated and installed by the tank manufacturer.

### 2.5 TANK PROTECTIVE COATINGS

#### 2.5.1 Interior Surfaces

Coat 100 percent of a metal tank's interior surfaces including all metal piping and metal appurtenances in accordance with the manufacturer's standard coating system as modified herein.

##### 2.5.1.1 Certifications of Coating Contractors

All Contractors and Subcontractors that perform surface preparation or coating application must be certified to SSPC QP 3 and SSPC QS 1 prior to contract award, and must remain certified while accomplishing any surface preparation or coating application. Painting contractors and painting subcontractors must remain so certified for the duration of the project. If a contractor or subcontractor certification expires, the firm is not allowed to perform any work until the certification is reissued. Requests for extension of time due to delay as a result of an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in certification status. Notify the Contracting Officer of all scheduled or unannounced on site audits from SSPC and provide a copy of all audit reports.

#### 2.5.2 Exterior Surfaces, Aboveground Tanks

Protect the exterior surfaces of each aboveground tank in accordance with Section 09 90 00 PAINTS AND COATINGS as modified herein.

##### 2.5.2.1 Certifications of Coating Contractors

All Contractors and Subcontractors that perform surface preparation or coating application must be certified to SSPC QP 3 and SSPC QS 1 prior to contract award, and must remain certified while accomplishing any surface preparation or coating application. Painting contractors and painting subcontractors must remain so certified for the duration of the project. If a contractor or subcontractor certification expires, the firm is not allowed to perform any work until the certification is reissued. Requests for extension of time due to delay as a result of an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in certification status. Notify the Contracting Officer of all scheduled or unannounced on site

audits from SSPC and provide a copy of all audit reports.

### 2.5.3 Tank Labeling

Tank must be labelled with the following information at a minimum:

- a. Product Stored and Tank Capacity (Per MIL-STD-161).
- b. Tank Number and Facility Number.
- c. NFPA 704 Diamond Hazmat Label.
- d. "No Smoking" on Class 1 tanks.
- e. "Confined Space" on tanks with manhole/ladder hatch.

## 2.6 TANK COMPONENTS

### 2.6.1 Tank Manhole

Tank manholes must have an internal diameter of 760 mm. Provide each manhole with a matching flanged watertight manhole cover. Manhole covers must be UL listed, be constructed of pressed or mild steel, and include a UL listed gasket.

### 2.6.2 Tank Piping Penetrations

Provide a welded-in-place flanged pipe nozzle for each tank piping connection. All unused or spare tank piping penetrations must be sealed with steel flanges.

### 2.6.3 Tank Striker Plates

Provide an interior striker plate under each tank manhole and pipe connection. Each plate must be a minimum of 6 mm in thickness, be larger in diameter than the tank penetration, fit the curvature of the tank bottom, and be completely coated in the same fashion as the interior tank bottom coating. Each plate must be welded to the tank bottom at the factory (full circumference connection). The welds must be non-destructive tested using the appropriate means.

### 2.6.4 Manual Gauging/Sampling Hatch

Provide a combination gauging and sampling hatch assembly. The assembly must include a bronze top-seal type adapter with a corresponding locking type cap (adapter and cap both externally-mounted to the top of the tank)

### 2.6.5 Tank Ladder

Provide interior tank ladders constructed of either fiberglass or steel. If steel, coat the ladder in the same fashion as the tank interior. The two stringers must be a minimum 10 mm thick and a minimum 50 mm wide. The rungs must be a minimum 20 mm rod on 300 mm centers. Members of the ladder must be securely affixed. Ladder must be of sufficient length to extend from the bottom of the tank to the top surface of the tank. Ladder must be rigidly connected to the tank bottom in accordance with the tank manufacturer's standard. Ladder must be connected to the top of the tank with pipe guides or slip bars to accommodate expansion of the two stringers.

## 2.6.6 Tank Venting

### 2.6.6.1 Atmospheric Vent

Provide atmospheric, updraft type cap. Cap must be constructed of aluminum or carbon steel. Cap must have an internal brass or bronze insect screen, minimum 40-mesh. Cap must prevent rain, snow, or ice from entering the vent piping.

### 2.6.6.2 Pressure Vent

Tank vent outlet must be equipped with pressure-vacuum breather vent, aluminum construction with weather hood and with fluoroelastomer (FKM, Viton) pallet seat inserts, high density screens, stainless steel internals, with pressure relief setting, vacuum relief setting, and venting capacity per tank manufacturer.

### 2.6.6.3 Emergency Vent

Vent must be the normally-closed, UL listed type that vents outward and upward. Vent must conform with NFPA 30 and UL 142 and must be sized by the tank manufacturer. Provide vent with the Liters per second (L/s) rating permanently labeled on the vent's exterior. For double wall or protected type tanks, provide a second emergency vent to protect the interstitial space.

## 2.7 INDEPENDENT LEVEL ALARM SYSTEM

Provide an independent level alarm system that will monitor 4 programmable liquid level setpoints. The system must delineate between each individual setpoint as well as each individual tank. The system must produce an audible and visible alarm in the event of monitoring an alarm condition. Mechanically-actuated float assemblies must be field adjustable. The system must be totally independent of the tank gauging system.

### 2.7.1 Setpoints

Configure the alarm system's 4 setpoints in accordance with the following:

- a. High Level Setpoint. Produce an alarm condition when a tank's liquid level rises above 90 percent capacity.
- b. High-High Level Setpoint. Produce an alarm condition when a tank's liquid level rises above 95 percent capacity.
- c. Low Level Setpoint. Produce an alarm condition when a tank's liquid level drops below 15 percent capacity.
- d. Low-Low Level Setpoint. Produce an alarm condition when a tank's liquid level drops below 5 percent capacity.

### 2.7.2 Independent Level Alarm Control Panel

Install the control panel for the alarm system in a standard industrial enclosure. Panel doors must swing left or right.

#### 2.7.2.1 Audible Alarm

Panel must have external speakers that produce a buzzer sound of 70 decibels or greater in the event of a detected alarm condition.

#### 2.7.2.2 Visual Alarm

Panel must have a visual alarm that illuminates in the event of a detected alarm condition. The visual alarm must include either individual lights for each alarm condition or must include a single light and a liquid crystal display (LCD) panel that displaces information regarding each alarm condition.

#### 2.7.2.3 Acknowledge Switch

Panel must have a manual acknowledge switch that will deactivate the audible alarm. The acknowledge switch must not deactivate subsequent audible alarms unless depressed manually again for each occurrence. Under no circumstance must this acknowledgement switch extinguish the visual alarms until the alarm condition has been corrected. The acknowledge switch must be an integral component located on the front of the control panel. The switch must be either a key switch or push button.

#### 2.7.2.4 Test Pushbutton

Panel must have a manual test pushbutton that will enable operators to verify that the panel is powered, and the visual and audible alarms are working properly.

### 2.8 TANK GAUGES

#### 2.8.1 Stick Gauge

For each tank, provide 2 wooden stick gauges. Gauge length must allow the measurement of the entire level of fuel in the corresponding tank. Gauges must be compatible with the fuel to be measured (no swelling or damage from fuel contact). Provide gauge with non-sparking caps on each end. Mark gauges in m and mm. The smallest unit of measure on the gauge must be 1 mm.

#### 2.8.2 Tank Strapping Table

Provide 2 certified strapping tables (calibration charts) for each tank. One of the tables must indicate the liquid contents in L for each 1 mm of tank depth. Strapping table volumes for all tanks 19,000 L and larger must be determined using physical measurements and not calculated values. For each tank, provide an electronic media file of each strapping table.

#### 2.8.3 Mechanical Clock Gauge

Gauge must be the level sensing, mechanically actuated type that provides the tank level readout in a sealed glass cap contained in a gauge box. Gauge must be accurate to plus or minus 6 mm and must measure the liquid level over the full range of a tank's height. Gauge must have vapor tight seals to prevent condensation from fogging the viewing glass.

## 2.9 ACCESSORIES

### 2.9.1 Concrete Anchor Bolts

Concrete anchors must be in accordance with ASTM F1554, hot-dipped galvanized.

### 2.9.2 Bolts and Studs

Carbon steel bolts and studs must be in accordance with ASTM A307, Grade B, hot-dipped galvanized. Stainless steel bolts and studs that be in accordance with ASTM A193/A193M, Grade 8.

### 2.9.3 Nuts

Carbon steel nuts must be in accordance with ASTM A563, Grade A, hex style, hot-dipped galvanized. Stainless steel nuts must be in accordance with ASTM A194/A194M, Grade 8.

### 2.9.4 Washers

Provide flat circular washers under each bolt head and each nut. Washer materials must be the same as the connecting bolt and nut. For ASTM F1554 concrete anchors, use ASTM F436, Type 1, hot-dipped galvanized washers. For ASTM A307 bolts and studs, carbon steel washers must be in accordance with ASTM F844, hot-dipped galvanized. Stainless steel washers must be in accordance with ASTM A194/A194M, Grade 8.

### 2.9.5 Polytetrafluoroethylene (PTFE) Tape

Tape must be in accordance with ASTM D3308.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Handle storage tanks with extreme care to prevent damage during placement and install in accordance with the manufacturer's installation instructions and NFPA 30, NFPA 30A, or NFPA 31 as applicable. Inspect the exterior surface of each tank for obvious visual damage prior to and during the placement of each storage tank. Repair surface damage to a storage tank according to manufacturer's requirements before proceeding with the system installation. Provide the termination of fill lines within a tank with an antisplash deflector. Provide nylon dielectric bushings on pipe connections to a steel tank.

Where anchors are required (either aboveground or underground), provide submittals for tank anchorage system materials and tank anchorage calculations.

#### 3.1.1 Aboveground Storage Tank

Install aboveground storage tanks in accordance with STI 700-50-5007 (STI R912) except as modified herein. Place tank that is equal to or greater than 18,900 L on a 3 mm per 30 mm slope with the fill point at the low end and the vent connection and issue pump at the high end. Place tank that is less than 18,900 L on a level surface.

#### 3.1.1.1 Steel Aboveground Storage Tank Handling

Store, handle, and place externally coated steel tanks with care and in a manner that will minimize damage to the coating and will not reduce its protective value. Place coated tanks in position carefully and with a minimum amount of handling. Repair any damaged tank coating in accordance with the appropriate UL or STI standard (UL 1746, STI 020-50-1000, or UL 58 ). Do not move the tank unless it is empty.

#### 3.1.1.2 Steel Aboveground Tank Installation Procedures

Tanks should be secured to the associated tank pad per tank manufacturer's recommendations using fasteners installed through the tank saddle base plate.

#### 3.1.2 System Components

Properly level, align, and secure system components in place in accordance with manufacturer's instructions. Provide supports for system components, appurtenances, and pipe as required. Install anchors, bolts, nuts, washers, and screws where required for securing the work in place. Sizes, types, and spacings of anchors and bolts not indicated or specified must be as required for proper installation.

### 3.2 FIELD QUALITY CONTROL

#### 3.2.1 Aboveground Storage Tank Tightness Tests

Perform tightness tests on each aboveground storage tank prior to making piping connections. Perform testing in accordance with STI 700-50-5007 (STI R912) except as modified herein. Gauges used to monitor the tests must have a scale with a maximum limit of 103 kPa. Repair leaks discovered during the tightness tests in accordance with tank manufacturer's instructions. Following any repair, re-test the tank until the tank successfully passes the testing requirements of this paragraph.

#### 3.2.2 Tank Manufacturer's Tests

In addition to the tests required herein, perform any additional tests (i.e., leak tests, cathodic protection verification tests, etc.) on each storage test that is required by the tank manufacturer's written test procedures. Manufacturer's tests that are redundant to tests already required by this specification will only be performed once per tank. Repair all leaks discovered during the tests in accordance with manufacturer's instructions. Following tank repairs, re-test the tank until the tank successfully passes the manufacturer's testing requirements.

#### 3.2.3 Tank Inspection Reports

Prior to system commissioning, a STI SP001 certified inspector must perform a Formal External Inspection on the completed aboveground tank in accordance with STI SP001 and deliver a full report to the Contracting Officer. The report must include a record of ultrasonic thickness measurements (UTMs), exclusive of the coating, of each single wall aboveground tank shell. The report must include the tank dataplate information and photograph of the tank data plate. Provide electronic copies of the tank inspection reports to Service Headquarters, Service Control Points, and DLA-Energy. The paper and electronic copies of the report and UTMs must be provided to the Contracting Officer for filing

with the tank's "as-built drawings." Refer to Section 01 45 00 QUALITY CONTROL for STI SP001 Inspector's Certification requirements.

### 3.3 DEMONSTRATIONS

Conduct a training session for designated Government personnel in the operation and maintenance procedures related to the system components and systems specified herein. Include pertinent safety operational procedures in the session as well as physical demonstrations of the routine maintenance operations. Provide instructors who are familiar with the system components and systems, both operational and practical theories, and associated routine maintenance procedures. The training session must consist of a total of hours of normal working time and must start after the system is functionally completed, but prior to final system acceptance. Submit a letter, at least 14 working days prior to the proposed training date, scheduling a proposed date for conducting the onsite training.

### 3.4 Tank Fill Tests

Tank fill tests must not be performed until after the flushing, cleaning, and adjusting requirements in accordance with Section 33 08 53 AVIATION FUEL DISTRIBUTION SYSTEM START-UP. For the tank fill tests, initially fill each storage tank with fuel in order to verify the tank level alarm system operates properly and the tank overfill protection device functions as designed. Stop filling each tank immediately once the overfill device operates. Do not overfill any storage tank more than the 98 percent level. Drain the system below the low liquid level setpoint to verify operation of the low level alarm. Correct and retest any problems with the level alarm system or the overfill device until each operate as specified herein. During the tests, verify that all tank gauges are calibrated and operating appropriately.

### 3.5 FIELD PAINTING

Painting required for surfaces not otherwise specified must be field painted in accordance with Section 09 90 00 PAINTING, GENERAL. Do not paint stainless steel and aluminum surfaces. Do not coat system components or components provided with a complete factory coating. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

-- End of Section --



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## SECTION 33 57 55

## FUEL SYSTEM COMPONENTS (NON-HYDRANT)

**11/18, CHG 1: 11/20**

## PART 1 GENERAL

## 1.1 SUMMARY

This section defines the requirements for system components as related to a non-hydrant fuel distribution system. Provide the entire fuel distribution system as a complete and fully operational system. Size, select, construct, and install equipment and system components to operate together as a complete system. Substitutions of functions specified herein will not be acceptable. Coordinate the work of the system manufacturer's service personnel during construction, testing, calibration, and acceptance of the system. System components and piping specified herein must be designed to handle a working pressure of 1965 kPa for carbon steel systems at 38 deg C. Components specified herein must be compatible with the fuel to be handled. Components to be suitable for outside, unsheltered location, and to function normally in ambient temperatures between -15 degrees F and 30 degrees F.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN PETROLEUM INSTITUTE (API)

API RP 540 (1999; R 2004) Electrical Installations in Petroleum Processing Plants

API RP 2003 (2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B40.100 (2022) Pressure Gauges and Gauge Attachments

## ASTM INTERNATIONAL (ASTM)

ASTM C827/C827M (2023) Standard Test Method for Change in Height at Early Ages of Cylindrical Specimens of Cementitious Mixtures

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book

IEEE 1100 (2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                      |
|----------|----------------------------------------------------------------------|
| NFPA 70  | (2023; ERTA 1 2024; TIA 24-1) National Electrical Code               |
| NFPA 77  | (2024; ERTA 1 2023) Recommended Practice on Static Electricity       |
| NFPA 407 | (2022; TIA 24-2) Standard for Aircraft Fuel Servicing                |
| NFPA 780 | (2023) Standard for the Installation of Lightning Protection Systems |

## SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

|             |                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------|
| SAE AMS3275 | (2009; Rev C) Sheet, Acrylonitrile Butadiene (NBR) Rubber and Non-Asbestos Fiber Fuel and Oil Resistant |
|-------------|---------------------------------------------------------------------------------------------------------|

## U.S. DEPARTMENT OF DEFENSE (DOD)

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| MIL-STD-130 | (2007; Rev N; Change 1 2012) Identification Marking of U.S. Military Property                                               |
| MIL-STD-161 | (2005; Rev G; Notice 1 2010) Identification Methods for Bulk Petroleum Products Systems Including Hydrocarbon Missile Fuels |

## U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|               |                                                                                |
|---------------|--------------------------------------------------------------------------------|
| CID A-A-50696 | (2016; Rev D) Reels, Static Discharge, Grounding, 50 and 75 Foot Cable Lengths |
|---------------|--------------------------------------------------------------------------------|

## 1.3 ADMINISTRATIVE REQUIREMENTS

Submit detail drawings consisting of illustrations, schedules, performance charts, instructions, brochures, diagrams, and other information to illustrate the requirements and operation of the system components and systems. Provide the drawings as one package with the design analysis. Shop fabrication drawings must include type of material, configuration, thickness, and necessary details of construction of the steel tank and vault. Shop drawings must also show the steel grating and supports. Submit Manufacturer's Catalog Data and Certificates of Compliance. Operation and maintenance information must be submitted for the system components items or systems listed in PART 2. Automatic pump controls must include step-by-step procedures required for system startup, operation, and shutdown.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-01 Preconstruction Submittals

Automated And Manual Data Collection Output Examples; G

## SD-02 Shop Drawings

Grounding and Bonding; G

## SD-03 Product Data

Pressure Gages; G

Differential Pressure Gauge; G

Automatic Fill Port; G

Automatic Fuel Polisher; G

## SD-07 Certificates

System Supplier; G

## SD-10 Operation and Maintenance Data

## 1.5 QUALITY ASSURANCE

Submit the following data for approval:

- a. Certification stating that the System Supplier has provided and installed at least five Programmable Logic Control (PLC)-based pump control systems in the last five years, for automatic cycling of pumps based upon varying dispensing demands, utilizing multiple pumps. These systems must be for dispensing diesel.
- b. Certification that six systems have been successfully operated over the last three years and are currently in service.
- c. Project names, locations, system description, and items provided at these installations. Include user point-of-contact and current telephone numbers.

## 1.5.1 Material and Equipment Qualifications

Provide materials and system components that are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship. Materials and system components must have been in satisfactory commercial or industrial use for a minimum two years prior to bid opening. The two year period must include applications of the system components and materials under similar circumstances and of similar size. Materials and system components must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the two year period.

### 1.5.2 Nameplates

Attach nameplates to all specified system components, thermometers, gauges, and valves defined herein. List on each nameplate the manufacturer's name, address, contract number, component type or style, model or serial number, catalog number, capacity or size, and the system that is controlled. Construct plates of melamine plastic, 3 mm thick, UV resistant, black with white center core, matte finish surface and square corners. Install nameplates in prominent locations with nonferrous screws, nonferrous bolts, or permanent adhesive. Minimum size of nameplates must be 25 by 65 mm. Lettering must be the normal block style with a minimum 6 mm height. Accurately align all lettering on nameplates.

For plastic nameplates, engrave lettering into the white core.

### 1.6 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect system components and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

## PART 2 PRODUCTS

If gasoline is being handled, refer to 40 CFR Part 60 Subpart Kb and XX, 40 CFR Part 63 Subpart R, BBBB, and CCCCC for design, installation, and testing requirements.

### 2.1 MATERIALS

Materials of construction must be stainless steel, aluminum or nonferrous material except positive displacement meter case may be steel with electroless nickel plated internals coated to 0.075 mm thickness, or interior epoxy coating. No ferrous or zinc-coated material bronze, brass or other copper bearing alloys must be used in contact with the fuel. Do not install cast iron bodied valves or system components. Do not use aluminum valves.

#### 2.1.1 Types of Fuel

Components must be suitable for use with diesel. Components to be ANSI Class 150 (1920 kPa at 38 degrees C) unless noted otherwise. Components to be suitable for outside, unsheltered location, and to function normally in ambient temperatures between -15 degrees C and 30 degrees C.

#### 2.1.2 Composition of Materials

Materials in contact with the fuel must be noncorrosive. No zinc-coated metals, brass, bronze, iron, lead or lead alloys, copper or copper alloys, or other light metal alloys containing more than 4 percent copper must be used in contact with the fuel.

#### 2.1.3 Gaskets

Gaskets must be in accordance with Section 33 52 10 FUEL SYSTEMS PIPING (SERVICE STATION).

#### 2.1.4 Bolts and Nuts

Bolts and nuts must be in accordance with Section 33 52 10 FUEL SYSTEMS

## PIPING (SERVICE STATION).

## 2.1.5 Flanges

Flanges and flanged end system components must be in accordance with Section 33 52 10 FUEL SYSTEMS PIPING (SERVICE STATION).

## 2.1.6 Nitrile Butadiene (Buna-N)

Provide Buna-N material that conforms to SAE AMS3275.

## 2.1.7 Acrylonitrile Butadiene Rubber (NBR)

Provide NBR material that conforms to SAE AMS3275.

## 2.2 SYSTEM COMPONENTS AND MATERIAL

## 2.2.1 General

All items of system components and material must be new and of the best quality used for the purpose in commercial practice and must be products of reputable manufacturers. Each major component of the system components must have the manufacturer's name, address and catalog number on a plate securely affixed in a conspicuous place. The nameplate of a distributing agent only will not be acceptable. The gears, couplings, projecting set screws, keys and other rotating parts located so that any person may come in close proximity thereto must be fully enclosed or properly guarded. System Components, assemblies and parts must be marked for identification in accordance with MIL-STD-130 and MIL-STD-161. Pump and filter vessel numbers must be as indicated on the drawings. In addition, filter vessels must include element numbers and the date of the next element change. Identification tags made of brass, stainless steel, or engraved anodized aluminum, indicating valve number and normally open (NO) or normally closed (NC) must be installed on valves. Tags must be 35 mm minimum diameter, and marking must be stamped or engraved. Indentations must be black, for reading clarity. Tags must be attached to valves with No 12 AWG, copper wire, stainless or aluminum hanging wires, or chrome-plated beaded chain designed for that purpose.

## 2.2.2 System Supplier

Since the pump control system, including but not limited to pump control panel, , transmitters, flow switches, fueling system pumps, all field instrumentation, and control valves with all hardware and software, is an integrated system it must be furnished by a single systems supplier regularly engaged in the supplying of these system components. System Supplier must be a company whose regular, normal, and primary business is representing manufacturers in the distribution and start-up of aviation fueling facilities, and have no affiliation with the Contractor other than as a seller to the Contractor. Supplier must provide all system components and appurtenances regardless of manufacture, be a factory authorized certified representative, and be responsible to the Contractor for satisfactory operation of the entire system, and must oversee the installation of the system components. Substitutions of functions specified will not be acceptable. The Contractor and the System Supplier must be present at the system commissioning, and must coordinate and schedule the work during construction, testing, calibration, and acceptance of the system. The System Supplier must be on-site with their mechanical and control personnel to supervise and assist the contractor

during pre-commissioning check-out of the mechanical systems and control systems, initial fuel receipt, initial filing, hydrostatic testing, pigging, flushing, cleaning, system component tests, performance testing and all training for the owner's representatives. The System Supplier must be responsible to the Contractor for scheduling all Contractor, Sub-contractor, and manufacturer's service personnel during system start-up and final commissioning.

## 2.3 ELECTRICAL

Motors, manual or automatic motor control system components except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

### 2.3.1 Grounding and Bonding

Ground and bond in accordance with NFPA 70, NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

## 2.4 PRESSURE GAGES

Pressure gages must conform to ASME B40.100 with metal cases and 100 mm diameter white dials. Gages must be bottom connected, without back flanges. A pulsation dampener, adjustable to the degree of dampening required, must be provided for each gage. Range of gages must be as indicated. A ball valve must be provided for each pressure gage. Gages must have all parts immersed in silicone oil. Gages must be labeled with the calibration date.

### 2.4.1 Quick Disconnect

If indicated on drawings provide quick disconnect on pressure gauge. Quick disconnects for pressure gauges must be double shut-off, dry-break design, 316 stainless steel construction, with Fluorocarbon (Viton) seals, minimum working pressure of 6.89 mPa at 38 degrees C, with 12.7 mm female NPT threaded connections for both coupler and adapter, manufactured in accordance with ISO 7241, Series B. The quick disconnect assembly must consist of a coupler, half to be connected to the pressure snubber under the pressure gauge, and a nipple/adapter half to be connected above the pressure gauge isolation ball valve. The nipple/adapter is to be provided with an aluminum dust cap to protect the fitting when the gauge is removed.

## 2.5 DIFFERENTIAL PRESSURE GAUGE

The gauge must consist of a spring-supported, corrosion resistant piston moving inside a glass cylinder, with high pressure applied on top of the piston and low pressure applied below it. Under a differential pressure of 206.8 kPa, leakage past the piston must not exceed 120 drops per minute. The cylinder and flanges must be stainless steel with Viton O-ring seals. The high pressure inlet of the gauge must have a 10-micron pleated paper filter and the low pressure connection must have a fine mesh stainless steel strainer. The gauge must have an operating pressure of 2068 kPa. Differential pressure range of the gauge through approximately 75 mm of

piston movement must be 0-2068 kPa with an accuracy of plus 34.5 kPa, calibrated linearly with one PSI scale graduations. High and low pressure connections must be 1/4-inch NPT female with a stainless steel bar stock valve at each connection. Construction of the gauge must be such that a 3-valve manifold is not necessary. If only one bar stock valve is closed, the gauge must not be damaged by up to 206.8 kPa differential pressure in either direction. A pressure gauge must be attached to the differential pressure gauge to indicate the high pressure and have a range of 2068 kPa.

## 2.6 GROUNDING CABLE AND CLAMP

Grounding system must conform to CID A-A-50696, Type II.

## 2.7 AUTOMATIC FILL PORT

Provide a stand-alone automatic fill port packaged with a tank gauge system. Fill port must automatically prevent the overflow of the receiving tank and automatically cutoff fuel flow in event of fire. Provide quick disconnect hose coupling with dust plug, hand pump for spill containment with cutoff and check valve, sump with drain, and stainless steel grounding lug. Construct system components to be chemically compatible with fuel to be handled.

Tank gauge system must be the mechanically or electronically actuated type that can continuously monitor a tank's usable liquid level storage capacity. The system must provide a digital readout of the tank's usable liquid level in terms of mm and L. The system must be accurate to plus or minus 2 mm. Panels must be a NEMA 4X enclosure as defined by NEMA 250 and shall comply with CEI EN 60079 series. Panel doors must swing left or right and be key or padlock lockable. The panel must also have external controls to allow operators to toggle between information on the readout without having to open the panel Display and input control must be accessible while fill port is in operation. Controls must be able to receive an external alarm signal from the electronic monitoring/alarm panel in Sections 33 01 50.31 LEAK DETECTION FOR FUELING SYSTEM.

### 2.7.1 Indication

Indication can be dedicated LEDs or via panel display unless specifically indicated. Provide indication as follows.

- a. Dedicated System Power LED.
- b. Tank Full with visual alarm LED.
- c. High Level with audible and visual alarm LED.
- d. Tank Level.
- e. Tank leak with audible and visual alarm LED.
- f. Liquid in Sump.
- G. Leak detection in alarm.

### 2.7.2 Input

Provide user input as follows.

- a. Dedicated System On-Off.
- b. Fill Port Valve Open / Close.
- c. Momentary audible alarm silence.



## 2.8 AUTOMATIC FUEL POLISHER

Provide a packaged stand-alone automatic fuel polisher with a particle filtration stage and a fuel/water separation stage. Particle filtration stage must remove particles greater than 4 micron and be able to maintain UNI ISO 4406 18/16/13 fuel particle characterization. System must be able to maintain water and sediment of less than 0.05 percent by volume.

Automation controls must have a cycle timer to alternate polishing and tank settling durations. Cycle timer must be adjustable by hours for all cycles up to 24 hours per cycle. Provide with a positive displacement pump capable of polishing fuel at a minimum of 20 liters per minute installed or as designated on the contract drawings. Provide motor controller in accordance with NEMA ICS 2, general purpose, class A, with enclosure.

Provide a single tank mount assembly for fuel pickup, supply and return. Fuel pickup nozzle must be connected to the mount assembly via steel pipe and rest on the striker plate below the tank flange without impeding flow. Construct system components to be chemically compatible with the fuel to be handled. Panel must be a NEMA 4X enclosure as defined by NEMA 250 and shall comply to CEI EN 60079 series. Panel doors must swing left or right and be key or padlock lockable. Controls must be able to receive an external alarm signal from the electronic monitoring/ alarm panel in Section 33 01 50.31 LEAK DETECTION FOR FUELING SYSTEM.

### 2.8.1 Indication

Indication can be dedicated LEDs or via panel display unless specifically indicated. Provide indication as follows.

- a. Dedicated system power LED/
- b. Pump power.
- c. fault.
- d. Fuel/water filter maintenance. Must be initiated by pressure and/or vacuum sensing and not by timer.
- e. Particle filter maintenance. Must be initiated by pressure and or vacuum sensing and not by timer.
- f. Liquid in sump.
- g. Water collector full.
- h. No Fuel Flow. Must be initiated by a flow switch.
- i. Leak detection in alarm

### 2.8.2 Input

Provide user input as follows.

- a. Dedicated System On-Off.
- b. Momentary Pump On.
- c. Momentary System Reset.
- d. Pump On Time.
- e. Pump Off Time.
- f. Emergency Stop.

## PART 3 EXECUTION

### 3.1 GENERAL

#### 3.1.1 Installation

Install equipment and components in position, true to line, level and plumb, and measured from established benchmarks or reference points. Follow manufacturer's recommended practices for system components installation. Provide required clearances between equipment components, system components, apparatus, and accessories requiring normal servicing or maintenance to be accessible.

#### 3.1.2 Anchoring

Anchor system components in place. Check alignment of anchor bolts before installing system components and clean-out associated sleeves. Do not cut bolts because of misalignment. Notify Contracting Officer of errors and obtain the Contracting Officer's acceptance before proceeding with corrections. Cut anchor bolts of excess length to the appropriate length without damage to threads. Where anchor bolts or like devices have not been installed, provide appropriate self-drilling type anchors for construction condition.

#### 3.1.3 Grouting

System components which are anchored to a pad is to be grouted in place. Before setting system components in place and before placing grout, clean surfaces to be in contact with grout, including fasteners and sleeves. Remove standing water, debris, oil, rust, and coatings which impair bond. Clean contaminated concrete by grinding. Clean metal surfaces of mill scale and rust by hand or power tool methods. Provide necessary formwork for placing and retaining grout. Grout to be non-metallic, non-shrink, fluid precision grout of a hydraulic cementitious system with graded and processed silica aggregate, Portland cement, shrinkage compensating agents, plasticizing and water reducing agents; free of aluminum powder agents, oxidizing agents and inorganic accelerators, including chlorides; proportioned, pre-mixed and packaged at factory with only the addition of water required at the project site. Grouting must be in accordance with ASTM C827/C827M. Perform all grouting in accordance with system components manufacturer's and grout manufacturer's published specifications and recommendations.

#### 3.1.4 Leveling and Aligning

Level and align system components in accordance with respective manufacturer's published data. Do not use anchor bolt, jack-nuts or wedges to support, level or align system components. Install only flat shims for leveling system components. Place shims to fully support system components. Wedging is not permitted. Shims to be fabricated flat carbon steel units of surface configuration and area not less than system components bearing surface. Shims to provide for full system components support. Shim to have smooth surfaces and edges, free from burrs and slivers. Flame or electrode cut edges not acceptable.

#### 3.1.5 Precautions

Special care must be taken to ensure that system components and materials are stored properly to prevent damage and maintain cleanliness, and that

the completed system is free of rocks, sand, dirt, and foreign objects. Take the following steps to insure these conditions.

- a. System components brought to the site and not stored inside, must be stored on blocks or horses at least 450 mm above ground.
- b. Visual inspection must be made of each piece of system components to ensure that it is clean prior to installation.
- c. The open ends of system components must be closed when work with that piece of system components is not in progress.

### 3.2 POSTED OPERATING INSTRUCTIONS

For each designated system or system components item, provide instructions for guidance of operating and maintenance personnel. Following approval of content, prepare these instructions in a form and scale that will be readily legible when displayed in appropriate locations, to be designated by the Contracting Officer and meet the following requirements:

#### 3.2.1 Each System

For each system, include diagrams of system components, piping, wiring and control. Define control sequences.

#### 3.2.2 Each Tank

For each tank provide a P.E. stamped certified tank calibration chart in 1/16-inch increments reading in gallons, except for tanks less than 5,000 gallons.

#### 3.2.3 Each Item

For each system components item, include starting, adjustment, operation, lubrication, safety precautions and shut-down procedures. Identify procedures to be performed in event of system components failure. Provide other instructions recommended by the manufacturer.

#### 3.2.4 Diagrams

Provide a professionally prepared isometric piping diagram of the fueling system apparatus. Diagram must be 914 by 1370 mm and must be color coded to match PCP color diagrams. Diagram must show the entire facility and must include all system components and the operational sequences of all system components with equipment numbers displayed. Diagram must show all valves along with the valve numbers shown on the drawings and listed as normally open/closed. It must be wall mounted under glass.

#### 3.2.5 Volume of Fuel

Provide a certified system inventory of fuel in the pipe, tank, pumphouse, etc. The piping will show length of pipe, size of pipe, L/s, and total L. Verify during initial fill.

### 3.3 DEMONSTRATIONS

Conduct a training session for designated Government personnel in the operation and maintenance procedures related to the system components/systems specified herein. Include pertinent safety operational

procedures in the session as well as physical demonstrations of the routine maintenance operations. Furnish instructors who are familiar with the installation/system components/systems, both operational and practical theories, and associated routine maintenance procedures. The training session must consist of a total of 4 hours of normal working time and must start after the system is functionally completed, but prior to final system acceptance. Submit a letter, at least 14 working days prior to the proposed training date, scheduling a proposed date for conducting the on-site training.

-- End of Section --

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## SECTION 33 71 01

## OVERHEAD TRANSMISSION AND DISTRIBUTION

**05/19, CHG 1: 11/19**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.11 (2022) Forged Fittings, Socket-Welding and Threaded

## AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA C25 (2003) Sawn Crossarms - Preservative Treatment by Pressure Processes

## ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A475 (2022) Standard Specification for Metallic-Coated Steel Wire Strand

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B2 (2013) Standard Specification for Medium-Hard-Drawn Copper Wire

ASTM B3 (2013; R 2024) Standard Specification for Soft or Annealed Copper Wire

ASTM B8 (2023) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM B230/B230M (2022) Standard Specification for Aluminum 1350-H19 Wire for Electrical Purposes

|                 |                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| ASTM B231/B231M | (2023) Standard Specification for Concentric-Lay-Stranded Aluminum 1350 Conductors                                            |
| ASTM B398/B398M | (2022) Standard Specification for Aluminum-Alloy 6201-T81 Wire for Electrical Purposes                                        |
| ASTM B399/B399M | (2023) Standard Specification for Concentric-Lay-Stranded Aluminum-Alloy 6201-T81 Conductors                                  |
| ASTM D1654      | (2008; R 2016; E 2017) Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments |

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

|             |                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------|
| IEEE 100    | (2000; Archived) The Authoritative Dictionary of IEEE Standards Terms                                 |
| IEEE 404    | (2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V |
| IEEE C2     | (2023) National Electrical Safety Code                                                                |
| IEEE C37.42 | (2016) Specifications for High-Voltage (> 1000 V) Fuses and Accessories                               |
| IEEE C62.11 | (2020) Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1kV)         |

## INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

|          |                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------|
| NETA ATS | (2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems |
|----------|--------------------------------------------------------------------------------------------------|

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|                          |                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------|
| ANSI C29.3               | (2015; R 2022) American National Standard for Wet Process Porcelain Insulators - Spool Type                |
| ANSI C29.5               | (2022) Wet-Process Porcelain Insulators (Low and Medium Voltage Pin Type)                                  |
| NEMA WC 74/ICEA S-93-639 | (2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy        |
| NEMA/ANSI C29.7          | (1996; 2002) American National Standard for Wet Process Porcelain Insulators - High-Voltage Line Post Type |

## U.S. DEPARTMENT OF AGRICULTURE (USDA)

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| RUS 202-1          | (2004) List of Materials Acceptable for Use on Systems of RUS Electrification Borrowers |
| RUS Bull 1728H-701 | (1993) Wood Crossarms (Solid and Laminated), Transmission Timbers and Pole Keys         |

## UNDERWRITERS LABORATORIES (UL)

|              |                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------|
| UL 6         | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel                                  |
| UL 467       | (2022) UL Standard for Safety Grounding and Bonding Equipment                                       |
| UL 486A-486B | (2018; Reprint Jul 2023) UL Standard for Safety Wire Connectors                                     |
| UL 510       | (2020; Dec 2022) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape |

## 1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section with additions and modifications specified herein.

## 1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, must be as defined in IEEE 100.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-03 Product Data

Conductors

Insulators

Cutouts

Surge Arresters

Guy Strand

Anchors

## SD-05 Design Data



## Power-Installed Screw Foundations

### SD-06 Test Reports

Wood Crossarm Inspection Report

Field Test Plan

Field Quality Control

Ground Resistance Test Reports

Medium-Voltage Preassembled Cable Test

Sag and Tension Test

Low-Voltage Cable Test

Acceptance Checks and Tests

### SD-07 Certificates

Wood Crossarms; G

### SD-09 Manufacturer's Field Reports

Operation and Maintenance Manuals

### SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals, Data Package 5

## 1.5 QUALITY ASSURANCE

### 1.5.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory except of the local governing code when more stringent requirements are specified or indicated, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of the local governing code and IEEE C2 unless more stringent requirements are specified or indicated.

### 1.5.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2-years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

#### 1.5.2.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

#### 1.5.2.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site must not be used, unless specified otherwise.

#### 1.5.3 Ground Resistance Test Reports

Submit the measured ground resistance of grounding system. When testing grounding electrodes and grounding systems, identify each grounding electrode and each grounding system for testing. Include the test method and test setup (i.e. pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

#### 1.5.4 Wood Crossarm Inspection Report

Furnish an inspection report from an independent inspection agency, approved by the Contracting Officer, stating that offered products comply with applicable AWP and RUS standards. The RUS approved Quality Mark "WQC" on each crossarm will be accepted, in lieu of inspection reports, as evidence of compliance with applicable AWP treatment standards.

##### 1.5.4.1 Field Test Plan

Provide a proposed field test plan 20 days prior to testing the installed system. No field test must be performed until the test plan is approved. The test plan must consist of complete field test procedures including tests to be performed, test equipment required, and tolerance limits.

#### 1.6 OPERATIONS AND MAINTENANCE DATA

Provide operation and maintenance manuals for systems in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provides basic data relating to the design, operation, and maintenance of the electrical distribution system.

##### 1.6.1 Additions to Operations and Maintenance Data

In addition to requirements of Data Package 5, include the following in the operation and maintenance manuals provided:

- a. Assembly and installation drawings
- b. Prices for spare parts and supply list
- c. Date of purchase

#### 1.7 DELIVERY, STORAGE, AND HANDLING

Devices and equipment must be visually inspected by the Contractor when received and prior to acceptance from conveyance. Protect stored items from the environment in accordance with the manufacturer's published

instructions. Replace damaged items.

## 1.8 WARRANTY

The equipment items must be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND EQUIPMENT

Consider materials specified herein or shown on contract drawings which are identical to materials listed in RUS 202-1 as conforming to requirements. Provide equipment and component items, not hot-dip galvanized or porcelain enamel finished, with corrosion-resistant finishes which must withstand 480 hours of exposure to the salt spray test specified in ASTM B117 without loss of paint or release of adhesion of the paint primer coat to the metal surface in excess of 1.6 mm from the test mark. Provide the described test mark and test evaluation in accordance with ASTM D1654 with a rating of not less than 7 in accordance with TABLE 1, (procedure A). Coat cut edges or otherwise damaged surfaces of hot-dip galvanized sheet steel or mill galvanized sheet steel with a zinc rich paint conforming to the manufacturer's standard.

### 2.2 CROSSARMS AND BRACKETS

#### 2.2.1 Wood Crossarms

Conform to RUS Bull 1728H-701. Pressure treat crossarms with pentachlorophenol, chromated copper arsenate (CCA), or ammoniacal copper arsenate (ACA). Treatment must conform to AWWA C25. Crossarms must be solid wood, distribution type, and a 6.4 mm 45 degree chamfer on all top edges. Cross-sectional area minimum dimensions must be 108.0 mm in height by 82.6 mm in depth in accordance with IEEE C2 for Grade B construction. Crossarms must be 2.4 m in length, except that 3.1 m crossarms must be used for crossarm-mounted banked single-phase transformers or elsewhere as indicated. Crossarms must be machined, chamfered, trimmed, and bored for stud and bolt holes before pressure treatment. Factory drilling must be provided for pole and brace mounting, for four pin or four vertical line-post insulators, and for four suspension insulators, except where otherwise indicated or required. Drilling must provide required climbing space and wire clearances. Crossarms must be straight and free of twists to within 2.5 mm per 304.8 mm of length. Bend or twist must be in one direction only.

#### 2.2.2 Crossarm Braces

Provide flat steel as indicated. Provide braces with 965 mm span with 2440 mm crossarms.

### 2.3 HARDWARE

Hardware must be hot-dip galvanized in accordance with ASTM A153/A153M and ASTM A123/A123M.

## 2.4 INSULATORS

Provide wet-process porcelain insulators which are radio interference free.

- a. Line post type insulators: NEMA/ANSI C29.7, Class 57-1L.
- c. Spool insulators: ANSI C29.3, Class 57-1L.
- e. Pin insulators: ANSI C29.5, Class 55-3.

## 2.5 OVERHEAD CONDUCTORS, CONNECTORS AND SPLICES

Conductors of bare copper, aluminum (AAC) or aluminum alloy (AAAC) of sizes and types indicated to match existing conditions.

### 2.5.1 Solid Copper

ASTM B1, ASTM B2, and ASTM B3, hard-drawn, medium-hard-drawn, and soft-drawn, respectively. ASTM B8, stranded.

### 2.5.2 Aluminum (AAC)

ASTM B230/B230M and ASTM B231/B231M.

### 2.5.3 Aluminum Alloy (AAAC)

ASTM B398/B398M or ASTM B399/B399M.

### 2.5.4 Connectors and Splices

Connectors and splices must be of copper alloys for copper conductors, aluminum alloys for aluminum-composition conductors, and a type designed to minimize galvanic corrosion for copper to aluminum-composition conductors. Aluminum-composition, aluminum-composition to copper, and copper-to-copper must comply with UL 486A-486B.

## 2.6 GUY STRAND

ASTM A475, high-strength, Class A or B, galvanized strand steel cable. Guy strand must be 9.5 mm in diameter with a minimum breaking strength of 68,500 Newton. Provide guy terminations designed for use with the particular strand and developing at least the ultimate breaking strength of the strand.

## 2.7 ROUND GUY MARKERS

Vinyl or PVC material, white colored, 2440 mm long and shatter resistant at sub-zero temperatures.

### 2.7.1 Guy Attachment

Thimble eye guy attachment.

## 2.8 ANCHORS AND ANCHOR RODS

Anchors must present holding area indicated on drawings as a minimum. Anchor rods must be triple thimble-eye, 19 mm diameter by 2440 mm long. Anchors and anchor rods must be hot dip galvanized.

### 2.8.1 Screw Anchors

Screw type anchors having a manufacturer's rating of not less than 10,850 Newton in loose to medium sand/clay soil, Class 6 and extra heavy pipe rods conforming to ASTM A53/A53M, Schedule 80, and couplings conforming to ASME B16.11.

## 2.9 GROUNDING AND BONDING

### 2.9.1 Driven Ground Rods

Provide cone pointed copper-clad steel ground rods conforming to UL 467 not less than 19 mm in diameter by 3.1 m in length. Sectional type rods may be used for rods 6.1 m or longer.

### 2.9.2 Grounding Conductors

ASTM B3. Provide soft drawn copper wire ground conductors a minimum No. 4 AWG. Ground wire protectors must be PVC. Keep ground conductors straight and short. Minimize bends in all ground connections.

### 2.9.3 Grounding Connections

UL 467. Exothermic weld or compression connector.

## 2.10 SURGE ARRESTERS

IEEE C62.11, metal oxide, polymeric-housed, surge arresters arranged for crossarm mounting. RMS voltage rating must be 9 kV. Arresters must be Distribution class.

## 2.11 FUSED CUTOUTS

Open type fused cutouts rated 100 amperes and 31,000 amperes symmetrical interrupting current at 15 kV ungrounded, conforming to IEEE C37.42. Type K fuses conforming to IEEE C37.42 with ampere ratings equal to 150 percent of the transformer full load rating. Open link type fuse cutouts are not acceptable.

## 2.12 CONDUIT RISERS AND CONDUCTORS

The riser shield must be PVC containing a PVC back plate and PVC extension shield or a rigid galvanized steel conduit, as indicated, and conforming to UL 6. Provide conductors and terminations as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

## 2.13 ELECTRICAL TAPES

Tapes must be UL listed for electrical insulation and other purposes in wire and cable splices. Terminations, repairs and miscellaneous purposes, electrical tapes must comply with UL 510.

## 2.14 CAULKING COMPOUND

Compound for sealing of conduit risers must be of a puttylike consistency workable with hands at temperatures as low as 2 degrees C, must not slump at a temperature of 150 degrees C, and must not harden materially when exposed to air. Compound must readily caulking or adhere to clean surfaces of the materials with which it is designed to be used. Compound

must have no injurious effects upon the workmen or upon the materials.

## 2.15 NAMEPLATES

### 2.15.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable. Equipment containing liquid-dielectrics must have the type of dielectric on the nameplate.

## 2.16 SOURCE QUALITY CONTROL

# PART 3 EXECUTION

## 3.1 INSTALLATION

Provide overhead pole line installation conforming to requirements of IEEE C2 for Grade B construction of overhead lines in light loading districts and the local governing code for overhead services. Provide material required to make connections into existing system and perform excavating, backfilling, and other incidental labor. Consider street, alleys, roads and drives "public." Pole configuration must be as indicated.

### 3.1.1 Anchors and Guys

Place anchors in line with strain. Indicate the length of the guy lead (distance from base of pole to the top of the anchor rod).

#### 3.1.1.1 Setting Anchors

Set anchors in place with anchor rod aligned with, and pointing directly at, guy attachment on the pole with the anchor rod projecting 150 to 230 mm out of ground to prevent burial of rod eye.

#### 3.1.1.2 Backfilling Near Anchors

Backfill plate, expanding, concrete, or cone type anchors with tightly tamped coarse rock 610 mm immediately above anchor and then with tightly tamped earth filling remainder of hole.

Backfill plate anchors with tightly tamped earth for full depth of hole.

#### 3.1.1.3 Screw Anchors

Install screw anchors by torquing with boring machine.

#### 3.1.1.4 Swamp Anchors

Install swamp anchors by torquing with boring machine or wrenches, adding sections of pipe as required until anchor helix is fully engaged in firm soil.

#### 3.1.1.5 Rock Anchors

Install rock anchors minimum depth 305 mm in solid rock.

### 3.1.1.6 Guy Installation

Provide guys where indicated, with loads and strengths as indicated, and wherever conductor tensions are not balanced, such as at angles, corners and dead-ends. Where single guy will not provide the required strength, provide two or more guys. Where guys are wrapped around poles, at least two guy hooks must be provided. Provide pole shims where guy tension exceeds 27,000 Newtons. Provide guy clamps 152 mm in length with three 16 mm bolts, or offset-type guy clamps, or approved guy grips at each guy terminal. Securely clamp plastic guy marker to the guy or anchor at the bottom and top of marker. Complete anchor and guy installation, dead end to dead end, and tighten guy before wire stringing and sagging is begun on that line section. Provide strain insulators at a point on guy strand 2435 mm minimum from the ground and 1825 mm minimum from the surface of pole. Effectively ground and bond guys to the system neutral.

### 3.1.2 Hardware

Provide hardware with washer against wood and with nuts and lock nuts applied wrench tight. Provide locknuts on threaded hardware connections. Locknuts must be M-F style and not palnut style.

### 3.1.3 Grounding

Unless otherwise indicated, grounding must conform to IEEE C2 and the local governing code. Pole grounding electrodes must have a resistance to ground not exceeding 25 ohms. When work in addition to that indicated or specified is directed in order to obtain specified ground resistance, provisions of the contract covering changes must apply.

#### 3.1.3.1 Grounding Electrode Installation

Install grounding electrodes as follows:

- a. Driven rod electrodes - Unless otherwise indicated, locate ground rods approximately 900 mm out from base of the pole and drive into the earth until the tops of the rods are approximately 300 mm below finished grade. Evenly spaced multiple rods at least 3 m apart and connected together 600 mm below grade with a minimum No. 6 bare copper conductor.

#### 3.1.3.2 Grounding Electrode Conductors

On multi-grounded circuits, as defined in IEEE C2, provide a single continuous vertical grounding electrode conductor. Bond neutrals, surge arresters, and equipment grounding conductors to this conductor. For single-grounded or ungrounded systems, provide a grounding electrode conductor for the surge arrester and equipment grounding conductors and a separate grounding electrode conductor for the secondary neutrals. Staple grounding electrode conductors to wood poles at intervals not exceeding 600 mm. On metal poles, a preformed galvanized steel strap, 15.9 mm wide by 0.853 mm minimum by length, secured by a preformed locking method standard with the manufacturer, must be used to support a grounding electrode conductor installation on the pole and spaced at intervals not exceeding 1.5 m with one band not more than 75 mm from each end of the vertical grounding electrode conductor. Size grounding electrode conductors as

indicated. Connect secondary system neutral conductors directly to the transformer neutral bushings, then connected with a neutral bonding jumper between the transformer neutral bushing and the vertical grounding electrode conductor as indicated. Bends greater than 45 degrees in grounding electrode conductor are not permitted.

#### 3.1.3.3 Grounding Electrode Connections

Make above grade grounding connections on pole lines by exothermic weld or by using a compression connector. Make below grade grounding connections by exothermic weld. Make exothermic welds strictly in accordance with manufacturer's written recommendations. Welds which have puffed up or which show convex surfaces indicating improper cleaning, are not acceptable. No mechanical connectors are required at exothermic weldments. Compression connectors must be type that uses a hydraulic compression tool to provide correct pressure. Provide tools and dies recommended by compression connector manufacturer. An embossing die code or similar method must provide visible indication that a connector has been fully compressed on ground wire.

#### 3.1.3.4 Grounding and Grounded Connections

- a. Where no primary or common neutral exists, bond surge arresters and frames of equipment operating at over 750 volts together and connected to a dedicated primary grounding electrode.
- b. Where no primary or common neutral exists, bond transformer secondary neutral bushing, secondary neutral conductor, and frames of equipment operating at under 750 volts together and connected to a dedicated secondary grounding electrode.
- c. When a primary or common neutral exists, the neutral must be connected to a grounding electrode. Transformer secondary neutral bushing and frames of equipment operating at under 750 volts must be bonded together and connected to a common neutral and to a common grounding electrode.

#### 3.1.3.5 Protective Molding

Protect grounding conductors which are run on surface of wood poles by PVC molding extending from ground line throughout communication and transformer spaces.

### 3.1.4 CONDUCTOR INSTALLATION

#### 3.1.4.1 Line Conductors

Unless otherwise indicated, install conductors in compliance with IEEE C2 Grade B requirements and in accordance with revised manufacturer's approved tables of sags and tensions. Handle conductors with care necessary to prevent nicking, kinking, gouging, abrasions, sharp bends, cuts, flattening, or otherwise deforming or weakening conductor or any damage to insulation or impairing its conductivity. Remove damaged sections of conductor and splice conductor. Conductors must be paid out with the free end of conductors fixed and cable reels portable, except where terrain or obstructions make this method unfeasible. Bend radius for any insulated conductor must not be less than the applicable NEMA specification recommendation. Conductors must not be drawn over rough or rocky ground, nor around sharp bends. When installed by machine power,



conductors must be drawn from a mounted reel through stringing sheaves in straight lines clear of obstructions. Initial sag and tension must be checked by the Contractor, in accordance with the manufacturer's approved sag and tension charts, within an elapsed time after installation as recommended by the manufacturer.

#### 3.1.4.2 Connectors and Splices

Conductor splices, as installed, must exceed ultimate rated strength of conductor and must be of type recommended by conductor manufacturer. No splice must be permitted within 3050 mm of a support. Connectors and splices must be mechanically and electrically secure under tension and must be of the nonbolted compression type. The tensile strength of any splice must be not less than the rated breaking strength of the conductor. Splice materials, sleeves, fittings, and connectors must be noncorrosive and must not adversely affect conductors. Aluminum-composition conductors must be wire brushed and an oxide inhibitor applied before making a compression connection. Connectors which are factory-filled with an inhibitor are acceptable. Inhibitors and compression tools must be of types recommended by the connector manufacturer. Primary line apparatus taps must be by means of hot line clamps attached to compression type bail clamps (stirrups). Low-voltage connectors for copper conductors must be of the solderless pressure type. Noninsulated connectors must be smoothly taped to provide a waterproof insulation equivalent to the original insulation, when installed on insulated conductors. On overhead connections of aluminum and copper, the aluminum must be installed above the copper.

#### 3.1.4.3 Conductor-To-Insulator Attachments

Conductors must be attached to insulators by means of clamps, shoes or tie wires, in accordance with the type of insulator. For insulators requiring conductor tie-wire attachments, tie-wire sizes must be as specified in TABLE I.

| TABLE I - TIE-WIRE REQUIREMENTS |                                     |
|---------------------------------|-------------------------------------|
| CONDUCTOR<br>Copper (AWG)       | TIE WIRE<br>Soft-Drawn Copper (AWG) |
| 6                               | 8                                   |
| 4 and 2                         | 6                                   |
| 1 through 3/0                   | 4                                   |
| 4/0 and larger                  | 2                                   |
| AAC, AAAC, or ACSR (AWG)        | AAAC OR AAC (AWG)                   |
| Any size                        | 6 or 4                              |

#### 3.1.4.4 Armor Rods

Provide armor rods for AAC, AAAC, and ACSR conductors. Armor rods must be installed at supports, except armor rods will not be required at primary dead-end assemblies if aluminum or aluminum-lined zinc-coated steel clamps

are used. Lengths and methods of fastening armor rods must be in accordance with the manufacturer's recommendations. For span lengths of less than 61 m, flat aluminum armor rods may be used. Flat armor rods, not less than 762.0 micrometers by 6.4 mm must be used on No. 1 AWG AAC and AAAC and smaller conductors and on No. 5 AWG ACSR and smaller conductors. On larger sizes, flat armor rods must be not less than 1.3 by 7.6 mm. For span lengths of 61 m or more, preformed round armor rods must be used.

#### 3.1.4.5 Ties

Provide ties on pin insulators tight against conductor and insulator and ends turned down flat against conductor so that no wire ends project.

#### 3.1.4.6 Low-Voltage Insulated Cables

Support low-voltage cables on clevis fittings using spool insulators. Provide dead-end clevis fittings and suspensions insulators where required for adequate strength. Dead-end construction must provide a strength exceeding the rated breaking strength of the neutral messenger. Provide clevis attachments with not less than 15.9 mm through-bolts. Secondary racks may be used when installed on wood poles and where the span length does not exceed 61 m. Secondary racks must be two-, three-, or four-wire, complete with spool insulators. Racks must meet strength and deflection requirements for heavy-duty steel racks, and must be rounded and smooth to avoid damage to conductor insulation. Each insulator must be held in place with a 15.9 mm button-head bolt equipped with a nonferrous cotter pin, or equivalent, at the bottom. Attach racks for dead-ending four No. 4/0 AWG or four larger conductors to poles with three 15.9 mm through-bolts. Attach other secondary racks to poles with at least two 15.9 mm through-bolts. Minimum vertical spacing between conductors must not be less than 200 mm.

#### 3.1.4.7 Reinstalling Conductors

Existing conductors to be reinstalled or resagged must be strung to "final" sag table values indicated for the particular conductor type and size involved.

#### 3.1.4.8 New Conductor Installation

String new conductors to "initial" sag table values recommended by the manufacturer for conductor type and size of conductor and ruling span indicated.

#### 3.1.4.9 Fittings

Dead end fittings must conform to written recommendations of conductor manufacturer and must develop full ultimate strength of conductor.

#### 3.1.4.10 Aluminum Connections

Make aluminum connections to copper or other material using only splices, connectors, lugs, or fittings designed for that specific purpose. Keep a copy of manufacturer's instructions for applying these fittings at job site for use of the inspector.

#### 3.1.5 Risers

Secure galvanized steel conduits on poles by two hole galvanized steel

pipe straps spaced as indicated and within 910 mm of any outlet or termination. Ground metallic conduits. Secure PVC riser shields on poles as indicated.

### 3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

### 3.3 FIELD QUALITY CONTROL

#### 3.3.1 General

Perform field testing in the presence of the Contracting Officer. The Contractor must notify the Contracting Officer 14 days prior to conducting tests. The Contractor must furnish materials, labor, and equipment necessary to conduct field tests. The Contractor must perform tests and inspections recommended by the manufacturer unless specifically waived by the Contracting Officer. The Contractor must maintain a written record of tests which includes date, test performed, personnel involved, devices tested, serial number and name of test equipment, and test results. Field reports will be signed and dated by the Contractor.

#### 3.3.2 Safety

The Contractor must provide and use safety devices such as rubber gloves, protective barriers, and danger signs to protect and warn personnel in the test vicinity. The Contractor must replace any devices or equipment which are damaged due to improper test procedures or handling.

#### 3.3.3 Medium-Voltage Preassembled Cable Test

After installation, prior to connection to an existing system, and before the operating test, the medium-voltage preassembled cable system must be given a high potential test. Apply direct-current voltage on each phase conductor of the system by connecting conductors at one terminal and connecting grounds or metallic shieldings or sheaths of the cable at the other terminal for each test. Prior to the test, the cables must be isolated by opening applicable protective devices and disconnecting equipment. The method, voltage, length of time, and other characteristics of the test for initial installation must be in accordance with NEMA WC 74/ICEA S-93-639 for the particular type of cable installed, and must not exceed the recommendations of IEEE 404 for cable joints unless the cable and accessory manufacturers indicate higher voltages are acceptable for testing. Should any cable fail due to a weakness of conductor insulation or due to defects or injuries incidental to the installation or because of improper installation of cable, cable joints, terminations, or other connections, the Contractor must make necessary repairs or replace cables as directed. Repaired or replaced cables shall be retested.

#### 3.3.4 Low-Voltage Cable Test

For underground secondary or service laterals from overhead lines, the low-voltage cable, complete with splices, must be tested for insulation resistance after the cables are installed, in their final configuration, ready for connection to the equipment, and prior to energization. The test voltage must be 500 volts dc, applied for one minute between each

conductor and ground and between all possible combinations of conductors in the same trench, duct, or cable, with other conductors in the same trench, duct, or conduit. The minimum value of insulation must be:

$R \text{ in megohms} = (\text{rated voltage in kV} + 1) \times 304,800 / (\text{length of cable in meters})$

Repair each cable failing this test or replace. The repaired cable must then be retested until failures have been eliminated.

### 3.3.5 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

#### 3.3.5.1 Overhead-Type Distribution Transformers

##### a. Visual and mechanical inspection

- (1) Compare equipment nameplate information with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermographic survey is not required.
- (4) Perform specific inspections and mechanical tests as recommended by manufacturer.
- (5) Verify correct equipment grounding.

##### b. Electrical tests

- (2) Perform insulation-resistance tests.
- (3) Perform continuity test.

#### 3.3.5.2 Pole Top Interrupter Switch

##### a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate information with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify appropriate equipment grounding.
- (4) Perform mechanical operator tests in accordance with manufacturer's instructions.
- (5) Verify correct blade alignment, blade penetration, travel stops, arc interrupter operation, and mechanical operation.

##### b. Electrical Tests

- (1) Perform insulation-resistance tests.
- (2) Perform dc over-potential tests.
- (3) Perform contact-resistance tests across each switch blade.

#### 3.3.5.3 Grounding System

##### a. Visual and mechanical inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

##### b. Electrical tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.
- (2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e. pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

#### 3.3.6 Devices Subject to Manual Operation

Each device subject to manual operation must be operated at least three times, demonstrating satisfactory operation each time.

#### 3.3.7 Follow-Up Verification

Upon completion of acceptance checks and tests, the Contractor must show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function.

-- End of Section --

## SECTION 33 71 02

## UNDERGROUND ELECTRICAL DISTRIBUTION

08/21

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M (2019; Errata 2022) Building Code Requirements for Structural Concrete & Commentary

ACI MNL-66 (2020) ACI Detailing Manual

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)

AEIC CS8 (2013) Specification for Extruded Dielectric Shielded Power Cables Rated 5 Through 46 kV

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B3 (2013; R 2024) Standard Specification for Soft or Annealed Copper Wire

ASTM B8 (2023) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM B800 (2005; R 2021) Standard Specification for 8000 Series Aluminum Alloy Wire for Electrical Purposes-Annealed and Intermediate Tempers

ASTM B801 (2018; R 2023) Standard Specification for Concentric-Lay-Stranded Conductors of 8000 Series Aluminum Alloy for Subsequent Covering or Insulation

ASTM C32 (2023) Standard Specification for Sewer

and Manhole Brick (Made from Clay or Shale)

|            |                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C309  | (2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete                                                                 |
| ASTM C478M | (2018) Standard Specification for Precast Reinforced Concrete Manhole Sections (Metric)                                                                 |
| ASTM C857  | (2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures                                      |
| ASTM C990M | (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants (Metric) |
| ASTM F512  | (2019; R 2024) Standard Specification for Smooth-Wall Poly (Vinyl Chloride) (PVC) Conduit and Fittings for Underground Installation                     |

#### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

|                      |                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IEEE 48              | (2020) Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV |
| IEEE 81              | (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System                                                                                                       |
| IEEE 386             | (2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV                                                                                                                |
| IEEE 400.2           | (2013) Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)                                                                                                                         |
| IEEE 404             | (2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V                                                                                                                 |
| IEEE C2              | (2023) National Electrical Safety Code                                                                                                                                                                                |
| IEEE Stds Dictionary | (2009) IEEE Standards Dictionary: Glossary of Terms & Definitions                                                                                                                                                     |

#### INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

|          |                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------|
| NETA ATS | (2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems |
|----------|--------------------------------------------------------------------------------------------------|

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|                               |                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| ANSI C119.1                   | (2023) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts                             |
| ANSI/NEMA WC 71/ICEA S-96-659 | (2014; R 2022) Standard for Nonshielded Cables Rated 2001-5000 Volts for use in the Distribution of Electric Energy     |
| NEMA RN 1                     | (2005; R 2013) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit |
| NEMA TC 2                     | (2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit                                                         |
| NEMA TC 6 & 8                 | (2020) Standard for Polyvinyl Chloride (PVC) Plastic Utilities Duct for Underground Installations                       |
| NEMA TC 9                     | (2020) Standard for Fittings for Polyvinyl Chloride (PVC) Plastic Utilities Duct for Underground Installation           |
| NEMA WC 74/ICEA S-93-639      | (2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy                     |

## TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

|         |                                                                                 |
|---------|---------------------------------------------------------------------------------|
| TIA-758 | (2012b) Customer-Owned Outside Plant Telecommunications Infrastructure Standard |
|---------|---------------------------------------------------------------------------------|

## U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|               |                                                                                  |
|---------------|----------------------------------------------------------------------------------|
| CID A-A-60005 | (Basic; Notice 2) Frames, Covers, Gratings, Steps, Sump And Catch Basin, Manhole |
|---------------|----------------------------------------------------------------------------------|

## UNDERWRITERS LABORATORIES (UL)

|        |                                                                                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------------|
| UL 6   | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel                                                              |
| UL 44  | (2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables                                            |
| UL 83  | (2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables                                        |
| UL 94  | (2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances |
| UL 467 | (2022) UL Standard for Safety Grounding                                                                                         |



## and Bonding Equipment

|              |                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------|
| UL 486A-486B | (2018; Reprint Jul 2023) UL Standard for Safety Wire Connectors                                               |
| UL 510       | (2020; Dec 2022) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape           |
| UL 514A      | (2013; Reprint Jun 2022) UL Standard for Safety Metallic Outlet Boxes                                         |
| UL 514B      | (2012; Reprint Mar 2024) UL Standard for Safety Conduit, Tubing and Cable Fittings                            |
| UL 651       | (2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings |
| UL 854       | (2020; Reprint Jul 2024) Standard for Service-Entrance Cables                                                 |
| UL 1072      | (2006; Reprint May 2024) Medium-Voltage Power Cables                                                          |
| UL 1242      | (2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel                |

## 1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

## 1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.
- b. In the text of this section, the words conduit and duct are used interchangeably and have the same meaning.
- c. In the text of this section, "medium voltage cable splices," and "medium voltage cable joints" are used interchangeably and have the same meaning.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Precast Underground Structures; G

SD-03 Product Data

- Medium Voltage Cable
- Medium Voltage Cable Joints
- Medium Voltage Cable Terminations
- Precast Concrete Structures; G
- Sealing Material
- Pulling-In Irons
- Manhole Frames and Covers
- Handhole Frames and Covers
- Cable Supports (racks, arms and insulators)

#### SD-06 Test Reports

- Medium Voltage Cable Qualification and Production Tests
- Field Acceptance Checks and Tests; G
- Arc-proofing Test for cable fireproofing tape

#### SD-07 Certificates

- Cable splicer/terminator
- Cable Installer Qualifications

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Precast Underground Structures

Submittal required for each type used. Provide calculations and drawings for precast manholes and handholes bearing the seal of a registered professional engineer including:

- a. Material description (i.e.,  $f'_c$  and  $F_y$ )
- b. Manufacturer's printed assembly and installation instructions
- c. Design calculations
- d. Reinforcing shop drawings in accordance with ACI MNL-66
- e. Plans and elevations showing opening and pulling-in iron locations and details

#### 1.5.2 Certificate of Competency for Cable Splicer/Terminator

Submit certification of the qualification of the cable splicer/terminator for approval, 30 days before splices or terminations are to be made in medium voltage (5 kV to 35 kV) cables. Include the training, and experience of the individual on the specific type and classification of cable to be provided under this contract. Indicate that the individual has had three or more years recent experience splicing and terminating

medium voltage cables. List a minimum of three splices/terminations that have been in operation for more than one year. In addition, the individual may be required to perform a dummy or practice splice/termination in the presence of the Contracting Officer, before being approved as a qualified cable splicer. If that additional requirement is imposed, the Contractor must provide short sections of the approved types of cables along with the approved type of splice/termination kit, and detailed manufacturer's instructions for the cable to be spliced. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for certification of an alternate cable splicer.

#### 1.5.3 Cable Installer Qualifications

Provide at least one onsite person in a supervisory position with a documentable level of competency and experience to supervise all cable pulling operations. Provide a resume showing the cable installers' experience in the last three years, including a list of references complete with points of contact, addresses and telephone numbers. Cable installer must demonstrate experience with a minimum of three medium voltage cable installations. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for an alternate qualified cable installer.

#### 1.5.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of IEEE C2 and the local governing code unless more stringent requirements are specified or indicated.

#### 1.5.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

##### 1.5.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

##### 1.5.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site

are not acceptable, unless specified otherwise.

## PART 2 PRODUCTS

### 2.1 CONDUIT, DUCTS, AND FITTINGS

#### 2.1.1 Rigid Metal Conduit

UL 6.

##### 2.1.1.1 Rigid Metallic Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 15.75 kV per mm at 60 Hz, and tensile strength must be minimum 25 MPa.

#### 2.1.2 Intermediate Metal Conduit

UL 1242.

##### 2.1.2.1 Intermediate Metal Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 15.75 kV per mm at 60 Hz, and tensile strength must be minimum 25 MPa.

#### 2.1.3 Plastic Conduit for Direct Burial and Riser Applications

UL 651 and NEMA TC 2, EPC-40 or EPC-80 as indicated.

#### 2.1.4 Plastic Duct for Concrete Encasement

Provide Type EB-20 per UL 651, ASTM F512, and NEMA TC 6 & 8.

#### 2.1.5 Duct Sealant

UL 94, Class HBF. Provide high-expansion urethane foam duct sealant that expands and hardens to form a closed, chemically and water resistant, rigid structure. Sealant must be compatible with common cable and wire jackets and capable of adhering to metals, plastics and concrete. Sealant must be capable of curing in temperature ranges of 2 degrees C to 35 degrees C. Cured sealant must withstand temperature ranges of -29 degrees C to 93 degrees C without loss of function.

#### 2.1.6 Fittings

##### 2.1.6.1 Metal Fittings

UL 514B.

##### 2.1.6.2 PVC Conduit Fittings

UL 514B, UL 651.

##### 2.1.6.3 PVC Duct Fittings

NEMA TC 9.

#### 2.1.6.4 Outlet Boxes for Steel Conduit

Outlet boxes for use with rigid or flexible steel conduit must be cast-metal cadmium or zinc-coated if of ferrous metal with gasketed closures and must conform to UL 514A.

### 2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES

Insulated conductors must be rated 600 volts and conform to the requirements of the local governing code, including listing requirements. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Service entrance conductors must conform to UL 854, type USE.

#### 2.2.1 Conductor Types

Cable and duct sizes indicated are for copper conductors and THHN/THWN unless otherwise noted. Conductors No. 10 AWG and smaller must be solid. Conductors No. 8 AWG and larger must be stranded. Conductors No. 6 AWG and smaller must be copper. Conductors No. 4 AWG and larger may be either copper or aluminum, at the Contractor's option. Do not substitute aluminum for copper if the equivalent aluminum conductor size would exceed 500 kcmil. When the Contractor chooses to use aluminum for conductors No. 4 AWG and larger, the Contractor must: increase the conductor size to have the same ampacity as the copper size indicated; increase the conduit and pull box sizes to accommodate the larger size aluminum conductors in accordance with the local governing code; ensure that the pulling tension rating of the aluminum conductor is sufficient; relocate equipment, modify equipment terminations, resize equipment, and resolve to the satisfaction of the Contracting Officer problems that are direct results of the use of aluminum conductors in lieu of copper.

#### 2.2.2 Conductor Material

Unless specified or indicated otherwise or required by the local governing code, wires in conduit, other than service entrance, must be 600-volt, Type THWN/THHN conforming to UL 83 or Type XHHW or RHW conforming to UL 44. Copper conductors must be annealed copper complying with ASTM B3 and ASTM B8. Aluminum conductors must be Type AA-8000 aluminum conductors complying with ASTM B800 and ASTM B801, and must be of an aluminum alloy listed or labeled by UL as "component aluminum-wire stock (conductor material). Type 1350 is not acceptable. Intermixing of copper and aluminum conductors in the same raceway is not permitted.

#### 2.2.3 Jackets

Provide multiconductor cables with an overall PVC outer jacket.

#### 2.2.4 Cable Marking

Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout the cable length.

Identify each cable by means of a fiber, laminated plastic, or non-ferrous metal tags in each manhole, handhole, junction box, and each terminal. Each tag must contain the following information; cable type, conductor size, circuit number, circuit voltage, cable destination and phase identification.

Color code conductors. Provide conductor identification within each enclosure where a tap, splice, or termination is made. Conductor identification must be by color-coded insulated conductors, plastic-coated self-sticking printed markers, colored nylon cable ties and plates, heat shrink type sleeves, or colored electrical tape. Properly identify control circuit terminations. Color must be green for grounding conductors and white for neutrals; except where neutrals of more than one system are installed in same raceway or box, other neutrals may be white with a different colored (not green) stripe for each. Color of ungrounded conductors in different voltage systems are as follows:

- a. 208/120 volt, three-phase
  - (1) Phase A - black
  - (2) Phase B - red
  - (3) Phase C - blue
- b. 480/277 volt, three-phase
  - (1) Phase A - brown
  - (2) Phase B - orange
  - (3) Phase C - yellow

## 2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS

Provide a uniform compression over the entire conductor contact surface. Use solderless terminal lugs on stranded conductors.

- a. For use with copper conductors: UL 486A-486B.
- b. For use with aluminum conductors: UL 486A-486B. For connecting aluminum to copper, connectors must be the circumferentially compressed, metallurgically bonded type.

## 2.4 LOW VOLTAGE SPLICES

Provide splices in conductors with a compression connector on the conductor and by insulating and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with ANSI C119.1.

### 2.4.1 Heat Shrinkable Splice

Provide heat shrinkable splice insulation by means of a thermoplastic adhesive sealant material applied in accordance with the manufacturer's written instructions.

### 2.4.2 Cold Shrink Rubber Splice

Provide a cold-shrink rubber splice which consists of EPDM rubber tube which has been factory stretched onto a spiraled core which is removed during splice installation. The installation must not require heat or flame, or any additional materials such as covering or adhesive. It must be designed for use with inline compression type connectors, or indoor,

outdoor, direct-burial or submerged locations.

## 2.5 MEDIUM VOLTAGE CABLE

Cable (conductor) sizes are designated by American Wire Gauge (AWG) and Thousand Circular Mils (Kcmil). Conductor and conduit sizes indicated are for copper conductors unless otherwise noted. Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout cable length. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Provide single conductor type cables unless otherwise indicated.

### 2.5.1 Cable Configuration

Provide Type MV cable, conforming to NEMA WC 74/ICEA S-93-639 and UL 1072. Provide cables manufactured for use in duct applications. Cable must be rated 15 kV with 133 percent insulation level.

### 2.5.2 Conductor Material

Provide concentric-lay-stranded, Class B conductors. Provide aluminum alloy Type AA-8000 aluminum conductors complying with ASTM B800 and ASTM B801 or soft drawn copper cables complying with ASTM B3 and ASTM B8 for regular concentric and compressed stranding.

### 2.5.3 Insulation

Provide ethylene-propylene-rubber (EPR) insulation conforming to the requirements of ANSI/NEMA WC 71/ICEA S-96-659.

### 2.5.4 Shielding

Cables rated for 2 kV and above must have a semiconducting conductor shield, a semiconducting insulation shield, and an overall copper tape shield for each phase.

### 2.5.5 Neutrals

Neutral conductors must be copper or aluminum, employing the same insulation and jacket materials as phase conductors, except that a 600-volt insulation rating is acceptable.

### 2.5.6 Jackets

Provide cables with a PVC jacket. Provide PVC jackets with a separator that prevents contact with underlying semiconducting insulating shield.

## 2.6 MEDIUM VOLTAGE CABLE TERMINATIONS

IEEE 48 Class 1; of the molded elastomer, prestretched elastomer, or heat-shrinkable elastomer. Acceptable elastomers are track-resistant silicone rubber or track-resistant ethylene propylene compounds, such as ethylene propylene rubber or ethylene propylene diene monomer. Separable insulated connectors may be used for apparatus terminations, when such apparatus is provided with suitable bushings. Provide terminations, where required, with mounting brackets suitable for the intended installation and with grounding provisions for the cable shielding, metallic sheath, or armor. Provide terminations in a kit, including: skirts, stress control

terminator, ground clamp, connectors, lugs, and complete instructions for assembly and installation. Terminations must be the product of one manufacturer, suitable for the type, diameter, insulation class and level, and materials of the cable terminated. Do not use separate parts of copper or copper alloy in contact with aluminum alloy parts in the construction or installation of the terminator.

#### 2.6.1 Cold-Shrink Type

Terminator must be a one-piece design, utilizing the manufacturer's latest technology, where high-dielectric constant (capacitive) stress control is integrated within a skirted insulator made of silicone rubber. Termination must not require heat or flame for installation. Termination kit must contain all necessary materials (except for the lugs). Design termination for installation in low or highly contaminated indoor and outdoor locations and must resist ultraviolet rays and oxidative decomposition.

#### 2.6.2 Heat Shrinkable Type

Terminator must consist of a uniform cross section heat shrinkable polymeric construction stress relief tubing and environmentally sealed outer covering that is nontracking, resists heavy atmospheric contaminants, ultra violet rays and oxidative decomposition. Provide heat shrinkable sheds or skirts of the same material. Design termination for installation in low or highly contaminated indoor or outdoor locations.

#### 2.6.3 Separable Insulated Connector Type

IEEE 386. Provide connector with steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material. Provide connectors of the loadbreak or deadbreak type as indicated, of suitable construction for the application and the type of cable connected, and that include cable shield adaptors. Provide external clamping points and test points. Do not use separable connectors in manholes/handholes.

- a. 200 Ampere loadbreak connector ratings: Voltage: 15 kV, 95 kV BIL. Short time rating: 10,000 rms symmetrical amperes.
- b. 600 Ampere deadbreak connector ratings: Voltage: 15 kV, 95 kV BIL. Short time rating: 25,000 rms symmetrical amperes. Connectors must have 200 ampere bushing interface for surge arresters.

### 2.7 MEDIUM VOLTAGE CABLE JOINTS

Provide joints (splices) in accordance with IEEE 404 suitable for the rated voltage, insulation level, insulation type, and construction of the cable. Joints must be certified by the manufacturer for waterproof, submersible applications. Upon request, supply manufacturer's design qualification test report in accordance with IEEE 404. Connectors for joint must be tin-plated electrolytic copper, having ends tapered and having center stops to equalize cable insertion.

#### 2.7.1 Heat-Shrinkable Joint

Consists of a uniform cross-section heat-shrinkable polymeric construction with a linear stress relief system, a high dielectric strength insulating material, and an integrally bonded outer conductor layer for shielding.



Replace original cable jacket with a heavy-wall heat-shrinkable sleeve with hot-melt adhesive coating.

#### 2.7.2 Cold-Shrink Rubber-Type Joint

Joint must be of a cold shrink design that does not require any heat source for its installation. Splice insulation and jacket must be of a one-piece factory formed cold shrink sleeve made of black EPDM rubber. Splice should be packaged three splices per kit, including complete installation instructions.

### 2.8 TAPE

#### 2.8.1 Insulating Tape

UL 510, plastic insulating tape, capable of performing in a continuous temperature environment of 80 degrees C.

#### 2.8.2 Buried Warning and Identification Tape

Provide detectable tape in accordance with Section 31 00 00 EARTHWORK.

#### 2.8.3 Fireproofing Tape

Provide tape composed of a flexible, conformable, unsupported intumescent elastomer. Tape must be not less than 0.762 mm thick, noncorrosive to cable sheath, self-extinguishing, noncombustible, adhesive-free, and must not deteriorate when subjected to oil, water, gases, salt water, sewage, and fungus.

### 2.9 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of 890 N.

### 2.10 GROUNDING AND BONDING

#### 2.10.1 Driven Ground Rods

Provide copper-clad steel ground rods conforming to UL 467 not less than 19 mm in diameter by 3.1 m in length. Sectional type rods may be used for rods 6.2 m or longer.

#### 2.10.2 Grounding Conductors

Stranded-bare copper conductors must conform to ASTM B8, Class B, soft-drawn unless otherwise indicated. Solid-bare copper conductors must conform to ASTM B1 for sizes No. 8 and smaller. Insulated conductors must be of the same material as phase conductors and green color-coded, except that conductors must be rated no more than 600 volts. Aluminum is not acceptable.

### 2.11 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE. In addition, provide concrete for encasement of underground ducts with 20 MPa minimum 28-day compressive strength. Concrete associated with electrical work for other than encasement of underground ducts must be 30 MPa minimum 28-day compressive strength unless specified

otherwise.

## 2.12 UNDERGROUND STRUCTURES

Provide precast concrete underground structures or standard type cast-in-place manhole types as indicated, conforming to ASTM C857 and ASTM C478M. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable.

### 2.12.1 Precast Concrete Structures, Risers and Tops

Precast concrete underground structures may be provided in lieu of cast-in-place subject to the requirements specified below. Precast units must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes.

#### 2.12.1.1 General

Precast concrete structures must have the same accessories and facilities as required for cast-in-place structures. Likewise, precast structures must have plan area and clear heights not less than those of cast-in-place structures. Concrete materials and methods of construction must be the same as for cast-in-place concrete construction, as modified herein. Slope in floor may be omitted provided precast sections are poured in reinforced steel forms. Concrete for precast work must have a 28-day compressive strength of not less than 30 MPa. Structures may be precast to the design and details indicated for cast-in-place construction, precast monolithically and placed as a unit, or structures may be assembled sections, designed and produced by the manufacturer in accordance with the requirements specified. Structures must be identified with the manufacturer's name embedded in or otherwise permanently attached to an interior wall face.

#### 2.12.1.2 Design for Precast Structures

ACI 318M. In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

- a. Angle of Internal Friction ( $\phi$ ) = 0.523 rad
- b. Unit Weight of Soil (Dry) = 1760 kg/m<sup>3</sup>, (Saturated)  
= 2080 kg/m<sup>3</sup>
- c. Coefficient of Lateral Earth Pressure ( $K_a$ ) = 0.33
- d. Ground Water Level = 915 mm below ground elevation
- e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact.

Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading per AASHTO HB-17.

- f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 26,700 N.
- g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for that particular component.
- h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

#### 2.12.1.3 Construction

Provide a uniform thickness for structure top, bottom, and wall not less than 150 mm. Thin-walled knock-out panels for designed or future duct bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the precaster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 305 mm vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 150 mm from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 4 bars around window openings. Provide drain sumps a minimum of 305 mm in diameter and 100 mm deep for precast structures.

#### 2.12.1.4 Joints

Provide tongue-and-groove joints on mating edges of precast components. Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip conforming to ASTM C990M. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

#### 2.12.2 Manhole Frames and Covers

Provide cast iron frames and covers for manholes conforming to CID A-A-60005. Cast the words "ELECTRIC" or "TELECOMMUNICATIONS" in the top face of power and telecommunications manhole covers, respectively.

### 2.12.3 Handhole Frames and Covers

Frames and covers of steel must be welded by qualified welders in accordance with standard commercial practice. Provide rolled-steel floor plate covers having an approved antislip surface. Hinges must be of stainless steel with bronze hinge pin, 125 by 125 mm by approximately 4.75 mm thick, without screw holes, and must be for full surface application by fillet welding. Hinges must have nonremovable pins and five knuckles. The surfaces of plates under hinges must be true after the removal of raised antislip surface, by grinding or other approved method.

### 2.12.4 Brick for Manhole Collar

Provide sewer and manhole brick conforming to ASTM C32, Grade MS.

## 2.13 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

Zinc coat the metal portion of racks and arms after fabrication.

### 2.13.1 Cable Rack Stanchions

The wall bracket or stanchion must be 100 mm by approximately 38 mm by 4.76 mm channel steel, or 100 mm by approximately 25 mm glass-reinforced nylon with recessed bolt mounting holes, 1220 mm long (minimum) in manholes. Space slots for mounting cable rack arms at 200 mm intervals.

### 2.13.2 Rack Arms

Cable rack arms must be steel or malleable iron or glass reinforced nylon and must be of the removable type. Rack arm length must be a minimum of 200 mm and a maximum of 305 mm.

### 2.13.3 Insulators

Insulators for metal rack arms must be dry-process glazed porcelain. Insulators are not required for nylon arms.

## 2.14 CABLE TAGS IN MANHOLES

Provide polyethylene tags for each power cable located in manholes. Do not provide handwritten letters. The first position on the power cable tag denotes the voltage. The second through sixth positions on the tag identifies the circuit. The next to last position denotes the phase of the circuit and include the Greek "phi" symbol. The last position denotes the cable size. As an example, a tag could have the following designation: "11.5 NAS 1-8(Phase A)500," denoting that the tagged cable is on the 11.5kV system circuit number NAS 1-8, underground, Phase A, sized at 500 kcmil.

### 2.14.1 Polyethylene Cable Tags

Provide tags of polyethylene having an average tensile strength of 22.4 MPa; and that are 2 millimeter thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 77 degrees C. Provide 1.3 mm (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag, having a minimum loop tensile strength of 778.75 N and black block letters, numbers, and symbols 25 mm high on a yellow background. Letters, numbers, and symbols must not fall off or

change positions regardless of the cable tags' orientation.

## 2.15 SOURCE QUALITY CONTROL

### 2.15.1 Arc-Proofing Test for Cable Fireproofing Tape

Manufacturer must test one sample assembly consisting of a straight lead tube 305 mm long with a 65.5 mm outside diameter, and a 3.175 mm thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions. The arc and fireproofing tape must withstand extreme temperature of a high-current fault arc 13,000 degrees K for 70 cycles as determined by using an argon directed plasma jet capable of constantly producing and maintaining an arc temperature of 13,000 degrees K. Temperature (13,000 degrees K) of the ignited arc between the cathode and anode must be obtained from a dc power source of 305 (plus or minus 5) amperes and 20 (plus or minus 1) volts. Direct the arc toward the sample assembly accurately positioned 5 (plus or minus 1) millimeters downstream in the plasma from the anode orifice by fixed flow rate of argon gas (0.18 g per second). Test each sample assembly at three unrelated points. Start time for tests must be taken from recorded peak current when the specimen is exposed to the full test temperature. Surface heat on the specimen prior to that time must be minimal. The end point is established when the plasma or conductive arc penetrates the protective tape and strikes the lead tube. Submittals for arc-proofing tape must indicate that the test has been performed and passed by the manufacturer.

### 2.15.2 Medium Voltage Cable Qualification and Production Tests

Results of AEIC CS8 qualification and production tests as applicable for each type of medium voltage cable.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of the local governing code and IEEE C2 as applicable. In addition to these requirements, install telecommunications in accordance with TIA-758.

### 3.2 CABLE INSPECTION

Inspect each cable reel for correct storage positions, signs of physical damage, and broken end seals prior to installation. If end seal is broken, remove moisture from cable prior to installation in accordance with the cable manufacturer's recommendations.

### 3.3 CABLE INSTALLATION PLAN AND PROCEDURE

Obtain from the manufacturer an installation manual or set of instructions which addresses such aspects as cable construction, insulation type, cable diameter, bending radius, cable temperature limits for installation, lubricants, coefficient of friction, conduit cleaning, storage procedures, moisture seals, testing for and purging moisture, maximum allowable pulling tension, and maximum allowable sidewall bearing pressure. Prepare a checklist of significant requirements and submit along with the manufacturer's instructions in accordance with SUBMITTALS. Install cable strictly in accordance with the cable manufacturer's recommendations and the approved installation plan.

Calculations and pulling plan must include:

- a. Site layout drawing with cable pulls identified in numeric order of expected pulling sequence and direction of cable pull.
- b. List of cable installation equipment.
- c. Lubricant manufacturer's application instructions.
- d. Procedure for resealing cable ends to prevent moisture from entering cable.
- e. Cable pulling tension calculations of all cable pulls.
- f. Cable percentage conduit fill.
- g. Cable sidewall bearing pressure.
- h. Cable minimum bend radius and minimum diameter of pulling wheels used.
- i. Cable jam ratio.
- j. Maximum allowable pulling tension on each different type and size of conductor.
- k. Maximum allowable pulling tension on pulling device.

#### 3.4 UNDERGROUND FEEDERS SUPPLYING BUILDINGS

Terminate underground feeders supplying building at a point 1525 mm outside the building and projections thereof, except that conductors must be continuous to the terminating point indicated. Coordinate connections of the feeders to the service entrance equipment with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide PVC, Type EPC-40 conduit from the supply equipment to a point 1525 mm outside the building and projections thereof. Protect ends of underground conduit with plastic plugs until connections are made.

Encase the underground portion of the conduit in a concrete envelope and bury as specified for underground duct with concrete encasement.

#### 3.5 UNDERGROUND STRUCTURE CONSTRUCTION

Provide standard type cast-in-place construction as specified herein and as indicated, or precast construction as specified herein. Horizontal concrete surfaces of floors must have a smooth trowel finish. Cure concrete by applying two coats of white pigmented membrane forming-curing compound in strict accordance with the manufacturer's printed instructions, except that precast concrete may be steam cured. Curing compound must conform to ASTM C309. Locate duct entrances and windows in the center of end walls (shorter) and near the corners of sidewalls (longer) to facilitate cable racking and splicing. Covers for underground structures must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Manhole locations, as

indicated, are approximate. Coordinate exact manhole locations with other utilities and finished grading and paving.

### 3.5.1 Cast-In-Place Concrete Structures

Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

### 3.5.2 Precast Concrete Construction

Set commercial precast structures on 150 mm of level, 90 percent compacted granular fill, 19 mm to 25 mm size, extending 305 mm beyond the structure on each side. Compact granular fill by a minimum of four passes with a plate type vibrator. Installation must additionally conform to the manufacturer's instructions.

### 3.5.3 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors. Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor, locate the irons a minimum of 150 mm from the edge of the sump, and in the walls, locate the irons within 150 mm of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 150 mm of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 150 mm clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 915 mm length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 75 mm from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 100 mm and must be designed to withstand a minimum pulling-in load of 26,700 N. Hot-dip galvanize irons after fabrication.

### 3.5.4 Cable Racks, Arms and Insulators

Cable racks, arms and insulators must be sufficient to accommodate the cables. Space racks in power manholes not more than 915 mm apart, and provide each manhole wall with a minimum of two racks. Space racks in signal manholes not more than 420 mm apart with the end rack being no further than 305 mm from the adjacent wall. Methods of anchoring cable racks are as follows:

- a. Provide a 15 mm diameter by 125 mm long anchor bolt with 75 mm foot cast in structure wall with 50 mm protrusion of threaded portion of bolt into structure. Provide 15 mm steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating

immediately prior to installing nuts.

- b. Provide concrete channel insert with a minimum load rating of 1192 kg per meter. Insert channel must be steel of the same length as "vertical rack channel;" and cast flush in structure wall. Provide 15 mm steel nuts in channel insert to receive 15 mm diameter by 75 mm long steel, square head anchor bolts.
- c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 365 kg load rating. Provide 15 mm diameter by 75 mm long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with suitable coating immediately prior to installing bolts.

### 3.5.5 Field Painting

Clean cast-iron frames and covers not buried in concrete or masonry of mortar, rust, grease, dirt and other deleterious materials, and coat with bituminous paint.

## 3.6 UNDERGROUND CONDUIT AND DUCT SYSTEMS

### 3.6.1 Requirements

Run conduit in straight lines except where a change of direction is necessary. Provide numbers and sizes of ducts as indicated. Provide a 4/0 AWG bare copper grounding conductor above medium-voltage distribution duct banks. Bond bare copper grounding conductor to ground rings (loops) in all manholes and to ground rings (loops) at all equipment slabs (pads). Route grounding conductor into manholes with the duct bank (sleeving is not required). Ducts must have a continuous slope downward toward underground structures and away from buildings, laid with a minimum slope of 75 mm per 30 m. Depending on the contour of the finished grade, the high-point may be at a terminal, a manhole, a handhole, or between manholes or handholes. Terminate all PVC conduit end points in utility holes, switching cabinets, transform handholes and buildings with end bells. The bell end of the conduits that enter manholes and handholes must be flush with the wall.

Perform changes in ductbank direction as follows:

- a. Short-radius manufactured 90-degree duct bends may be used only for pole or equipment risers, unless specifically indicated as acceptable.
- b. The minimum manufactured bend radius must be 450 mm for ducts of less than 80 mm diameter, and 900 mm for ducts 80 mm or greater in diameter.
- c. As an exception to the bend radius required above, provide field manufactured longsweep bends having a minimum radius of 7.6 m for a change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections allowed for use in field manufactured longsweep bend: 30 degrees.

### 3.6.2 Treatment

Keep ducts clean of concrete, dirt, or foreign substances during construction. Make field cuts requiring tapers with proper tools and



match factory tapers. Use a coupling recommended by the duct manufacturer whenever an existing duct is connected to a duct of different material or shape. Store ducts to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Thoroughly clean ducts before being laid. Store plastic ducts on a flat surface and protected from the direct rays of the sun.

### 3.6.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes 75 mm and larger, draw a flexible testing mandrel approximately 305 mm long with a diameter less than the inside diameter of the conduit through the conduit. After which, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs. For conduit sizes less than 75 mm, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

### 3.6.4 Multiple Conduits

Separate multiple conduits by a minimum distance of 75 mm, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of 300 mm. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 3050 mm of conduit assembly.

### 3.6.5 Conduit Plugs and Pull Rope

Provide new conduit indicated as being unused or empty with plugs on each end. Plugs must contain a weephole or screen to allow water drainage. Provide a plastic pull rope having 915 mm of slack at each end of unused or empty conduits.

### 3.6.6 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 610 mm below finished grade. Provide not less than 75 mm clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 75 mm, fill and tamp level with original bottom with sand or earth free from particles, that would be retained on a 6.25 mm sieve. The first 150 mm layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 75 to 150 mm layers. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

#### 3.6.6.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 75 mm concrete cover around ducts. Extend concrete encasement at least 1525 mm beyond the edges of paved areas and roads, and 3660 mm beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 610 mm below finished grade.

### 3.6.7 Duct Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than 450 mm below finished grade, except under roads and pavement, concrete envelope must be not less than 610 mm below finished grade. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and provide at least 75 mm of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 75 mm. Before pouring concrete, anchor duct bank assemblies, prevent floating during concrete pouring by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly. Provide steel reinforcing in the concrete envelope as indicated. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

#### 3.6.7.1 Connections to Manholes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 300 mm in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

#### 3.6.7.2 Connections to Existing Underground Structures

For duct bank connections to existing structures, break the structure wall out to the dimensions required and preserve steel in the structure wall. Cut steel and extend into the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

#### 3.6.7.3 Connections to Existing Concrete Pads

For duct bank connections to concrete pads, break an opening in the pad out to the dimensions required and preserve steel in pad. Cut the steel and extend into the duct bank envelope. Chip out the opening in the pad to form a key for the duct bank envelope.

#### 3.6.7.4 Connections to Existing Ducts

Where connections to existing duct banks are indicated, excavate the banks to the maximum depth necessary. Cut off the banks and remove loose concrete from the conduits before new concrete-encased ducts are installed. Provide a reinforced concrete collar, poured monolithically with the new duct bank, to take the shear at the joint of the duct banks. Abandon in place those no longer used ducts and cables which do not interfere with the work.

#### 3.6.7.5 Partially Completed Duct Banks

During construction wherever a construction joint is necessary in a duct bank, prevent debris such as mud, and, and dirt from entering ducts by providing suitable conduit plugs. Fit concrete envelope of a partially

completed duct bank with reinforcing steel extending a minimum of 610 mm back into the envelope and a minimum of 610 mm beyond the end of the envelope. Provide one No. 4 bar in each corner, 75 mm from the edge of the envelope. Secure corner bars with two No. 3 ties, spaced approximately 305 mm apart. Restrain reinforcing assembly from moving during concrete pouring.

#### 3.6.7.6 Removal of Ducts

Where duct lines are removed from existing underground structures, close the openings to waterproof the structure. Chip out the wall opening to provide a key for the new section of wall.

#### 3.6.8 Duct Sealing

Seal all electrical penetrations for radon mitigation, maintaining integrity of the vapor barrier, and to prevent infiltration of air, insects, and vermin.

### 3.7 CABLE PULLING

Test existing duct lines with a mandrel and thoroughly swab out to remove foreign material before pulling cables. Pull cables down grade with the feed-in point at the manhole or buildings of the highest elevation. Use flexible cable feeds to convey cables through manhole opening and into duct runs. Do not exceed the specified cable bending radii when installing cable under any conditions, including turnups into switches, transformers, switchgear, switchboards, and other enclosures. Cable with tape shield must have a bending radius not less than 12 times the overall diameter of the completed cable. If basket-grip type cable-pulling devices are used to pull cable in place, cut off the section of cable under the grip before splicing and terminating.

#### 3.7.1 Cable Lubricants

Use lubricants that are specifically recommended by the cable manufacturer for assisting in pulling jacketed cables.

### 3.8 CABLES IN UNDERGROUND STRUCTURES

Do not install cables utilizing the shortest path between penetrations, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support on brackets and cable insulators. Support cable splices in underground structures by racks on each side of the splice. Locate splices to prevent cyclic bending in the spliced sheath. Install cables at middle and bottom of cable racks, leaving top space open for future cables, except as otherwise indicated for existing installations. Provide one spare three-insulator rack arm for each cable rack in each underground structure.

#### 3.8.1 Cable Tag Installation

Install cable tags in each manhole as specified, including each splice. Tag wire and cable provided by this contract. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes.

### 3.9 CONDUCTORS INSTALLED IN PARALLEL

Group conductors such that each conduit of a parallel run contains one Phase A conductor, one Phase B conductor, one Phase C conductor, and one neutral conductor.

### 3.10 LOW VOLTAGE CABLE SPLICING AND TERMINATING

Make terminations and splices with materials and methods as indicated or specified herein and as designated by the written instructions of the manufacturer. Do not allow the cables to be moved until after the splicing material has completely set. Make splices in underground distribution systems only in accessible locations such as manholes, handholes, or aboveground termination pedestals.

#### 3.10.1 Terminating Aluminum Conductors

- a. Use particular care in making up joints and terminations. Remove surface oxides by cleaning with a wire brush or emery cloth. Apply joint compound to conductors, and use UL-listed solid aluminum connectors for connecting aluminum conductors. When connecting aluminum to copper conductors, use connectors specifically designed for this purpose.
- b. Terminate aluminum conductors to copper bus either by: (1) in line splicing a copper pigtail to the aluminum conductor (copper pigtail must have a ampacity at least that of the aluminum conductor); or (2) using a circumferential compression type, aluminum bodied terminal lug UL listed for AL/CU and steel Belleville spring washers, flat washers, bolts, and nuts. Belleville spring washers must be cadmium-plated hardened steel. Install the Belleville spring washers with the crown up toward the nut or bolt head, with the concave side of the Belleville bearing on a heavy-duty, wide series flat washer of larger diameter than the Belleville. Tighten nuts sufficient to flatten Belleville and leave in that position. Lubricate hardware with joint compound prior to making connection. Wire brush and apply joint compound to conductor prior to inserting in lug.
- c. Terminate aluminum conductors to aluminum bus by using all-aluminum nuts, bolts, washers, and lugs. Wire brush and apply inhibiting compound to conductor prior to inserting in lug. Lubricate hardware with joint compound prior to making connection; if bus contact surface is unplated, scratch-brush and coat with joint compound (without grit).

### 3.11 MEDIUM VOLTAGE CABLE TERMINATIONS

Make terminations in accordance with the written instruction of the termination kit manufacturer.

### 3.12 MEDIUM VOLTAGE CABLE JOINTS

Provide power cable joints (splices) suitable for continuous immersion in water. Make joints only in accessible locations in manholes or handholes by using materials and methods in accordance with the written instructions of the joint kit manufacturer.

#### 3.12.1 Joints in Shielded Cables

Cover the joined area with metallic tape, or material like the original

cable shield and connect it to the cable shield on each side of the splice. Provide a bare copper ground connection brought out in a watertight manner and grounded to the manhole grounding loop as part of the splice installation. Ground conductors, connections, and rods must be as specified elsewhere in this section. Wire must be trained to the sides of the enclosure to prevent interference with the working area.

### 3.13 CABLE END CAPS

Cable ends must be sealed at all times with coated heat shrinkable end caps. Cables ends must be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps must remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

### 3.14 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES

Fireproof (arc proof) wire and cables which will carry current at 2200 volts or more in underground structures.

#### 3.14.1 Fireproofing Tape

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Install tape in accordance with manufacturer's instructions.

### 3.15 GROUNDING SYSTEMS

The local governing code and IEEE C2, except provide grounding systems with a resistance to solid earth ground not exceeding 25 ohms.

#### 3.15.1 Grounding Electrodes

Provide cone pointed driven ground rods driven full depth plus 150 mm, installed to provide an earth ground of the appropriate value for the particular equipment being grounded. If the specified ground resistance is not met, provide an additional ground rod in accordance with the requirements of the local governing code (placed not less than 6 feet from the first rod). Should the resultant (combined) resistance exceed the specified resistance, measured not less than 48 hours after rainfall, notify the Contracting Officer immediately.

#### 3.15.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Tools and dies must be as recommended by the manufacturer. An embossing die code or other standard method must provide visible indication that a connector has

been adequately compressed on the ground wire.

### 3.15.3 Grounding Conductors

Provide bare grounding conductors, except where installed in conduit with associated phase conductors. Ground cable sheaths, cable shields, conduit, and equipment with #6 AWG. Ground other noncurrent-carrying metal parts and equipment frames of metal-enclosed equipment. Ground metallic frames and covers of handholes and pull boxes with a braided, copper ground strap with equivalent ampacity of #6 AWG. Provide direct connections to the grounding conductor with 600 v insulated, full-size conductor for each grounded neutral of each feeder circuit, which is spliced within the manhole.

### 3.15.4 Ground Cable Crossing Expansion Joints

Protect ground cables crossing expansion joints or similar separations in structures and pavements by use of approved devices or methods of installation which provide the necessary slack in the cable across the joint to permit movement. Use stranded or other approved flexible copper cable across such separations.

### 3.15.5 Manhole Grounding

Loop a #4/0 AWG grounding conductor around the interior perimeter, approximately 305 mm above finished floor. Secure the conductor to the manhole walls at intervals not exceeding 914 mm. Connect the conductor to the manhole grounding electrode with #4/0 AWG conductor. Connect all incoming #4/0 grounding conductors to the ground loop adjacent to the point of entry into the manhole. Bond the ground loop to all cable shields, metal cable racks, and other metal equipment with a minimum 6 AWG conductor.

### 3.15.6 Fence Grounding

Provide grounding for fences with a ground rod at each fixed gate post and at each corner post. Drive ground rods until the top is 305 mm below grade. Attach a No. 4 AWG copper conductor, by exothermic weld to the ground rods and extend underground to the immediate vicinity of fence post. Lace the conductor vertically into 305 mm of fence mesh and fasten by two approved bronze compression fittings, one to bond wire to post and the other to bond wire to fence. Bond each gate section to its gatepost by a 3 by 25 mm flexible braided copper strap and ground post clamps. Clamps must be of the anti-electrolysis type.

## 3.16 EXCAVATING, BACKFILLING, AND COMPACTING

Provide in accordance with the local governing code and Section 31 00 00 EARTHWORK.

### 3.16.1 Reconditioning of Surfaces

#### 3.16.1.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct. Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface

with the same quantity and formula of seed as that used in the original seeding, and provide topsoiling, fertilizing, liming, seeding, sodding, sprigging, or mulching.

#### 3.16.1.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists, restore such surface treatment or pavement the same thickness and in the same kind as previously existed, except as otherwise specified, and to match and tie into the adjacent and surrounding existing surfaces. Make repairs as specified in Section 32 13 13.06 PORTLAND CEMENT CONCRETE PAVEMENT FOR ROADS AND SITE FACILITIES.

### 3.17 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

#### 3.17.1 Concrete Slabs (Pads) for Equipment

Unless otherwise indicated, the slab must be at least 200 mm thick, reinforced with a 152 mm by 152 mm - MW19 by MW19 (6 by 6 - W2.9 by W2.9) mesh, placed uniformly 100 mm from the top of the slab. Place slab on a 150 mm thick, well-compacted gravel base. Top of concrete slab must be approximately 100 mm above finished grade with gradual slope for drainage. Edges above grade must have 15 mm chamfer. Slab must be of adequate size to project at least 200 mm beyond the equipment.

Stub up conduits, with bushings, 50 mm into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

#### 3.17.2 Sealing

When the installation is complete, seal all conduit and other entries into the equipment enclosure with an approved sealing compound. Seals must be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

### 3.18 FIELD QUALITY CONTROL

#### 3.18.1 Performance of Field Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

##### 3.18.1.1 Medium Voltage Cables

Perform tests after installation of cable, splices, and terminators and before terminating to equipment or splicing to existing circuits.

##### a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.

- (3) Inspect for proper shield grounding, cable support, and cable termination.
- (4) Verify that cable bends are not less than ICEA or manufacturer's minimum allowable bending radius.
- (5) Inspect for proper fireproofing.
- (6) Visually inspect jacket and insulation condition.
- (7) Inspect for proper phase identification and arrangement.

b. Electrical Tests

- (1) Perform a shield continuity test on each power cable by ohmmeter method. Record ohmic value, resistance values in excess of 10 ohms per 1000 feet of cable must be investigated and justified.
- (2) Perform acceptance test on new cables before the new cables are connected to existing cables and placed into service, including terminations and joints. Perform maintenance test on complete cable system after the new cables are connected to existing cables and placed into service, including existing cable, terminations, and joints. Tests must be very low frequency (VLF) alternating voltage withstand tests in accordance with IEEE 400.2. VLF test frequency must be 0.05 Hz minimum for a duration of 60 minutes using a sinusoidal waveform. Test voltages must be as follows:

| CABLE RATING AC TEST VOLTAGE for<br>ACCEPTANCE TESTING |                |
|--------------------------------------------------------|----------------|
| 5 kV                                                   | 10kV rms(peak) |
| 8 kV                                                   | 13kV rms(peak) |
| 15 kV                                                  | 20kV rms(peak) |
| 25 kV                                                  | 31kV rms(peak) |
| 35 kV                                                  | 44kV rms(peak) |

| CABLE RATING AC TEST VOLTAGE for<br>MAINTENANCE TESTING |                |
|---------------------------------------------------------|----------------|
| 5 kV                                                    | 7kV rms(peak)  |
| 8 kV                                                    | 10kV rms(peak) |
| 15 kV                                                   | 16kV rms(peak) |
| 25 kV                                                   | 23kV rms(peak) |



| CABLE RATING AC TEST VOLTAGE for<br>MAINTENANCE TESTING |                |
|---------------------------------------------------------|----------------|
| 35 kV                                                   | 33kV rms(peak) |

#### 3.18.1.2 Low Voltage Cables, 600-Volt

Perform tests after installation of cable, splices and terminations and before terminating to equipment or splicing to existing circuits.

##### a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Verify tightness of accessible bolted electrical connections.
- (4) Inspect compression-applied connectors for correct cable match and indentation.
- (5) Visually inspect jacket and insulation condition.
- (6) Inspect for proper phase identification and arrangement.

##### b. Electrical Tests

- (1) Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of approximately 1000 volts dc for one minute.
- (2) Perform continuity tests to insure correct cable connection.

#### 3.18.1.3 Grounding System

##### a. Visual and mechanical inspection

Inspect ground system for compliance with contract plans and specifications.

##### b. Electrical tests

Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

### 3.18.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days advance notice of the dates and times of checking and testing.

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## SECTION 33 82 00

## TELECOMMUNICATIONS OUTSIDE PLANT (OSP)

04/06

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM B1    | (2013) Standard Specification for Hard-Drawn Copper Wire                                                                                                                  |
| ASTM B8    | (2023) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft                                                                   |
| ASTM D709  | (2017) Standard Specification for Laminated Thermosetting Materials                                                                                                       |
| ASTM D1557 | (2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft <sup>3</sup> ) (2700 kN-m/m <sup>3</sup> ) |

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

|          |                                                                       |
|----------|-----------------------------------------------------------------------|
| IEEE 100 | (2000; Archived) The Authoritative Dictionary of IEEE Standards Terms |
| IEEE C2  | (2023) National Electrical Safety Code                                |

## INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

|               |                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| ICEA S-87-640 | (2023) Optical Fiber Outside Plant Communications Cable; 4th Edition                                                           |
| ICEA S-98-688 | (2012) Broadband Twisted Pair Telecommunication Cable, Aircore, Polyolefin Insulated, Copper Conductors Technical Requirements |
| ICEA S-99-689 | (2012) Broadband Twisted Pair Telecommunication Cable Filled, Polyolefin Insulated, Copper Conductors Technical Requirements   |

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|             |                                                                                                |
|-------------|------------------------------------------------------------------------------------------------|
| ANSI C62.61 | (1993) American National Standard for Gas Tube Surge Arresters on Wire Line Telephone Circuits |
|-------------|------------------------------------------------------------------------------------------------|

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National  
Electrical Code

## SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 6/NACE No.3 (2007) Commercial Blast Cleaning

## TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-455-46A (1990) FOTP-46 Spectral Attenuation  
Measurement for Long-Length, Graded-Index  
Optical Fibers

TIA-455-78-B (2020c) FOTP-78 Optical Fibres - Part  
1-40: Measurement Methods and Test  
Procedures - Attenuation

TIA-455-107 (1999a) FOTP-107 Determination of  
Component Reflectance or Link/System  
Return Loss using a Loss Test Set

TIA-472D000 (2007b) Fiber Optic Communications Cable  
for Outside Plant Use

TIA-492AAAA (2009b) 62.5-um Core Diameter/125-um  
Cladding Diameter Class 1a Graded-Index  
Multimode Optical Fibers

TIA-492AAAB (2009a) 50-Um Core Diameter/125-Um  
Cladding Diameter Class IA Graded-Index  
Multimode Optical Fibers

TIA-492CAAA (1998; R 2002) Detail Specification for  
Class IVa Dispersion-Unshifted Single-Mode  
Optical Fibers

TIA-492E000 (1996; R 2002) Sectional Specification for  
Class IVd Nonzero-Dispersion Single-Mode  
Optical Fibers for the 1550 nm Window

TIA-526-7 (2015a; R 2022) Measurement of Optical  
Power Loss of Installed Single-Mode Fiber  
Cable Plant, Adoption of IEC 61280-4-2  
edition 2: Fibre-Optic Communications  
Subsystem Test Procedures - Part 4-2:  
Installed Cable Plant - Single-Mode  
Attenuation and Optical Return Loss  
Measurement

TIA-526-14 (2023d) OFSTP-14A Optical Power Loss  
Measurements of Installed Multimode Fiber  
Cable Plant

TIA-568.1 (2020e) Commercial Building  
Telecommunications Infrastructure Standard

TIA-568.2 (2018d) Balanced Twisted-Pair

|                 |                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | Telecommunications Cabling and Components Standards                                                                                                        |
| TIA-568.3       | (2022e) Optical Fiber Cabling Components Standard                                                                                                          |
| TIA-569         | (2019e; Add 1 2022) Telecommunications Pathways and Spaces                                                                                                 |
| TIA-590         | (1997a) Standard for Physical Location and Protection of Below Ground Fiber Optic Cable Plant                                                              |
| TIA-606         | (2021d) Administration Standard for Telecommunications Infrastructure                                                                                      |
| TIA-607         | (2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises                                                                  |
| TIA-758         | (2012b) Customer-Owned Outside Plant Telecommunications Infrastructure Standard                                                                            |
| TIA/EIA-455     | (1998b) Standard Test Procedure for Fiber Optic Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components |
| TIA/EIA-455-204 | (2000) Standard for Measurement of Bandwidth on Multimode Fiber                                                                                            |
| TIA/EIA-598     | (2014D; Add 2 2018) Optical Fiber Cable Color Coding                                                                                                       |

## U.S. DEPARTMENT OF AGRICULTURE (USDA)

|                    |                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------|
| RUS 1755           | Telecommunications Standards and Specifications for Materials, Equipment and Construction |
| RUS Bull 345-50    | (1979) Trunk Carrier Systems (PE-60)                                                      |
| RUS Bull 345-65    | (1985) Shield Bonding Connectors (PE-65)                                                  |
| RUS Bull 345-72    | (1985) Filled Splice Closures (PE-74)                                                     |
| RUS Bull 345-83    | (1979; Rev Oct 1982) Gas Tube Surge Arrestors (PE-80)                                     |
| RUS Bull 1751F-630 | (1996) Design of Aerial Plant                                                             |
| RUS Bull 1751F-640 | (1995) Design of Buried Plant, Physical Considerations                                    |
| RUS Bull 1751F-643 | (2002) Underground Plant Design                                                           |
| RUS Bull 1751F-815 | (1979) Electrical Protection of Outside Plant                                             |

RUS Bull 1753F-201 (1997) Acceptance Tests of  
Telecommunications Plant (PC-4)

RUS Bull 1753F-401 (1995) Splicing Copper and Fiber Optic  
Cables (PC-2)

#### UNDERWRITERS LABORATORIES (UL)

UL 83 (2017; Reprint Mar 2020) UL Standard for  
Safety Thermoplastic-Insulated Wires and  
Cables

UL 497 (2001; Reprint Jul 2022) UL Standard for  
Safety Protectors for Paired Conductor  
Communication Circuits

UL 510 (2020; Dec 2022) UL Standard for Safety  
Polyvinyl Chloride, Polyethylene and  
Rubber Insulating Tape

## 1.2 RELATED REQUIREMENTS

[ Section 27 10 00, BUILDING TELECOMMUNICATIONS CABLING SYSTEM, ][Section  
33 71 01, OVERHEAD TRANSMISSION AND DISTRIBUTION, and[Section 33 71 02,  
UNDERGROUND ELECTRICAL DISTRIBUTION] apply to this section with additions  
and modifications specified herein.

## ]1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms  
used in this specification shall be as defined in TIA-568.1, TIA-568.2,  
TIA-568.3, TIA-569, TIA-606, and IEEE 100 and herein.

### 1.3.1 Campus Distributor (CD)

A distributor from which the campus backbone cabling emanates.  
(International expression for main cross-connect - (MC).)

### 1.3.2 Entrance Facility (EF) (Telecommunications)

An entrance to the building for both private and public network service  
cables (including antennae) including the entrance point at the building  
wall and continuing to the entrance room or space.

### 1.3.3 Entrance Room (ER) (Telecommunications)

A centralized space for telecommunications equipment that serves the  
occupants of a building. Equipment housed therein is considered distinct  
from a telecommunications room because of the nature of its complexity.

### 1.3.4 Building Distributor (BD)

A distributor in which the building backbone cables terminate and at which  
connections to the campus backbone cables may be made. (International  
expression for intermediate cross-connect - (IC).)

### 1.3.5 Pathway

A physical infrastructure utilized for the placement and routing of

telecommunications cable.

#### 1.4 SYSTEM DESCRIPTION

The telecommunications outside plant consists of cable, conduit, manholes, poles, etc. required to provide signal paths from the closest point of presence to the new facility, including free standing frames or backboards, interconnecting hardware, terminating cables, lightning and surge protection modules at the entrance facility. The work consists of providing, testing and making operational cabling, interconnecting hardware and lightning and surge protection necessary to form a complete outside plant telecommunications system for continuous use.[ The telecommunications contractor must coordinate with the NMCI contractor concerning layout and configuration of the EF telecommunications and OSP. The telecommunications contractor may be required to coordinate work effort for access to the EF telecommunications and OSP with the NMCI contractor.]

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are [for Contractor Quality Control approval.][for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.] Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Telecommunications Outside Plant; G[, [\_\_\_\_]]

Telecommunications Entrance Facility Drawings; G[, [\_\_\_\_]]

In addition to Section 01 33 00 SUBMITTAL PROCEDURES, provide shop drawings in accordance with paragraph SHOP DRAWINGS.

##### SD-03 Product Data

Wire and Cable; G[, [\_\_\_\_]]

Cable Splices, and Connectors; G[, [\_\_\_\_]]

Closures; G[, [\_\_\_\_]]

Building Protector Assemblies; G[, [\_\_\_\_]]

Protector Modules; G[, [\_\_\_\_]]

Cross-Connect Terminal Cabinets; G[, [\_\_\_\_]]

[ Spare Parts; G[, [\_\_\_\_]]

] Submittals shall include the manufacturer's name, trade name, place of manufacture, and catalog model or number. Submittals shall also include applicable federal, military, industry, and technical society publication references. Should manufacturer's data require supplemental information for clarification, the supplemental information shall be submitted as specified in paragraph REGULATORY REQUIREMENTS and as required for certificates



in Section 01 33 00 SUBMITTAL PROCEDURES.

#### SD-06 Test Reports

Pre-installation Tests; G[, [\_\_\_\_]]

Acceptance Tests; G[, [\_\_\_\_]]

Outside Plant Test Plan; G[, [\_\_\_\_]]

#### SD-07 Certificates

Telecommunications Contractor Qualifications; G[, [\_\_\_\_]]

Key Personnel Qualifications; G[, [\_\_\_\_]]

Minimum Manufacturer's Qualifications; G[, [\_\_\_\_]]

#### SD-08 Manufacturer's Instructions

Building Protector Assembly Installation; G[, [\_\_\_\_]]

Cable Tensions; G[, [\_\_\_\_]]

Fiber Optic Splices; G[, [\_\_\_\_]]

Submit instructions prior to installation.

#### SD-09 Manufacturer's Field Reports

Factory Reel Test Data; G[, [\_\_\_\_]]

#### SD-10 Operation and Maintenance Data

Telecommunications Outside Plant (OSP), Data Package 5; G[, [\_\_\_\_]]

Commercial off-the-shelf manuals shall be provided for operation, installation, configuration, and maintenance of products provided as a part of the telecommunications outside plant (OSP). Submit operations and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein not later than[ 2][\_\_\_\_] months prior to the date of beneficial occupancy. In addition to requirements of Data package 5, include the requirements of paragraphs TELECOMMUNICATIONS OUTSIDE PLANT SHOP DRAWINGS and TELECOMMUNICATIONS ENTRANCE FACILITY DRAWINGS.

#### SD-11 Closeout Submittals

Record Documentation; G[, [\_\_\_\_]]

In addition to other requirements, provide in accordance with paragraph RECORD DOCUMENTATION.

### 1.6 QUALITY ASSURANCE

#### 1.6.1 Shop Drawings

Include wiring diagrams and installation details of equipment indicating

proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals shall include the nameplate data, size, and capacity. Submittals shall also include applicable federal, military, industry, and technical society publication references.

#### 1.6.1.1 Telecommunications Outside Plant Shop Drawings

Provide Outside Plant Design in accordance with TIA-758, RUS Bull 1751F-630 for aerial system design, and RUS Bull 1751F-643 for underground system design. Provide T0 shop drawings that show the physical and logical connections from the perspective of an entire campus, such as actual building locations, exterior pathways and campus backbone cabling on plan view drawings, major system nodes, and related connections on the logical system drawings in accordance with TIA-606. Drawings shall include wiring and schematic diagrams for fiber optic and copper cabling and splices, copper conductor gauge and pair count, fiber pair count and type, pathway duct and innerduct arrangement, associated construction materials, and any details required to demonstrate that cable system has been coordinated and will properly support the switching and transmission system identified in specification and drawings.[ Provide Registered Communications Distribution Designer (RCDD) approved drawings of the telecommunications outside plant.][ Update existing telecommunication Outside Plant T0 drawings to include information modified, deleted or added as a result of this installation in accordance with TIA-606.] The telecommunications outside plant (OSP) shop drawings shall be included in the operation and maintenance manuals.

#### 1.6.1.2 Telecommunications Entrance Facility Drawings

[Provide T3 drawings for EF Telecommunications in accordance with TIA-606 that include telecommunications entrance facility plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and[ cabinet][, rack][, backboard][ and] wall elevations. Drawings shall show layout of applicable equipment including incoming cable stub or connector blocks, building protector assembly, outgoing cable connector blocks, patch panels and equipment spaces and cabinet/racks. Drawings shall include a complete list of equipment and material, equipment rack details, proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearance for maintenance and operation. Drawings may also be an enlargement of a congested area of T1 or T2 drawings.][Provide T3 drawings for EF Telecommunications as specified in the paragraph TELECOMMUNICATIONS SPACE DRAWINGS of Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEMS.] The telecommunications entrance facility shop drawings shall be included in the operation and maintenance manuals.

#### 1.6.2 Telecommunications Qualifications

Work under this section shall be performed by and the equipment shall be provided by the approved telecommunications contractor and key personnel. Qualifications shall be provided for: the telecommunications system contractor, the telecommunications system installer, the supervisor (if different from the installer), and the cable splicing and terminating

personnel. A minimum of 30 days prior to installation, submit documentation of the experience of the telecommunications contractor and of the key personnel.

#### 1.6.2.1 Telecommunications Contractor Qualifications

The telecommunications contractor shall be a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor shall demonstrate experience in providing successful telecommunications systems that include outside plant and broadband cabling within the past 3 years. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for the telecommunications contractor. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems in accordance with TIA-758 within the past 3 years.

#### 1.6.2.2 Key Personnel Qualifications

Provide key personnel who are regularly and professionally engaged in the business of the application, installation and testing of the specified telecommunications systems and equipment. There may be one key person or more key persons proposed for this solicitation depending upon how many of the key roles each has successfully provided. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems within the past 3 years.

Cable splicing and terminating personnel assigned to the installation of this system or any of its components shall have training in the proper techniques and have a minimum of 3 years experience in splicing and terminating the specified cables. Modular splices shall be performed by factory certified personnel or under direct supervision of factory trained personnel for products used.

Supervisors and installers assigned to the installation of this system or any of its components shall have factory or factory approved certification from each equipment manufacturer indicating that they are qualified to install and test the provided products.

Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for each of the key personnel. Documentation for each key person shall include at least two successful system installations provided that are equivalent in system size and in construction complexity to the telecommunications system proposed for this solicitation. Include specific experience in installing and testing telecommunications outside plant systems, including broadband cabling, and provide the names and locations of at least two project installations successfully completed using[ optical fiber and] copper telecommunications cabling systems. All of the existing telecommunications system installations offered by the key persons as successful experience shall have been in successful full-time service for at least 18 months prior to the issuance date for this solicitation. Provide the name and role of the key person, the title, location, and completed installation date of the referenced project, the referenced project owner point of contact information including name, organization, title, and telephone number, and generally, the referenced project description including system size and construction complexity.

Indicate that all key persons are currently employed by the telecommunications contractor, or have a commitment to the telecommunications contractor to work on this project. All key persons shall be employed by the telecommunications contractor at the date of issuance of this solicitation, or if not, have a commitment to the telecommunications contractor to work on this project by the date that the bid was due to the Contracting Officer.

Note that only the key personnel approved by the Contracting Officer in the successful proposal shall do work on this solicitation's telecommunications system. Key personnel shall function in the same roles in this contract, as they functioned in the offered successful experience. Any substitutions for the telecommunications contractor's key personnel requires approval from The Contracting Officer.

#### 1.6.2.3 Minimum Manufacturer's Qualifications

Cabling, equipment and hardware manufacturers shall have a minimum of[ 3][\_\_\_\_\_] years experience in the manufacturing, assembly, and factory testing of components which comply with, TIA-568.1, TIA-568.2 and TIA-568.3. In addition, cabling manufacturers shall have a minimum of[ 3][\_\_\_\_\_] years experience in the manufacturing and factory testing of cabling which comply with ICEA S-87-640, ICEA S-98-688, and ICEA S-99-689.

#### 1.6.3 Outside Plant Test Plan

Prepare and provide a complete and detailed test plan for field tests of the outside plant including a complete list of test equipment for the[ copper conductor][ and][ optical fiber] cables, components, and accessories for approval by the Contracting Officer. Include a cut-over plan with procedures and schedules for relocation of facility station numbers without interrupting service to any active location. Submit the plan at least[ 30][\_\_\_\_\_] days prior to tests for Contracting Officer approval. Provide outside plant testing and performance measurement criteria in accordance with TIA-568.1 and RUS Bull 1753F-201. Include procedures for certification, validation, and testing that includes fiber optic link performance criteria.

#### 1.6.4 Standard Products

Provide materials and equipment that are standard products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and shall be the manufacturer's latest standard design that has been in satisfactory commercial or industrial use for at least[ 2][ 1] year[s] prior to bid opening. The[ 2][ 1]-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the[ 2][ 1]-year period. Products supplied shall be specifically designed and manufactured for use with outside plant telecommunications systems. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

##### 1.6.4.1 Alternative Qualifications

Products having less than a[ 2][ 1]-year field service record will be

acceptable if a certified record of satisfactory field operation for not less than[ 6000][ 3000] hours, exclusive of the manufacturers' factory or laboratory tests, is provided.

#### 1.6.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site shall not be used, unless specified otherwise.

#### 1.6.5 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of NFPA 70 unless more stringent requirements are specified or indicated.

##### 1.6.5.1 Independent Testing Organization Certificate

In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate shall state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

#### 1.7 DELIVERY, STORAGE, AND HANDLING

Ship cable on reels in [152][305][\_\_\_\_\_] meter ([500][1000][\_\_\_\_\_] feet) length with a minimum overage of 10 percent. Radius of the reel drum shall not be smaller than the minimum bend radius of the cable. Wind cable on the reel so that unwinding can be done without kinking the cable. Two meters of cable at both ends of the cable shall be accessible for testing. Attach permanent label on each reel showing length, cable identification number, cable size, cable type, and date of manufacture. Provide water resistant label and the indelible writing on the labels. Apply end seals to each end of the cables to prevent moisture from entering the cable. Reels with cable shall be suitable for outside storage conditions when temperature ranges from minus 40 degrees C to plus 65 degrees C, with relative humidity from 0 to 100 percent. Equipment, other than cable, delivered and placed in storage shall be stored with protection from weather, humidity and temperature variation, dirt and dust, or other contaminants in accordance with manufacturer's requirements.

#### 1.8 MAINTENANCE

##### 1.8.1 Record Documentation

Provide the activity responsible for telecommunications system maintenance and administration a single complete and accurate set of record documentation for the entire telecommunications system with respect to this project.

[ Provide T5 drawings including documentation on cables and termination hardware in accordance with TIA-606. T5 drawings shall include schedules to show information for cut-overs and cable plant management, patch panel

layouts, cross-connect information and connecting terminal layout as a minimum. T5 drawings shall be provided[ in hard copy format][ on electronic media using Windows based computer cable management software.][ A licensed copy of the cable management software including documentation, shall be provided.][ Update existing record documentation to reflect campus distribution T0 drawings and T3 drawing schedule information modified, deleted or added as a result of this installation.] Provide the following T5 drawing documentation as a minimum:

- a. Cables - A record of installed cable shall be provided in accordance with TIA-606. The cable records shall [include only the required data fields][include the required data fields for each cable and complete end-to-end circuit report for each complete circuit from the assigned outlet to the entry facility ]in accordance with TIA-606. Include manufacture date of cable with submittal.
- b. Termination Hardware - Provide a record of installed patch panels, cross-connect points, campus distributor and terminating block arrangements and type in accordance with TIA-606. Documentation shall include [only ]the required data fields[ as a minimum] in accordance with TIA-606.

][Provide record documentation as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

#### ][1.8.2 Spare Parts

In addition to the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA, provide a complete list of parts and supplies, with current unit prices and source of supply, and a list of spare parts recommended for stocking. Spare parts shall be provided no later than the start of field testing.

#### ][1.9 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

### PART 2 PRODUCTS

#### 2.1 MATERIALS AND EQUIPMENT

Products supplied shall be specifically designed and manufactured for use with outside plant telecommunications systems.

#### 2.2 TELECOMMUNICATIONS ENTRANCE FACILITY

##### 2.2.1 Building Protector Assemblies

Provide self-contained[ 5 pin][ screw type] unit supplied with a field cable stub factory connected to protector socket blocks to terminate and accept protector modules for [\_\_\_\_\_] pairs of outside cable. Building protector assembly shall have interconnecting hardware for connection to interior cabling at full capacity. Provide manufacturers instructions for building protector assembly installation. Provide copper cable interconnecting hardware as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

### 2.2.2 Protector Modules

Provide in accordance with UL 497[ three][ two]-electrode gas tube or solid state type[ 5 pin][ screw type] rated for the application. Provide gas tube protection modules in accordance with RUS Bull 345-83 and shall be[ heavy duty, A>10kA, B>400, C>65A][ maximum duty, A>20kA, B>1000, C>200A] where A is the maximum single impulse discharge current, B is the impulse life and C is the AC discharge current in accordance with ANSI C62.61. The gas modules shall shunt high voltage to ground, fail short, and be equipped with an external spark gap and heat coils in accordance with UL 497. Provide the number of surge protection modules equal to the number of pairs of exterior cable of the building protector assembly.

### 2.2.3 Fiber Optic Terminations

Provide fiber optic cable terminations as specified in 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

## 2.3 CLOSURES

### 2.3.1 Copper Conductor Closures

#### 2.3.1.1 Aerial Cable Closures

Provide cable closure assembly consisting of a frame with clamps, a lift-off polyethylene cover, cable nozzles, and drop wire rings. Closure shall be suitable for use on Figure 8 cables. Closures shall be free breathing and suitable for housing[ straight-through type][ branch type][ of the type indicated] splices of non-pressurized communications cables and shall be sized as indicated. The closure shall be constructed with ultraviolet resistant PVC.

#### 2.3.1.2 Underground Cable Closures

- a. Aboveground: Provide aboveground closures constructed of[ not less than 14 gauge steel][ ultraviolet resistant PVC] and acceptable for[ pole][ stake] mounting in accordance with RUS 1755.910. Closures shall be sized and contain a marker as indicated. Covers shall be secured to prevent unauthorized entry.
- b. Direct burial: Provide buried closure suitable for enclosing a straight, butt, and branch splice in a container into which can be poured an encapsulating compound. Closure shall have adequate strength to protect the splice and maintain cable shield electrical continuity in the buried environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure. Provide filled splice cases in accordance with RUS Bull 345-72.
- c. In vault or manhole: Provide underground closure suitable to house a straight, butt, and branch splice in a protective housing into which can be poured an encapsulating compound. Closure shall be of suitable thermoplastic, thermoset, or stainless steel material supplying structural strength necessary to pass the mechanical and electrical requirements in a vault or manhole environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure. Provide filled splice cases in accordance

with RUS Bull 345-72.

## 2.3.2 Fiber Optic Closures

### 2.3.2.1 Aerial

Provide aerial closure that is free breathing and suitable for housing splice organizer of non -pressurized cables. Closure shall be constructed from heavy PVC with ultraviolet resistance.

### 2.3.2.2 Direct Burial

Provide buried closure suitable to house splice organizer in protective housing into which can be poured an encapsulating compound. Closure shall have adequate strength to protect the splice and maintain cable shield electrical continuity, when metallic, in buried environment. Encapsulating compound shall be reenterable and shall not alter chemical stability of the closure.

### 2.3.2.3 In Vault or Manhole

Provide underground closure suitable to house splice organizer in a protective housing into which can be poured an encapsulating compound. Closure shall be of thermoplastic, thermoset, or stainless steel material supplying structural strength necessary to pass the mechanical and electrical requirements in a vault or manhole environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure.

## 2.4 PAD MOUNTED CROSS-CONNECT TERMINAL CABINETS

Provide in accordance with RUS 1755.910 and the following:

- a. Constructed of 14 gauge steel or [\_\_\_\_\_].
- b. Equipped with a double set of hinged doors with closed-cell foam weatherstripping. Doors shall be locked and contain a marker as indicated.
- c. Equipped with spool spindle bracket, mounting frames, binding post log,[ and] jumpering instruction label[, and load coil mounting provisions].
- d. Complete with cross connect modules to terminate number of pairs as indicated.
- e. Sized as indicated.

## 2.5 CABLE SPLICES, AND CONNECTORS

### 2.5.1 Copper Cable Splices

Provide[ multipair,[ foldback][ in-line]][ single pair,[ in-line][ butt][ box tap]] splices of a moisture resistant,[ two][ three][\_\_\_\_\_] -wire [ insulation displacement] connector held rigidly in place to assure maximum continuity in accordance with RUS Bull 1753F-401. Cables greater than 25 pairs shall be spliced using multipair splicing connectors, which accommodate 25 pairs of conductors at a time. Provide correct connector size to accommodate the cable gauge of the supplied cable.



### [2.5.2 Copper Cable Splice Connector

Provide splice connectors with a polycarbonate body and cap and a tin-plated brass contact element. Connector shall accommodate 22 to 26 AWG solid wire with a maximum insulation diameter of 1.65 mm. Fill connector with sealant grease to make a moisture resistant connection, in accordance with RUS Bull 1753F-401.

### ][2.5.3 Fiber Optic Cable Splices

Provide fiber optic cable splices and splicing materials for[ fusion][ mechanical] methods at locations shown on the construction drawings. The splice insertion loss shall be 0.3 dB maximum when measured in accordance with TIA-455-78-B using an Optical Time Domain Reflectometer (OTDR). Splices shall be designed for a return loss of 40.0 db max for single mode fiber when tested in accordance with TIA-455-107. Physically protect each fiber optic splice by a splice kit specially designed for the splice.

### ][2.5.4 Fiber Optic Splice Organizer

Provide splice organizer suitable for housing fiber optic splices in a neat and orderly fashion. Splice organizer shall allow for a minimum of 1 m of fiber for each fiber within the cable to be neatly stored without kinks or twists. Splice organizer shall accommodate individual strain relief for each splice and allow for future maintenance or modification, without damage to the cable or splices. Provide splice organizer hardware, such as splice trays, protective glass shelves, and shield bond connectors in a splice organizer kit.

### ]2.5.5 Shield Connectors

Provide connectors with a stable, low-impedance electrical connection between the cable shield and the bonding conductor in accordance with RUS Bull 345-65.

## 2.6 CONDUIT

Provide conduit as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

## 2.7 PLASTIC INSULATING TAPE

UL 510.

## 2.8 WIRE AND CABLE

### 2.8.1 Copper Conductor Cable

Solid copper conductors, covered with an extruded solid insulating compound. Insulated conductors shall be twisted into pairs which are then stranded or oscillated to form a cylindrical core. For special high frequency applications, the cable core shall be separated into compartments. Cable shall be completed by the application of a suitable core wrapping material, a corrugated copper or plastic coated aluminum shield, and an overall extruded jacket. Telecommunications contractor shall verify distances between splice points prior to ordering cable in specific cut lengths. Gauge of conductor shall determine the range of numbers of pairs specified; 19 gauge (6 to 400 pairs), 22 gauge (6 to 1200

pairs), 24 gauge (6 to 2100 pairs), and 26 gauge (6 to 3000 pairs). Copper conductor shall conform to the following:

#### 2.8.1.1 Underground

Provide filled cable meeting the requirements of ICEA S-99-689 and [ RUS 1755.390] [RUS 1755.890].

#### 2.8.1.2 Aerial

Provide filled cable meeting the requirements of[ ICEA S-99-689][ ICEA S-98-688], and RUS 1755.390 except that it shall be suitable for aerial installation and shall be Figure 8 distribution wire with 26,700 N Class A galvanized steel or 26,700 N aluminum-clad steel strand.

#### 2.8.1.3 Screen

Provide screen-compartmental core cable filled cable meeting the requirements of ICEA S-99-689 and RUS 1755.390.

#### 2.8.2 Fiber Optic Cable

Provide[ single-mode, 8/125-um, 0.10 aperture 1310 nm fiber optic cable in accordance with TIA-492CAAA][ single-mode, 8/125-um, 0.10 aperture 1550 nm fiber optic cable in accordance with TIA-492E000][ and ][ multimode 62.5/125-um, 0.275 aperture fiber optic cable in accordance with TIA-492AAAA][ multimode 50/125-um, 0.275 aperture fiber optic cable in accordance with TIA-492AAAB], TIA-472D000, and ICEA S-87-640 including any special requirements made necessary by a specialized design. Provide[ 12][ \_\_\_\_] optical fibers[ as indicated]. Fiber optic cable shall be specifically designed for outside use with loose buffer construction. Provide fiber optic color code in accordance with TIA/EIA-598

#### 2.8.2.1 Strength Members

Provide[ central][ non-central],[ non-metallic][ metallic] strength members with sufficient tensile strength for installation and residual rated loads to meet the applicable performance requirements in accordance with ICEA S-87-640. The strength member is included to serve as a cable core foundation to reduce strain on the fibers, and shall not serve as a pulling strength member.

#### [2.8.2.2 Shielding or Other Metallic Covering

Provide[ copper, copper alloy or copper and steel laminate][ copper and stainless steel, coated stainless steel or bare low carbon steel][ bare aluminum or coated aluminum],[ single][ dual] tape covering or shield in accordance with ICEA S-87-640.

#### ]2.8.2.3 Performance Requirements

Provide fiber optic cable with optical and mechanical performance requirements in accordance with ICEA S-87-640.

#### 2.8.3 Grounding and Bonding Conductors

Provide grounding and bonding conductors in accordance with RUS 1755.200, TIA-607, IEEE C2, and NFPA 70. Solid bare copper wire meeting the requirements of ASTM B1 for sizes No. 8 AWG and smaller and stranded bare

copper wire meeting the requirements of ASTM B8, for sizes No. 6 AWG and larger. Insulated conductors shall have 600-volt, Type TW insulation meeting the requirements of UL 83.

## 2.9 T-SPAN LINE TREATMENT REPEATERS

Provide as indicated. Repeaters shall be pedestal mounted with non-pressurized housings, sized as indicated and shall meet the requirements of RUS Bull 345-50.

## 2.10 POLES AND HARDWARE

Provide poles and hardware as specified in Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION.

## 2.11 CABLE TAGS IN MANHOLES, HANDHOLES, AND VAULTS

Provide tags for each telecommunications cable or wire located in manholes, handholes, and vaults. Cable tags shall be[ stainless steel][ or][ polyethylene] and labeled[ as indicated][ in accordance with TIA-606 ]. Handwritten labeling is unacceptable.

### [2.11.1 Stainless Steel

Provide stainless steel, cable tags 41.25 mm in diameter 1.58 mm thick minimum, and circular in shape. Tags shall be die stamped with numbers, letters, and symbols not less than 6.35 mm high and approximately 0.38 mm deep in normal block style.

### ]2.11.2 Polyethylene Cable Tags

Provide tags of polyethylene that have an average tensile strength of 22.4 MPa; and that are two millimeter thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 77 degrees C. Provide 1.3 mm (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag. Ties shall have a minimum loop tensile strength of 778.75 N. The cable tags shall have black block letters, numbers, and symbols 25 mm high on a yellow background. Letters, numbers, and symbols shall not fall off or change positions regardless of the cable tags' orientation.

## ]2.12 BURIED WARNING AND IDENTIFICATION TAPE

Provide fiber optic media marking and protection in accordance with TIA-590. Provide color, type and depth of tape as specified in paragraph BURIED WARNING AND IDENTIFICATION TAPE in Section 31 00 00, EARTHWORK.

## 2.13 GROUNDING BRAID

Provide grounding braid that provides low electrical impedance connections for dependable shield bonding in accordance with RUS 1755.200. Braid shall be made from flat tin-plated copper.

## 2.14 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be

acceptable.

## 2.15 FIELD FABRICATED NAMEPLATES

Provide laminated plastic nameplates in accordance with ASTM D709 for each patch panel, protector assembly, rack, cabinet and other equipment or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, 3 mm thick, white with [black][ ] center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be 25 by 65 mm. Lettering shall be a minimum of 6.35 mm high normal block style.

## 2.16 TESTS, INSPECTIONS, AND VERIFICATIONS

### 2.16.1 Factory Reel Test Data

Test 100 percent OTDR test of FO media at the factory in accordance with TIA-568.1 and TIA-568.3. Use TIA-526-7 for single mode fiber and TIA-526-14 Method B for multi mode fiber measurements. Calibrate OTDR to show anomalies of 0.2 dB minimum. Enhanced performance filled OSP copper cables, referred to as Broadband Outside Plant (BBOSP), shall meet the requirements of ICEA S-99-689. Enhanced performance air core OSP copper cables shall meet the requirements of ICEA S-98-688. Submit test reports, including manufacture date for each cable reel and receive approval before delivery of cable to the project site.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install all system components and appurtenances in accordance with manufacturer's instructions IEEE C2, NFPA 70, and as indicated. Provide all necessary interconnections, services, and adjustments required for a complete and operable telecommunications system.

#### 3.1.1 Contractor Damage

Promptly repair indicated utility lines or systems damaged during site preparation and construction. Damages to lines or systems not indicated, which are caused by Contractor operations, shall be treated as "Changes" under the terms of the Contract Clauses. When Contractor is advised in writing of the location of a nonindicated line or system, such notice shall provide that portion of the line or system with "indicated" status in determining liability for damages. In every event, immediately notify the Contracting Officer of damage.

#### 3.1.2 Cable Inspection and Repair

Handle cable and wire provided in the construction of this project with care. Inspect cable reels for cuts, nicks or other damage. Damaged cable shall be replaced or repaired to the satisfaction of the Contracting Officer. Reel wraps shall remain intact on the reel until the cable is ready for placement.

#### 3.1.3 Direct Burial System

Installation shall be in accordance with RUS Bull 1751F-640. Under railroad tracks, paved areas, and roadways install cable in conduit

encased in concrete. Slope ducts to drain. Excavate trenches by hand or mechanical trenching equipment. Provide a minimum cable cover of 610 mm below finished grade. Trenches shall be not less than 155 mm wide and in straight lines between cable markers. Do not use cable plows. Bends in trenches shall have a radius of not less than [915][\_\_\_\_\_] mm. Where two or more cables are laid parallel in the same trench, space laterally at least 78 mm apart. When rock is encountered, remove it to a depth of at least 78 mm below the cable and fill the space with sand or clean earth free from particles larger than 6 mm. Do not unreel and pull cables into the trench from one end. Cable may be unreeled on grade and lifted into position. Provide color, type and depth of warning tape as specified in paragraph BURIED WARNING AND IDENTIFICATION TAPE in Section 31 00 00 EARTHWORK.

#### 3.1.3.1 Cable Placement

- a. Separate cables crossing other cables or metal piping from the other cables or pipe by not less than [78][\_\_\_\_\_] mm of well tamped earth. Do not install circuits for communications under or above traffic signal loops.
- b. Cables shall be in one piece without splices between connections except where the distance exceeds the lengths in which the cable is furnished.
- c. Avoid bends in cables of small radii and twists that might cause damage. Do not bend cable and wire in a radius less than 10 times the outside diameter of the cable or wire.
- d. Leave a horizontal slack of approximately 915 mm in the ground on each end of cable runs, on each side of connection boxes, and at points where connections are brought aboveground. Where cable is brought aboveground, leave additional slack to make necessary connections.

#### 3.1.3.2 Identification Slabs [Markers]

Provide a marker at each change of direction of the cable, over the ends of ducts or conduits which are installed under paved areas and roadways and over each splice. Identification markers shall be of concrete, approximately 508 mm square by 155 mm thick.

#### 3.1.3.3 Backfill for Rocky Soil

When placing cable in a trench in rocky soil, the cable shall be cushioned by a fill of sand or selected soil at least 53 mm thick on the floor of the trench before placing the cable or wire. The backfill for at least 103 mm above the wire or cable shall be free from stones, rocks, or other hard or sharp materials which might damage the cable or wire. If the buried cable is placed less than 610 mm in depth[, a protective cover of [metal][ concrete] shall be used].

#### 3.1.4 Cable Protection

Provide direct burial cable protection in accordance with NFPA 70 and as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Galvanized conduits which penetrate concrete (slabs, pavement, and walls) shall be PVC coated and shall extend from the first coupling or fitting outside either side of the concrete minimum of 155 mm per 305 mm burial depth beyond the edge of the surface where cable protection is required;

all conduits shall be sealed on each end. Where additional protection is required, cable may be placed in galvanized iron pipe (GIP) sized on a maximum fill of 40 percent of cross-sectional area, or in concrete encased 103 mm PVC pipe. Conduit may be installed by jacking or trenching. Trenches shall be backfilled with earth and mechanically tamped at 155 mm lift so that the earth is restored to the same density, grade and vegetation as adjacent undisturbed material.

#### 3.1.4.1 Cable End Caps

Cable ends shall be sealed at all times with coated heat shrinkable end caps. Cables ends shall be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps shall remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

#### 3.1.5 Underground Duct

Provide underground duct and connections to existing[ manholes,][ handholes,][ concrete pads,][ and][ existing ducts] as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION with any additional requirements as specified herein.

#### 3.1.6 Reconditioning of Surfaces

Provide reconditioning of surfaces as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

#### 3.1.7 Penetrations

Caulk and seal cable access penetrations in walls, ceilings and other parts of the building. Seal openings around electrical penetrations through fire resistance-rated wall, partitions, floors, or ceilings in accordance with Section 07 84 00 FIRESTOPPING.

#### 3.1.8 Cable Pulling

Test duct lines with a mandrel and swab out to remove foreign material before the pulling of cables. Avoid damage to cables in setting up pulling apparatus or in placing tools or hardware. Do not step on cables when entering or leaving the manhole. Do not place cables in ducts other than those shown without prior written approval of the Contracting Officer. Roll cable reels in the direction indicated by the arrows painted on the reel flanges. Set up cable reels on the same side of the manhole as the conduit section in which the cable is to be placed. Level the reel and bring into proper alignment with the conduit section so that the cable pays off from the top of the reel in a long smooth bend into the duct without twisting. Under no circumstances shall the cable be paid off from the bottom of a reel. Check the equipment set up prior to beginning the cable pulling to avoid an interruption once pulling has started. Use a cable feeder guide of suitable dimensions between cable reel and face of duct to protect cable and guide cable into the duct as it is paid off the reel. As cable is paid off the reel, lubricate and inspect cable for sheath defects. When defects are noticed, stop pulling operations and notify the Contracting Officer to determine required corrective action. Cable pulling shall also be stopped when reel binds or does not pay off freely. Rectify cause of binding before resuming pulling operations.

Provide cable lubricants recommended by the cable manufacturer. Avoid bends in cables of small radii and twists that might cause damage. Do not bend cable and wire in a radius less than 10 times the outside diameter of the cable or wire.

#### 3.1.8.1 Cable Tensions

Obtain from the cable manufacturer and provide to the Contracting Officer, the maximum allowable pulling tension. This tension shall not be exceeded.

#### 3.1.8.2 Pulling Eyes

Equip cables 32 mm in diameter and larger with cable manufacturer's factory installed pulling-in eyes. Provide cables with diameter smaller than 32 mm with heat shrinkable type end caps or seals on cable ends when using cable pulling grips. Rings to prevent grip from slipping shall not be beaten into the cable sheath. Use a swivel of 19 mm links between pulling-in eyes or grips and pulling strand.

#### 3.1.8.3 Installation of Cables in Manholes, Handholes, and Vaults

Do not install cables utilizing the shortest route, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support cables on brackets and cable insulators at a maximum of 1220 mm. In existing manholes, handholes, and vaults where new ducts are to be terminated, or where new cables are to be installed, modify the existing installation of cables, cable supports, and grounding as required with cables arranged and supported as specified for new cables. Identify each cable with corrosion-resistant embossed metal tags.

#### 3.1.9 Aerial Cable Installation

Pole installation shall be as specified in Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION. Where physical obstructions make it necessary to pull distribution wire along the line from a stationary reel, use cable stringing blocks to support wire during placing and tensioning operations. Do not place ladders, cable coils, and other equipment on or against the distribution wire. Wire shall be sagged in accordance with the data shown. Protect cable installed outside of building less than 2.5 meters above finished grade against physical damage.

##### 3.1.9.1 Figure 8 Distribution Wire

Perform spiraling of the wire within 24 hours of the tensioning operation. Perform spiraling operations at alternate poles with the approximate length of the spiral being 4575 mm. Do not remove insulation from support members except at bonding and grounding points and at points where ends of support members are terminated in splicing and dead-end devices. Ground support wire at poles to the pole ground.

##### 3.1.9.2 Suspension Strand

Place suspension strand as indicated. Tension in accordance with the data indicated. When tensioning strand, loosen cable suspension clamps enough to allow free movement of the strand. Place suspension strand on the road side of the pole line. In tangent construction, point the lip of the suspension strand clamp toward the pole. At angles in the line, point the suspension strand clamp lip away from the load. In level construction

place the suspension strand clamp in such a manner that it will hold the strand below the through-bolt. At points where there is an up-pull on the strand, place clamp so that it will support strand above the through-bolt. Make suspension strand electrically continuous throughout its entire length, bond to other bare cables suspension strands and connect to pole ground at each pole.

### 3.1.9.3 Aerial Cable

Keep cable ends sealed at all times using cable end caps. Take cable from reel only as it is placed. During placing operations, do not bend cables in a radius less than 10 times the outside diameter of cable. Place temporary supports sufficiently close together and properly tension the cable where necessary to prevent excessive bending. In those instances where spiraling of cabling is involved, accomplish mounting of enclosures for purposes of loading, splicing, and distribution after the spiraling operation has been completed.

### 3.1.10 Cable Splicing

#### 3.1.10.1 Copper Conductor Splices

Perform splicing in accordance with requirements of RUS Bull 1753F-401 except that direct buried splices and twisted and soldered splices are not allowed. Exception does not apply for pairs assigned for carrier application.

#### 3.1.10.2 Fiber Optic Splices

Fiber optic splicing shall be in accordance with manufacturer's recommendation and shall exhibit an insertion loss[ not greater than 0.2 dB for fusion splices][ not greater than 0.4 db for mechanical splices].

#### 3.1.11 Surge Protection

All cables and conductors, except fiber optic cable, which serve as communication lines through off-premise lines, shall have surge protection installed at each end which meet the requirements of RUS Bull 1751F-815.

### 3.1.12 Grounding

Provide grounding and bonding in accordance with RUS 1755.200, TIA-607, IEEE C2, and NFPA 70. Ground exposed noncurrent carrying metallic parts of telephone equipment, cable sheaths, cable splices, and terminals.

#### 3.1.12.1 Telecommunications Master Ground Bar (TMGB)

The TMGB is the hub of the basic telecommunications grounding system providing a common point of connection for ground from outside cable, CD, and equipment. Establish a TMGB for connection point for cable stub shields to connector blocks and CD protector assemblies as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 3.1.12.2 Incoming Cable Shields

Shields shall not be bonded across the splice to the cable stubs. Ground shields of incoming cables in the EF Telecommunications to the TMGB.



### 3.1.12.3 Campus Distributor Grounding

- a. Protection assemblies: Mount CD protector assemblies directly[ on the telecommunications backboard][ in the telecommunications [rack][cabinet]]. Connect assemblies mounted on each vertical frame with No. 6 AWG copper conductor to provide a low resistance path to TMGB.
- [ b. TMGB connection: Connect TMGB to TGB with copper conductor with a total resistance of less than 0.01 ohms.

### ]3.1.13 Cut-Over

All necessary transfers and cut-overs, shall be accomplished by the telecommunications contractor.

### ]3.2 LABELING

#### 3.2.1 Labels

Provide labeling for new cabling and termination hardware located within the facility in accordance with TIA-606. Handwritten labeling is unacceptable. Stenciled lettering for cable and termination hardware shall be provided using[ thermal ink transfer process][ laser printer][\_\_\_\_\_].

#### 3.2.2 Cable Tag Installation

Install cable tags for each telecommunications cable or wire located in manholes, handholes, and vaults including each splice.[ Tag only new wire and cable provided by this contract.][ Tag new wire and cable provided under this contract and existing wire and cable which are indicated to have splices and terminations provided by this contract.] The labeling of telecommunications cable tag identifiers shall be[ as indicated][ in accordance with TIA-606].[ Tag legend shall be as indicated.] Do not provide handwritten letters. Install cable tags so that they are clearly visible without disturbing any cabling or wiring in the manholes, handholes, and vaults.

#### 3.2.3 Termination Hardware

Label patch panels, distribution panels, connector blocks and protection modules using color coded labels with identifiers in accordance with TIA-606.

### 3.3 FIELD APPLIED PAINTING

[Provide ferrous metallic enclosure finishes in accordance with the following procedures. Ensure that surfaces are dry and clean when the coating is applied. Coat joints and crevices. Prior to assembly, paint surfaces which will be concealed or inaccessible after assembly. Apply primer and finish coat in accordance with the manufacturer's recommendations.][Provide ferrous metallic enclosure finishes as specified in Section 09 90 00 PAINTS AND COATINGS.]

#### [3.3.1 Cleaning

Clean surfaces in accordance with SSPC SP 6/NACE No.3.

## ][3.3.2 Priming

Prime with a two component polyamide epoxy primer which has a bisphenol-A base, a minimum of 60 percent solids by volume, and an ability to build up a minimum dry film thickness on a vertical surface of 0.127 mm. Apply in two coats to a total dry film thickness of 0.127 to 0.2 mm.

## ][3.3.3 Finish Coat

Finish with a two component urethane consisting of saturated polyester polyol resin mixed with aliphatic isocyanate which has a minimum of 50 percent solids by volume. Apply to a minimum dry film thickness of 0.05 to 0.076 mm. Color shall be the manufacturer's standard.

## ]3.4 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

## 3.5 FIELD QUALITY CONTROL

Provide the Contracting Officer[ 10][\_\_\_\_\_] working days notice prior to[ each][\_\_\_\_\_] test. Provide labor, equipment, and incidentals required for testing. Correct defective material and workmanship disclosed as the results of the tests. Furnish a signed copy of the test results to the Contracting Officer within 3 working days after the tests for each segment of construction are completed. Perform testing as construction progresses and do not wait until all construction is complete before starting field tests.

## 3.5.1 Pre-Installation Tests

Perform the following tests on cable at the job site before it is removed from the cable reel. For cables with factory installed pulling eyes, these tests shall be performed at the factory and certified test results shall accompany the cable.

## 3.5.1.1 Cable Capacitance

Perform capacitance tests on at least 10 percent of the pairs within a cable to determine if cable capacitance is within the limits specified.

## 3.5.1.2 Loop Resistance

Perform DC-loop resistance on at least 10 percent of the pairs within a cable to determine if DC-loop resistance is within the manufacturer's calculated resistance.

## 3.5.1.3 Pre-Installation Test Results

Provide results of pre-installation tests to the Contracting Officer at least[ 5][\_\_\_\_\_] working days before installation is to start. Results shall indicate reel number of the cable, manufacturer, size of cable, pairs tested, and recorded readings. When pre-installation tests indicate that cable does not meet specifications, remove cable from the job site.

### 3.5.2 Acceptance Tests

Perform acceptance testing in accordance with RUS Bull 1753F-201 and as further specified in this section. Provide personnel, equipment, instrumentation, and supplies necessary to perform required testing. Notification of any planned testing shall be given to the Contracting Officer at least [ 14 ][ \_\_\_\_ ] days prior to any test unless specified otherwise. Testing shall not proceed until after the Contractor has received written Contracting Officer's approval of the test plans as specified. Test plans shall define the tests required to ensure that the system meets technical, operational, and performance specifications. The test plans shall define milestones for the tests, equipment, personnel, facilities, and supplies required. The test plans shall identify the capabilities and functions to be tested. Provide test reports in booklet form showing all field tests performed, upon completion and testing of the installed system. Measurements shall be tabulated on a pair by pair or strand by strand basis.

#### 3.5.2.1 Copper Conductor Cable

Perform the following acceptance tests in accordance with TIA-758:

- a. Wire map (pin to pin continuity)
- b. Continuity to remote end
- c. Crossed pairs
- d. Reversed pairs
- e. Split pairs
- f. Shorts between two or more conductors

#### 3.5.2.2 Fiber Optic Cable

Test fiber optic cable in accordance with TIA/EIA-455 and as further specified in this section. Two optical tests shall be performed on all optical fibers: Optical Time Domain Reflectometry (OTDR) Test, and Attenuation Test. In addition, a Bandwidth Test shall be performed on all multimode optical fibers. These tests shall be performed on the completed end-to-end spans which include the near-end pre-connectorized single fiber cable assembly, outside plant as specified, and the far-end pre-connectorized single fiber cable assembly.

- a. OTDR Test: The OTDR test shall be used to determine the adequacy of the cable installations by showing any irregularities, such as discontinuities, micro-bendings or improper splices for the cable span under test. Hard copy fiber signature records shall be obtained from the OTDR for each fiber in each span and shall be included in the test results. The OTDR test shall be measured in both directions. A reference length of fiber, [ 20 ][ \_\_\_\_ ] m minimum, used as the delay line shall be placed before the new end connector and after the far end patch panel connectors for inspection of connector signature. Conduct OTDR test and provide calculation or interpretation of results in accordance with TIA-526-7 for single-mode fiber and TIA-526-14 for multimode fiber. Splice losses shall not exceed 0.3 db.
- b. Attenuation Test: End-to-end attenuation measurements shall be made on

all fibers, in both directions, using a[ 850][ 1300][ 1310][ 1550] nanometer light source at one end and the optical power meter on the other end to verify that the cable system attenuation requirements are met in accordance with[ TIA-455-46A for multimode][ and][ TIA-526-7 for single-mode] fiber optic cables. The measurement method shall be in accordance with TIA-455-78-B. Attenuation losses shall not exceed 0.5 db/km at 1310 nm and 1550 nm for single-mode fiber. Attenuation losses shall not exceed 5.0 db/km at 850 nm and 1.5 db/km at 1300 nm for multimode fiber.

- c. Bandwidth Test: The end-to-end bandwidth of all multimode fiber span links shall be measured by the frequency domain method. The bandwidth shall be measured in both directions on all fibers. The bandwidth measurements shall be in accordance with TIA/EIA-455-204.

### 3.5.3 Soil Density Tests

- [ a. Determine soil-density relationships for compaction of backfill material in accordance with ASTM D1557, Method D.
- ][b. Determine soil-density relationships as specified for soil tests in Section 31 00 00 EARTHWORK.
- ] -- End of Section --

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